GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

CITY CIRCLE - JAIPUR

BASIC SCHEDULE OF RATES 2019

(EFFECTIVE FROM 18th June, 2019)

- GENERAL BASIC RATES
- BUILDINGS WORKS
- SANITARY WORKS
- HORTICULTURE WORKS
- MISCELLANEOUS CIRCULARS

AVAILABLE WITH

EXECUTIVE ENGINEER,
PWD CITY DIVISION – II,
JAIPUR

Price : **Rs. 600.00**

PREFACE

The basic schedule of Rate 2019 for building works has been prepared as a revision of BSR-2018 for PWD City Circle, Jaipur. The Chief Engineer (Building) PWD Rajasthan has approved the same vide his letter No. CE(Building)/SEB/EE-1/BSR/2019/D-170 dated 18-06-2019 Standard pattern of nomenclature for various items of building works has been adopted in preparation of the BSR-2019. It contains almost all items required for building works. Various items are based on standard PWD specification and Bureau of Indian Standard Codes as amended/ revised from time to time and should be read accordingly. It has been prepared on the basis of present prevailing market rates in Jaipur.

Due care has been taken while preparing this BSR-2019 yet it is possible that some typical or other errors might have left. Suggestions for improvement and correction if any, in this Basic Schedule of Rate of Building Works for Jaipur city are always welcome.

I would like to convey my sincere thanks to Shri Akhilesh Gupta, Executive Engineer Cum T.A to S.E. PWD City Circle and Mrs. Harshita Soni, AEn, City Circle, Jaipur for their suggestions and devotion in preparing this BSR 2019 for building works of PWD City Circle, Jaipur.

(Sunil Gupta)
Superintending Engineer,
P.W.D. City Circle,
Jaipur

OFFICE OF THE CHIEF ENGINEER, P.W.D., RAJASTHAN, JAIPUR.

No-CE(Building)/SEB/EE-1/BSR/2019/ D- 170 Date- 18 .06.2019

Superintending Engineer,
Public Works Department
Circle- Jaipur

Sub:- Approval of BSR-2019 of building works for PWD

Ref: - Your Office Letter no. SE/ D-545 dated 28-05-2019

Sir,

In context to your referred letter on above cited subject, the draft of BSR-2019 submitted by you along with relevant documents has been considered and approved with certain modifications by BSR Approval Task force on dated 17-06-2019. The approval is granted in exercise of the powers delegated to the undersigned under Clause 32 of the Schedule of Powers and CE&Addl.Secretary, PWD office order No.830 dt. 6-2-2018.

Supdt. Engineer, PWD Circle Jaipur will be responsible for ensuring compliance of meeting minutes (attached herewith) & that the modifications decided in the draft BSR in the Task force meeting are actually incorporated in the building BSR-2019 before printing BSR.

You are requested to publish the approved BSR at the earliest and submit hard as well as soft copy for use & record of this office. It is enjoined upon all the concerned to point out the errors and omissions if any to this office.

The BSR(Buildings)-2019 of Circle - Jaipur will be come in force with immediate effect and this new BSR will not affect any past contracts/ agreements.

Your faithfully

(Anil Nepalia)
Chief Engineer (Bldg.),
PWD, Rajasthan Jaipur.

Encl.: as above

No-CE(Building)/SEB/EE-1/BSR/2019/ D- 170 Date- 18 .06.2019

Copy submitted to the following for information & necessary action please:-

1. PS to Addl. Chief Secretary, PWD, Rajasthan, Jaipur.
2. PS to Secretary, PWD, Rajasthan, Jaipur.
3. PS to Chief Engineer & Addl. Secretary, PWD Rajasthan, Jaipur.
4. Chief Engineer (Bldg)/Road/PMGSY/NH/SS/QC, PWD, Rajasthan, Jaipur.
5. The Managing Director, RSRDCC, Setu Bhawan, Jhalana Doongaree, Jaipur
6. Financial Advisor, PWD, Rajasthan, Jaipur.
7. The Addl. Chief Engineer, PWD Zone-I – Jaipur
8. The Superintending Engineer, PWD, Circle - Jaipur

(L.K. Gupta)
Supdt. Engineer (Building),
PWD Rajasthan, Jaipur

OFFICE OF THE SUPERINTENDING ENGINEER, P.W.D. CITY CIRCLE, JAIPUR

No.:- 102 DATE :20.06.2019

OFFICE ORDER

The Basic Schedule of Rates 2019 (Building Works), P.W.D. City Circle, Jaipur is approved by Chief Engineer (Building) vide his letter No. CE(Building)/SEB/EE-1/BSR/2019/D-170 dated 18.06.2019 under clause 32 of schedule of power.

The Basic Schedule of Rates-2019 (Building Works), P.W.D. City Circle, Jaipur will come in force with immediate effect and this new BSR-2019 will not affect any past contracts/agreements.

Although every care has been taken in preparation of this BSR even then it is enjoined upon all the concerned to please point out the errors and omission if any, observed in this BSR.

(Sunil Gupta)
Superintending Engineer,
P.W.D. City Circle,
Jaipur

No. Date :

Copy Submitted/ forwarded to the following for information and necessary action:-

1. The Chief Engineer cum Additional Secretary, PWD, Rajasthan, Jaipur.
2. The Chief Engineer (NH /Building/PMGSY/SS) PWD Rajasthan, Jaipur.
3. The Managing Director RSRDCC Setu Bhawan Jhalana Doongari Jaipur.
4. The Housing Commissioner Rajasthan Housing Board, Jyoti Nagar, Jaipur.
5. The Chief Engineer (Civil) Rajasthan Rajya Vidhut Prasaran Nigam Ltd. Jaipur.
6. The Administrator, Rajashtan State Agriculture Marketing Board, Pant Bhawan Jaipur
7. The Accountant General, Audit/Inspection Rajasthan Jaipur.
8. The Chief General Manager (P) RIICO, Udyog Bhawan Jaipur.
9. The Commissioner Jaipur Development Authority, Jaipur.
10. The C.E.O. Jaipur Nagar Nigam, Jaipur.
11. The Addl. Chief Engineer, PWD, Zone ____________________(All)
12. The Superintending Engineer, PWD City Circle /Rural Circle Jaipur
13. The TA to SE, PWD City circle Jaipur
14. The Executive Engineer, PWD Dn. City Dn. I/II/III//Const/New Delhi.

(Sunil Gupta)
Superintending Engineer,
P.W.D. City Circle,
Jaipur

INPUTS

CODE	NAME OF MATERIALS	UNIT	Final BSR Rate 2019
	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1	Hire charges of Coaltar Boiler 900 to 1400 litres	Day	788.00
2	Hire charges of Concrete Mixer 0.14 cubic metre	Day	1050.00
3	Hire charges of Diesel Road Roller - 8 to 10 tonne	Day	1575.00
4	Production cost of concrete by batch mix plant.	cum	263.00
5	Hire charges of Diesel Truck - 9 tonne	Day	1260.00
6	Hire charges of Spraying machine including electric charges	Day	210.00
7	Hire charges of Coaltar Sprayer	Day	368.00
8	Hire charges of Barber green, drying, mixing and Asphalt Plant, with accessories, capacity 30/45 tonne	Day	8085.00
9	Pumping charges of concrete including Hire charges of pump, piping work & accessories etc.	Cum	92.00
10	Hire charges of Derrickmonkey rope	Day	578.00
11	Hire charges of Pumpset of capacity 4000 litres/hour.	Day	1103.00
12	Vibrator(Needle type 40mm)	Day	630.00
13	Machine for rubbing of floors **with driver & car**	Day	525.00
14	Front end loader	Day	6930.00
16	Mastic Cooker	Day	578.00
17	Hire and running charges of tipper	Day	1260.00
18	Hire and running charges of loader.	Day	1050.00
19	Hand Grinder For mirror polish	Day	210.00
20	Hydraulic Excavator (3D) with driver and fuel.	Day	7875.00
21	Pin vibrator	Day	525.00
22	Surface Vibrator	Day	525.00
24	Hire and running charges of hydraulic piling rig with power unit etc. including complete accessories and shifting at site.	per day	33600.00
25	Hire and running charges of light crane.	per day	2625.00
26	Hire and running charges of bentonite pump.	per day	5250.00
27	Hire and running charges of vibrating pile driving hammer complete with power unit and accessories .	per day	33600.00
28	Hire and running charges of crane 20 tonne capacity.	per day	5775.00
29	Carriage of concrete by transit mixer.	km/ cum	26.00
30	Generator 250 KVA.	per day	10500.00
33	Paint applicator.	per day	578.00
37	Mobile crane.	per day	5775.00
38	Tractor with ripper attachment.	per day	1155.00
39	Tractor with trolley .	per day	1050.00
40	Air compressor 250 cfm with two leads for pneumatic cutters/ hammers.	day	1575.00
41	Joint cutting machine with 2-3 blades	per day	788.00
42	C.C .batch mix plant.	day	80850.00
43	Road sweeper	day	630.00
45	Slip form paver with sensor.	day	11550.00
46	Water tanker 5000 litr capacity.	day	1050.00
47	Concrete joint cutting machine.	day	1050.00
48	Texturing machine.	day	578.00
49	**Production Cost of concrete for weigh batcher, operator charges & other charges for designing**	**cum**	**50.00**
50	**Water tanker 6 kL capacity**	**hour**	**250.00**

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
CODE	**Labour**	**UNIT**	
100	Bandhani	Day	300.00
101	Bhisti	Day	350.00
102	Blacksmith 1st class	Day	500.00
103	Blacksmith 2nd class	Day	400.00
111	Carpenter 1st class	Day	500.00
112	Carpenter 2nd class	Day	400.00
113	Chowkidar	Day	350.00
114	Beldar	Day	350.00
115	Coolie	Day	300.00
116	Fitter (grade 1)	Day	500.00
117	Assistant Fitter or 2nd class Fitter	Day	400.00
119	Glazier	Day	500.00
122	Mason (for plaster of paris work) 1st class	Day	600.00
123	Mason (brick layer) 1st class	Day	500.00
124	Mason (brick layer) 2nd class	Day	400.00
125	Mason (for plain stone work) 2nd class)	Day	500.00
126	Mason (for ornamental stone work) 1st class	Day	800.00
127	Driver (for Road Roller, Concrete Mixer, Truck etc.)	Day	600.00
128	Mate	Day	500.00
130	Mistry	Day	500.00
131	Painter	Day	450.00
132	Rock Excavator	Day	800.00
133	Rock Breaker	Day	800.00
134	Rock Hole Driller	Day	800.00
135	Stone Chiseller	Day	500.00
138	Sprayer (for bitumen, tar etc.)	Day	500.00
139	Skilled Beldar (for floor rubbing etc.)	Day	400.00
141	White Washer	Day	400.00
155	Mason (average)	Day	500.00
156	Carpenter (average)	Day	500.00
157	Operator (Pile/ Special machine)	Day	800.00
159	Skilled torch operator for laying tack	Day	500.00
160	**Sweeper**	**Day**	500.00
161	**Helper**	**Day**	350.00
162	**Dhobi**	**Day**	450.00
163	**Tailor/Upholsterer**	**Day**	400.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
CODE	**NAME OF MATERIALS**	**UNIT**	
222	Seam bolts and nuts 6 mm dia and 25 mm long	10 Nos	53.00
223	Non - Asbestos fibre cement corrugated sheet 6mm thick.	sqm	236.00
224	Non - Asbestos fibre cement close fitting adjustable ridge.	metre	284.00
225	Non - Asbestos fibre cement corrugate serrated adjustable ridge.	metre	242.00
226	Non - Asbestos fibre cement plain wing adjustable ridge.	metre	263.00
227	Non - Asbestos fibre cement unserrated adjustable ridge for hips.	metre	273.00
228	Non - Asbestos fibre cement corrugated appron piece.	metre	210.00
229	Non - Asbestos fibre cement eaves filler piece.	each	137.00
230	Non - Asbestos fibre cement north light curves.	metre	273.00
231	Non - Asbestos fibre cement ventilator curves.	each	347.00
232	Non - Asbestos fibre cement barge boards 6 mm thick.	metre	294.00
233	Non - Asbestos fibre cement ridge finial .	pair	131.00
234	Non - Asbestos fibre cement special north light curves.	each	368.00
235	Non - Asbestos fibre cement S type louvers.	each	221.00
236	Non - Asbestos multi purpose fibre cement board 6mm thick.	sqm	252.00
237	Non - Asbestos multi purpose fibre cement board 8mm thick.	sqm	315.00
285	Brick Aggregate (Single size) : 63 mm nominal size	cum	441.00
286	Brick Aggregate (Single size) : 50 mm nominal size	cum	478.00
287	Brick Aggregate (Single size) : 40 mm nominal size	cum	504.00
291	Stone Aggregate (Single size) : 63 mm nominal size	cum	788.00
292	Stone Aggregate (Single size) : 50 mm nominal size	cum	840.00
293	Stone Aggregate (Single size) : 40 mm nominal size	cum	893.00
294	Stone Aggregate (Single size) : 25 mm nominal size	cum	893.00
295	Stone Aggregate (Single size) : 20 mm nominal size	cum	945.00
296	Stone Aggregate (Single size) : 12.5 mm nominal size	cum	998.00
297	Stone Aggregate (Single size) : 10 mm nominal size	cum	998.00
298	Stone Aggregate (Single size) : 06 mm nominal size	cum	945.00
302	Safeda ballies 125 mm diameter	metre	53.00
304	Bajri	cum	893.00
305	Bamboo 25 mm dia 2.5 metre long	score	**450.00**
308	Bhusa	quintal	315.00
309	Paving bitumen S-90 of approved quality	tonne	50400.00
310	Bitumen emulsion	tonne	44100.00
312	Bitumen grade PMB - 40	M.T.	50400.00
313	Blown type petroleum bitumen of penetration 85/25 of approved quality	tonne	50400.00
314	Bitumen hot sealing compound : grade A	kilogram	29.00
316	Bitumen solution primer of approved quality	litre	68.00
317	Premoulded joint filler 12 mm thick	sqm	473.00
318	Bitumen felt fibre base (vegetable or animal):Type 2 grade 1	sqm	105.00
322	Bitumen felt :Type 3 grade 1	sqm	105.00
324	Coal Tar	litre	63.00
325	Blasting powder	kilogram	37.00
326	Blasting fuse (fuse wire)	each	16.00
328	White face insulating board:12 mm thick	sqm	368.00
332	Natural colour insulating board:12 mm thick	sqm	273.00
336	Flame retardent face insulating board: 12 mm thick	sqm	420.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
338	Flame retardent face insulating, Impregnated fibre board 12 mm thick	sqm	525.00
339	Flame retardent face insulating, Impregnated fibre board 18 mm thick	sqm	739.00
340	Flame retardent face insulating, Impregnated fibre board 25 mm thick	sqm	1016.00
341	Flat pressed 3 layer particle board (medium density) Grade I :12 mm thick	sqm	315.00
346	Extra for veneered particle board with : Teak veneering on one side and commercial veneered on other side	sqm	231.00
347	Extra for veneered particle board with : Commercial veneering on both sides	sqm	131.00
348	Extra for veneered particle board with : Teak veneering on both sides	sqm	473.00
349	SILFIL (Armour Bound) 20 mm thick	sqm	247.00
350	SILFIL (Armour Bound) 25 mm thick	sqm	310.00
351	SILFIL (Armour Bound) 30 mm thick	sqm	368.00
352	SILFIL (Armour Bound) 40 mm thick	sqm	494.00
353	Double side tape for fixing SILFIL (Density not less 30 kg./cum.)	sqm	63.00
362	Brick bats	cum	420.00
364	Wire brush	each	26.00
365	Soft brush	each	23.00
367	Portland Cement	tonne	5460.00
368	White Cement	tonne	15750.00
369	Shree binder	tonne	368.00
370	Coal (steam)	quintal	3150.00
373	Cramp Gun metal 25x6x300 mm	each	68.00
378	Brass butt hinges (light/ordinary type) : 125x70x4 mm	10 Nos	609.00
379	Brass butt hinges (light/ordinary type) : 100x70x4 mm	10 Nos	410.00
380	Brass butt hinges (light/ordinary type) : 75x40x2.5 mm	10 Nos	221.00
381	Brass butt hinges (light/ordinary type) : 50x40x2.5 mm	10 Nos	121.00
382	Brass butt hinges (heavy type) : 125x85x5.5 mm(.70)kg	10 Nos	1943.00
383	Brass butt hinges (heavy type) : 100x85x5.5 mm(.56)kg	10 Nos	1733.00
384	Brass butt hinges (heavy type) :75x65x4.0 mm(.20)kg	10 Nos	630.00
385	Brass parliamentary hinges 150x125x27x5 mm	10 Nos	2100.00
386	Brass parliamentary hinges 125x125x27x5 mm	10 Nos	1890.00
387	Brass parliamentary hinges 100x125x27x5 mm	10 Nos	1575.00
388	Brass parliamentary hinges75x100x20x3.2 mm	10 Nos	1260.00
389	Brass single acting spring hinges 150 mm	each	263.00
390	Brass single acting spring hinges 125 mm	each	210.00
391	Brass single acting spring hinges 100 mm	each	147.00
392	Brass double acting spring hinges 150 mm	each	431.00
393	Brass double acting spring hinges 125 mm	each	315.00
394	Brass double acting spring hinges 100 mm	each	252.00
400	Brass tower bolt (barrel type) 250x10 mm	each	168.00
401	Brass tower bolt (barrel type) 200x10 mm	each	137.00
402	Brass tower bolt (barrel type) 150x10 mm	each	105.00
403	Brass tower bolt (barrel type) 100x10 mm	each	74.00
404	Brass flush bolt 250 mm	each	116.00
405	Brass flush bolt 150 mm	each	95.00
406	Brass flush bolt 100 mm	each	63.00
408	Brass handles 125 mm with plate 175x32 mm	each	126.00
409	Brass handles 100 mm with plate 150x32 mm	each	116.00
410	Brass handles75 mm with plate 125x32 mm	each	84.00
411	Brass door latch 300x16x5 mm (0.380 kg)	each	116.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
412	Brass door latch 250x16x5 mm (0.350 kg)	each	110.00
413	Brass mortice latch and lock 100x65 mm with6 levers and a pair of brass lever handles	each	578.00
414	Brass mortice latch 100x65mm with a pair of brass lever handles	each	683.00
417	Brass 150 mm floor door stopper (0.357kg)	each	147.00
418	Brass hard drawn hooks and eyes 300 mm	10 Nos	609.00
419	Brass hard drawn hooks and eyes 250 mm	10 Nos	557.00
420	Brass hard drawn hooks and eyes 200 mm	10 Nos	525.00
421	Brass hard drawn hooks and eyes 150 mm	10 Nos	494.00
422	Brass hard drawn hooks and eyes 100 mm	10 Nos	420.00
423	Brass casement window fastener	Each	37.00
424	Brass casement stays (straight peg type) 300 mm weighing not less than 0.33 kg	each	89.00
425	Brass casement stays (straight peg type) 250 mm weighing not less than 0.28 kg	Each	74.00
426	Brass casement stays (straight peg type) 200 mmweighing not less than 0.24 kg	each	63.00
427	Brass quadrant stays 300 mm	each	105.00
428	Brass fanlight catch	10 Nos	158.00
429	Brass fanlight pivot	10 Nos	210.00
430	Brass chain with hook for fan light catch	each	21.00
431	Brass hasps and staples (safety type) 300 mm	10 Nos	630.00
432	Brass hasps and staples (safety type) 115 mm	10 Nos	525.00
433	Brass hasps and staples (safety type)90 mm	10 Nos	420.00
438	Brass Night latch	each	462.00
442	Brass helical spring 150 mm	each	252.00
444	Brass curtain rod 20 mm dia 1.25 mm thick	metre	86.00
445	Brass curtain rod 25 mm dia 1.25 mm thick	metre	100.00
446	Brass brackets (curtain rods) 20 mm	each	37.00
447	Brass cupboard knob or ward robe knob 50 mm	each	29.00
449	Brass screws 50 mm	100 Nos	368.00
450	Brass screws 40 mm	100 Nos	273.00
451	Brass screws 30 mm	100 Nos	210.00
452	Brass screws 25 mm	100 Nos	179.00
453	Brass screws 20 mm	100 Nos	97.00
524	Chromium plated Brass butt hinges (lheavy) type 75x65x4 .0 mm (200gms)	10 Nos	651.00
525	Chromium plated Brass butt hinges (light/ordinary) type 125x70x4 mm	10 Nos	625.00
526	Chromium plated Brass butt hinges (light/ordinary) type 100x70x4 mm	10 Nos	557.00
527	Chromium plated Brass butt hinges (light/ordinary) type 75x40x2.5 mm	10 Nos	205.00
528	Chromium plated Brass butt hinges (light/ordinary) type 50x40x2.5 mm	10 Nos	147.00
555	Chromium plated Brass handles 125 mm with plate 175 x32 mm	Each	131.00
556	Chromium plated Brass handles 100 mm with plate 150 x 32 mm	Each	105.00
557	Chromium plated Brass handles 75mm with plate 125x32 mm	Each	95.00
558	Chromium plated Brass mortice latch and lock 100x65 mm with6 levers and a pair of brass lever handles	each	467.00
568	Chromium plated brass casement window fastner	each	79.00
569	Chromium plated Brass casement stays (straight peg type) 300 mmweighing not less than 0.33 kg	each	105.00
570	Chromium plated Brass casement stays (straight peg type) 250 mmweighing not less than 0.28 kg	each	95.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
571	Chromium plated Brass casement stays (straight peg type) 200 mmweighing not less than 0.24 kg	each	79.00
583	Chromium plated Brass Night latch	each	599.00
584	Chromium plated Brass Wardrobe Knobe 50 mm	each	525.00
585	Chromium plated Brass screws 50 mm	100 Nos	389.00
586	Chromium plated Brass screws 40 mm	100 Nos	294.00
587	Chromium plated Brass screws 30 mm	100 Nos	231.00
588	Chromium plated Brass screws 25 mm	100 Nos	189.00
589	Chromium plated Brass screws 20 mm	100 Nos	116.00
590	Chromium plated Brass curtain rod 12 mm dia 1.25mm thick	metre	189.00
591	Chromium plated Brass curtain rod 20 mm dia 1.25mm thick	metre	263.00
592	Chromium plated Brass curtain rod 25 mm dia 1.25mm thick	metre	420.00
594	Bright finished or black enamelled mild steel butt hinges 125x65x2.12 mm	10 Nos	131.00
595	Bright finished or black enamelled mild steel butt hinges 100x58x1.90 mm	10 Nos	105.00
596	Bright finished or black enamelled mild steel butt hinges75x47x1.70 mm	10 Nos	74.00
597	Bright finished or black enamelled mild steel butt hinges50x37x1.50 mm	10 Nos	53.00
608	Nickel plated mild steel piono hinges 1 mm thick 35 mm wide	metre	32.00
635	Bright finished or black enamelled mild steel screws 50 mm	100 Nos	68.00
637	Bright finished or black enamelled mild steel screws 40 mm	100 Nos	35.00
638	Bright finished or black enamelled mild steel screws 30 mm	100 Nos	42.00
639	Bright finished or black enamelled mild steel screws 25 mm	100 Nos	32.00
640	Bright finished or black enamelled mild steel screws 20 mm	100 Nos	17.00
641	Bright finished or black enamelled mild steel bolts and nuts 50x6 mm	each	6.00
642	Oxidised mild steel butt hinges 125x65x2.12 mm	10 Nos	126.00
643	Oxidised mild steel butt hinges 100x58x1.90 mm	10 Nos	95.00
644	Oxidised mild steel butt hinges75x47x1.70 mm	10 Nos	63.00
645	Oxidised mild steel butt hinges50x37x1.50 mm	10 Nos	47.00
646	Oxidised mild steel parliamentary hinges150x125x27x2.8 mm	10 Nos	368.00
647	Oxidised mild steel parliamentary hinges 125x125x27x2.8 mm	10 Nos	294.00
648	Oxidised mild steel parliamentary hinges 100x125x27x2.8 mm	10 Nos	231.00
649	Oxidised mild steel parliamentary hinges 75x100x20x2.24 mm	10 Nos	179.00
650	Oxidised mild steel single acting spring hinges 150 mm	each	95.00
651	Oxidised mild steel single acting spring hinges 125 mm	each	86.00
652	Oxidised mild steel single acting spring hinges 100 mm	each	74.00
653	Oxidised mild steel double acting spring hinges 150 mm	each	97.00
654	Oxidised mild steel double acting spring hinges 125 mm	each	89.00
655	Oxidised mild steel double acting spring hinges 100 mm	each	76.00
656	Nickel plated mild steel piono hinges 1 mm thick 35 mm wide	metre	32.00
660	Oxidised mild steel sliding door bolt 300x16 mm	each	68.00
661	Oxidised mild steel sliding door bolt 250x16 mm	each	53.00
662	Oxidised mild steel door latch 300x20x6 mm	each	37.00
663	Oxidised mild steel door latch 250x20x6 mm	each	32.00
664	Oxidised mild steel tower bolt (barrel type) 250x10 mm	each	32.00
665	Oxidised mild steel tower bolt (barrel type) 200x10 mm	each	26.00
666	Oxidised mild steel tower bolt (barrel type) 150x10 mm	each	23.00
667	Oxidised mild steel tower bolt (barrel type) 100x10 mm	each	19.00
668	Oxidised mild steel handles 125 mm	each	11.00
669	Oxidised mild steel handles 100 mm	each	8.00
670	Oxidised mild steel handles75 mm	each	6.00
679	Oxidised mild steel hasps and staples(safety type) 150 mm	10 Nos	105.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
680	Oxidised mild steel hasps and staples(safety type) 115 mm	10 Nos	84.00
681	Oxidised mild steel hasps and staples(safety type)90 mm	10 Nos	63.00
682	Oxidised mild steel screws 50 mm	100 Nos	50.00
683	Oxidised mild steel screws 40 mm	100 Nos	40.00
684	Oxidised mild steel screws 30 mm	100 Nos	30.00
685	Oxidised mild steel screws 25 mm	100 Nos	19.00
686	Oxidised mild steel screws 20 mm	100 Nos	17.00
687	Anodised Aluminium butt hinges 125x75x4 mm	10 Nos	609.00
688	Anodised Aluminium butt hinges 125x63x4 mm	10 Nos	578.00
689	Anodised Aluminium butt hinges 100x75x4 mm	10 Nos	551.00
690	Anodised Aluminium butt hinges 100x63x3.2 mm	10 Nos	494.00
691	Anodised Aluminium butt hinges 100x63x4 mm	10 Nos	504.00
692	Anodised Aluminium butt hinges 75x63x4 mm	10 Nos	357.00
693	Anodised Aluminium butt hinges 75x63x3.2 mm	10 Nos	315.00
694	Anodised Aluminium butt hinges 75x45x3.2 mm	10 Nos	168.00
696	Anodised Aluminium sliding door bolt 300x16 mm	each	173.00
697	Anodised Aluminium sliding door bolt 250x16 mm	each	137.00
698	Anodised Aluminium tower bolt (barrel type)300x10 mm	10 Nos	683.00
699	Anodised Aluminium tower bolt (barrel type)250x10 mm	10 Nos	578.00
700	Anodised Aluminium tower bolt (barrel type)200x10 mm	10 Nos	473.00
701	Anodised Aluminium tower bolt (barrel type)150x10 mm	10 Nos	368.00
702	Anodised Aluminium tower bolt (barrel type)100x10 mm	10 Nos	273.00
703	Anodised Aluminium handles 125 mm with plate 175 x 32 mm	10 Nos	473.00
704	Anodised Aluminium handles 100 mm with plate 150 x 32 mm	10 Nos	368.00
705	Anodised Aluminium handles 75mm with plate 125 x 32 mm	10 Nos	263.00
706	Anodised Aluminium kicking plate 50 cm long100x3.15 mm	each	116.00
713	Block board construction flush door with teak wood ply on both faces 35 mm thick	sqm	1523.00
714	Block board construction flush door with teak wood ply on both faces 30 mm thick	sqm	1470.00
715	Block board construction flush door with teak wood ply on both faces 25 mm thick	sqm	1418.00
717	Block board construction flush door with commercial ply on both faces 35 mm thick	sqm	1103.00
718	Block board construction flush door with commercial ply on both faces 30 mm thick	sqm	1050.00
719	Block board construction flush door with commercial ply on both faces 25 mm thick	sqm	998.00
720	Block board construction flush door with teak wood ply on one faces 35 mm thick	sqm	1313.00
721	Block board construction flush door with teak wood ply onone faces 30 mm thick	sqm	1155.00
722	Block board construction flush door with teak wood ply on one faces 25 mm thick	sqm	1103.00
752	Block board construction flush door lipping	sqm of door area	137.00
753	Square vision panel in Block board construction flush door	sqm of door area	105.00
754	Circular vision panel in Block board construction flush door	Sqm of doo	158.00
755	Decorative type Louvers in Block board construction flush door	sqm of door area	368.00
757	Rebate cutting in Block board construction flush door	sqm of door area	74.00
759	Decorative plywood 4 mm	sqm	420.00
761	Fuel wood	quintal	420.00
763	Glue	kilogram	263.00
765	Hessian cloth	sqm	17.00
768	Cement Concrete Jali 50 mm thick	sqm	226.00
769	Cement Concrete Jali 40 mm thick	sqm	194.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
770	Cement Concrete Jali 30 mm thick	sqm	168.00
771	Kerosene oil	litre	37.00
772	Synthetic colour	Kg	116.00
773	Unslaked lime	quintal	315.00
774	Khameera	Kg	84.00
775	Dehradun white lime	quintal	1575.00
776	Satna lime	quintal	1260.00
777	Dry hydrated lime (factory made)	quintal	1260.00
784	Marble dust/ powder	cum	735.00
785	Marble chips upto 4mm and downsize White & black	quintal	168.00
788	Marble chips large size above 4 mm White & black	quintal	168.00
789	Jhingra	cum	53.00
790	MS hold fast	Nos	26.00
810	Moorum	cum	525.00
811	Mud (dry)	cum	105.00
815	Dry distemper	kilogram	37.00
816	Oil bound washable distemper/ Acrylic distemper	kilogram	53.00
818	Linseed oil (double boiled)	litre	158.00
820	Cement primer	litre	47.00
821	Distemper primer	litre	84.00
823	Pink primer (for wood)	litre	116.00
826	Aluminium paint	litre	273.00
827	Acid proof paint (chocolate or black)	litre	147.00
828	Anticorrosive bituminous paint (black)	litre	105.00
829	Black japan	litre	158.00
830	Enamel paint	litre	263.00
831	Floor enamel paint in all shades except green	litre	231.00
833	Synthetic enamel paint in black or chocolate shade	litre	236.00
834	Synthetic enamel paint in all shades except black or chocolate shade	litre	252.00
835	Plastic emulsion paint	litre	221.00
845	Roofing paint for iron sheets in red colour	litre	158.00
850	White lead	kilogram	84.00
851	Water proofing cement paint	kilogram	53.00
855	Wax polish (ready made)	kilogram	305.00
856	Ordinary varnish	litre	74.00
857	Superior copal varnish	litre	126.00
858	Superior spar varnish	litre	263.00
859	Oil type wood preservative	litre	74.00
863	Putty for wood work	kilogram	53.00
865	Pig lead	kilogram	121.00
868	Premixed super white gypsum plaster.	kg	11.00
869	Plaster of Paris	kilogram	7.00
870	Plug	each	6.00
871	White cement based putty	kilogram	23.00
873	Copper pins 6 mm dia 7.5 cm long	each	11.00
874	Black colour dark shade pigment	kilogram	63.00
875	Red, chocolate, orange, buff or yellow (red oxide of iron) light shade pigment	kilogram	53.00
876	Green or blue medium shade pigment	kilogram	53.00
886	Standard holdar bat clamps for sand cast iron or cast iron pipes 150 mm dia	each	26.00
887	Gur	Kg	37.00
888	Methi	Kg	42.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
889	Hemp	Kg	28.00
890	Water for wetting of bedding sand	kl	47.00
966	Sand Cast iron plain shoe 150 mm dia	each	797.00
967	Copper plate	kilogram	273.00
969	Pully 25 mm dia	each	21.00
973	Rolling shutter made of 80x1.25 mm machine rolled laths	sqm	1412.00
974	Top cover for rolling shutters	metre	630.00
975	27.5 cm long wire spring grade no 2 for rolling shutters	each	223.00
976	Ball bearing for rolling shutters	each	394.00
977	Extra for mechanical devices chain and cranked operation for operatingrolling shutters : exceeding 10.00 sq.m and upto 16.80 sq.m area of door	sqm	788.00
978	Extra for mechanical devices chain and cranked operation for operating rolling shutters : exceeding 16.80 sq.m area of door	sqm	840.00
979	Royalty for good earth	cum	26.00
980	Royalty for sludge	cum	95.00
982	Coarse sand (zone III)	cum	1155.00
983	Fine sand (zone IV)	cum	1155.00
992	Galvanised steel plain sheets	quintal	6300.00
994	Standard quality hard board sheet 3 mm thick	sqm	263.00
996	Standard quality hard board sheet 4.5 mm thick	sqm	315.00
999	Shellac	kilogram	945.00
1000	Spirit	litre	58.00
1001	Spunyarn	kilogram	89.00
1002	Mild steel round bar 12 mm dia and below	quintal	4095.00
1003	Mild steel round bar above 12 mm dia	quintal	4095.00
1004	Average rate of Mild steel round bars for reinforcements	quintal	4095.00
1005	Twisted steel / deformed bars	quintal	4725.00
1006	Mild steel square bars	quintal	4095.00
1007	Structurals such as tees,angles channels and R.S. joists	quintal	4410.00
1008	Flats upto 10 mm in thickness	quintal	4095.00
1009	Flats exceeding 10 mm in thickness	quintal	4200.00
1010	Mild steel plates	quintal	4410.00
1013	Mild steel sheets for tanks	quintal	4410.00
1015	Mild steel expanded metal 20x60 mm strands	sqm	273.00
1016	TMT bar	quintal	4725.00
1019	Mild steel hooks	each	29.00
1020	Mild steel rivets	quintal	4725.00
1021	Hard drawn steel wire fabric	sqm	446.00
1022	Galvanised steel bolts & nuts 6 mm dia and 25 mmlong round head with slots	10 Nos	21.00
1023	Galvanised steel J or L hooks 8 mm dia	10 Nos	126.00
1024	Galvanised steel bolts & nuts 10 mm dia and 125 mm long round head with slots	each	11.00
1025	Mild stel bolts 6 mm dia and 25 mm long with hexagonal head	10 Nos	23.00
1026	Black steel barbed wire	quintal	6300.00
1028	Straining bolts	each	63.00
1029	Galvanised steel barbed wire	quintal	7350.00
1030	Galvanised steel turn buckles	each	13.00
1031	Galvanised steel bolts & nuts 10 mm dia and 27 cm long both sides threaded with 4 galvanised steel nuts	each	16.00
1032	Galvanised steel bolts 10 mm dia and 7 cm long with nuts	each	11.00
1034	Bolts and nuts upto 300 mm in length	quintal	5775.00
1035	Bolts and nuts above 300 mm in length	quintal	6300.00
1036	Iron pintels including welded pin	each	37.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1037	Binding wire	Kg	68.00
1038	Wire mesh (rabbit)	Sqm	194.00
1143	Steel beading	metre	23.00
1145	Aluminium Plain Strip edging 38x12x3 mm	metre	105.00
1148	Glass strip 3 mm thick 20 mm deep	metre	9.00
1149	Glass strip 4 mm thick40 mm deep	metre	15.00
1151	Boundry stone top chisel dressed 15x15x90 cm	each	74.00
1154	Through and bond stone	100 Nos	1155.00
1157	Stone for masonary work	cum	709.00
1158	Stone for pitching 15 cm x 22.5 cm	cum	446.00
1159	Stone dust	cum	473.00
1160	Red sand stone block	10 cudm	63.00
1161	White/Pink sand stone block	10 cudm	74.00
1163	White sand stone slab 75 mm thick (un-dressed)	sqm	347.00
1164	Red sand stone slab 40 mmthick (un-dressed)	sqm	630.00
1165	White sand stone slab 40 mm thick (un-dressed)	sqm	683.00
1166	Red sand stone slab 30 mmthick (un-dressed)	sqm	305.00
1167	Stone chajja patti	sqm	630.00
1168	Kota stone slab 20 mm to 25 mm thick (semi-polished)	sqm	284.00
1169	Kotastone slab 40mm thick (rough chisled)	sqm	226.00
1170	Katla stone pati 80 mm thick	sqm	1523.00
1171	Katla stone pati 50 mm thick	sqm	1470.00
1172	Stone slab	sqm	394.00
1173	Kota stone slab minimum 30mm thick (semi-polished)	sqm	394.00
1174	Red sand stone slab 45 mm and 50 mm thick (un-dressed)	sqm	683.00
1175	White sand stone slab 45 mm and 50 mm thick (un-dressed)	sqm	683.00
1177	Stone grit 6 mm and down size or pea sized gravel	cum	893.00
1179	Crushed stone 2.36 mm to 12.5 mm size	cum	893.00
1182	Surkhi	cum	473.00
1183	Shesham wood	10 cudm	714.00
1184	Nigeria wood	10 cudm	814.00
1185	Ghana wood	10 cudm	814.00
1186	Superior class teak wood such as Dandeli, Balarshah or Malabar in planks	10 cudm	1260.00
1187	First class teak wood in scantling	10 cudm	998.00
1188	First class teak wood in planks	10 cudm	1050.00
1189	Second class teak wood in scantling	10 cudm	630.00
1190	Second class teak wood in planks	10 cudm	683.00
1191	Ivory coast	10 cudm	987.00
1192	Chir wood	10 cudm	341.00
1194	Second class deodar wood in planks	10 cudm	672.00
1196	First class kail wood in planks	10 cudm	357.00
1197	Second class kail wood in scantling	10 cudm	263.00
1198	Second class kail wood in planks	10 cudm	284.00
1199	Sal wood in scantling	10 cudm	452.00
1201	Precast terrazo tiles 22 mm thick (light shade)	sqm	143.00
1202	Precast terrazo tiles 22 mm thick(medium shade)	sqm	134.00
1203	Precast terrazo tiles 22 mm thick (dark shade)	sqm	126.00
1204	Precast terrazo tiles 20 mm thick for roof	sqm	150.00
1207	G.I. Limpet washer	100 Nos	26.00
1208	Bitumen washer	100 Nos	23.00
1209	G.I. plain washer thick	100 Nos	35.00
1210	G.I. plain washer thin	100 Nos	26.00
1211	G.I. plain washer for seam bolts	100 Nos	26.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1213	Water proofing materials	kilogram	32.00
1214	Welding by gas plant	cm	2.00
1215	Welding by electric plant	cm	2.00
1216	Whiting	quintal	735.00
1218	Spigot for standard jointing	kilogram	53.00
1219	Wire nails	kilogram	58.00
1220	Wire mesh (rabbit)	sqm	194.00
1221	20 mm dia holding down bolts	quintal	5250.00
1222	Mild steel sheets with bolts and nuts to rest on pintels	each	95.00
1224	Hard drawn steel wire	quintal	5250.00
1225	Mild steel flat strap fitting	quintal	5040.00
1227	Chequered terrazo tiles 22 mm thick(light shade)	sqm	179.00
1228	Chequered terrazo tiles 22 mm thick(medium shade)	sqm	168.00
1229	Chequered terrazo tiles 22 mm thick (dark shade)	sqm	168.00
1231	Extra for selected planks of second class teakwood	10 cudm	105.00
1234	Aluminium Plain Strip edging 57x12x3 mm	metre	147.00
1235	Diesel oil	litre	58.00
1237	Cutting marble or sand stone slab upto 50 mm thick by mechanical device	metre	11.00
1238	Extra for selected planks of first class teakwood	10 cudm	158.00
1241	Commercial LPG in cylinder.	kg	95.00
1301	Bleaching powder	quintal	1838.00
1304	Surface box for stop cock	each	89.00
1305	Surface box for sluice valve	each	179.00
1307	Surface box for water meter	each	294.00
1309	C.I. bracket for wash basin and sinks	pair	95.00
1314	C.P.brass chain with 32 mm dia rubber plug	each	68.00
1315	C.P.brass chain with 40 mm dia rubber plug	each	116.00
1330	Clamps and M.S. stays including bolts and nuts for 100 mm pipe	each	89.00
1331	M.S.Holder bat clamp of approved design for 100 mm S.C.I. pipe	each	42.00
1332	M.S.Holder bat clamp of approved design for75 mm S.C.I. pipe	each	37.00
1334	Clamps and M.S. stays including bolts and nuts for 50 mm pipe	each	32.00
1335	Clamps and M.S. stays including bolts and nuts for 75 mm pipe	each	37.00
1336	Clearing eye with chain and lid 100 mm dia	each	47.00
1337	Clearing eye with chain and lid 150 mm dia	each	53.00
1339	Brass bib-cock 15 mm dia	each	331.00
1340	Brass bib-cock 20 mm dia	each	410.00
1342	Brass stop-cock 15 mm dia	each	362.00
1343	Brass stop-cock 20 mm dia	each	490.00
1350	Mosquito proof coupling of approved design	each	29.00
1352	C.I. cover and frame 300x300 mm inside	each	337.00
1353	C.I.cover without frame 300x300mm inside i.e.cover of 4.50 kg	each	216.00
1354	Rectangular cover 455x610 mm with frame (low duty)	each	1771.00
1355	Rectangular cover 455x610mm without frame (low duty)	each	1073.00
1356	500 mm dia cover with frame (medium duty)	each	5429.00
1357	500 mm dia cover without frame (medium duty)	each	2705.00
1358	**ferrow cement cover and frame 300x300 mm inside**	**each**	**210.00**
1360	C.I.mouth, brass ferrule 15 mm dia	each	126.00
1361	C.I.mouth, brass ferrule 20 mm dia	each	168.00
1362	C.I.mouth, brass ferrule 25 mm dia	each	210.00
1363	Vitreous china foot rests 250x130x30 mm	pair	158.00
1364	C.I. grating 100x100 mm	each	37.00
1366	C.I. grating 150x150 mm	each	63.00
1367	C.I. grating 180x180 mm	each	84.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1368	**C.I. grating 225x225 mm**	**each**	**95.00**
1369	S.C.I. gully or nahani grating 100 mm dia	each	37.00
1373	Rubber insertions for80 mm dia pipe joints	each	12.00
1374	Rubber insertions for 100 mm dia pipe joints	each	15.00
1375	Rubber insertions for 125 mm dia pipe joints	each	16.00
1376	Rubber insertions for 150 mm dia pipe joints	each	19.00
1377	Rubber insertions for 200 mm dia pipe joints	each	25.00
1378	Rubber insertions for 250 mm dia pipe joints	each	38.00
1379	Rubber insertions for 300 mm dia pipe joints	each	48.00
1380	Rubber insertions for 350 mm dia pipe joints	each	57.00
1381	Rubber insertions for 400 mm dia pipe joints	each	84.00
1382	Rubber insertions for 450 mm dia pipe joints	each	105.00
1383	Rubber insertions for 500 mm dia pipe joints	each	135.00
1384	Rubber insertions for 600 mm dia pipe joints	each	150.00
1391	bevelling/teak wood lipping of mirror 60x45cm	each	74.00
1392	Mirror of superior make glass 60x45 cm	Each	231.00
1393	Mirror of superior make 4mm thick	Sqm	819.00
1396	Vitrous china pedestal for wash basin	each	1386.00
1397	Pig lead	kilogram	121.00
1402	Squatting Pan (Indian type) white glazed vitreous China Ist quality W.C. Pan Size 450 mm.	each	309.00
1403	Squatting Pan (Indian type) white glazed vitreous China Ist quality W.C. Pan Size 510 mm.	each	383.00
1404	Squatting Pan (Indian type) white glazed vitreous China Ist quality W.C. Pan Size 580 mm.	each	420.00
1405	100 vitrious china P or S Trap	each	231.00
1406	Indian type white glazed vitreous china Ist quality W.C. orissa pan Size 530x410mm.	each	1143.00
1407	Indian type white glazed vitreous china Ist quality W.C. orissa pan Size 580x440 mm.	each	2147.00
1408	Carriage and fixing of material (Seat Cover etc.)	each	2.00
1409	European type white glazed vitreous china Ist quality W.C. pan	each	1706.00
1410	European type white glazed vitreous china Ist quality Double symphonic W.C.	each	4620.00
1411	European type white glazed vitreous china Ist quality W.C. Wall Mounting.	each	2583.00
1412	European type white glazed vitreous china Ist quality W.C. Wall Mounting.(extended)	each	2888.00
1413	European type white glazed vitreous china Ist quality W.C. Wall Mounting (syphonic)	each	3402.00
1414	White glazed Conversion Bend for W.C. from P trap to S trap	each	116.00
1415	White Vitreous China Double Syphonic European W.C.with mounted W.V.C. flushing cistern of of 10 litre capacity	each	7691.00
1416	WVC Urinal Flat Back (small) size 440x265x315 mm.	each	725.00
1417	WVC Urinal Flat Back (large) or half stall size 590x375x390 mm.	each	2310.00
1418	WVC Urinal Mini stall size 440x315x320 mm.	each	971.00
1420	Dome Waste coupling for urinal 32 mm	each	121.00
1421	Iron bracket/Plugs & Screws for urinal installation	pair	42.00
1422	W.V.C. Urinal Partition plate size 630x315	each	756.00
1423	Marble Urinal Partition slab both sides polished size 900x600mm.Makrana II quality marble having light spots.	each	630.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1424	Marble Urinal Partition slab both sides polished size 900x600mm. Makrana II quality marble having light spots.	each	578.00
1425	Marble Urinal Partition slab both sides polished size 900x600mm. Makrana 'Adanga' or Raj Nagar ISt Quality White or Andhi Indi-Italian marble.	each	525.00
1426	W.V.C. Half-Round Channel Size 100 mm dia.	Mtr	399.00
1427	W.V.C. Half-Round Channel Size 150 mm dia.	Mtr	536.00
1428	High Level Flushing Cistern of 10 Litres capacity	each	895.00
1429	32mm G.I. Flush pipe	each	208.00
1430	over flow pipe 185cm	each	138.00
1431	brackets	each	63.00
1432	Low level Flushing Cistern of 10 litres capacity : WVC with C.P. brass bend.	each	1676.00
1433	Low level Flushing Cistern of 10 litres capacity : PVC with PVC bend as per IS : 7231	each	840.00
1434	Low level Flushing Cistern of 10 litres capacity : PVC with PVC bend and superior internal fittings	each	1680.00
1435	Low level Flushing Cistern of 10 litres capacity : WVC for symphonic EWC.	each	1730.00
1436	Low level Flushing Cistern of 10 litres capacity : PVC with CP brass long band.	each	1029.00
1437	Automatic Flushing Cistern	each	971.00
1438	bend	each	131.00
1439	G.I. Flush pipe 'B' class (Concealed) 25 mm.	each	263.00
1440	G.I. Flush pipe 'B' class (Concealed) 32 mm.	each	326.00
1441	M.S. clamps	each	42.00
1442	Flush pipe PVC exposed (heavy) 32 mm.	each	74.00
1443	Flush pipe GI (Sheet) hot galvanized exposed (10 Litres).	each	105.00
1444	WVC (10 litres) low-level flushing cistern with cover.	each	662.00
1445	WVC (10 litres) low level flushing cistern cover only.	each	231.00
1446	WVC siphon with plunger plate only.	each	147.00
1447	WVC Auto cistern with cover only.	each	347.00
1448	WVC Auto cistern cover only.	each	105.00
1449	C.P. brass Urinal Flush pipe set For 4 Urinals.	each	919.00
1450	C.P. brass Urinal Flush pipe set For 3 Urinals.	each	693.00
1451	C.P. brass Urinal Flush pipe set For 2 Urinals.	each	520.00
1452	C.P. brass Urinal Flush pipe set For 1 Urinals.	each	320.00
1453	C.P. brass Urinal Spreader for stall urinal	each	273.00
1454	Synthetic (PTMT) Urinal Spreader for stall urinal	each	221.00
1455	Dome Waste coupling for urinal 40 mm	each	168.00
1456	Plastic Syphon of approved make for automatic flush tank.	each	179.00
1457	Copper tube Syphon for auto flush tank.	each	378.00
1458	C.P. Brass flush Valve Lever type with Elbow 32 mm dia.	each	2300.00
1459	C.P. Brass flush Valve Push type with Elbow 32 mm dia.	each	2419.00
1460	flush Bend	each	420.00
1461	C.P. Brass flush Valve Concealed type for 112 mm.	each	2814.00
1462	C.P. Brass flush Valve for 225 mm	each	2919.00
1464	S & S.C.I.standard specials upto 300 mm dia (heavy class)	quintal	3049.00
1466	S & S.C.I.standard specials over 300 mm dia (heavy class)	quintal	3297.00
1468	Flanged C.I. standard specials upto 300 mm dia(heavy class)	quintal	5488.00
1470	Flanged C.I. standard specials over 300 mm dia(heavy class)	quintal	6063.00
1472	Casing pipe 100 mm dia	metre	371.00
1532	Flush pipe with union spreaders and clamps all in C.P. brass for single stall	each	258.00
1533	Flush pipe with union spreaders and clamps all in C.P. brass for double stall	each	423.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1534	Flush pipe with union spreaders and clamps all in C.P. brass for range of three stall	each	527.00
1535	Flush pipe with union spreaders and clamps all in C.P. brass for range of four stall	each	590.00
1540	Flush pipe and spreaders G.I.for single set of one squatting plate urinal	each	175.00
1541	Flush pipe and spreaders G.I.for range of two squatting plates urinal	each	257.00
1542	Flush pipe and spreaders G.I.for range of three squatting plates urinal	each	323.00
1543	Flush pipe and spreaders G.I.for range of four squatting plates urinal	each	406.00
1545	G.I. pipes 15 mm dia (B Class)	metre	114.00
1546	G.I. pipes 20 mm dia (B Class)	metre	149.00
1547	G.I. pipes 25 mm dia (B Class)	metre	210.00
1548	G.I. pipes 32 mm dia (B Class)	metre	265.00
1549	G.I. pipes 40 mm dia (B Class)	metre	310.00
1550	G.I. pipes 50 mm dia (B Class)	metre	440.00
1551	G.I. pipes 65 mm dia (B Class)	metre	515.00
1552	G.I. pipes 80 mm dia (B Class)	metre	660.00
1555	G.I. back (jam) nuts25 mm dia	each	6.00
1559	G.I. back (jam) nuts65 mm dia	each	20.00
1608	G.I. tees (equal) 25 mm	each	48.00
1612	G.I. tees (equal) 65 mm	each	273.00
1614	G.I. inlet connection	each	64.00
1615	S.C.I. soil, waste and vent single socketed pipe 1.80 metres long:50mm dia	each	945.00
1616	S.C.I. soil, waste and vent single socketed pipe 1.80 metres long:75mm dia	each	1271.00
1617	S.C.I. soil, waste and vent single socketed pipe 1.80 metres long: 100mm dia	each	1617.00
1618	S.C.I. soil, waste and vent single socketed pipe 1.80 metres long: 150mm dia	each	2940.00
1620	S.C.I. plain bend75mm dia	each	313.00
1621	S.C.I. plain bend 100mm dia	each	460.00
1622	S.C.I. plain bend 150mm dia	each	965.00
1624	S.C.I. bend with access door 75mm dia	each	380.00
1625	S.C.I. bend with access door 100mm dia	each	520.00
1627	S.C.I. plain single equal junctions75x75x75 mm dia	each	260.00
1628	S.C.I. plain single equal junctions100x100x100 mm dia	each	336.00
1630	S.C.I. single equal junctions75x75x75 mm dia with access door.	each	285.00
1631	S.C.I. single equal junctions 100x100x100 mm dia with access door.	each	378.00
1633	S.C.I. plain double equal junctions 75x75x75x75 mm dia	each	376.00
1634	S.C.I. plain double equal junctions100x100x100x100 mm dia	each	500.00
1636	S.C.I. double equal junctions75x75x75x75 mm dia with access door.	each	423.00
1637	S.C.I. double equal junctions 100x100x100x100 mm dia with access door.	each	530.00
1639	Slotted cowl (terminal guard)75 mm dia	each	139.00
1640	Slotted cowl (terminal guard) 100 mm dia	each	193.00
1641	G.I. Union 15 mm nominal bore	each	46.00
1642	G.I. Union 20 mm nominal bore	each	79.00
1643	G.I. Union 25 mm nominal bore	each	100.00
1644	G.I. Union 32 mm nominal bore	each	168.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
1645	G.I. Union 40 mm nominal bore	each	203.00
1646	G.I. Union 50 mm nominal bore	each	318.00
1647	G.I. Union 65 mm nominal bore	each	656.00
1648	G.I. Union 80mm nominal bore	each	735.00
1649	Polyethylene water storage tank with cover and suitable locking arrangement	per litre	5.00
1653	Sand cast iron S&S plain single unequal junctions : 100x100x75 mm dia	each	366.00
1656	Sand cast iron S&S single unequal junctions: 100x100x75 mm dia with access door.	each	407.00
1659	Sand cast iron S&S plain double unequal junctions : 100x100x75x75 mm dia	each	538.00
1662	Sand cast iron S&S double unequal junctions: 100x100x75x75 mm dia with access door.	each	575.00
1666	Sand cast iron heel rest bend75mm dia	each	209.00
1667	Sand cast iron heel rest bend 100mm dia	each	251.00
1669	S.C.I. single equal invert branch of required degree75x75x75 mm dia	each	303.00
1670	S.C.I. single equal invert branch of required degree 100x100x100 mm dia	each	407.00
1672	S.C.I. double equal invert branch of required degree 75x75x75x75 mm dia	each	399.00
1673	S.C.I. double equal invert branch of required degree 100x100x100x100 mm dia	each	536.00
1674	S.C.I. single unequal invert branch of required degree100x100x75 mm dia	each	485.00
1677	S.C.I. double unequal invert branch of required degree 100x100x75x75 mm dia	each	582.00
1682	S.C.I. door pieces 75 mm dia	each	268.00
1683	S.C.I. door pieces 100 mm dia	each	370.00
1685	S.C.I. collar 75 mm dia	each	104.00
1686	S.C.I. collar 100 mm dia	each	140.00
1687	Unplasticised P.V.C.connection pipe with brass union 30 cm long 15 mm bore	each	23.00
1688	Unplasticised P.V.C.connection pipe with brass union 30 cm long 20 mm bore	each	32.00
1689	Unplasticised P.V.C.connection pipe with brass union 45 cm long 15 mm bore	each	32.00
1690	Unplasticised P.V.C.connection pipe with brass union 45 cm long 20 mm bore	each	44.00
1693	S.C.I. hand pump	each	572.00
1700	R.C.C. pipes NP2 class 100 mm dia	meter	431.00
1701	R.C.C. pipes NP2 class 150 mm dia	meter	473.00
1702	R.C.C. pipes NP2 class 250 mm dia	meter	630.00
1703	R.C.C. pipes NP2 class 300 mm dia	meter	851.00
1704	R.C.C. pipes NP2 class 450 mm dia	meter	1302.00
1705	R.C.C. pipes NP2 class 500 mm dia	meter	1449.00
1706	R.C.C. pipes NP2 class 600 mm dia	meter	1943.00
1707	R.C.C. pipes NP2 class 700 mm dia	meter	2394.00
1709	R.C.C. pipes NP2 class 800 mm dia	meter	3150.00
1710	R.C.C. pipes NP2 class 900 mm dia	meter	3948.00
1711	R.C.C. pipes NP2 class 1000 mm dia	meter	4515.00
1712	R.C.C. pipes NP2 class 1100 mm dia	meter	5355.00
1713	R.C.C. pipes NP2 class 200 mm dia	meter	557.00
1714	R.C.C. collarsNP2 class 100 mm dia	each	25.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1715	R.C.C. collarsNP2 class 150 mm dia	each	32.00
1716	R.C.C. collarsNP2 class 250 mm dia	each	51.00
1717	R.C.C. collarsNP2 class 300 mm dia	each	64.00
1718	R.C.C. collarsNP2 class 450 mm dia	each	96.00
1719	R.C.C. collarsNP2 class 500 mm dia	each	111.00
1720	R.C.C. collarsNP2 class 600 mm dia	each	140.00
1721	R.C.C. collarsNP2 class 700 mm dia	each	155.00
1723	R.C.C. collarsNP2 class 800 mm dia	each	222.00
1724	R.C.C. collarsNP2 class 900 mm dia	each	272.00
1725	R.C.C. collarsNP2 class 1000 mm dia	each	320.00
1726	R.C.C. collarsNP2 class 1100 mm dia	each	368.00
1727	R.C.C. collarsNP2 class 1200 mm dia	each	427.00
1728	R.C.C. collarsNP2 class 200 mm dia	each	45.00
1854	Stoneware pipes grade A (60 cm long) 100 mm dia	each	78.00
1855	Stoneware pipes grade A (60 cm long) 150 mm dia	each	133.00
1856	Stoneware pipes grade A (60 cm long) 200 mm dia	each	196.00
1857	Stoneware pipes grade A (60 cm long) 230 mm dia	each	231.00
1858	Stoneware pipes grade A (60 cm long) 250 mm dia	each	266.00
1859	Stoneware pipes grade A (60 cm long) 300 mm dia	each	404.00
1860	Stoneware pipes grade A (60 cm long) 400 mm dia	each	635.00
1863	Fire clay kitchen sink: 600x450x250 mm	each	3098.00
1864	Fire clay kitchen sink: 600x450x200 mm	each	2940.00
1865	White vitreous china laboratry sink 530x430x180 mm.	each	1328.00
1866	White vitreous china laboratry sink 500x350x150 mm.	each	1571.00
1867	Mosaic fine polished kitchen sink size 600x450x200 mm.	each	462.00
1868	Mosaic fine polished kitchen sink size 450x450x200 mm.	each	318.00
1869	Mosaic fine polished kitchen sink size 600 x 450 x 200 mm with 600 x 450mm drain board	each	635.00
1871	White vitreous china laboratry sink450x300x150 mm	each	1328.00
1872	White vitreous china laboratry sink600x450x200 mm	each	1956.00
1873	Drain Board WVC (I.S.: 2556 Mark) size 530 x 450mm	each	896.00
1874	Drain Board Mosaic fine polished size 600 x 450	each	173.00
1875	White plastic seat (solid)with lid C.P.brass hinges and rubber buffers	each	350.00
1876	Black plastic seat (solid) with lid C.P.brass hinges and rubber buffers	each	318.00
1878	Shower rose C.P.brass for 15 to 20 mm inlet 100 mm dia	each	36.00
1879	Shower rose C.P.brass for 15 to 20 mm inlet 150 mm dia	each	50.00
1881	Spun yarn	kilogram	68.00
1882	Strainer brass 40 mm dia 1.5 metre long	each	53.00
1885	15 mm C.P.brass tap	each	130.00
1889	C.P.brass toilet paper holder of standard size	each	89.00
1891	C.I. trap for standard urinal with vent arm with operating and other couplings in C.P.brass: 50 mm dia	each	159.00
1893	C.I. trap for standard urinal with vent arm with operating and other couplings in C.P.brass: 80 mm dia	each	201.00
1894	CI P trap	each	737.00
1895	C.P.brass trap40 mm dia	each	126.00
1896	100 mm S.C.I. trap with vent heel	each	331.00
1897	100 mm S.C.I. trap with 100 mm inlet and 100 mm outlet	each	280.00
1898	100 mm S.C.I. trap with 100 mm inlet and75 mm outlet	each	208.00
1900	S.W. gully trap P type 100x100 mm	each	69.00
1902	S.W. gully trap P type 150x100 mm	each	110.00
1903	**S.W. gully trap P type 225x150 mm**	**each**	190.00
1904	S.W. gully trap P type 180x150 mm	each	204.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1913	Vitros china lipped front urinal	each	483.00
1915	Vitros china squatting plate urinal Size 450x350mm	each	756.00
1922	H.P. or L.P. ball valve with polythene floats: 15 mm dia	each	222.00
1923	H.P. or L.P. ball valve with polythene floats: 20 mm dia	each	331.00
1924	H.P. or L.P. ball valve with polythene floats: 25 mm dia	each	381.00
1927	Brass full way valve with C.I. wheel (screwed end) 25 mm dia	each	328.00
1928	Brass full way valve with C.I. wheel (screwed end) 32 mm dia	each	415.00
1929	Brass full way valve with C.I. wheel (screwed end) 40 mm dia	each	461.00
1930	Brass full way valve with C.I. wheel (screwed end) 50 mm dia	each	674.00
1931	Brass full way valve with C.I. wheel (screwed end) 65 mm dia	each	979.00
1932	Brass full way valve with C.I. wheel (screwed end) 80 mm dia	each	1621.00
1933	Gunmetal non-return valve-horizontal (screwed end) 25 mm dia	each	333.00
1934	Gunmetal non-return valve-horizontal (screwed end) 32 mm dia	each	426.00
1935	Gunmetal non-return valve-horizontal (screwed end) 40 mm dia	each	573.00
1936	Gunmetal non-return valve-horizontal (screwed end) 50 mm dia	each	864.00
1937	Gunmetal non-return valve-horizontal (screwed end) 65 mm dia	each	1515.00
1938	Gunmetal non-return valve-horizontal (screwed end) 80 mm dia	each	2151.00
1940	C.I.sluice valve (with caps) class I : 100 mm dia	each	2401.00
1941	C.I.sluice valve (with caps) class I : 125 mm dia	each	3004.00
1942	C.I.sluice valve (with caps) class I : 150 mm dia	each	3608.00
1943	C.I.sluice valve (with caps) class I : 200 mm dia	each	6867.00
1944	C.I.sluice valve (with caps) class I : 250 mm dia	each	10228.00
1945	C.I.sluice valve (with caps) class I : 300 mm dia	each	12603.00
1947	Vitreous china flat back wash basin 630x450 mm	each	742.00
1949	Vitreous china angle back wash basin 600x480 mm	each	686.00
1950	Vitreous china angle back wash basin 400x400 mm	each	441.00
1951	C.P. brass waste coupling 32 mm	each	91.00
1952	C.P. brass waste coupling 40 mm	each	146.00
1953	Vitreous china indian type w.c. pan size 580 mm	each	424.00
1954	Vitreous china orrisa type w.c. pan size 580 mm	each	1023.00
1955	Vitreous china pedestal type water closet	each	810.00
1956	Bolts and nuts 16 mm dia 60 mm long	each	11.00
1957	Bolts and nuts 16 mm dia 65 mm long	each	11.00
1958	Bolts and nuts 20 mm dia 65 mm long	each	15.00
1959	Bolts and nuts 20 mm dia 70 mm long	each	15.00
1960	Bolts and nuts 20 mm dia 75 mm long	each	16.00
1961	Bolts and nuts 20 mm dia 80 mm long	each	19.00
1962	Bolts and nuts 24 mm dia 85 mm long	each	27.00
1963	Bolts and nuts 24 mm dia 90 mm long	each	34.00
1964	Bolts and nuts 27 mm dia 100 mm long	each	38.00
1965	White vitreous china dual purpose closet (Anglo Indian W.C.) suitable for use as sequatting pan or European type water closet as per manufacturer's specifications	each	2373.00
1966	C.P. brass waste coupling 50 mm	each	196.00
1967	SS waste coupling 125 mm	each	231.00
1968	P.V.C Waste pipe with C.P nut 32 mm	each	127.00
1969	P.V.C Waste pipe with C.P. nut 40 mm dia	each	139.00
1970	P.V.C Waste pipe with PVC nut 32 mm	each	58.00
1971	P.V.C Waste pipe with C.P nut 40 mm	each	139.00
1972	Waste pipe with G.I(B class) 25mm	each	139.00
1973	Waste pipe with G.I(B class) 32mm	each	191.00
1974	Waste pipe with G.I(B class) 40 mm	each	313.00
1979	Vitreous china foot rests 250x125x25 mm	pair	116.00
1980	Flyash	cum	11.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
1984	F.P.S. bricks tile class designation 100	1,000 Nos	4200.00
1986	Modular bricks class designation 75	1,000 Nos	5775.00
2200	Carriage of steam coal	tonne	0.00
2201	Carriage of Bricks	1,000 Nos	0.00
2202	Carriage of Stone aggregate below 40 mm nominal size	cum	0.00
2203	Carriage of Coarse sand	cum	0.00
2204	Carriage of Timber	cum	0.00
2205	Carriage of Steel	tonne	105.00
2206	Carriage of Stone aggregate 40 mm nominal size and above	cum	0.00
2207	Carriage of Brick tiles	1,000 Nos	0.00
2208	Carriage of Lime	cum	0.00
2209	Carriage of Cement	tonne	0.00
2210	Carriage of Shree binder	tonne	0.00
2211	Carriage of Tar bitumen	tonne	0.00
2215	Carriage of Soling stone & masonary stone	cum	0.00
2216	Carriage of Stone blocks white & red sand stone & kota stone slab	tonne	0.00
2224	Carriage of S.W. pipes100 mm dia	100 metre	0.00
2225	Carriage of S.W. pipes150 mm dia	100 metre	0.00
2226	Carriage of S.W. pipes200 mm dia	100 metre	0.00
2227	Carriage of S.W. pipes230 mm dia	100 metre	0.00
2228	Carriage of S.W. pipes250 mm dia	100 metre	0.00
2229	Carriage of S.W. pipes300 mm dia	100 metre	0.00
2241	Carriage of Good earth	cum	0.00
2242	Carriage of Dump manure	cum	0.00
2260	Carriage of Brick aggregate	cum	0.00
2261	Carriage of Fine sand (1 part badarpur sand : 2 parts jamuna sand)	cum	0.00
2262	Flyash	cum	0.00
2264	Carriage of Rubbish	cum	0.00
2265	Carriage of Moorum	cum	0.00
2266	Carriage of Surkhi	cum	0.00
2267	Carriage of Stone dust	cum	0.00
2268	Carriage of Marble dust and marble chips	cum	0.00
2271	Carriage of G.I. pipes below 100 mm dia	tonne	0.00
2273	Carriage of A.C.sheet and accessories	tonne	0.00
2275	Carriage of R.C.C. pipes 100 mm dia	100 metre	0.00
2281	Carriage of R.C.C. pipes 150 mm dia	100 metre	0.00
2287	Carriage of R.C.C. pipes 250 mm dia	100 metre	0.00
2290	Carriage of R.C.C. pipes 300 mm dia	100 metre	0.00
2299	Carriage of R.C.C. pipes 450 & 500 mm dia	100 metre	0.00
2302	Carriage of G.I.sheet and accessories	tonne	0.00
2303	Carriage of R.C.C. pipes 600, 700, 750 & 800 mm dia	100 metre	0.00
2308	Carriage of Plaster of paris	tonne	0.00
2309	Carriage of Cast iron fittings	tonne	0.00
2311	Carriage of Red bajri	cum	0.00
2314	Carriage of Barbed wire	tonne	0.00
2317	Carriage of Sludge	cum	0.00
2319	Carriage of Spun iron S & S pipes 100 mm dia	100 metre	0.00
2320	Carriage of Spun iron S & S pipes 125 mm dia	100 metre	0.00
2321	Carriage of Spun iron S & S pipes 150 mm dia	100 metre	0.00
2322	Carriage of Spun iron S & S pipes 200 mm dia	100 metre	0.00
2323	Carriage of Spun iron S & S pipes 250 mm dia	100 metre	0.00
2324	Carriage of Spun iron S & S pipes 300 mm dia	100 metre	0.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
2325	Carriage of Spun iron S & S pipes 350 mm dia	100 metre	0.00
2326	Carriage of Spun iron S & S pipes 400 mm dia	100 metre	0.00
2327	Carriage of Spun iron S & S pipes 450 mm dia	100 metre	0.00
2328	Carriage of Spun iron S & S pipes 500 mm dia	100 metre	0.00
2329	Carriage of Spun iron S & S pipes 600mm dia	100 metre	0.00
2330	Carriage of C.I. pipes 500 mm dia	100 metre	0.00
2331	Carriage of R.C.C. pipes 900 mm dia	100 metre	0.00
2332	Carriage of R.C.C. pipes 1000 mm dia	100 metre	0.00
2333	Carriage of R.C.C. pipes 1100 mm dia	100 metre	0.00
2334	Carriage of R.C.C. pipes 1200 mm dia	100 metre	0.00
2335	Carriage of Jamuna sand	cum	0.00
2341	Carriage of Pig lead	tonne	0.00
2342	Carriage of solvent/ Diesel.	quintal	0.00
2343	Carriage of ductile iron pipes (k7) 100 mm dia	100 metre	0.00
2344	Carriage of cast iron pipes 150 mm dia	100 metre	0.00
2345	Carriage of cast iron pipes 200 mm dia	100 metre	0.00
2346	Carriage of cast iron pipes 250 mm dia	100 metre	0.00
2347	Carriage of cast iron pipes 300 mm dia	100 metre	0.00
2348	Carriage of cast iron pipes 350 mm dia	100 metre	0.00
2349	Carriage of cast iron pipes 400 mm dia	100 metre	0.00
2350	Carriage of cast iron pipes 450 mm dia	100 metre	0.00
2351	Carriage of cast iron pipes 500 mm dia	100 metre	0.00
2352	Carriage of cast iron pipes 600 mm dia	100 metre	0.00
2353	Carriage of cast iron pipes 700 mm dia	100 metre	0.00
2355	Carriage of cast iron pipes 800 mm dia	100 metre	0.00
2356	Carriage of cast iron pipes 900 mm dia	100 metre	0.00
2357	Carriage of cast iron pipes 1000 mm dia	100 metre	0.00
2391	Strips-Aluminium fluted 3.15mm thick and 150mm wide	metre	294.00
2392	Strips Aluminium fluted 3.15mm thick and 200mm wide metre	Metre	392.00
2393	5mm thick Acrylic sheet (150mm wide)	Metre	47.00
2406	Float glass sheet of nominal thickness 4 mm (weight not less than 10kg/sqm).	sqm	284.00
2407	Float glass sheet of nominal thickness 5 mm.(weight not less than 13.50 kg/sqm).	sqm	473.00
2408	Float glass sheet of nominal thickness 8 mm	sqm	840.00
2412	Ply wood 5 ply with commercial ply on both faces 6 mm thick	sqm	515.00
2447	Hollock ballies 125 mm diameter	metre	53.00
2449	Oxidised mild steel pull bolt lock (locking bolt) of size 85 mm x 42 mm with screws,bolts,nuts and washers complete	each	42.00
2451	Brass cupboard lock 6 levers (best make of approved quality) 40 mm size	each	74.00
2452	Brass cupboard lock 6 levers (best make of approved quality) 50 mm size	each	105.00
2453	Brass cupboard lock 6 levers (best make of approved quality) 65 mm size	each	126.00
2454	Brass cupboard lock 6 levers (best make of approved quality) 75 mm size	each	137.00
2455	Brass hanging type door stopper 150 mm	each	74.00
2456	Hydraulic door closer bottle type M.S. body with necessary accessories and screws complete	each	714.00
2459	Anodised Aluminium hanging type door stopper	each	21.00
2464	Anodised Aluminium pull bolt lock (locking bolt) of size 85 mmx42 mm with screws,bolts,nuts and washers complete	each	44.00
2465	Anodised Aluminium Casement stay 250 mm	each	53.00
2466	Hollock wood in scantling	10 cudm	420.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
2467	Chromium plated Brass pull bolt lock (locking bolt) of size 85 mmx42 mm with screws,bolts,nuts and washers complete	each	168.00
2468	Nickled Chromium Brass cupboard lock 40 mm size	each	63.00
2469	Nickled Chromium Brass cupboard lock 50 mm size	each	79.00
2470	Nickled Chromium Brass cupboard lock 65 mm size	each	95.00
2471	Nickled Chromium Brass cupboard lock 75 mm size	each	116.00
2480	Ply wood 5 ply with teak ply on both faces 9 mm thick	sqm	935.00
2481	Ply wood 5 ply with teak ply on one face and commercial ply on another face 9 mm thick	sqm	772.00
2482	6 mm thick commercial ply	sqm	223.00
2483	9 mm thick commercial ply	sqm	441.00
2484	9 mm teak ply BWR one side	sqm	772.00
2485	9 mm teak ply BWR both side	sqm	935.00
2486	12 mm thick commercial ply	sqm	452.00
2487	12 mm teak ply BWR one side	sqm	866.00
2488	12mm teak ply BWR both side	sqm	1050.00
2489	0.6 mm mica	sqm	210.00
2490	0.8 mm mica	sqm	315.00
2491	1 mm mica	sqm	420.00
2492	4 mm thick commercial ply	sqm	247.00
2493	4 mm teak ply BWR one side	sqm	420.00
2494	6 mm teak ply BWR one side	sqm	515.00
2495	12 mm thick Acoustical Board	sqm	536.00
2496	12 mm thick one side laminated board Practical board (Phenol bonded)	sqm	725.00
2497	50 mm glass whool	sqm	315.00
2498	Hassian cloth with fibre glass	sqm	147.00
2500	Extra for selected planks of second class deodarwood	10 cudm	105.00
2504	Kiln seasoning of timber	cum	919.00
2505	Hollock wood in planks	10 cudm	452.00
2602	F.P.S. bricks class designation 75	1,000 Nos	4200.00
2603	F.P.S. bricks class designation 50	1,000 Nos	3990.00
2704	Aluminium Strip 40 mm wide and 2 mm thick	kilogram	263.00
2710	White marble makranasecondquality plain veined stone pieces for crazy flooring	quintal	158.00
2750	8 mm thick granite stone tiles (mirror polished of all shades)	sqm	704.00
2751	8 mm thick marble tiles (polished) Raj Nagar	sqm	378.00
2901	Stone Aggregate (Single size) : 100 mm nominal size	cum	683.00
2902	Stone Aggregate (Single size) : 80 mm nominal size	cum	735.00
2903	Stone chippings/ screenings 4.75 mm nominal size	cum	987.00
2904	Stone chippings/ screenings 150 micron nominal size	cum	987.00
2908	Over burnt (Jhama) Brick Aggregate: 120 mm to 40 mm size	cum	368.00
2909	Over burnt (Jhama) Brick Aggregate: 90 mm to 40 mm size	cum	420.00
2910	Stone chippings/ screenings 12.5/ 13.2 mm nominal size	cum	735.00
2911	Stone chippings/ screenings 10/ 11.2 mm nominal size	cum	735.00
2914	Solvent	kilogram	32.00
2916	Paving Asphalt 80/100 penetration	tonne	47250.00
3002	Polyvinyle chloride sheet 400 micron thick	sqm	34.00
3004	Stone ware spouts 100 mm dia 60 cm long	each	37.00
3050	Galvanised steel corrugated sheets	quintal	6300.00
3213	Vitreous china Surgeon type wash basin of size 660x460 mm	each	1132.00
3228	600x120 mm glass shelf with anodised aluminium angle frame, C.P. brass brackets and guard rail of standard size	each	131.00
3229	Vitreous china flat back wash basin 550x400 mm	each	907.00
3311	C.I.sluice valve (with caps) class II : 100 mm dia	each	2478.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
3314	C.I.sluice valve (with caps) class II : 125 mm dia	each	3131.00
3317	C.I.sluice valve (with caps) class II : 150 mm dia	each	3812.00
3320	C.I.sluice valve (with caps) class II : 200 mm dia	each	7369.00
3321	C.I.sluice valve (with caps) class II : 250 mm dia	each	10418.00
3326	C.I.sluice valve (with caps) class II : 300 mm dia	each	13023.00
3617	C.P.brass union 40 mm dia	each	165.00
3620	C.C.I.(spun) socketed soil, waste and vent pipe 1.80 metres long:100mm dia	each	1243.00
3621	C.C.I.(spun) socketed soil, waste and vent pipe 1.80 metres long:75mm dia	each	1075.00
3624	S.C.I. S&S bends with access door100mm dia	each	318.00
3625	S.C.I. S&S bends with access door75mm dia	each	228.00
3628	S.C.I. S&S bend100mm dia	each	242.00
3629	S.C.I. S&S bend75mm dia	each	182.00
3634	S.C.I. S&S heel rest sanitary bend 100mm dia	each	296.00
3635	S.C.I. S&S heel rest sanitary bend 75mm dia	each	258.00
3640	S.C.I. S&S single equal junctions100x100x100 mm	each	466.00
3641	S.C.I. S&S single equal junctions75x75x75 mm	each	351.00
3644	S.C.I. S&S single equal junctions with access door 100x100x100 mm	each	504.00
3645	S.C.I. S&S single equal junctions with access door 75x75x75 mm	each	373.00
3650	S.C.I. S&S double equal junctions100x100x100x100 mm	each	593.00
3651	S.C.I. S&S double equal junctions75x75x75x75 mm	each	474.00
3654	S.C.I. S&S double equal junctions with access door 100x100x100x100 mm.	each	637.00
3655	S.C.I. S&S double equal junctions with access door 75x75x75x75 mm.	each	508.00
3660	S.C.I. S&S single unequal junctions100x100x75 mm	each	586.00
3664	S.C.I. S&S single unequal junctions with access door 100x100x75 mm	each	665.00
3670	S.C.I. S&S double unequal junctions100x100x75x75 mm	each	812.00
3674	S.C.I. S&S double unequal junctions with access door 100x100x75x75 mm	each	889.00
3681	S.C.I. S&S single equal invert branch of required degree 100x100x100 mm dia	each	414.00
3682	S.C.I. S&S single equal invert branch of required degree 75x75x75 mm dia	each	314.00
3685	S.C.I. S&S double equal invert branch of required degree100x100x100x100 mm dia	each	526.00
3686	S.C.I. S&S double equal invert branch of required degree 75x75x75x75 mm dia	each	424.00
3690	S.C.I. S&S single unequal invert branch of required degree100x100x75 mm dia	each	536.00
3695	S.C.I. S&S double unequal invert branch of required degree100x100x75x75 mm dia	each	729.00
3699	S.C.I. S&S, 75 mm offset for75 mm dia pipe	each	228.00
3707	S.C.I. S&S, 150 mm offset for75 mm dia pipe	each	284.00
3708	S.C.I. S&S, 150 mm offset for100 mm dia pipe	each	390.00
3712	S.C.I. S&S, 114 mm offset for75 mm dia pipe	each	279.00
3713	S.C.I. S&S, 114 mm offset for100 mm dia pipe	each	368.00
3716	S.C.I. S&S, 152 mm offset for75 mm dia pipe	each	349.00
3717	S.C.I. S&S, 152 mm offset for100 mm dia pipe	each	459.00
3728	S.C.I. S&S door pieces 100 mm dia	each	292.00
3729	S.C.I. S&S door pieces 75 mm dia	each	224.00
3733	S.C.I. S&S, Slotted Cowl (Terminal Guard) 100 mm	each	196.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
3734	S.C.I. S&S, Slotted Cowl (Terminal Guard) 75 mm	each	181.00
3738	S.C.I. S&S, collars 100 mm	each	205.00
3739	S.C.I. S&S, collars 75 mm	each	146.00
3746	S.C.I. S&S, 76 mm offset for75 mm dia pipe	each	161.00
3747	S.C.I. S&S, 76 mm offset for100 mm dia pipe	each	279.00
3749	Vitreous china toilet paper holder of standard size	each	113.00
3860	560 mm dia cover with frame (Heavy duty)	each	2625.00
3861	560 mm dia cover without frame (Heavy duty)	each	2310.00
4006	Pressed steel door frames (mild steel sheet 1.25mm) Profile "A"	metre	263.00
4007	Pressed steel door frames (mild steel sheet 1.25mm) Profile "B"	metre	294.00
4008	Pressed steel door frames (mild steel sheet 1.25mm) Profile "C"	metre	326.00
4009	Mild steel tubes hot finished welded type	kilogram	53.00
4010	Mild steel tubes hot finished seamless type	kilogram	58.00
4011	Mild steel tubes electric resistant or inductionbutt welded	kilogram	74.00
4012	Circular C.I. Box for ceiling fan	each	53.00
4013	Pully 40 mm dia	each	32.00
4014	Ready made steel door with necessary hinges, lugs and glazing clips excluding other fittings & their fixing	sqm	1943.00
4201	Aluminium primer	litre	89.00
4202	Red oxide Zinc chromate primer	litre	79.00
4203	Copper acetate	kilogram	315.00
4204	Hydrochloric acid	kilogram	32.00
4205	Copper chloride	kilogram	326.00
4206	Copper nitrate	kilogram	221.00
4207	Ammonium chloride	kilogram	16.00
5001	Mobil oil	litre	131.00
6001	White marble slab Makrana second quality plain veined 18 mm thick	sqm	2100.00
6007	Pink marble slab plain 18mm thick	sqm	735.00
6010	Udaypur green marble slab plain 18mm thick	sqm	693.00
6019	Black Zebra marble slab plain 18mm thick	sqm	578.00
6501	Sand zone V ()	cum	231.00
6502	Sand as per Table 1500. 5	cum	840.00
7001	Brass 100mm mortice latch and lock with6 levers without pair of handles	each	263.00
7003	Pair of Anodised Aluminium lever handles for 100mm mortice latch and lock	each	247.00
7004	Vitreous china flat back wash basin 450x300 mm	each	578.00
7005	Vitreous china 10 litres low level cistern without fittings	each	951.00
7006	Vitreous china 10 litres low level cistern with fittings	each	1605.00
7008	F.P.S. clay fly ash bricks class designation 75	1,000 Nos	4200.00
7009	Gypsum board	sqm	289.00
7010	Ceiling sections	metre	53.00
7011	Perimetre channel	metre	32.00
7012	Intermediate channel	metre	55.00
7013	Ceiling angle	metre	21.00
7014	Connecting clips	each	6.00
7015	Soffit cleat	each	4.00
7016	Joint filler	kilogram	26.00
7017	Joint finisher	kilogram	29.00
7018	Joint tape roll	roll	147.00
7019	Dash fastner/ chemical fastener	each	13.00
7020	All drive screws (for gypsum board)	100 Nos	53.00
7021	Primer (for gypsum board)	litre	116.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7022	Chlorpyriphos 20% E.C. / Lindane 20% E.C.	litre	200.00
7023	Chromium plated brackets (curtain rods)	each	6.00
7024	Acid Proof cement	tonne	8925.00
7027	M.S. Butt hinges 125x90x4 mm	10 Nos	105.00
7029	Galvanised wire mesh of average width of aperture 1.4 mm and nominal dia. of wire 0.63 mm	sqm	179.00
7032	Frosted glass sheet of nominal thickness 4 mm (weighing not less than 10 kg/sqm)	sqm	420.00
7034	Nickle plated M.S. pipe 20 mm dia.	metre	66.00
7035	Nickle plated M.S. Brackets for curtain rod 20 mm	each	4.00
7036	Nickle plated M.S. Brackets for curtain rod 25 mm	each	5.00
7040	Oxidised mild steel screws 35 mm	100 Nos	37.00
7042	Mild steel conduit pipe (heavy type) ISI marked-20 mm dia.	metre	55.00
7043	Mild steel conduit pipe (heavy type) ISI marked-25 mm dia.	metre	63.00
7044	Rolling shutters of 80x0.90 mm laths	sqm	809.00
7045	Rolling shutters of 80x1.2 mm laths	sqm	924.00
7046	Top cover of Rolling shutters 0.90 mm thick	metre	420.00
7047	Top cover of Rolling shutters 1.20 mm thick	metre	473.00
7048	Rawl plug 50 mm (designation 10 no.)	each	8.00
7049	Teak wood lipping of size 25x3 mm in pelmets	metre	21.00
7055	Flat pressed 3 layer and graded particle board (medium density) Grade 1 conforming to IS : 3087 - 18 mm thick	sqm	378.00
7056	Aluminium tee channel (heavy duty) with rollers and stop end	metre	44.00
7059	Aluminium hanging floor door stopper with twin rubber & stopper	each	55.00
7060	Hydraulic door closer tubular type Aluminium section body	each	1155.00
7063	Oxidised M.S.casement stay (straight peg type) 300 mm not less than 0.33 kg	each	19.00
7064	Oxidised M.S.casement stay (straight peg type) 250 mm not less than 0.28 kg	each	16.00
7065	Oxidised M.S.casement stay (straight peg type) 200 mm not less than 0.24 kg	each	15.00
7068	Extra for providing grilled rolling shutters with 8 mm dia M.S. rod	sqm	268.00
7070	Chequered precast cement concrete tiles 22mm thick using marble chips of size 6mm - Light shade using white cement	sqm	134.00
7071	White marble Raj Nagar plain 20 mm thick (slab area 0.10 sqm to 0.20 sqm)	sqm	578.00
7077	Acid and alkali resistant tiles 300x300 mm size, 10 mm thick	10 Nos	578.00
7087	S.C.I. Tee 150 mm	each	735.00
7090	Expanded polystyrene type N- Normal	sqm	147.00
7091	Expanded polystyrene type - SE	sqm	179.00
7092	Stainless steel kitchen sink Size 18x16x6 inches	each	2895.00
7093	Stainless steel kitchen sink Size 22x18x7 inches	each	3449.00
7094	Stainless steel kitchen sink Size 24x18x8 inches	each	5891.00
7095	Stainless steel kitchen sink - with drain board bowl depth 250 mm.	each	6225.00
7096	Stainlees steel kitchen sink - with drain board 510 x 1040mm bowl depth 225 mm.	each	5844.00
7097	Stainlees steel kitchen sink - with drain board 510 x 1040mm bowl depth 200 mm.	each	5209.00
7098	Stainless steel kitchen sink - with drain board 510x1040mm bowl depth 178 mm	each	3303.00
7099	Stainless steel kitchen sink Size 37x18x7 inches with drain board	each	6965.00
7100	Stainless steel kitchen sink Size 45x20x8 inches with drain board	each	9841.00
7101	Stainless steel kitchen sink - without drain board 610x510mm bowl depth 200 mm	each	3430.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7102	Stainless steel kitchen sink - without drain board 610x460mm bowl depth 200 mm.	each	3176.00
7103	Stainless steel kitchen sink - without drain board 470x420mm bowl depth 178 mm	each	2033.00
7104	Coloured Orissa pattern W.C. pan 580x440 mm	each	3220.00
7105	Coloured Pedestal type W.C. pan (European type)	each	2559.00
7106	Coloured Vitreous china 10 lit. low level cistern	each	1612.00
7107	Coloured (other than black) solid P.V.C. seat in European W.C. pan	each	389.00
7108	Non-breakable of other approved make Black.seat in European W.C. pan	each	231.00
7109	Non-breakable of other approved make White.seat in European W.C. pan	each	252.00
7110	Non-breakable of other approved make Coloured.seat in European W.C. pan	each	294.00
7111	C.P. lugs for toilet seat cover.		84.00
7112	Circular shape 500 mm dia Mirror with Plastic moulded frame	each	444.00
7113	Rectangular shape 500x400 mm Mirror with Plastic moulded frame	each	420.00
7114	Ovel shape 450x350 mm (outer dimensions) Mirror with Plastic moulded frame	each	473.00
7115	Rectangular shape 1500x450 mm Mirror with Plastic moulded frame	each	806.00
7116	Hard board 6 mm thick	sqm	192.00
7117	Semi Rigid PVC waste pipe for sink and wash basin 32 mm dia with length not less than 700 mm i/c PVC waste fittings	each	58.00
7118	Semi Rigid PVC waste pipe for sink and wash basin 40 mm dia with length not less than 700 mm i/c PVC waste fittings	each	68.00
7119	Flexible (coil shaped) PVC waste pipe for sink and wash basin 32 mm dia with length not less than 700 mm i/c PVC waste fittings	each	42.00
7120	Flexible (coil shaped) PVC waste pipe for sink and wash basin 40 mm dia with length not less than 700 mm i/c PVC waste fittings	each	47.00
7123	Coloured High density polythylene/ poly propylene 10 lit. (full flush) capacity controlled low level flushing cistern with fittings	each	803.00
7126	White Vitreous china 10 lit. (full flush) capacity controlled low level flushing cistern with all fittings	each	1090.00
7127	Coloured Vitreous china 10 lit. (full flush) capacity controlled low level flushing cistern with all fittings	each	1623.00
7128	S.W. intercepting trap 100 mm dia	each	221.00
7129	S.W. intercepting trap 150 mm dia	each	305.00
7130	Rectangular shape 600x450 mm precast R.C.C. manhole cover with frame - L.D. - 25	each	1155.00
7131	Square shape 350x350 mm precast R.C.C. manhole cover with frame - L.D. - 25	each	788.00
7132	Circular shape 450 mm dia precast R.C.C. manhole cover with frame - L.D. - 25	each	945.00
7133	Rectangular shape 500x500 mm precast R.C.C. manhole cover with frame - M.D. - 10	each	2100.00
7134	Circular shape 500 mm dia precast R.C.C. manhole cover with frame - M.D. - 10	each	2100.00
7135	Circular shape 560 mm dia precast R.C.C. manhole cover with frame - H.D. - 20	each	2625.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7136	Circular shape 560 mm dia precast R.C.C. manhole cover with frame - E.H.D. - 35	each	2993.00
7137	Factory made 35 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of 12 mm thick plain type-I, medium density flat pressed three layer, graded particle board (FPT-I) as per IS:3087-1985 bonded with BWP type synthe	sqm	1733.00
7139	Factory made 35 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of 12 mm thick both sides prelaminated type-I, medium density flat pressed three layer, graded particle board (FPT-I) as per IS:3087-1985 bonded w	sqm	1869.00
7143	Factory made 35 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of 12 mm thick one side prelaminated type-I, and other side balancing lamination, medium density flat pressed three layer, graded particle board (	sqm	1815.00
7151	Factory made 30 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of sheet glass using 10 kg/ sqm glass panes	sqm	1490.00
7154	Factory made 35 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of galvanised wire gauge with average width of aperture 1.4 mm on both directions with wire of dia 0.63 mm	sqm	1534.00
7155	Factory made 30 mm thick shutters with laminated veneer lumber styles rails as per TADS IS:1995 and panels of galvanised wire gauge with average width of aperture 1.4 mm on both directions with wire of dia 0.63 mm	sqm	1328.00
7157	Laminated veneer lumber confirming to TADSS IS:1995 manufactured in factory in frames of doors, windows	10 cudm	651.00
7181	C.I. pile shoe	kilogram	48.00
7182	M.S. clamps for pile shoe	kilogram	42.00
7183	Bentonite	tonne	3360.00
7184	Oxidised M.S. safety chain (weighing not less than 450 gms) for door	each	63.00
7187	C.I. grating 150 mm dia. (Weighing not less than 440 gm)	each	47.00
7188	U-PVC pipes (working pressure 4 kg / cm^2) Single socketed pipe 75 mm dia.	metre	84.00
7189	U-PVC pipes (working pressure 4 kg / cm^2) Single socketed pipe 110 mm dia.	metre	150.00
7190	U-PVC pipes (working pressure 4 kg / cm^2) Rubber (Seal) Ring 75 mm dia.	each	21.00
7191	U-PVC pipes (working pressure 4 kg / cm^2) Rubber (Seal) Ring 110 mm dia.	each	26.00
7192	UPVC coupler for UPVC drainage pipes 75 mm	each	38.00
7193	UPVC coupler for UPVC drainage pipes 110 mm	each	50.00
7194	UPVC pushfit coupler (single) 75 mm thick	each	60.00
7195	UPVC pushfit coupler (single) 110 mm thick	each	91.00
7196	UPVC single equal Tee (with door) 75x75x75 mm	each	105.00
7197	UPVC single equal Tee (with door) 110x110x110 mm	each	146.00
7198	UPVC single equal Tee (with door) 75x75x75 mm	each	127.00
7199	UPVC single equal Tee (with door) 110x110x110 mm	each	204.00
7208	UPVC bend 87.5o 75 mm bend	each	64.00
7209	UPVC bend 87.5o 110 mm bend	each	107.00
7212	UPVC plain shoe 75 mm bend	each	133.00
7213	UPVC plain shoe 110 mm bend	each	250.00
7214	UPVC pipe clip 75 mm bend	each	19.00
7215	UPVC pipe clip 110 mm bend	each	38.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
7231	Resin Bonded Glass wool 16 kg/m³ 50 mm thick	sqm	147.00
7232	Resin Bonded Glass wool 24 kg/m³ 50 mm thick	sqm	221.00
7233	Fibre glass tissue reinforcement Type II Grade I	sqm	84.00
7236	Precast chequered cement tiles 22 mm thick Dark shade using ordinary cement	sqm	150.00
7237	Precast chequered cement tiles 22 mm thick medium shade using 50% white cement, 50% ordinary cement	sqm	231.00
7239	Epoxy paint	litre	368.00
7240	Fire retardant paint	litre	441.00
7241	Melamine polish	litre	341.00
7242	Expoxy primer for steel	lt	2048.00
7243	Expoxy harder and resin	kg	1129.00
7244	Table rubbed polished stone 18 mm thick (75x50cm) Agaria Marble stone - 18 mm thick	sqm	1864.00
7245	Table rubbed polished stone 18mm thick (75x50cm) Granite stone - 18mm thick	sqm	2153.00
7246	Verticle load testing (INITIAL) of piles in accordance with IS : 2911 (Part-IV) including installation of loading platform and prepration of pile head or construction of test cap and dismantling of test cap after test etc. complete as per specification a	per test	24675.00
7247	Verticle load testing (INITIAL) of piles in accordance with IS : 2911 (Part-IV) including installation of loading platform and prepration of pile head or construction of test cap and dismantling of test cap after test etc. complete as per specification &	per test	34650.00
7248	Verticle load testing (INITIAL) of piles in accordance with IS : 2911 (Part-IV) including installation of loading platform and prepration of pile head or construction of test cap and dismantling of test cap after test etc. complete as per specification &	per test	47250.00
7249	Cyclic verticle load testing of piles in accordance with IS : 2911 (Part-IV) including prepration of pile head etc. for Single pile upto 50 tonne capacity	per test	14700.00
7250	Cyclic verticle load testing of piles in accordance with IS : 2911 (Part-IV) including prepration of pile head etc. forSingle pile above 50 tonne capacity pile and upto 100 tonne capacity pile	per test	22890.00
7251	Cyclic verticle load testing of piles in accordance with IS : 2911 (Part-IV) including prepration of pile head etc. forGroup of two piles upto 50 tonne capacity each	per test	26565.00
7252	Lateral load testing of single pile in accordance with IS : 2911 part -IV for determining safe allowable lateral load on pile. Upto 50 tonne capacity	per test	14700.00
7253	Lateral load testing of single pile in accordance with IS : 2911 part -IV for determining safe allowable lateral load on pile. Above 50 tonne capacity	per test	23625.00
7254	Hardening compound	litre	44.00
7255	Road marking paint (spirit based)	litre	137.00
7256	Superior quality road marking paint	litre	210.00
7257	C.P. Brass bibcock 15 mm	each	473.00
7258	C.P. Brass long nose bibcock 15 mm	each	835.00
7259	C.P. Brass long body bibcock 15 mm	each	641.00
7260	C.P. Brass stop cock (concealed) 15 mm	each	907.00
7261	C.P. Brass angle valve 15 mm	each	420.00
7266	Pressed clay tiles	1,000 Nos	11307.00
7267	Plain ceiling tiles (BWP type phenol formaldehyde synthetic resin bonded) (600x600x12 mm)	each	126.00
7268	Semi perforated ceiling tiles (600x600x12 mm)	each	114.00
7269	25 mm thick particle board	sqm	499.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
7270	30 mm thick prelaminated flush door shutter	sqm	893.00
7271	IInd class teak wood lipping 25 mm wide x 12 mm thick	metre	43.00
7272	25 mm thick melamine faced prelaminated three layer particle board	sqm	851.00
7295	Jhunjhunu / Jalore (Red/Choclate/Black/Pink Colour) 18 mm thick slab, upto 1500 cm2	sqm	630.00
7296	Jhunjhunu / Jalore (Red/Choclate/Black/Pink Colour) 18 mm thick slab, above 1500 upto 3600 cm2	sqm	945.00
7297	Jhunjhunu / Jalore (Red/Choclate/Black/Pink Colour) 18 mm thick slab, above 3600 cm2	sqm	1155.00
7298	Granite South Red 18 mm thick slab, upto 1500 cm2	sqm	924.00
7299	Granite South Red 18 mm thick slab, above 1500 upto 3600 cm2	sqm	1470.00
7300	Granite South Red 18 mm thick slab, above 3600 cm2	sqm	2825.00
7301	South Zed Black 18 mm thick slab, upto 1500 cm2	sqm	840.00
7302	South Zed Black 18 mm thick slab, above 1500 upto 3600 cm2	sqm	1680.00
7303	South Zed Black 18 mm thick slab, above 3600 cm2	sqm	2147.00
7304	Good quality South Black 18 mm thick slab, upto 1500 cm2	sqm	630.00
7305	Good quality South Black 18 mm thick slab, above 1500 upto 3600 cm2	sqm	1155.00
7306	Aluminium T or L sections	kilogram	210.00
7307	For flush door shutters Extra for providing teak veneering on one side instead of commercial veneering	sqm	420.00
7308	Good quality South Black 18 mm thick slab, above 3600 cm2	sqm	1890.00
7309	Paving Asphalt 60/70 penetration	tonne	51450.00
7312	Expandable fastner with plastic sleeve and M.S. screws. 25 mm long	each	11.00
7313	Expandable fastner with plastic sleeve and M.S. screws. 32 mm long	each	12.00
7314	Expandable fastner with plastic sleeve and M.S. screws. 40 mm long	each	13.00
7315	Expandable fastner with plastic sleeve and M.S. screws. 50 mm long	each	14.00
7318	Plasticizer / super plasticizer	kilogram	42.00
7319	Wall form panel 1250x500 mm	each	756.00
7320	Tie bolt 12 mm dia 100 mm length	each	42.00
7321	Tie bolt 12 mm dia 150 mm length	each	53.00
7322	Tie bolt 20 mm dia 150 mm length	each	74.00
7323	Tie bolt 20 mm dia 225 mm length	each	89.00
7324	Spring coil 12 mm	each	14.00
7325	Plastic cone 12 mm dia	each	16.00
7326	Corner angle 45x45x5 mm 1.50 m long	each	221.00
7327	100 mm channel shoulder 2.5 m long	each	788.00
7328	Double clip (bridge clip)	each	79.00
7329	Single clip	each	68.00
7330	M.S. tube 40 mm dia	metre	200.00
7331	Wall form panel 1250x450 mm	each	683.00
7332	Corner angle 45x45x5 m 2.50 m long	each	315.00
7333	Column clamp 450x1070 m	each	1103.00
7334	Prop 2 m (2-3.5m)	each	788.00
7335	Binding wire	kilogram	68.00
7338	Gun metal cramp	kilogram	368.00
7339	Stainless steel cramp	kilogram	378.00
7340	Stainless steel pin .	kg	163.00
7342	Adjustable span ESO+SI (2.35-3.40)	each	1785.00
7343	Adjustable telescopic prop 3 m (2.02-3.75 m)	each	1050.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
7344	Beam clamp 300-380 mm (450-1070 mm)	each set	420.00
7345	Prop 4 m	each	1155.00
7346	Double coupler	each	53.00
7347	Cadmium plated full threaded steel screws (30x4 mm dia.)	100 Nos	32.00
7348	Aluminium washer 2 mm thick 15 mm dia	100 Nos	11.00
7349	12 mm M.S. 'U' beading	metre	13.00
7354	Plastic encapsuled M.S. foot rest 30x20x15 cm	each	121.00
7358	Flushing Cistern P.V.C. 10 lts capacity (low level) (White) (with fittings, accessories and flush pipe)	each	793.00
7359	P.V.C. automatic flushing cistern 5 lts capacity	each	546.00
7361	P.V.C. automatic flushing cistern 10 lts capacity	each	590.00
7363	15 mm C.P. brass tap with elbow operation lever	each	788.00
7364	White glazed fire clay draining board 600x450x25 mm	each	496.00
7366	Glass reinforced Gyp sum (GRG) board 8.5 mm thick	sqm	263.00
7367	Galvanised M.S. sheet 0.5 mm thick pressed channel section of size 50x32 mm	metre	74.00
7369	Galvanised M.S. sheet 0.50 mm thick pressed stud. 48x34x36 mm	metre	79.00
7375	G.I. flush pipe and C.P. brass spreader including C.P. connecting pipe Single lipped urinal	each	447.00
7376	G.I. flush pipe and C.P. brass spreader including C.P. connecting pipe Range of two lipped urinals	each	1118.00
7377	G.I. flush pipe and C.P. brass spreader including C.P. connecting pipe Range of three lipped urinals	each	1360.00
7378	G.I. flush pipe and C.P. brass spreader including C.P. connecting pipe Range of four lipped urinals	each	1944.00
7379	White vitreous china clay half stall urinal flat back 580x380x350 mm or angle back 450x375x350 mm with waste fittings as per IS : 2556	each	1555.00
7380	Precast R.C.C. grating with frame 500x450 mm horizontal grating	each	746.00
7381	Precast R.C.C. grating with frame 450x100 mm vertical grating	each	350.00
7382	Bitumen emulsion rapid setting (R.S.) confirming to IS : 8887-1995	tonne	39900.00
7385	3 mm thick translucent white acrylic plastic sheet	sqm	641.00
7386	12 thick particle board ceiling tile	sqm	284.00
7388	Dash hold fastener 12.5 mm dia, 40 mm long with 6 mm dia bolt	each	40.00
7389	Anodising 15 microns on aluminium sections	kilogram	32.00
7390	Neoprin/EPDM rubber gasket	metre	37.00
7391	Anodising 25 microns on aluminium sections	kilogram	53.00
7392	Epoxy powder coating 50 microns on aluminium sections.	kilogram	53.00
7393	Polyester powder coating 50 microns on aluminium sections	kilogram	63.00
7394	Double action hydraulic floor spring with stainless steel cover plate	each	1155.00
7395	6 mm dia. G.I. adjustable hangers including clips (upto 1.2 m length)	each	32.00
7396	Double action hydraulic floor spring with brass cover plate	each	1575.00
7397	Base jack	each	231.00
7398	Challies	each	893.00
7399	Cup lock	each	95.00
7400	15 mm PTMT bib cock	each	172.00
7401	15 mm PTMT bib cockwith flange (fancy)	each	237.00
7402	15 mm PTMT bib cock long body with flange	each	286.00
7403	15 mm dia PTMT stop cock (male thread)	each	181.00
7405	20 mm dia. PTMT stop cock	each	202.00
7406	PTMT pillar cock	each	290.00
7407	PTMT push cock 15 mm dia.	each	141.00
7408	PTMT push cock 12 mm dia. 20 mm BSP	each	141.00
7409	PTMT grating 100 mm dia.	each	63.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7410	PTMT pillar cock (fancy) 15 mm foam flow.	each	302.00
7411	125 mm grating withwaste hole	each	71.00
7412	Rectangular type with openable circular lid 150 mm size 18 mm high with 100 mm dia. (110 gm)	each	114.00
7415	Double acting air valve 50 mm	each	3876.00
7416	Double acting air valve 80 mm	each	5692.00
7417	Double acting air valve 100 mm	each	7369.00
7418	Water meter (including testing charges) 80 mm	each	2287.00
7419	Water meter (including testing charges) 100 mm	each	3557.00
7420	Water meter (including testing charges) 150 mm	each	5082.00
7421	Water meter (including testing charges) 200 mm	each	5717.00
7422	Dirt box srainer 80 mm	each	3176.00
7423	Dirt box srainer 100 mm	each	4765.00
7424	Dirt box srainer 150 mm	each	6162.00
7425	Dirt box srainer 200 mm	each	8639.00
7426	Cat's eye	each	578.00
7427	Water stops Serrated with central bulb (225 mm wide, 8-11 mm thick)	metre	420.00
7428	Water stops Dumb bell with central bulb	metre	389.00
7429	Kickers	metre	399.00
7430	Wedge expansion hold fastner 1/4" or 6 mm	each	14.00
7431	Wedge expansion hold fastner 3/8" or 10 mm	each	16.00
7432	Wedge expansion hold fastner 1/2" or 12 mm	each	32.00
7439	8mm thick (mirror polished tiles machine cut edge) Raj Nagar white	sqm	551.00
7442	Wheel 75 mm dia. 40 mm wide	each	65.00
7443	Aluminium single cleat of size 30x32x3	each	14.00
7444	Aluminium grip strip of size 50x12x2	each	11.00
7445	25 mm prelaminated flush door both side decorative	sqm	788.00
7449	Aluminium U beading	kilogram	242.00
7451	Glass sheet (Pin headed) 4 mm thick	sqm	205.00
7452	Raj nagar I quality plain white marble (table rubbed and polished) 18 mm thick upto 1500 cm2	sqm	352.00
7453	Raj nagar plain I quality white marble (table rubbed and polished) 18 mm thick above 1500 and upto 3600 cm2	sqm	525.00
7454	Raj nagar plain I quality white marble (table rubbed and polished) 18 mm thick above 3600 cm2	sqm	630.00
7455	Makrana Adanga 15-18mm upto 1500 cm2	sqm	578.00
7456	Makrana Adanga 15-18mm above 1500 and upto 3600 cm2	sqm	683.00
7457	Makrana Adanga 15-18mm above 3600 cm2	sqm	893.00
7458	Keseriaji/ Abu Ambaji (Green Marble)15-18mm upto 1500 cm2	sqm	578.00
7459	Keseriaji/ Abu Ambaji (Green Marble)15-18mm above 1500 and upto 3600 cm2	sqm	683.00
7460	Keseriaji/ Abu Ambaji (Green Marble) 15-18mm above 3600 cm2	sqm	893.00
7461	Bhaislana,Abu,Kishangarh Zebra (Black Marble) 5-18mm upto 1500 cm2	sqm	446.00
7462	Bhaislana,Abu,Kishangarh Zebra (Black Marble) 15-18mm above 1500 and upto 3600 cm2	sqm	525.00
7463	Bhaislana,Abu,Kishangarh Zebra (Black Marble) 15-18mm above 3600 cm2	sqm	693.00
7466	Second class deodar teak wood lipping 30 mm widthx12mm	metre	34.00
7468	Veneered particle board with commercial veneering on both sides 12 mm thick	sqm	525.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
7477	Prelaminated particle board with one side decorative and other side balancing lamination, flat pressed 3 layer & graded (medium density) Grade I, Type II conforming to IS : 12823 (exterior grade)12 mm thick	sqm	735.00
7478	Prelaminated particle board with one side decorative and other side balancing lamination, flat pressed 3 layer & graded (medium density) Grade I, Type II conforming to IS : 12823 (exterior grade)18 mm thick	sqm	945.00
7479	Prelaminated particle board with one side decorative and other side balancing lamination, flat pressed 3 layer & graded (medium density) Grade I, Type II conforming to IS : 12823 (exterior grade)25 mm thick	sqm	1050.00
7480	Prelaminated particle board with both sides decorative lamination, flat pressed 3 layer & graded (medium density) Grade I, Type II conforming to IS : 12823 (exterior grade)12 mm thick	sqm	788.00
7485	Oxidised M. S. hinges finished with nickel plating 50 mm (Over all width)	metre	37.00
7486	Oxidised M. S. hinges finished with nickel plating 65 mm (Over all width)	metre	42.00
7491	PTMT Waste Coupling 31/32MM	Each	88.00
7492	PTMT Waste Coupling 38/40MM	Each	122.00
7493	PTMT Bottle Trap 31/32MM	Each	491.00
7494	PTMT Bottle Trap 38/40MM	Each	517.00
7495	PTMT Ball Cock 15mm Complete with Epoxy Coated Aluminium Road & H.D. Ball	Each	260.00
7496	PTMT Ball Cock 20mm Complete with Epoxy Coated Aluminium Road & H.D. Ball	Each	384.00
7497	PTMT Ball Cock 25mm Complete with Epoxy Coated Aluminium Road & H.D. Ball	Each	718.00
7498	PTMT Ball Cock 40mm Complete with Epoxy Coated Aluminium Road & H.D. Ball	Each	1197.00
7499	PTMT Ball Cock 50mm Complete with Epoxy Coated Aluminium Road & H.D. Ball	Each	1775.00
7500	PTMT Angle Stop cock with Flenge 15mm	Each	214.00
7501	PTMT Swiveling shower 15mm	Each	187.00
7503	PTMT Liquid Soap Container of 400ml capacity	Each	258.00
7504	PTMT Towel Ring 215xd200x37mm	Each	176.00
7505	PTMT Towel Rail (450MM)	Each	334.00
7506	PTMT Towel Rail (600MM)	Each	401.00
7507	PTMT Shelf 450x124x36mm	Each	454.00
7508	PTMT Urinal Spreader 15MM	Each	200.00
7509	PTMT Soap Dish/Holder 138x102x75mm	Each	176.00
7512	PTMT handle 125x34x24mm	Each	35.00
7513	PTMT handle 150x34x24mm	Each	38.00
7514	PTMT butt hinges 75x60x10mm	Each	46.00
7515	PTMT butt hinges 100x75x10mm	Each	61.00
7516	PTMT Tower bolt 152x42x18mm	Each	77.00
7517	PTMT Tower bolt 202x42x18mm	Each	91.00
7518	PTMT door catcher 72x42mm	Each	30.00
7519	Synthetic Material(PTMT) of approved quality/make size 525 x90 x 75mm	Each	176.00
7520	Synthetic Material(PTMT) of approved quality/make size 675 x90 x 75mm	Each	197.00
7521	Synthetic Material (PTMT) Towel Ring of approved quality/make.	Each	176.00
7552	Coir veneered board 4mm thick	sqm	324.00
7553	Coir veneered board 6mm thick	sqm	427.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7555	Coir veneered board 12mm thick	sqm	759.00
7556	Coir veneered board 18mm thick	sqm	1136.00
7651	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 100mm dia	Metre	836.00
7652	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 150mm dia	Metre	1243.00
7653	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 200mm dia	Metre	1713.00
7654	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 250mm dia	Metre	2214.00
7655	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 300mm dia	Metre	2843.00
7656	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 350mm dia	Metre	3335.00
7657	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 400mm dia	Metre	4781.00
7658	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 450mm dia	Metre	5304.00
7659	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 500mm dia	Metre	7115.00
7660	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 600mm dia	Metre	7972.00
7661	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 700mm dia	Metre	10848.00
7662	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 750mm dia	Metre	12354.00
7663	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 800mm dia	Metre	12705.00
7664	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 900mm dia	Metre	14818.00
7665	Ductile Iron class K - 9 pipe Conforming to I.S. 8329 - 1000mm dia	Metre	16675.00
7666	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 100mm dia	Each	32.00
7668	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 150mm dia	Each	44.00
7669	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 200mm dia	Each	79.00
7670	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 250mm dia	Each	87.00
7671	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 300mm dia	Each	125.00
7672	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 350mm dia	Each	156.00
7673	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 400mm dia	Each	315.00
7674	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 450mm dia	Each	349.00
7675	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 500mm dia	Each	361.00
7676	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 600mm dia	Each	441.00
7677	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 700mm dia	Each	699.00
7678	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 750mm dia	Each	823.00
7679	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 800mm dia	Each	921.00
7680	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 900mm dia	Each	1231.00
7681	Ruber Gaskets Conforming to I.S 5382 of S.B.R quality 1000mm dia	Each	1510.00
7682	Ductile Iron K - 12 specials suitale for push on jointing upto 600mm dia	Quintal	11356.00
7683	Ductile Iron K - 12 specials suitale for push on jointing over 600mm dia	Quintal	17040.00
7684	Ductile Iron specials suitable for mechanical jointing as per I.S. 9523 - upto 600mm dia	Quintal	11943.00
7685	Ductile Iron Specials suitable for mechancial jointing as per I.S. 9523 over 600mm dia	Quintal	18090.00
7686	Ductile Iron Pipe Class K-9 flanges and welding 100mm dia	Metre	2399.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7687	Ductile Iron Pipe Class K-9 flanges and welding 150 dia	metre	3319.00
7688	Ductile Iron Pipe Class K-9 flanges and welding 200mm dia	Metre	4305.00
7689	Ductile Iron Pipe Class K-9 flanges and welding 250mm dia	Metre	5715.00
7690	Ductile Iron Pipe Class K-9 flanges and welding 300mm dia	metre	7354.00
7691	Ductile Iron Pipe Class K-9 flanges and welding 350mm dia	Metre	9160.00
7692	Ductile Iron Pipe Class K-9 flanges and welding 400mm dia	Metre	10942.00
7693	Ductile Iron Pipe Class K-9 flanges and welding 450mm dia	Metre	13229.00
7694	Ductile Iron Pipe Class K-9 flanges and welding 500mm dia	Metre	16040.00
7695	Ductile Iron Pipe Class K-9 flanges and welding 600mm dia	Metre	21757.00
7696	Ductile Iron Pipe Class K-9 flanges and welding 700mm dia	Metre	26839.00
7697	S&S Centrifugally (Spun) C.I. Pipe class 75mm dia (300 cm long)	Each	2515.00
7698	S&S Centrifugally (Spun) C.I. Pipe class 100mm dia (300 cm long)	Each	3092.00
7699	S&S Centrifugally (Spun) C.I. Pipe class 150mm dia (300 cm long)	Each	5649.00
7700	S&S Centrifugally (Spun) C.I. Pipe class LA 200mm dia	Metre	1749.00
7701	S&S Centrifugally (Spun) C.I. Pipe class LA 250mm dia	Metre	2486.00
7702	S&S Centrifugally (Spun) C.I. Pipe class LA 300mm dia	Metre	3306.00
7703	S&S Centrifugally (Spun) C.I. Pipe class LA 350mm dia	metre	4078.00
7704	S&S Centrifugally (Spun) C.I. Pipe class LA 400mm dia	Metre	4994.00
7705	S&S Centrifugally (Spun) C.I. Pipe class LA 450mm dia	Metre	5976.00
7706	S&S Centrifugally (Spun) C.I. Pipe class LA 500mm dia	Metre	7282.00
7707	S&S Centrifugally (Spun) C.I. Pipe class LA 600mm dia	Metre	9619.00
7708	S&S Centrifugally (Spun) C.I. Pipe Specials as per IS 1538 suitable for lead jointing upto 300mm dia	Quintal	4485.00
7709	S&S Centrifugally (Spun) C.I. Pipe Specials as per IS 1538 suitable for lead jointing over 300mm dia	Quintal	5067.00
7710	S&S Centrifugally (Spun) C.I. Pipe specials suitable for mechanical joint as per I.S. 13382 upto 300mm dia	Quintal	6782.00
7711	S&S Centrifugally (Spun) C.I. Pipe Specials suitabe for mechanical joint as per IS 13382 over 300mm dia	Quintal	7090.00
7712	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 100mm dia	Metre	1242.00
7713	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 150mm dia	Metre	1919.00
7714	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 200mm dia	Metre	2658.00
7715	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 250mm dia	Metre	3557.00
7716	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 300mm dia	Metre	4522.00
7717	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 350mm dia	Metre	5784.00
7718	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 400mm dia	metre	7361.00
7719	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 450mm dia	Metre	9689.00
7720	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 500mm dia	metre	12657.00
7721	Screwed double flanged centrigually cast (spun) C.I. Pipe of Class B conforming to I.S. 1536, - 600mm dia	Metre	16506.00
7722	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 100mm dia	Metre	758.00
7723	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 150mm dia	Metre	1106.00
7724	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 200mm dia	Metre	1572.00
7725	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 250mm dia	Metre	2127.00
7726	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 300mm dia	Metre	3010.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7727	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 350mm dia	Metre	3466.00
7728	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 400mm dia	Metre	4137.00
7729	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 450mm dia	Metre	4906.00
7730	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 500mm dia	Metre	5790.00
7731	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 600mm dia	metre	7637.00
7732	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 700mm dia	Metre	9482.00
7733	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 800mm dia	Metre	13220.00
7734	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 900mm dia	metre	16260.00
7735	Ductile Iron Class K- 7 pipe conforming to I.S. 8329 - 1000mm dia	metre	19699.00
7736	Extruded burnt flyash clay sewer bricks conforming to I.S 4885 - 1988	1,000 Nos	4515.00
7737	Fly ash lime bricks (FALG Bricks) conforming to I.S. 12894-1989	1,000 Nos	2520.00
7738	Calcium Silicate Bricks machine moulded confirming to I.S. 4139 - 1989	1,000 Nos	4410.00
7739	Modified Bitumen Refinary produced CRMB - 55	Tonne	51450.00
7741	Modified Bitumen Refinary produced CRMB - 60	tonne	51450.00
7742	Bitumen emulsion medium setting (M.S.) confirming to IS : 8887-1995	tonne	42000.00
7750	Toilet shelf W.V.C. Size 300mm	Each	312.00
7751	Toilet shelf W.V.C. Size 500mm	Each	376.00
7752	Toilet shelf Synthetic Material with tumbler size 470 x 114 x 36mm	Each	315.00
7753	Toilet shelf Hard Plastic of superior quality size 590x140mm	Each	410.00
7754	Toilet shelf Acrylic shelf on C.P. brass casted brackets & guard rail of superior quality size 550x125mm	Each	714.00
7755	Toilet Glass Shelf with edges rounded off Anodised Aluminium angle frame & C.P. brass brackets & guard rail size 600x120mm	Each	263.00
7756	C.P. brass Towel Rail elbow type with concealed screws size 450mm (Heavy duty).	Each	368.00
7757	C.P. brass Towel Rail elbow type with concealed screws size 600 mm (Heavy duty).	Each	420.00
7758	C.P. brass towel rail with brackets 450 x 20mm.	Each	208.00
7759	C.P. brass towel rail with brackets 600 x 20mm.	Each	219.00
7760	WVC (1stquality, ISI Marked) of approved make. Recessed towel Hanger size 108 x 108mm.	Each	63.00
7761	C.P. Brass Towel Ring revolving type	Each	158.00
7799	Broken ceramic tiles	Kg	6.00
7800	Ceramic Glazed Tiles Ist quality size 200 mm x 300 mm in white, gray, ivory, fume red brown, light green, light blue and other light shades	Sq.m.	268.00
7801	Ceramic Mat finished Tiles Ist quality size 300 mm x 300 mm in white, gray, ivory, fume red brown, light green, light blue and other light shades	Sq.m.	315.00
7802	Ceramic Glazed Tiles Ist quality 300 x 300 in all shades designs except White, Ivory, Grey, Fuem Red Brown etc.	Sq.m.	320.00
7803	Rectified Ceramic Glazed Tiles Ist quality 300x300mm or more in all shades designs White, Ivory, Grey, Fuem Red Brown etc.	Sq.m.	538.00
7804	Rectified Ceramic Glazed Tiles Ist quality 300x300mm or more in all shades designs except White, Ivory, Grey, Fuem Red Brown etc.	Sq.m.	662.00
7805	Salem Stainless steel AISI - 304 (18/8) Orrisa pattern W.C. pan 724mm X 578mm	each	4257.00
7806	Salem Stainless steel AISI - 304 (18/8) Round basin 405mm X 355mm	each	1906.00
7807	Salem Stainless steel AISI - 304 (18/8) Wash basin 530mm X 345mm	each	1716.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
7808	Centrifugally cast (spun) iron S&S 100 mm inlet and 100 mm outlet	each	393.00
7809	Centrifugally cast (spun) iron S&S 100 mm inlet and 75 mm outlet	each	424.00
7810	75 mm wide decorative border	Rm	105.00
7811	Motive 200x200 mm	Each	100.00
7812	Motive 200x300 mm	Each	121.00
7813	Motive 300x300 mm	Each	158.00
7814	Motive 300x450 mm	Each	184.00
7815	Motive 300x600 mm	Each	210.00
7825	Ceramic Glazed Vitrified Tiles Size : 600mm x 1200mm	Sqm	929.00
7826	Ceramic Glazed Vitrified Tiles Size : 800mm x 1200mm	Sqm	1233.00
7827	Ceramic Glazed Vitrified Tiles Size : 196mm x 1215mm	Sqm	1232.00
7828	Ist Qulality MAT and glossy finished ceramic tiles Size : 250mm x 375mm	Sqm	288.00
7829	Ist Qulality MAT and glossy finished ceramic tiles Size : 300mm x 450mm	Sqm	356.00
7830	Ist Qulality MAT and glossy finished ceramic tiles Size : 300mm x 600mm	Sqm	428.00
7831	Ist Qulality MAT and glossy finished ceramic tiles Size : 250mm x 1000mm	Sqm	583.00
7832	Ist quality heavy duty Vitrified polished digital tile size:298mmX 298mm	Sqm	393.00
7833	Ist quality heavy duty Vitrified polished digital tile size:300mmX450 mm	Sqm	394.00
7834	Ist quality heavy duty Vitrified polished digital tile size:600mmX600 mm	Sqm	630.00
7850	Agaria White marble slab plain 18mm thick	sqm	1271.00
7857	P.T.M.T. Grating square slit 150mm	each	130.00
7858	P.T.M.T. Urinal cock 15mm dia	each	141.00
7859	P.T.M.T. Bib cock with nozzle 15mm	each	183.00
7861	P.T.M.T. Stop cock (concealed) 15mm	each	250.00
7862	15 mm nominal bore and 30 cm length PVC connection pipe with P.T.M.T. Nuts	each	61.00
7863	15 mm nominal bore and 45 cm length PVC connection pipe with P.T.M.T. Nuts	each	76.00
7864	P.T.M.T. extension nipple 15mm	each	38.00
7865	P.T.M.T. extension nipple 20mm	each	46.00
7866	P.T.M.T. extension nipple 25mm	each	68.00
7900	Modular bricks of class designation 100	1,000 Nos	4158.00
7901	Machine moulded perforated FPS bricks of class designation 125	1,000 Nos	4751.00
7902	Machine moulded modular perforated bricks of class designation 125	1,000 Nos	4704.00
7903	Machine moulded FPS bricks of class designation 125	1,000 Nos	4463.00
7904	Machine moulded tile bricks of class designation 125	1,000 Nos	4200.00
8001	24 mm thick Factory made shutters with frame, rails and panels of PVC extruded sections in white, grey or wooden finish	sqm	1355.00
8002	30 mm thick Factory made shutters with frame, rails and panels of PVC extruded sections in white, grey or wooden finish	sqm	1412.00
8003	Factory made PVC rigid foam panelled shutter	sqm	1695.00
8004	Factory made PVC rigid foam panelled shutter as per IS : 4020	sqm	1811.00
8006	Factory made PVC rigid foam sheet 1mm thick	sqm	165.00
8007	Factory made PVC rigid foam sheet 5mm thick	sqm	698.00
8008	Factory made prelaminated PVC rigid foam sheet 5mm thick	sqm	838.00
8010	48mmX40mmX1.5mm thick Factory made door frame of PVC extruded sections in white, grey or wooden finish	metre	172.00
8011	Factory made door frame PVC extruded sheet i/c carriage	metre	210.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8012	Adhesive solvent cement	kg	116.00
8100	Powder coated M.S. butt hinges 100mm X58mmX1.9mm	10 Nos	89.00
8200	A.P.P. modified polymeric felt (two layers) 1.5 mm thick	sqm	68.00
8201	A.P.P. modified polymeric felt (two layers) 2 mm thick	sqm	126.00
8203	A.P.P. modified 2 mm thick membrance reinforced with glass fibre matt	sqm	221.00
8204	A.P.P. modified 3 mm thick membrance reinforced with glass fibre matt	sqm	284.00
8205	A.P.P. modified 3 mm thick membrance reinforced with polyster matt	sqm	341.00
8206	Bitumen primer for bitumen membrance	litre	95.00
8207	Geotextile 120 gsm membrance	sqm	34.00
8210	Stainless steel screws 50 mm	100 Nos	277.00
8211	Stainless steel screws 40 mm	100 Nos	221.00
8212	Stainless steel screws 30 mm	100 Nos	134.00
8214	Stainless steel screws 20 mm	100 Nos	91.00
8215	Stainless steel butt hinges 125x64x1.9 mm IS : 12817 marked	10 Nos	284.00
8216	Stainless steel butt hinges 100x58x1.9 mm IS : 12817 marked	10 Nos	189.00
8217	Stainless steel butt hinges 75x47x1.8 mm IS : 12817 marked	10 Nos	158.00
8218	Stainless steel butt hinges 50x37x1.5 mm IS : 12817 marked	10 Nos	137.00
8219	Stainless steel butt hinges (heavy weight) 125x64x2.5 mm IS : 12817 marked	10 Nos	441.00
8220	Stainless steel butt hinges (heavy weight) 100x60x2.5 mm IS : 12817 marked	10 Nos	389.00
8221	Stainless steel butt hinges (heavy weight) 75x50x2.5 mm IS : 12817 marked	10 Nos	336.00
8222	M.S. heavy weight but hinges 125x90x4.0mm IS : 1341 marked.	10 10 Nos	299.00
8223	M.S. heavy weight butt hinges 100x75x3.5 mm IS: 1341 marked	10 10 Nos.	153.00
8224	M.S. heavy weight butt hinges 75x60x3.1 mm IS: 1341 marked	10 10 Nos	89.00
8225	M.S. heavy weight butt hinges 50x40x2.5 mm IS : 1341 marked	10 10 Nos	74.00
8300	1216 mm PE-AL-PE Composit pressure pipe	Metre	82.00
8301	1620 mm PE-AL-PE Composit pressure pipe	Metre	92.00
8302	2025 mm PE-AL-PE Composit pressure pipe	Metre	119.00
8303	2532 mm PE-AL-PE Composit pressure pipe	Metre	170.00
8304	3240 mm PE-AL-PE Composit pressure pipe	Metre	219.00
8305	4050 mm PE-AL-PE Composit pressure pipe	Metre	315.00
8501	Polymer modified cementation coating	kilogram	263.00
8502	Fibre glass cloth	sqm	42.00
8504	Multi surface paint	litre	457.00
8505	Acrylic exterior paint	litre	420.00
8506	Premium Acrylic exterior paint	litre	233.00
8507	Textured exterior paint	litre	336.00
8508	Primer for cement paint	litre	158.00
8509	Special Primer	litre	189.00
8510	Metal Primer	litre	126.00
8520	Edge blocks 60 mx2	cum	95.00
8521	**inter–locking block , M-30, 60 mm thick** ,Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	578.00
8522	**inter–locking block , M-30, 60 mm thick**,Category ‘B’ Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	sqm	551.00
8523	**inter–locking block , M-30, 60 mm thick** ,Category ‘C’not denated on any its faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	sqm	525.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8524	**inter–locking block , M-35, 60 mm thick** ,Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	709.00
8525	**inter–locking block , M-35, 60 mm thick** ,Category B Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	683.00
8526	**inter–locking block , M-35, 60 mm thick** ,Category C Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	609.00
8527	**inter–locking block , M-40, 80 mm thick** ,Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	824.00
8528	**inter–locking block , M-40, 80 mm thick** ,Category B Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	798.00
8529	**inter–locking block , M-40, 80 mm thick** ,Category C Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	746.00
8530	**inter–locking block , M-50, 100 mm thick** ,Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	1024.00
8531	**inter–locking block , M-50, 100 mm thick** ,Category B Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	998.00
8532	**inter–locking block , M-50, 100 mm thick** ,Category C Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	945.00
8533	**inter–locking block , M-55, 120 mm thick** ,Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	1213.00
8534	**inter–locking block , M-55, 120 mm thick** ,Category B Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	1187.00
8535	**inter–locking block , M-55, 120 mm thick** ,Category C Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	sqm	1134.00
8536	25 mm thick Non slippery reflective type designer paving tile ,M 40 grade	sqm	788.00
8537	25 mm thick Non slippery reflective type designer paving tile ,M 30 grade	sqm	683.00
8589	Calcium Silicate tegular edged celling tiles	Sqm	901.00
8590	Galvanised Steel main Tee ceiling section Size	Each	126.00
8591	Galvanised Steel perimeter wall Angle Size 24 x	Each	126.00
8592	Galvanised Steel intermediate cross T section	each	126.00
8593	Galvanised Steel intermediate cross T section	each	126.00
8595	Wooden screws with plastic rawl plugs 35mm x8mm	each	126.00
8596	Galvanised iron 8mm Outer Diameter M-6 dash fastener 50 mm long	each	40.00
8611	Main T ceiling sections 24x38x0.3 mm (3 metre long)	each	206.00
8612	Perimeter wall angle 21x21x0.3mm (3 metre long)	each	129.00
8613	Intermediate cross T-section 24x25x0.3mm (1.2 mtrs long)	each	79.00
8614	Intermediate cross T-section 24x25x0.3mm (0.6 mtrs long)	each	37.00
8615	Hanger rod 0.5 mm thick	each	11.00
8616	Adjustment clip	each	5.00
8617	Soffit cleat	each	3.00
8618	Dash fastener 6 mm dia 50 mm long	each	11.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
8619	alvanised iron L-Shape level Adjuster	each	15.00
8620	Vitrified floor tile 50x50 cm	sqm	463.00
8621	Vitrified floor tile 60x60 cm	sqm	566.00
8622	Vitrified floor tile 80x80 cm	sqm	766.00
8623	Vitrified floor tile 100x100 cm	sqm	809.00
8625	Poly propylene- Random - Co - Polymer (PPR) pipes SDR 7.4 - 16 Outer dia	metre	47.00
8626	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 20mm Outer dia.	metre	63.00
8627	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 25mm Outer dia.	metre	105.00
8628	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 32mm Outer dia.	metre	168.00
8629	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 40mm Outer dia.	metre	252.00
8630	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 50mm Outer dia.	metre	389.00
8631	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 63mm Outer dia.	metre	592.00
8632	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 75mm Outer dia.	metre	730.00
8633	Poly propylene - Random - Co - polymer (PPR) pipes SDR 7.4 - 90mm Outer dia.	metre	1166.00
8634	Poly propylene - Random - Co - polymer (PPR) pipes SDR - 11 - 110mm Outer dia.	metre	1222.00
8635	Poly propylene - Random - Co - polymer (PPR) pipes SDR - 11- 160mm Outer dia.	metre	2577.00
8636	Chlorinated Polyvinyl - chloride (CPVC) pipe 15 mm outer dia.	metre	61.00
8637	Chlorinated Polyvinyl - chloride (CPVC) pipe 20 mm outer dia.	metre	76.00
8638	Chlorinated Polyvinyl - chloride (CPVC) pipe 25 mm outer dia.	metre	108.00
8639	Chlorinated Polyvinyl - chloride (CPVC) pipe 32 mm outer dia.	metre	151.00
8640	Chlorinated Polyvinyl - chloride (CPVC) pipe 40 mm outer dia.	metre	237.00
8641	Chlorinated Polyvinyl - chloride (CPVC) pipe 50 mm outer dia.	metre	342.00
8642	Chlorinated Polyvinyl - chloride (CPVC) pipe 62.5mm inner dia.	metre	950.00
8643	Chlorinated Polyvinyl - chloride (CPVC) pipe 75 mm inner dia.	metre	1250.00
8644	Chlorinated Polyvinyl - chloride (CPVC) pipe 100 mm inner dia.	metre	1733.00
8645	Chlorinated Polyvinyl - chloride (CPVC) pipe 150 mm inner dia.	metre	3019.00
8646	Silicon sealant.	cartridge	347.00
8647	Stainless steel screws 30mmx4mm	cent	525.00
8648	Intermediate cross T-section 24x25x0.3mm (0.6 mtrs long)	each	3780.00
8649	Stainless steel (SS 304 grade) adjustable friction window stay. 205 x 19mm	each	168.00
8650	Stainless steel (SS 304 grade) adjustable friction window stay 255 x 19mm	each	186.00
8651	Stainless steel (SS 304 grade) adjustable friction window stay. 355 x 19mm	each	240.00
8652	Stainless steel (SS 304 grade) adjustable friction window stay. 510 x 19mm	each	450.00
8653	Stainless steel (SS 304 grade) adjustable friction window stay. 710 x 19mm	each	826.00
8654	Masking tape.	metre	6.00
8655	Autoclaved aerated cement (AAC) blocks.	cum	2287.00
8656	Gypsum panel 666 X 500 X 100 mm size.	sqm	735.00
8657	Bonding Plaster for Gypsum Panel	kg	84.00
8658	Mechanised Autoclaved flyash lime bricks 100	1000 Nos.	4725.00
8659	Water proof ply 12mm thick.	Sqm.	1187.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8660	Aluminium casement window fastner (Anodised AC 15)	each	74.00
8661	Aluminium casement window fastner (powder coated).	each	80.00
8662	Aluminium casement window fastner (polyester powder coated).	each	95.00
8663	Aluminium round shape handle (anodised AC 15)	each	79.00
8664	Aluminium round shape handle (powder coated)	each	84.00
8665	Aluminium round shape handle (polyester powder coated).	each	84.00
8666	Stainless steel screws 25mm x4mm	cent	131.00
8667	UV stabilised 2 mm thick plain FRP sheet .	sqm	677.00
8668	UV stabilised 3 mm thick plain FRP sheet .	sqm	1015.00
8669	Manglore ridge tiles ....x.......mm 20mm thick.	each	37.00
8670	Manglore tiles ....x.......mm 20mm thick.	each	21.00
8671	Precoated galvanised iron profile sheet 0.50 mm TCT	sqm	394.00
8672	Precoated galvanised steel plain ridges.	metre	400.00
8673	Precoated galvanised steel flashings/aprons.	metre	400.00
8674	Precoated galvanised steel gutter	metre	437.00
8675	Precoated galvanised steel north light curves.	metre	437.00
8676	Precoated galvanised steel barge board.	metre	400.00
8677	Precoated galvanised steel crimp curve	sqm	509.00
8678	1mm thick 35mm wide bright finished stainless steel piano hinges .	metre	100.00
8683	Red sand stone gang saw cut 30mm thick.	sqm	415.00
8684	White sand stone gang saw cut 30mm thick.	sqm	714.00
8685	Delinator	each	777.00
8686	Precast C.C. Kerb stone M - 25	cum	4646.00
8687	Thermoplastic paint	kg	95.00
8688	Glass beads	kg	100.00
8689	Interlocking C.C. paver block (60 mm thick, M-30)	sqm	284.00
8690	High intensity retro - reflective sheet.	sqm	1785.00
8691	Punched tape concertina coil 600 m dia. 10m openable length (Total length 90m)	bundle	814.00
8692	RBT reinforced barbed wire.	metre	11.00
8693	Turn buckle and strengthening bolt.	each set	44.00
8694	Precast pavement slab 450 x 450 x 50mm (M - 30).	each	55.00
8695	Chain link fabric fancing mesh of size 50x50mm made of G.I. wire of dia. 3.15mm.	sqm	189.00
8696	Chain link fabric fancing mesh of size 75x75mm made of G.I. wire of dia. 3.15mm.	sqm	142.00
8697	Chain link fabric fancing mesh of size 100x100mm made of G.I. wire of dia. 3.15mm.	sqm	110.00
8698	Stainless steel cramps with nuts, bolts and washer for dry stone cladding .	each	254.00
8699	8 mm thick tapered edge calcium silicate board .	sqm	326.00
8700	10 mm thick calcium silicate board.	sqm	407.00
8701	Chain link fabric fancing mesh of size 150x150mm made of G.I. wire of dia. 3.15mm.	sqm	97.00
8703	Telescopic drawer channels 300mm long .	set	215.00
8704	Stainless steel roller for sliding arrangment in racks/ cupboards/ cabinets shutter .	each	26.00
8705	50mmX42mmX2mm thick Factory made door frame of PVC extruded sections in white, grey or wooden finish	metre	189.00
8706	25mm thick factory made PVC flash foor shutter i/c carriage.	sqm	2520.00
8707	Factory made glass reinforced plastic door frame 90x45 mm i/c carriage.	metre	410.00
8708	30 mm thick factory madeglass fiber reinforced plastic panel door shutter i/c carriage.	sqm	2940.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8710	Factory made solid PVC door frame 60 x 30mm i/c carriage.	metre	305.00
8711	28mm factory made solid PVC panel door shutteri/c carriage.	sqm	2961.00
8713	Fiber glass reinforced plastic chajja.	sqm	4883.00
8714	Magenatic catcher triple strip verticle type.	each	68.00
8715	Magenatic catcher double strip horizontal type.	each	42.00
8716	Mortice lock of approved make Godrej or equivalent with pair of CP handles	each	761.00
8717	12.5 mm thick Glass fibre reinforced Gypsum board .	sqm	210.00
8718	Almirah lock of approved make Godrej or equivalent	each	210.00
8719	2nd class teak wood lipping/ moulded beadibg or Taj beading of size 18X5mm	metre	28.00
8720	Ceiling sections 0.55 mm thick having a knurled web of 51.55mm and two flanges of 26mm each with lips of 10.55mm.	metre	63.00
8721	Perimeter channel having one flange of 20mm and another flange of 30mm with thickness of 0.55mm and web of length 27mm.	metre	76.00
8722	Nylon sleeves & wooden screws (40mm)	each	3.00
8723	Counter sunk ribbed head screw 25mm.	cent	95.00
8724	12mm thick marine plywood conforming to IS:710	sqm	1365.00
8725	12mm thick fire retardant plywood conforming to IS: 5509.	sqm	1155.00
8726	1.5mm thick decorative laminated sheet	sqm	630.00
8727	1.0mm thick decorative laminated sheet	sqm	473.00
8730	30 mm thick factory made glass fiber reinforced plastic flush door shutter	sqm	2625.00
8731	High polymer modified quickset tile adhesive.	per kg	43.00
8732	synthetic Fiber 6mm / 12mm	per pack of 125 gm	42.00
8733	Welded mesh 50x50x2.1 mm	sqm	170.00
8734	Welded mesh 50x25x2.1 mm	sqm	189.00
8735	Welded mesh 25x25x2.1 mm	sqm	215.00
8736	Expended metal size 20 - 25 mm size 18swg	sqm	194.00
8737	Steel pipe 80x40 16 gauge	Qtl	7277.00
8738	Steel pipe 50x25 18 gauge	Qtl	7350.00
8739	UV stabilised 5 mm thick plain FRP sheet .	sqm	1279.00
8740	Acoustical fine fissured tiles/ Mineral fibre high density tiles	Sqm	641.00
8741	Glass Fibre Reinforced Gypsum False ceiling tiles	Sqm	961.00
8742	Earthen pots	Each	11.00
8743	Disc for floater	Each	6300.00
8744	Trowel blades	Each	6300.00
8745	Filter mats	Each	12705.00
8746	Top mat	Each	37800.00
8747	Concrete Cutter diamond saw	Each	4830.00
8748	Designer polished CC tiles	Sqm	683.00
8749	8mm thick laminated floor	Sqm	1680.00
8750	89 mm wide having both side laminations	Rm	630.00
8751	transaction/adaptation profile	Rm	504.00
8752	Alumium grill (size 7.5mm x 6.0mm and opening of size 102mm x 99mm)	Sqm	630.00
8753	Y & H section	Rm	53.00
8754	Anodising 15 microns on aluminium sections for grill	Sqm	473.00
8755	Powder coating aluminum for grill	Sqm	525.00
8756	Polyester Powder coating for grill	Sqm	630.00
8757	SS wire gauge	Sqm	389.00
8758	Vertical blinds	Sqm	1082.00
8759	Horizontal blinds /Venation blind	Sqm	1082.00
8760	Drapery rod (20 gauge, 25.4 mm diameter)	Rm	189.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8761	Sun control film	sqm	315.00
8762	Polysulphide	Kg	624.00
8763	Primer	Lt	1250.00
8764	Backup rod	Rm	57.00
8765	Masking tape, thiner	Rm	50.00
8766	Door frame RCC Size 75x50 mm	Rm	63.00
8767	Door frame RCC Size 75 x 75mm	Rm	76.00
8768	Door frame RCC Size 100 x 75mm	Rm	82.00
8769	Door frame RCC Size 125 x 75mm	Rm	89.00
8770	Stone Door frame single paitam Size 75 x 60mm	Rm	79.00
8771	Stone Door frame single paitam Size 75 x 75mm	Rm	89.00
8772	Stone Door frame single paitam Size 100 x 60mm	Rm	105.00
8773	Stone Door frame single paitam Size 100x 75mm	Rm	137.00
8774	Stone Door frame single paitam Size 125 x 100mm	Rm	147.00
8775	Stone Door frame Double paitam Size 100x 75mm	Rm	121.00
8776	Stone Door frame Double paitam Size 125 x 100mm	Rm	158.00
8777	PVC sheet 1 mm	sqm	378.00
8778	PVC sheet 1.5 mm	sqm	446.00
8779	PVC sheet 2 mm	sqm	504.00
8780	Melamine polish	Lt	420.00
8781	Melamine polish	Lt	315.00
8782	PW 6250 (PANEL)	Sqm	504.00
8783	PW 928 (PERIMETER SECTION)	Rm	42.00
8784	M.S. PIPE(20x20MM)WITH PRIMER COATING	Rm	44.00
8785	PW 10150 (PANEL)	Sqm	714.00
8786	PW 1228 (PERIMETER SECTION)	Rm	62.00
8800	**Imidacloprid 30.5% EC**	**Ltr.**	**3500.00**
8801	**Termitubes**	**Rmt.**	**10.00**
8802	**Junction Box for antitermite work**	**Each**	**600.00**
8803	**Autoclave aerated blocks of Conforming to Grade-I**	**cum**	**2800.00**
8804	**polymer modified adhesive mortar**	**Kg.**	**13.50**
8805	**Hand Grinder for groove cutting**	**metre**	**300.00**
8806	**Heavy Duty Vitrified Polished Digital tiles 60x60 cm**	**sqm**	**745.00**
8807	**Heavy Duty Vitrified Polished Digital tiles 60x120 cm**	**sqm**	**1220.00**
8808	**Heavy Duty Vitrified glazed MAT tiles 60x60 cm**	**sqm**	**1016.00**
8809	**Heavy Duty Vitrified glazed MAT tiles 60x120 cm**	**sqm**	**1422.00**
8810	**Heavy Duty Vitrified Double Charged tiles 60x60 cm**	**sqm**	**813.00**
8811	**Heavy Duty Vitrified Double Charged tiles 80x80 cm**	**sqm**	**1016.00**
8812	**Heavy Duty Vitrified Double Charged tiles 100x100 cm**	**sqm**	**1151.00**
8813	**19mm thick BWP block board**	**sqm**	**968.00**
8814	**Front side finishing with 4mm thick teak veneer**	**sqm**	**860.00**
8815	**Outer Bidding 35mm x 12mm**	**Rmt.**	**70.00**
8816	**Brass Lock**	**each**	**200.00**
8817	**4.0mm thick veneer**	**sqm**	**800.00**
8818	**6.00mm thick veneer**	**sqm**	**900.00**
8819	**S.S. Rod pipe 30mm**	**Rmt.**	**656.00**
8820	**S.S. Rod pipe 30mm (Bracket)**	**Each**	**100.00**
8821	**Stainless steel (Grade-304)hollow section round/square tubes**	**Kg.**	**250.00**
8822	**Stainless steel bolts/square bar and plates**	**Kg.**	**120.00**
8823	**GI metal Tile Lay-in Plain Tegular edge global white color tiles of size 595x595mm and 0.5mm thick**	**Sqm**	**750.00**
8824	**Main T Ceiling sections 24x38x0.3mm (3 metre long)**	**Each**	**187.00**
8825	**Perimeter wall angle 24x24x0.3mm(3 metre long)**	**Each**	**118.00**

	BASIC RATES OF LABOUR/MATERIALS		
CODE	NAME OF MATERIALS	UNIT	
8826	Intermediate cross T-ection 24x25x0.3mm (1.2 m long) on grid for cut outs	Each	72.00
8827	Intermediate cross T-Section 24x25x0.3 mm (0.6 m long) on grid for cut outs	Each	34.00
8828	Hanger rod 4mm thick	Each	7.00
8829	Adjustment clip 85x30x0.8mm	Each	6.00
8830	Soffit cleat (Size 27x37x25x1.60mm)	Each	3.00
8831	Dash Hold fastener 12.5 mm dia 50 mm long	Each	38.00
8832	GI metal Tile Lay-in Perforated Tegular edge global white color tiles of size 595x595mm and 0.5mm thick	sqm	850.00
8833	Mineral Fibre Anti Bacterial Suspended beveled tegular edge ceiling tiles	Sqm	857.00
8834	G.I. Main Runner 15mm x 32mm 3000 mm Length 0.33 mm thick	Each	177.00
8835	G.I. Cross Tee 15mm x 32mm 1200 mm Length 0.33 mm thick	Each	74.00
8836	G.I. Cross Tee 15mm x 32mm 600 mm Length 0.33 mm thick	Each	37.00
8837	Mineral Fibre Acoustical Suspended beveled tegular edge ceiling tiles	Sqm	786.00
8838	Primer (10 Sqm. P.Ltr in to Coats)	Ltr.	130.00
8839	for putty Choke Mitty	kg	15.00
8840	Linseed oil	Ltr.	230.00
8841	Varnish	Ltr.	240.00
8842	Enamel Paint	Kg	250.00
8843	Valvet touch paint (5 Sqm. P.Ltr. For three coat)	Ltr.	475.00
8844	Primer (20 Sqm. In 1 Ltr. for one coat)	Ltr.	130.00
8845	Primer	Ltr.	260.00
8846	Duco Putty	kg	150.00
8847	Thinner	Ltr.	130.00
8848	Duco Paint	Ltr.	345.00
8849	Hire Charges of Spray Machine	Day	500.00
8850	Core cutting machine	day	15000.00
8851	Nano Technology 5mm Crack Filling High Strength Chemical	Liters	400.00
8852	Nano Technology Highly Hydrophobic Deep Penetrating Primer	Liters	480.00
8853	Fiber Mess Net Pasting on Roof	sqm.	32.00
8854	Nano Technology Oiliphobic Water Proof, Water Based, Breathable Coat	Liters	600.00
8855	Aerogel Green Based Thermal Insulation 1000 Micron Coating With Sri Value 102 or More	Liters	700.00
8856	Nano Technology Anti Static + Anti Stain+ Anti Skid+High Gloss Top Coat	Liters	300.00
8858	Nano Technology Degreasing Stone Cleaner, Non Acidec, Non Alcaline, Non Harmful For Cleaning of Stones, Without Damaging Stones	Liters	350.00
8859	Nano Technology Deep Penetrating Primer Deep Penetrating 2-3 MM Inside.	Liters	350.00
8860	Silicon Dioxide formulated with nano technology sol gel process, water based, UV Protection & Breathable Product- Giving Effects of water proof stone, Breathable, with no change in Stoe color, Absolutely Natural & Food Safe, No Coating visible on stone, only impregnation Material, Upto 5 mm Penetrating power	Liters	1200.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8861	Anti Stain Treatment with nano technology with Transparent matt or high shine protective coat formulated with pure silica, Flexible in Nature for Sealing stone Joints and Making Stones anti stain, Anti Slippery with Absolutely Natural Appearance, Avoides Stone Loosing and Giving Firm Grip to Stone Facades	Liters	700.00
8862	Dow coarning 789 Weather proofing silicon	cartage	100.00
8863	Silicon Structural adhesive	Mtr.	250.00
8864	Toughened Glass 6mm thick	Sqm	1400.00
8865	sundries Backup Rod, Masking tape, spacer tape etc.	L.S.	1.25
8866	Polycrete wall 125x125 (main member pillar)	Mtr.	645.00
8867	Polycrete wall 20x305 (Horizontal panel)	Mtr.	345.00
8868	Alumium grill section of size 7.5mm x 6.0mm and opening of size 75mmx72mm	Sqm	650.00
8869	Alumium grill section of size 7.5mm x 6.0mm and opening of size 50mmx48mm	Sqm	700.00
8870	Solution Required for Dry Cleaning of sofa sets	Ltr.	900.00
8871	Vacume Cleaner Shampoing/Drycleaning Machine	Hr.	12.00
8872	Tapestry cloth of Sofa set	Mtr.	300.00
8873	100mm thick Joint less P.U. foam	sqm	1260.00
8874	50mm thick Joint less P.U. foam	sqm	630.00
8875	25mm thick Joint less P.U. foam	sqm	315.00
8876	Cost of jata / Dori	L.S	1.25
8877	Cost of Sofa Wooden Leg	Each	75.00
8878	Synthetic Carpet Carpet including 5% Wastage	P.Sqm.	320.00
8879	Chiken wire mesh	P.Sqm.	40.00
8880	Premium Quality white glazed vitreous china W.C. orissa pan (IS:2556 Mark) Size 580x440 mm.	Each	1650.00
8881	Premium Quality white glazed vitreous china Universal (anglo-Indian type) WC Pan	Each	4590.00
8882	Premium Quality WVC Wall Hunged WC Pan	Each	6790.00
8883	Double Syphonic European W.C. with mounted W.V.C. flushing cisternand seat cover (complete set)	Each	13970.00
8884	premium qulaity WVC Urinal Flat Back (small) size 440x265x315 mm.	Each	1320.00
8885	premium qulaity WVC Urinal Flat Back (large) or half stall size 590x375x390 mm.	Each	4290.00
8886	premium qulaity W.V.C. Urinal Partition plate of size 630x315 mm	Each	1990.00
8887	premium qulaity WVC Wash basin Size 550 mm x 400 mm	Each	1340.00
8888	premium qulaity WVC Wash basin with C.I. brackets	Each	25.00
8889	premium qulaity WVC Wash basin Size 550 mm x 400 mm dia for counter top.	Each	2190.00
8890	Premium quality water closet Seat Cover Solid PVC (ISI marked) grade-I White for EWC.	Each	580.00
8891	Premium quality Low level Flushing Cistern of 10 litres capacity PVC with PVC bend as per IS : 7231	Each	1340.00
8892	Ornamental frame size 50x12 with 6mm Ply Board on Back Side	(3'+2')x2	40.00
8893	looking Mirror 5mm thick	(3'x2')	60.00
8894	Water Proof Ply	(3'x2')	28.00
8895	Hydraulic Seat Cover for Europeon WC	each	3240.00
8896	chair for wall hung W.C.	each	1200.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
8897	**Fixing Concealed cistern with Floor mounting frame and flush plate**	**each**	**6200.00**
8898	**New spindle for Bib cock/pillar cock**	**each**	**60.00**
8899	**Toilet corner Glass Shelf size 225x225mm with C.P. brass brackets**	**each**	**1200.00**
8900	**premium quality waste coupling 32mm half thread**	**Each**	**400.00**
8901	**Premium quality Bottle trap internal portion 32mm, size 250mm long**	**Each**	**1300.00**
8902	**Premium Towel rack 600mm long 260mm wide with lower hangers**	**Each**	**2450.00**
8903	**Premium quality Angular stop cock with wall flange**	**Each**	**1100.00**
8904	**Premium quality Two way Bib cock with wall flange**	**Each**	**1750.00**
8905	**PVC Tank Cover 300 Lt. capacity**	**Each**	**110.00**
8906	**PVC Tank Cover 500 Lt. capacity**	**Each**	**120.00**
8907	**PVC Tank Cover Above 500 Lt. capacity**	**Each**	**130.00**
8908	**C.I. Saddle pieces 100mm dia**	**Each**	**110.00**
8909	**Sluice valve 100mm dia**	**Each**	**3800.00**
8910	**DI Pipe (Ductile iron pipe) of 100 mm dia**	**Meter**	**705.00**
9000	Mechanised Autoclaved flyash lime bricks 75	1,000 Nos	4410.00
9001	uPVC Casement Window Frame 60x60mm with EPDM & 1.2mm thk G.I Reinforcement (BR60DA)	Meter	420.00
9004	uPVC Casement Window Fix Mullion 76x60mm with EPDM & 1.2mm thk G.I Reinforcement (SE76NDA)	Meter	499.00
9005	uPVC Single Glazing Bead 34x20 with EPDM (GB34DA)	Meter	63.00
9022	Multipoint Transmission Gear (ESPAG) with white finish powder coated handle	Per. Pc.	635.00
9023	5mm Thick Float Glass (ISI make)	Sq. Mtr.	509.00
9050	Labour & Manufacturing Charges (Cutting, Drilling, Welding, Trimming, Cleaning, Glass & Hardware Fixing & Packing)	Sq. Mtr.	572.00
9051	uPVC Window & Door Site Installation Labour (Unloading, assembling & fixing, silicon applicaton & cleaning)	Sq. Mtr.	347.00
9052	Bectokill chemical	Ltr	1549.00
9053	UV tube	Each	7980.00
9054	6 mm thick tapered edge calcium silicate board/tile	sqm	244.00
9055	6 mm thick calcium silicate perforated tile/board	sqm	420.00
9056	8 mm thick calcium silicate perforated tile/board	sqm	536.00
9100	C.P. Brass flush Valve Economy model 25 mm.	each	1304.00
9101	C.P. Brass flush Valve Economy model 32 mm.	each	1433.00
9102	C.P. Pillar Cock 15 mm nominal size	each	624.00
9103	Vitreous china flat back wash basin Size 510 mm x 440 mm	each	831.00
9104	Vitreous china flat back wash basin Size 580 mm x 430 mm	each	1000.00
9105	Vitreous china flat back wash basin Size 610 mm x 510 mm	each	1605.00
9106	Vitreous china flat back wash basin Size 400 mm x 400 mm corner	each	751.00
9107	Vitreous china counter top wash basin Size 410 dia	each	1271.00
9108	Vitreous china counter top wash basin Size 550 mm x 400 mm dia	each	1271.00
9109	Vitreous china counter top wash basin Size 630 mm x 500 mm dia	each	1409.00
9110	terrazzo (Mosaic) wash hand basin size 450 mm x 280 mm.	each	318.00
9111	terrazzo (Mosaic) wash hand basin size 550 mm x 400 mm.	each	381.00
9112	Bottle Trap C.P. Brass 32 mm	each	706.00
9113	Bottle Trap C.P. Brass 40 mm	each	813.00
9114	Bottle Trap Synthetic Material (PTMT) 32 mm.	each	413.00
9115	Bottle Trap Synthetic Material (PTMT) 40 mm.	each	466.00
9116	Bottle Trap PVC	each	98.00
9117	Grating Stainless Steel Sheet size 125mm dia.	each	47.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
9118	Grating Stainless Steel Sheet size 125mm dia. Heavy Quality	each	79.00
9119	Grating Synthetic Material(PTMT) of approved quality/make size 125mm dia	each	74.00
9120	Grating Synthetic Material(PTMT) of approved quality/make size 150mm dia	each	79.00
9121	Grating Square 150 x 150 x 8mm. (Anti Cockroach)	each	288.00
9122	Grating C.P. brass with frame (Heavy) & superior quality size 125mm dia.	each	101.00
9123	Grating C.P. brass with frame Light & superior quality size 125mm dia	each	47.00
9124	Grating Cast Iron.	each	63.00
9125	Liquid Soap Container C.P. brass	each	399.00
9126	Liquid Soap Container Synthetic Material	each	147.00
9127	Tooth Paste & Brush Holder C.P. brass	each	147.00
9128	Tooth Paste & Brush Holder C.P. brass (Heavy and Superior quality.)	each	322.00
9129	Tooth Paste & Brush Holder Stainless Steel	each	257.00
9130	Toilet Paper Holder with rod C.P. brass	each	257.00
9131	Toilet Paper Holder with rod C.P. brass heavy & Superior quality.	each	462.00
9132	Toilet Paper Holder with rod WVC (1stquality, ISI Marked) recessed size 150x150mm.	each	185.00
9133	Toilet Paper Holder with rod WVC (1stquality, ISI Marked) size 150x150mm .Exposed	each	143.00
9134	Soap Dish or Tray C.P. brass	each	252.00
9135	Soap Dish or Tray C.P. brass heavy and superior quality.	each	147.00
9136	Soap Dish or Tray WVC (1st Quality, ISI Marked) recessed size 150 x 150 mm.	each	171.00
9137	Soap Dish or Tray WVC (1st Quality, ISI Marked) recessed size 200 x100mm.	each	200.00
9138	Soap Tray WVC (1st Quality, ISI Marked) size 200 x100mm.	each	147.00
9139	Soap Dish WVC (1st Quality, ISI Marked) size 380mm.	each	200.00
9140	Twin- peg C.P. brass	each	58.00
9141	Twin- peg C.P. brass (heavy)Superior quality	each	110.00
9142	Twin- peg Acrylic	each	40.00
9151	Shower arm 15mm dia C.P.brass heavy & superior quality 150mm Projection.	each	158.00
9152	Shower arm 15mm dia C.P.brass heavy & superior quality 220mm Projection.	each	252.00
9153	Shower arm 15mm dia C.P.brass heavy & superior quality 300mm Projection.	each	299.00
9154	Shower arm 15mm dia C.P.brass heavy & superior quality 370mm Projection.	each	315.00
9155	Bath Shower C.P. brass fixed.	each	126.00
9156	Bath Shower C.P. brass of Heavy & superior quality 150mm.	each	294.00
9157	Bath Shower C.P. brass of Heavy & superior quality, revolving with adjustable key 150mm.	each	415.00
9158	Extension Pipe for bib cock C.P. brass of Heavy & Superior quality 50x15mm	each	63.00
9159	Extension Pipe for bib cock C.P. brass of Heavy & Superior quality 100x15mm	each	95.00
9160	Extension Pipe for bib cock C.P. brass of Heavy & Superior quality 150x15mm	each	142.00
9161	Acrylic Toilet accessories set consisting of Towel rail, shelf, towel ring, soap dish, paste & brush holder & twin peg	each	1155.00
9162	Jet spray for water closet with C.P. Copper Tube flange	each	347.00
9163	C.P.flange for 15mm dia taps.	each	16.00
9164	C.P. Health Faucet with 1Mtr. Long Tube & Hook	each	578.00

	BASIC RATES OF LABOUR/MATERIALS		
CODE	**NAME OF MATERIALS**	**UNIT**	
9165	C.P. brass Urinal Spreader for Large Urinal of heavy duty	each	567.00
9166	P.V.C. Cistern fittings (Syphon, Ball cock & Handle Set)	Set	231.00
9996	Carriage & sundries	L.S.	1.25
9997	Water charge		1.00%
9998	Contractor profit & overhead		10.00%
9999	Sundries	L.S.	1.25

CHAPTER : G-4

CARRIAGE OF MATERIALS

NOTE

1 Rates are for net quantities after deduction of voids
2 Part of Km beyond 1 Km. will be treated as under :
1.499 = 1.00 Km. ; 1.500 Km. = 2.00 Km
3 The rates are inclusive of loading and unloading
4 The rates are inclusive of stacking. If stacking is not required, no deduction is to be done.

Ch Code No.	Description	Unit	Rates										
			Up to 50 M	Add for each 50 M (upto 500 M)	For 500 M (0.5 Km.)	For 1 Km.	Add for each 1 Km beyond Ist Km (upto 5 Km.)	For 5 Km.	Add for each 1 Km beyond 5 Km (upto 10 Km.)	For 10 Km.	Add for each 1 Km beyond 10 Km (upto 20 Km.)	For 20 Km.	Add for each 1 Km beyond 20 Km
1	2	3	4	5	6	7	8	9	10	11	12	13	14
4.1	Earth, Sand, Lime, Morrum manure or sludge	Cum	23.00	3.50	54.00	64.00	7.00	92.00	6.50	124.00	6.00	181.00	5.40
4.2	Building Rubbish Stone metal (Grit and ballast etc.)	Cum	30.00	4.70	72.00	84.00	7.00	111.00	6.50	143.00	6.00	201.00	5.40
4.3	Stone for Masonry work & soling	Cum	44.00	5.40	90.00	105.00	7.00	132.00	6.50	164.00	6.00	221.00	5.40
4.4	Excavated rock	Cum	44.00	5.40	90.00	105.00	7.00	132.00	6.50	164.00	6.00	221.00	5.40
4.5	Bricks	1000 Nos	89.00	12.00	193.00	227.00	14.00	282.00	13.00	345.00	12.00	460.00	8.40
4.6	Brick Tiles	1000 Nos	36.00	7.00	98.00	115.00	13.00	167.00	13.00	230.00	12.00	345.00	7.00
4.7	Cement, Stone blocks pipes and other heavy materials	MT	25.00	3.50	56.00	66.00	8.40	98.00	8.40	138.00	7.00	207.00	4.20
4.8	Steel, Tar bitumen , Timber and steam coal	MT	23.00	3.00	50.00	59.00	13.00	110.00	13.00	173.00	12.00	288.00	4.80
4.9	Empty bitumen drums	10 Nos	38.00	3.00	65.00	77.00	4.80	95.00	4.20	115.00	3.50	150.00	2.40
4.10	**Carriage with care Precast cement concrete blocks like Dand & kerbs etc weighing**												
4.10.1	Upto 50 Kg.	Each	--	--	3.95	4.60	0.40	6.00	0.40	8.00	0.35	11.00	0.35
4.10.2	51 to 70 Kg.	Each	--	--	6.60	7.60	0.40	9.00	0.45	11.00	0.35	15.00	0.35
4.10.3	71 to 100 Kg.	Each	--	--	8.00	9.00	0.50	11.00	0.50	14.00	0.35	17.00	0.35
4.11	**R.C.C. Hume Pipe with collar**												
4.11.1	Dia 300 mm	Mtr.	--	--	24.00	28.00	0.70	31.00	0.60	33.00	0.35	37.00	0.35
4.11.2	Dia 600 mm	Mtr.	--	--	61.00	72.00	2.00	80.00	1.80	88.00	1.40	102.00	0.95
4.11.3	Dia 900 mm	Mtr.	--	--	127.00	154.00	550.00	176.00	4.80	198.00	3.50	231.00	3.00
4.11.4	Dia 1200mm	Mtr.	--	--	193.00	231.00	550.00	253.00	4.80	275.00	3.50	308.00	3.00

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	**4**
4.12	Loading in or unloading cement of the Railway wagons at siding and carrying the same from or into godowns adjacent to the siding, including stacking the same properly in rows upto any height as per direction of Engineer-in Charge, sweeping the wagons and screening the swept cement and filling in bags complete.	M.T.	35.00
4.13	Loading in or unloading from the Railway wagons and stacking as per the direction of Engineer-in-Charge		
4.13.1	Steel	M.T	44.00
4.13.2	G.I., C.I. R.C.C. or C.C. pipes upto 500mm dia and similar heavy materials.	M.T	35.00
4.13.3	Full bitumen drums	Each	12.00
4.13.4	Heavy materials where each piece or bundle, crate or case weight more than one tonne and R.C.C. C.I. and concrete pipes above 500 mm dia	M.T	45.00
Remark	The rates will be applicable in all cases whether material are unloaded on or loaded from railway siding or directly unloaded on or loaded from transport No deduction shall be made from carriage rates for such direct unloading or loading		
4.14	Extra for sorting the steel size-wise inside the store yard and stacking the same for measurement within the lead of 100 meters as directed by the Engineer-in-charge	M.T	24.00

CHAPTER G-5
HIRE CHARGES OF TENT ARTICLES

NOTE : Hire charges includes transporatation at site within Muncipal Limits, Loading, Unloading, Placing and removing complete.

		Final BSR Rate 2019			
Chapter Code No.	**Description**	**Unit**	**Cost (Rs.)**	**Rate (Rs.)**	
				1st Day	**Subsequent days (per Day)**
1	**2**	**3**	**4**	**5**	**6**
	Chairs :				
5.1	PVC chair	Each.	500.00	9.00	2.00
5.2	Steel pipe pedded chair	Each.	1200.00	30.00	10.00
	Tables :				
5.3	Wooden Table 6' x 3' (Babool).	Each.	800.00	25.00	7.00
5.4	Wooden Table **6' x 2'** (Babool).	Each.	750.00	30.00	8.00
5.5	Centre Table sunmica Top 2'x4'	Each.	1800.00	50.00	15.00
5.6	Peg Table Sunmica Top 1'x1'	Each.	800.00	15.00	5.00
	Floor Covering :				
5.7	Cotton Durry (Old).	Sqm.	18.00	2.00	0.60
5.8	Cotton Durry (New).	Sqm.	30.00	3.00	1.00
5.9.1	Non woven synthtic carpet (Red/Green) (New)	Sqm.	160.00	25.00	8.00
5.9.2	Non woven synthtic carpet (Red/Green) (old)	Sqm.	160.00	15.00	5.00
5.10.1	Woolen Carpet (New).	Sqm.	600.00	35.00	12.00

Chapter Code No.	Description	Unit	Cost (Rs.)	Rate (Rs.)	
				1st Day	Subsequent days (per Day)
5.10.2	Woolen Carpet (old).	Sqm.	600.00	24.00	8.00
5.11	White Cotton chader (Sheet) (Washed)	Sqm.	40.00	20.00	1.50
5.12	Pipe Pandal with covering ht **from** to 12 to 18 ft	Sqm		30.00	10.00
5.13	Curtain cloth (View cutter) made out of steel pipe frame covering with good quality white cloth to cut view including making holes and made necessary strengthening support against wind pressure complete .	Sqm		25.00	7.00
5.14	Kanat fixed with Tent. **(15'x6')**	Each.	850.00	40.00	15.00
5.15	Kanat fixed without Tent. **(15'x6')**	Each.	850.00	50.00	10.00
5.16	5mm Galvanised Iron (G.I) pipe 6 mtr. long with new silk flag.	Each.		72.00	33.00
5.17	Water Proof Tripal.	Sqm.		10.00	4.00
5.18	Wooden stage height 60cm.with steps complete	Sqm.		48.00	12.00
5.19	Wooden stage height 120 to 180 cm.with steps made of 50 mm dia iron pipe frame with adjustable height system and having smooth top wooden board complete .	Sqm.		150.00	50.00
5.20	Cotton mattress with white sheet (chader).	Each		24.00	10.00
5.21	Dunlop mattress with white chadar size 6'x3'.	Each.		40.00	12.00
5.22	Massand with white cover.	Each.		14.00	5.00
5.23	Speech Stand having Sunmica top.	Each.	2850.00	150.00	60.00
5.24	white cotton cloth cover for padded chair inculding ribbon	Each.		10.00	4.00

Chapter Code No.	Description	Unit	Cost (Rs.)	Rate (Rs.)	
				1st Day	Subsequent days (per Day)
	Barricading				
5.25.1	Barricading with Sal ballies as per design including tying with vertical post by coconut strings including digging out holes in all types of soil complete in all respect with two horizontal members height 1.2 m upto 1.5m above ground level and vertical supports upto 2.5 centre and upto 3 days period including removal and cleaning the site complete in all respect & including dressing of sides in all types of soil.	Rmt			30.00
5.25.2	**Barricading with M.S. Pipe as per design including tying with vertical post including digging out holes in all types of soil complete in all respect with two horizontal members height 1.2 m upto 1.5m above ground level and vertical supports upto 2.5 centre and upto 3 days period including removal and cleaning the site complete in all respect & including dressing of sides in all types of soil.**	**Rmt.**			**25.00**
5.26	Add extra for item No. **5.25.1 & 5.25.2** in barricading for each subsequent day after 3 days	Rmt			1.80
5.27	Making of holes in cement concrete/ Bitumenous road by power driven hammer for barricating work as per direction of Engineer in charge	Each			170.00
5.28	Hire charges for folded type two seater sofa set (cum bed) with good quality white cloth cover	Each		240.00	80.00
5.29	VIP Chair **with** cushion	Each		100.00	40.00
5.30	Add extra for hire charge for providing welded mesh size 25x25 mm (16SWG) over barricating (excluding hire charge of barricating)	Sqm		30.00	6.00
5.31	Water Proof Tent with pipe pandal supported with Bamboo/Balli **/MS Pipe frame.**	Sqm		90.00	30.00
5.32	**Water proofing canopy with ornamental frill, inner ceiling & curtains**				
5.32 .1	**Upto size 20'x20'**	**Sqm**		**175.00**	**50.00**
5.32 .2	**Above size 20'x20'**	**Sqm**		**250.00**	**80.00**

Chapter Code No.	Description	Unit	Cost (Rs.)	Rate (Rs.)	
				1st Day	Subsequent days (per Day)
5.33	Deluxe sofa				
5.33.1	Three Seater	Each		700.00	200.00
5.33.2	Two Seater	Each		500.00	150.00
5.33.3	Single Seater	Each		300.00	100.00
5.34	Chemical toilet with house keeping arrangement and providing liquid soap, tissues, comb, freshener etc.	Each		6000.00	1500.00
5.35	Water proof dome with truss steel strcture, inner ceiling & curtains				
5.35.1	60' span	Sqm		225.00	22.00
5.35.2	80' span	Sqm		250.00	25.00
5.35.3	90' span	Sqm		260.00	28.00

CHAPTER B-1
EARTH WORK

Note : The items in this Chapter will cover Earth Work for Building and Sanitary Work.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
1.1	Earth work in surface excavation not exceeding 30cm. in depth but exceeding 1.5 meter in width as well as10 Sqm on plan including disposal of excavated earth up to 50 Mtr. and lift up to 1.5 Mtr. disposed soil to be levelled and neatly dressed: All kinds of soil	Sqm	32.00
1.2	Earth work in rough excavation, banking excavated earth in layers not exceeding 20cm in depth, breaking clods watering, rolling each layer with 1/2 tonne roller, or stone or steel rammers and rolling every 3rd and top- most layer with power roller of minimum 8-10 tonne capacity and dressing up in embankment for roads, flood banks marginal banks and guide banks or filling up ground depressions, lead up to 50 Mtr. and lift up to 1.5 Mtr.: All kinds of soil	Cum	247.00
1.3	Banking excavated earth in layers not exceeding 20cm. in depth breaking clods watering, rolling each layer with 1/2 tonne roller, or stone or steel rammers, and rolling every 3rd and top- most layer with power roller of 8-10 tonne capacity and dressing up in embankments for roads, flood banks marginal banks, and guide banks etc. lead up to 50 Mtr. and lift up to 1.5 Mtr. All kinds of soils	Cum	155.00
1.4	Deduct for not rolling with power roller of 8 to 10 tonne capacity for banking excavated earth in layers not exceeding 20 cm in depth	Cum	2.60
1.5	Deduct for not watering the excavated earth for banking.	Cum	10.50
1.6	Earth work in excavation by mechanical means (Hydraulic excavator)/ manual means over areas (exceeding 30cm in depth. 1.5m in width as well as 10 **sqm** on plan) including disposal of excavated earth, lead upto 50m and lift upto 1.5 m , disposed earth to be levelled and neatly dressed: All kinds of soil	Cum	159.00
1.7	Earth work in excavation/ by mechanical means (Hydraulic Excavator)/ manual means over areas (exceeding 30 cm in depth, 1.5m in width as well as 10 **sqm** on plan) including disposal of excavated earth, lead upto 50 m and lift upto 1.5 m , disposed earth to be levelled and neatly dressed:		
1.7.1	Ordinary rock	Cum	228.00
1.7.2	Hard rock (requiring blasting) useful material 30%	Cum	373.00
1.7.3	Hard rock(blasting prohibited) useful material 30%	Cum	491.00

1.8	Earth work in excavation by mechanical means (Hydraulic Excavator)/ manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 **sqm** on plan) including dressing of sides and ramming of bottoms, lift upto 1.5 m, including taking out the excavated soil and depositing and refilling of jhiri with watering & ramming and disposal of surplus excavated soil as directed with in a lead of 50 meter. All kinds of soils	Cum	162.00
1.9	Excavation work by mechanical means (Hydraulic excavator) / manual means in foundation trenches or drains not exceeding 1.5 m in width or 10 **sqm** on plan including dressing of sides and ramming of bottoms lift upto 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed with a lead in a 50 meter including stacking of useful material if available :		
1.9.1	Ordinary rock	Cum	244.00
1.9.2	Hard rock (requiring blasting) useful material 30%	Cum	404.00
1.9.3	Hard rock(blasting prohibited) useful material 30%	Cum	501.00
1.10	Excavating trenches of required width for pipe cables, etc. including excavation for sockets, and dressing of sides, ramming of bottoms, depth up to 1.5 Mtr. including taking out the excavated soil, and then returning the soil as required in layers not exceeding 20cm in depth including consolidating each deposited layer by ramming, watering etc. and disposal of surplus excavated soil as directed within a lead of 50 Mtr.: All kinds of soil		
1.10.1	Pipes cables etc. not exceeding 80mm dia	Mtr.	105.00
1.10.2	Pipes cables, etc. exceeding 80mm dia but not exceeding 300mm dia.	Mtr.	170.00
1.10.3	Pipes, cables etc. exceeding 300mm dia but not exceeding 600mm dia.	Mtr.	267.00
1.11	Add extra for excavating trench for pipes, cables etc. in all kinds of soil for depth exceeding 1.5 Mtr. but not exceeding 3 Mtr.	Mtr.	1.41
1.12	Add extra for excavating trench for pipes, cables etc. in all kinds of soil for depth exceeding 3 Mtr. but not exceeding 4.5 Mtr.	Mtr.	3.61
1.13	Excavating trenches of required width for pipes cables etc. including excavation for sockets, depth up to 1.5 Mtr. including taking out the excavated materials, refilling the soil as required in layers not exceeding 20cm in depth including consolidating each deposited layers by ramming, watering etc., stacking serviceable material for measurements and disposal of unserviceable material as directed, with in a lead of 50 Mtr.		
1.13.1	ORDINARY ROCK: -		
1.13.1.1	Pipes cables etc. not exceeding 80mm dia	Mtr.	147.00
1.13.1.2	Pipes cables, etc. exceeding 80mm dia but not exceeding 300mm dia.	Mtr.	363.00
1.13.1.3	Pipes, cables etc. exceeding 300mm dia but not exceeding 600mm dia.	Mtr.	416.00
1.13.2	HARD ROCK (REQUIRING BLASTING) USEFUL MATERIAL 30% OF VOLUME EXCAVATED		
1.13.2.1	Pipes cables etc. not exceeding 80mm dia	Mtr.	221.00

1.13.2.2	Pipes cables, etc. exceeding 80mm dia but not exceeding 300mm dia.	Mtr.	549.00
1.13.2.3	Pipes, cables etc. exceeding 300mm dia but not exceeding 600mm dia.	Mtr.	633.00
1.13.3	HARD ROCK (BLASTING PROHIBITED) USEFUL MATERIAL 30% OF VOLUME EXCAVATED		
1.13.3.1	Pipes cables etc. not exceeding 80mm dia	Mtr.	268.00
1.13.3.2	Pipes cables, etc. exceeding 80mm dia but not exceeding 300mm dia.	Mtr.	664.00
1.13.3.3	Pipes, cables etc. exceeding 300mm dia but not exceeding 600mm dia.	Mtr.	766.00
1.14	Add extra for excavating trenches for pipes, cables, etc. in ordinary/hard rock exceeding 1.5 m in depth but not exceeding 3 m. (Rate is over corresponding basic item for depth upto 1.5 metre.)	Mtr.	1.10
1.15	Add extra for excavating trenches for pipes, cables, etc. in ordinary/hard rock exceeding 3m in depth but not exceeding 4.5 m. (Rate is over corresponding basic item for depth upto 1.5 metre.)	Mtr.	2.74
1.16	P & F Close timbering in trenches including strutting, shoring and packing cavities (wherever required) etc. Complete (Measurement to be taken of the face area timbered):		
1.16.1	Depth not exceeding 1.5 Mtr.	Sqm	86.00
1.16.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	90.00
1.16.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	99.00
1.17	P & F Close timbering in case of shafts, wells, soak pits, man-holes and the like including strutting, shoring and packing cavities (wherever required) etc complete (Measurement to be taken of the face area timbered):		
1.17.1	Depth not exceeding 1.5 Mtr.	Sqm	90.00
1.17.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	97.00
1.17.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	107.00
1.18	P & F Close timbering over area including strutting, shoring and packing cavities (wherever required) etc. complete (Measurement to be taken of the face area timbered):		
1.18.1	Depth not exceeding 1.5 Mtr.	Sqm	76.00
1.18.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	81.00
1.18.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	87.00
1.19	Add extra for planking, strutting and packing material for cavities (in close timbering) if required to be left permanently in position (face area of timber permanently left to be measured)	Sqm	1140.00
1.20	P & F of Open timbering in trenches including strutting, shoring complete (measurements to be taken of the face area timbered)		
1.20.1	Depth not exceeding 1.5 Mtr.	Sqm	43.00
1.20.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	45.00
1.20.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	49.00
1.21	P & F of Open timbering in case of shafts, wells, soak pits, manholes and the like including strutting, shoring complete (Measurement to be taken of the face are timbered):		
1.21.1	Depth not exceeding 1.5 Mtr.	Sqm	37.00

1.21.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	42.00
1.21.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	48.00
1.22	P & F Open timbering over area including strutting, shoring etc complete (Measurement to be taken of the face are timbered):		
1.22.1	Depth not exceeding 1.5 Mtr.	Sqm	27.00
1.22.2	Depth exceeding 1.5 Mtr. but not exceeding 3 Mtr	Sqm	29.00
1.22.3	Depth exceeding 3 Mtr. but not exceeding 4.5 Mtr	Sqm	35.00
1.23	Add extra for planking and strutting in open timbering if required to be left permanently in position (face area of timber permanently left to be measured)	Sqm	572.00
1.24	Add extra rate for quantities of works, executed :		
1.24.1	In or under water and /or liquid mud, including pumping out water as required	P.Mtr. depth	0.20
1.24.2	In or under foul position , including pumping out water as required	P.Mtr. depth	0.25
	Note for item no. 1.24:- The extra percentage rate is applicable in respect of each item but limited to quantities of work executed in these difficult conditions. The unit, namely, metre depth, to be considered for payment, shall be the depth measured from the sub soil water level up to the centre of gravity of the qty executed in difficult condition. The depth shall be reckoned correct to 0.10m, 0.05m or more shall be taken as 0.10m and less than 0.05m ignorel.		
1.25	Filling available excavated earth (excluding rock) in trenches, plinth side of foundation etc. in layers not exceeding 20 cm. in depth, consolidating each deposited layer by ramming and watering including lead up to 50 meter and with all lift.	Cum	58.00
1.26	Add extra for foundation/trenches/drains for every additional lift of 1.5 Mtr. or part thereof in		
1.26.1	**All** kinds of soil	Cum	31.00
1.26.2	Ordinary or hard rock.	Cum	55.00
1.27	Supplying and Filling in plinth with river sand under floors including watering ramming consolidating and dressing complete including cost of sand	Cum	931.00
1.28	Surface dressing of the ground including removing vegetation and inequalities not exceeding 15cm deep and disposal of rubbish lead up to 50 Mtr. and lift up to 1.5 Mtr.: All kinds of soil	Sqm	7.80
1.29	Ploughing the existing ground to a depth of 15 cm to 25 cm and watering the same: All kinds of soil	Sqm	7.80
1.30	Excavating holes upto 0.5 cum including getting out the excavated soil, then returning the soil as required in layers not exceeding 20 cm in depth, including consolidating each deposited layer by ramming, watering etc, disposing of surplus excavted soil; as directed within a lead of 50 m and lift to 1.5 m		
1.30.1	All kinds of soil	Cum	85.00
1.30.2	Ordinary rock	Cum	123.00
1.30.3	Hard rock (requiring blasting) useful material 30%	Cum	221.00

1.30.4	Hard rock(blasting prohibited) useful material 30%	Cum	269.00
1.31	Clearing jungle including uprooting of rank vegetation, grass, brush wood, trees and saplings of girth upto 30 cm measured at a height of 1 m above ground level and removal of rubbish upto a distance of 50 m outside the periphery of the area cleared.	Sqm	4.20
1.32	Clearing grass and removal of the rubbish upto a distance of 50 m outside the periphery of the area cleared.	Sqm	1.68
1.33	Felling trees of the girth (measured at a height of 1 m above ground level) including cutting of trunks and branches removing the roots and stacking of serviceable material and disposal of unserviceable material.		
1.33.1	Beyound 30 cm girth upto including 60 cm girth	Each	127.00
1.33.2	Beyound 60 cm girth upto including 120 cm girth	Each	564.00
1.33.3	Beyound 120 cm girth upto including 240 cm girth	Each	2614.00
1.33.4	Above 240 cm girth	Each	5244.00
1.34	Supplying chemical emulsion in sealed containers including delivery as specified. Chlorpyriphos/Lindane emulsifiable concentrate of 20%	litre	221.00
1.35	Diluting and injecting by hand operated pressure pumps chemical Emulsion timber ground treatment [TCGT] in ratio 1: 2 as per manufacturers specification for POST CONSTRUCTIONAL Anti-termite treatment (as per IS 6313) part - III 1971 as amended from time to time) (excluding the cost of chemical emulsion) :		
1.35.1	Along external wall where the apron is not provided using chemical emulsion @ 7.5 litres / sqm of the vertical surface of the substructure to a depth of 300mm including excavation channel along the wall & rodding etc. complete: With Chlorpyriphos/Lindane E.C. 20% with 1% concentration	metre	10.50
1.35.2	Along the external wall below concrete or masonry apron using chemical emulsion @ 2.25 litres per linear metre including drilling and plugging holes etc. With Chlorpyriphos/Lindane E.C. 20% with 1% concentration	metre	15.00
1.35.3	Treatment of soil under existing floors using chemical emulsion @ one litre per hole, 300 mm apart including drilling 12 mm diameter holes and plugging with cement mortar 1 :2 (1 cement : 2 Coarse sand) to match the existing floor. With Chlorpyriphos/Lindane E.C. 20% with 1% concentration	sqm	79.00
1.35.4	Treatment of existing masonry using chemical emulsion@ one litre per hole at 300mm interval including drilling holes at 45 degree and plugging them with cement mortar 1:2(1 cement : 2 coarse sand) to the full depth of the hole. With Chlorpyriphos/Lindane E.C. 20% with 1% concentration	metre	11.50
1.35.5	Treatment at points of contact of wood work by chemical emulsion Chlorpyriphos/Lindane (in oil or kerosene based solution) @ 0.5 litres per hole by drilling 6 mm dia holes at downward angle of 45 degree at 150mm centre to centre and sealing the same. With Chlorpyriphos/Lindane E.C. 20% with 1% concentration	metre	113.00

1.36	Diluting and injection Chloropyrifos Emulsifiable concentrate 20% with 1% concentration for PRE-CONSTRUCTIONAL Anti termite treatment as per IS 6313 part III as amended from time to time and creating a continuous chemical barrier under and around the column pits, wall trenches, basement excavation, top surface of plinth filling, junction of wall and floor along the external perimeter of building expansion joints, over the top surface of consolidated earth of which aproch is to be laid surrounding of pipes and conduits etc. complete as per specification (Plinth floor area only shall be measured for payment and excluding the cost of chemical emulsion)	sqm	43.00
1.37	Drilling borehole and conducting standard penetration test from ground level and upto the depth given below or upto refusal whichever occurs earlier as per IS:2131-1981 and IS 2132-1986.The work also includes the determination of Safe Bearing Bearing capacity as per IS:6403-1981 and IS:8009-1976 from disturrbed and undisturbed samples. The following properties should be also reported as per related IS codes .The whole work should be done as per direction given by Engineerand in-charge and upto satisfaction of Engineer in Charge.		
(i)	Grain Size Anaylsis		
(ii)	Atterberg's Limit		
(iii)	Bulk Density		
(iv)	Moisture Cotent		
(v)	Specific Gravity		
(vi)	Shear Parameters including Cohesion and Angle of Internal Friction		
(a)	Upto 6m depth	Each	7850.00
(b)	upto 10m depth	Each	10500.00
1.38	**Anti-termite treatment of existing floor and wall using Imidacloprid 30.5% EC non repellent, odorless, low dose, environimentally safe Termiticide with rapid action non corrosive stain free, by injecting the emulsion as per specification and plugging with cement mortar 1:2 (1 cement : 2 coarse sand) in the whole floor area. The pesticide emulsion of imidacloprid 30.5% EC mixed with water in the ration of 1:474 will be pumped by a motorized spray pump specified manner. The net floor and wall area treated will be measured for payment.**	**Sqm**	**82.00**

1.39	**Anti termite treatment of floor using Imidacloprid 30.5% EC non repellent, odorless, low dose, environmentally safe terniticide with rapid action non corrosive stain free, by drilling 12mm diameter holes 300mm apart injecting the emulsion as per specification and plugging with cement mortar 1:2 (1 cement : 2 coarse sand) in the whole floor area DPE 12mm dia termitube system with required number of junction boxes are to be installed. The termitube will run all along the perimeter wall both inside and outside. In the middle area the termitube to be installed at an interval of 1.00 Mtr. or less both ways as required with sufficient number of junction boxes to pump the emulsion. After successful testing of termitube, pesticide emulsion of imidacloprid 30.5% EC mixed with water in the ration of 1:474 will be pumped by a motorized spray pump from the junction box in specified manner. The plinth area of the area treated will be measured for payment.**	**Sqm**	**196.00**

CHAPTER B-2
MORTAR

NOTE :

1 Complete rate for mortar is inclusive of cost of material, T & P & cost of water with all leads and lifts involved.

2 Compressive strength of mortars shall be as per the relevant specifications.

3 Cement mortar shall be prepared by mechanical mixer only.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	CEMENT SAND MORTAR :		
2.1	Cement sand Mortar 1 : 2 (1 cement : 2 sand)	Cum	4559.00
2.2	Cement sand Mortar 1 : 3 (1 cement : 3 sand)	Cum	3768.00
2.3	Cement sand Mortar 1 : 4 (1cement : 4 sand)	Cum	3092.00
2.4	Cement sand Mortar 1 : 5 (1 cement : 5 sand)	Cum	2728.00
2.5	Cement sand Mortor 1 : 6 (1 cement : 6 sand)	Cum	2416.00
2.6	Cement sand Mortor 1 : 8 (1 cement : 8 sand)	Cum	2000.00
2.7	Cement sand Mortar 1:10 (1 cement :10 sand)	Cum	1775.00
	CEMENT MARBLE POWDER MORTAR :		
2.8	Cement marble powder mortar 1 : 2 (1-cement : 2-marble powder)	Cum	4487.00
2.9	Cement marble powder mortar 1 : 4 (1cement : 4-marble powder)	Cum	2647.00
2.10	White cement marble powder mortar 1 :2 (1 white cement : 2 marble powder	Cum	11077.00
2.11	White cement marble powder mortar 1:3 (1 white cement : 3 marble powder)	Cum	8610.00

2.12	White cement marble powder mortar 1:4 (1 white cement : 4 marble powder)	Cum	5610.00
	MUD MORTAR :		
2.13	Mud Mortar including specified ingredients.	Cum	207.00

CHAPTER B-3
CONCRETE WORK

NOTE :

1. The rates are for all lead of cement, ballast, sand, and grit etc. at site of work and are inclusive of laying and mixing, all T & P, mixer, vibrator with P.O.L, scaffolding etc. including cost of curing.
2. For all concrete work only coarse sand shall be used.
3. Concrete mix prepared by power-driven mechanical mixer shall be used.
4. Under exceptional conditions, where mechanical mixer and vibrator are not used (with the approval of Engineer-in-Charge) rates to be reduce by 15%, for mechanical mixer and 10 % for vibrator.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	**2**	**3**	**4**
	CEMENT CONCRETE (CAST-IN-SITU):		
3.1	Providing and laying in position cement concrete including curing, compaction etc. complete in specified grade excluding the cost of centering and shuttering - All work up to plinth level.		
3.1.1	M20 grade Nominal Mix 1: 1.5: 3 (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20mm nominal size).	Cum	4468.00
3.1.2	M15 grade Nominal Mix 1: 2: 4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size).	Cum	4204.00
3.1.3	M15 grade Nominal Mix 1: 2: 4 (1 cement : 2 coarse sand : 4 graded stone aggregate 40mm nominal size).	Cum	4131.00
3.1.4	M10 grade Nominal Mix 1: 3: 6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20mm nominal size).	Cum	3497.00
3.1.5	M10 grade Nominal Mix 1: 3: 6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size).	Cum	3392.00
3.1.6	1:4:8 (1 cement : 4 coarse sand : 8 graded stone aggregate 40 mm nominal size).	Cum	3002.00

3.1.7	1:5:10 (1 cement : 5 coarse sand : 10 graded stone aggregate 40mm nominal size).	Cum	2687.00
3.2	Providing and laying cement concrete including curing, compaction etc. complete in retaining walls, return walls, walls (any thickness) including attached pilasters, columns, piers, abutments, pillars, posts, struts, buttresses, string or lacing courses, parapets, coping, bed blocks, anchor blocks, plain window sills, fillets, levelling course etc up to floor five level excluding the cost of centering and shuttering.		
3.2.1	M20 grade Nominal Mix 1: 1.5: 3 (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20mm nominal size).	Cum	4955.00
3.2.2	M15 grade Nominal Mix 1: 2: 4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size).	Cum	4647.00
3.2.3	M15 grade Nominal Mix 1: 2: 4 (1 cement : 2 coarse sand : 4 graded stone aggregate 40mm nominal size).	Cum	4562.00
3.2.4	M10 grade Nominal Mix 1: 3: 6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20mm nominal size).	Cum	3927.00
3.2.5	M10 grade Nominal Mix 1: 3: 6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40mm nominal size).	Cum	3822.00
	CEMENT CONCRETE (PRE-CAST):		
3.3	Providing and fixing up to floor five level pre cast cement concrete string or lacing courses, copings, bed plates, anchor blocks, windows sills, shelves, louvers steps, staircases, etc. including carriage, hoisting and setting in position with cement mortar 1:3 (1 cement : 3 coarse sand) cost of required centering, shuttering and finishing smooth with 6 mm thick cement plaster 1:3 (1 cement : 3 fine sand) on exposed surface, including all T & P, curing etc. complete:		
3.3.1	M15 grade Nominal Mix 1: 2: 4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size).	Cum	6352.00

3.3.2	M10 grade Nominal Mix 1: 3: 6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20mm nominal size).	Cum	5632.00
3.4	Add Extra for concrete work in superstructure above floor V level for each four floors or part thereof.	Cum	371.00
3.5	Add Extra for laying concrete in or under water and/or liquid mud including cost of pumping or bailing out water and removing slush etc. complete.	Per Cum per Mtr. depth	279.00
Note :	The quantity will be calculated by multiplying the depth measured from the sub-soil water level up to centre of gravity of concrete under sub-soil water level with quantity of concrete in cum executed under sub-soil water. The depth of centre of gravity shall be reckoned correct to 0.1 m; 0.05 m or more shall be taken as 0.1 m and less than 0.05 m ignored.		
3.6	Add Extra for laying concrete in or under foul positions.	Cum	114.00
	DAMP- PROOF COURSE AND PRECASTCOPING		
3.7	Providing and laying damp-proof course with cement concrete grade M-150 (1 : 2 : 4) mortar prepared with 1% solution of water-proof compound complete as per specification .		
3.7.1	50mm thick.	Sqm	355.00
3.7.2	75mm thick.	Sqm	496.00
3.8	Providing & fixing precast cement concrete coping 1 : 2 : 4 mix 50mm thick complete as per specification :	Sqm	334.00
3.9	Cement, Concrete flooring/ cement plaster/ Cement Concret Road, Plain or RCC work & water retaining works providing and mixing Additive of Synthetic polypropylene Fibrillated mesh fibers free form any reprocess Olefins &comfirming to ASTM C 1116 Type III 4.1.3 Having grade 6mm/12mm **/18mm** melting point of 165 deg.C., strength>600 mpa Sp. Gravity:0.92g/cc, diameter 10-70 microns. It is required in specified ratio @ 0.25% (125 Gm. Fibers in 50 Kg. cement) for use. Mix it directly with contents in rotating site mixer direction of Engineer in charge with all leads and lifts.	Per Pack 125 gm.	60.00

3.10	**Cement, Concrete Flooring/Cement Plaster/Cement Comcrete Road, Plain or RCC work & water retaining works providing and mixing admixture of Synthetic Nylon-6 Fibres or polycaprolactam free from any reprocess Olefins & confirming to ASTM C 1116 Type III 4.1.3 Having grade 6mm / 12 mm /18 mm melting point 220 deg. C, Sp. Gravity:1.14 g/cc, diameter 10-40 microns. It is required in specific ratio @ 0.20% (100 gm Fibres in 50 Kg Cement) for use as specified by manufacture specification and direction of Engineer In-charges with all leads and lifts.**	**per 100 gram pack**	**55.00**

CHAPTER B-4
REINFORCED CONCRETE WORK

Note:

1 Rates for complete work laying, compaction, vibration, curing and all types of T & P, including P.O.L. etc and grinding and mixing of mortar.

2 The rates for centering and shuttering are for steel plates shuttering for complete work including centering, shuttering nails and other such items required. The shuttering shall be removed after required time as per instruction by Engineer-in-charge & IS 456:2000.

3 The Concrete for all grade shall be produced in conformation to the laid down procedures and all relevant clauses of IS 456 : 2000 **and further amendments.**

4 If any machinery like mechanical mixture machine, vibrator or mechanical mortar miller etc. is supplied by the Department, the hire charges as per department rate will be charged and transportation cost will be borne by the contractor.

5 No R.C.C. work shall be permitted without the use of mechanical mixer by hopper and vibrator.

6 Contractor at site shall arrange a stand by mechanical mixer and extra vibrator if more than 20 cum of RCC is to be done in a single day.

7 The rate for R.C.C. work is inclusive of cost of providing cement mortar covers Blocks. The rates for centering includes placement of cover blocks.

8 (a) For Design mix ,the producer / manufacturer / contractor shall obtain mix proportions of each grade of concrete from a recognized laboratory and submit detail to the Engineer-in-charge for approval. Concrete of any particular design mix and grade shall be produced / manufactured by weight batched hopper mechnical mixer with assigned capacity of mixing drum as directed by Engineer-in-charge.

8 (b) For Nominal mix concreting work,the coarse aggregates and fine aggregates will be measured by the standard boxes and cement by bag.The boxes should be designed for 1/2 bag,1 bag etc. of cement.

9 If any change required in the approved Job Mix design ingredient, Fresh Job Mix design shall be obtain by the contractor from approved lab and shall be got approved from Engineer-in-charge

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	CAST-IN-SITU CONCRETE:		
4.1	Providing and laying in position specified grade of cement concrete for all RCC structural elements upto plinth level including curing, compaction, finishing with rendering in cement sand mortar 1:3 (1 cement: 3 coarse sand) and making good the joints and cost of plastizers(if required) excluding the cost of centering, shuttering and reinforcement. M20 grade Nominal Mix / Design Mix	Cum	4485.00
4.2	Providing and laying in position specified grade of cement concrete for RCC structural elements upto floor five level including curing, compaction, finishing with rendering in cement sand mortar 1:3 (1 cement: 3 coarse sand) and making good the joints and cost of plastizers (if required) excluding the cost of centering, shuttering and reinforcement for Walls (any thickness) including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, piers, abutments, posts and struts etc. M20 grade Nominal Mix / Design Mix	Cum	5099.00
4.3	Providing and laying in position specified grade of cement concrete for RCC structural elements upto floor five level including curing, compaction, finishing with rendering in cement sand mortar 1:3 (1 cement: 3 coarse sand) and making good the joints and cost of plastizers(if required) excluding the cost of centering, shuttering and reinforcement for Beams, suspended floors, roofs, griders having slopes up to 15º, landings, balconies, shelves, chajjas, lintels, bands, plain windows sills, staircases and spiral staircases etc. M20 grade Nominal Mix / Design Mix	Cum	4714.00

4.4	Providing and laying in position specified grade of cement concrete for RCC structural elements upto floor five level including curing, compaction, finishing with rendering in cement sand mortar 1:3 (1 cement: 3 coarse sand) and making good the joints and cost of plastizers(if required) excluding the cost of centering, shuttering and reinforcement for Arches, arch ribs, domes, vaults, shells, folded plate, chimneys, shafts , roofs having slope more than 15º ,vertical and horizontal fins individually or forming box louvers, facias and eaves boards M20 grade Nominal Mix / Design Mix	Cum	5055.00
4.5	Providing and laying in position Ready mix concrete manufactured in fully automatic Batching Plant and transported to site in transit mixer for having continous agitated mixer, manufactured as per approved mix design of specified grade of RCC work including pumping of R.M.C. from transit mixer to site of laying , exculding the cost of centering, shuttering and reinforcement with all lead and lift including cost of admixtures in recommended portion as per IS 9103 to accelerate/ retard setting of concrete, improve workability without impairing strength and durability as per direction of Engineer in charge . All works upto floor V floor M20 grade Design Mix by using cement **as per codal** provision.	Cum	5535.00
4.6	Add extra for providing richer mixes respectively at all floor levels		
4.6.1	Providing M-25 grade concrete by using cement **as per codal provision** instead of M-20 grade design mix.	Cum	63.00
4.6.2	Providing M-30 grade concrete by using cement **as per codal provision** instead of M-20 grade design mix.	Cum	126.00
4.6.3	Providing M-35 grade concrete by using cement **as per codal provision** instead of M-20 grade design mix.	Cum	176.00
4.6.4	Providing M- 40 grade concrete by using cement **as per codal provision** instead of M-20 grade design mix.	Cum	220.00
4.7	Add extra for RCC work in superstructure above floor five level for each four floor or part thereof .	Cum	102.00
4.8	Deduct for using less cement quantity than the quantity as provided in the item of design mix concrete/RMC as arrived as per mix design.(on cement quantity)	Kg	6.00
	FORMK WORK		
4.9	Centering and Shuttering with plywood or steel sheets including strutting, propping bracing both ways and removal of formwork for foundation , footings, strap beam, raft , bases of columns etc.	Sqm.	143.00

4.10	Centering & shuttering with plywood or steel sheets including strutting, propping bracing both ways with steel props and removal of formwork for upto floor five level for :		
4.10.1	Walls (any thickness) including attached pilasters, buttresses plinth and string course.	Sqm	263.00
4.10.2	Suspended floors, roofs, landings, staircases, balconies, girders, cantilevers, bands, coping bed plates, anchor blocks, sills, chhajjas, lintel, beam, plinth beam etc.	Sqm	309.00
4.10.3	Columns, pillars, posts and struts etc.	Sqm.	341.00
4.10.4	Spiral staircases, chimneys shafts etc.	Sqm.	274.00
4.10.5	Arches, domes, vaults etc. up to 6 Mtr span.	Sqm.	799.00
4.10.6	Arches, domes, vaults etc exceeding 6 Mtr span	Sqm.	299.00
4.10.7	Vertical and horizontal fins individually or forming box louvers band, facias and eaves boards.	Sqm.	449.00
4.10.8	Add extra for shuttering in circular work (in respective centering and shuttering items)	Sqm	20%
4.10.9	Cornices and mouldings	Sqm	431.00
4.10.10	Grid roofing with panel area less than 2.25 Sqm. (contact area to be measured).	Sqm	368.00
4.10.11	Deduct for using wooden vertical member instead of steel props Note : The uses of wooden member is allowed in works which A & F sanction up to 25 lacs only.	Sqm	20%
4.11	Add extra for additional height in centring, shuttering where ever required with adequate bracing, propping etc. including cost of de-shuttering and decentring at all levels, over a height of 3.5 m, for every additional height of 1 metre or part thereof (Plan are is to be measured)	Sqm	155.00
4.12	Add extra for centering and shuttering work in superstructure above floor five level for each four floor or part thereof .	Sqm.	31.00
	STEEL REINFORCEMENT:		
4.13	Providing and fabricating reinforcement for R.C.C. work including straightening, cutting, bending, placing in position and binding (including cost of binding wire) all complete up to floor five level. **('Original producers' who manufacture billet directly from iron ores and roll the bilets to produce steel conforming to IS:1786)**		
4.13.1	Cold twisted deformed bars (IS : 1786).	Kg.	64.00
4.13.2	Hot rolled deformed bars (IS : 1139).	Kg.	64.00
4.13.3	Thermo-mechanically Treated bars (Conforming of relevent IS code)	Kg.	64.00

4.14	Labour charges for cutting, bending for fabrication and binding reinforcement (plain or tor/ribbed/TMT steel) as per drawing and design for R.C.C. and R.B. work including cost of binding wire with all lead and lift up to floor five level complete.	Kg.	10.50
4.15	Add extra for reinforcement in superstructure above floor five level for each four floor or part thereof .	Kg	2.60
	PRECAST CEMENT CONCRETE		
4.16	Providing and fixing of 200mm thick RCC precast panel made of M35 grade concrete with required steel, rebar chair, wire mesh, connection bolt, fastener and beveled washer with **erection** ready complete in all respect with all lead.	Sqm	3705.00
4.17	Providing and fixing of 150 mm thick RCC precast panel made of M35 grade concrete with required steel, rebar chair, wire mesh, connection bolt, fastener and beveled washer with **erection** ready complete in all respect with all lead.	Sqm	3497.00
4.18	Providing, hoisting and fixing up to floor five level precast reinforced concrete work in Nominal M20 grade mix including cost of required centering, shuttering, finishing, smooth with 6 mm thick cement plaster 1:3 (1 cement: 3 fine sand) on exposed surface complete excluding cost of reinforcement		
4.18.1	String courses, copings, bed plates, anchor blocks, plain window sills etc	Cum	6391.00
4.18.2	Small lintels not exceeding 1.5 m clear span	Cum	8330.00
4.18.3	Mouldings as in cornices, windows sills etc.	Cum	9329.00
4.18.4	Lintels beams and bressumers including setting in cement mortar 1:3 (1 cement: 3 coarse sand) aggregate 20mm nominal size).	Cum	7851.00
4.18.5	Beams, Columns, slabs, Shelves, Vertical & horizontal fins individually or forming box louvers setting in cement mortar 1:2 (1 cement: 2 sand) or grouts as per requirements	Cum	11851.00
4.19	Providing, hoisting and fixing up to floor five level precast cement concrete Jali 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 6mm nominal size) reinforced with 1.6 mm dia Mild steel wire including centring and shuttering, roughening cleaning, fixing and finishing in cement mortar 1 : 3 (1 cement : 3 fine sand) etc complete as per approved design excluding plastering of jambs, sills and soffits		
4.19.1	50mm thick	Sqm	573.00
4.19.2	40 mm thick	Sqm.	485.00

4.19.3	30mm thick	Sqm.	447.00
4.2	Add extra over for both side jali	Sqm.	20.00%
	ENCASING ROLLED STEEL SECTION		
4.21	Encasing rolled steel section in beams and columns, with M20 Nominal Mix cement concrete including centering and shuttering complete but excluding cost of reinforcement.	Cum	7691.00
4.22	Encasing rolled steel section in grillages with M20 Nominal Mix including centering and shuttering complete but excluding cost of expanded metal and hangers.	Cum	5097.00
4.23	Add Extra for providing and fixing expanded metal mesh of size 20x60mm and strands 3.25mm wide 1.6mm thick weighing 3.64 kg. per **sqm.** for encasing of rolled steel sections in beams, columns and grillages excluding cost of hangers.	Sqm.	346.00
4.24	Providing ,arranging and erecting 50mm dia M.S. pipe cup lock shuttering arrangement above the ground level upto the height given below. The girding arrangement of cup lock system should have 2.50 meter 3.00 meter long coupling pipes and 0.90mt long lacer pipe (Horizontal bracing pipes) reconnected in both ways so as to make a square grid of size 0.90x0.90 meter in plan. The Horizontal bracing and lacer pipes should have a maximum spacing of 1.50 meter c/c both ways. The bottom of the coupling pipe must rest over a firm support at ground. the shuttering plates of 900mmx600mm size duly braced by angle supports and should rest over 50mmx50mm size wooden logs supported by 75x40x6mm size channels and U-Jacks made of M.S.25mm dia steel bars with adjustable thread and lock arrangement placed at a spacing of 0.60 meter c/c. The whole arrangement should be duly shored from ground by U-Jacks and lacer pipes to prevent any lateral movement during casting of slabs. Additional support as per direction of engineer in charge are also to by provided wherever felt necessary. The work would include transportation of material erecting the whole grid structure, bracing both way, shoring, leveling of shuttering and		
(a)	Upto 6.63 meter height	Sqm.	958.00
(b)	Upto 9.14 meter height	Sqm.	1722.00

CHAPTER B-5
BRICK WORK

Note :

1 Fly ash bricks shall be invariably be used in building works except petty or repair works, where clay bricks can be used.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
5.1	Brick masonry with F.P.S. bricks of class designation 75 in foundation and plinth with bricks		
5.1.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Cum.	4063.00
5.1.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Cum.	3875.00
5.1.3	Mud mortar	Cum.	2968.00
5.2	Brick work with F.P.S. bricks of class designation 75 in superstructure above plinth level upto floor V level in all shapes and sizes in :		
5.2.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Cum.	4724.00
5.2.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Cum.	4536.00
5.2.3	Mud mortar	Cum.	3447.00
5.3	Add extra for Square or Rectangular Pillars in Superstructre brick work up to five storey		
5.3.1	Upto 45 cm length and all sides are free	Cum	297.00
5.3.2	Upto 45 cm length and two corners are free	Cum	148.00
5.3.3	Upto 90 cm length and all sides are free	Cum	253.00
5.3.4	Circular Pillars	Cum.	301.00
5.4	Brick work in plain arches in super structure upto five storey in cement mortar 1 : 3 including centering and shuttering complete span upto 6 Mtr. with Brick of class designation 75.	Cum.	6147.00
5.5	Brick work in gauged arches in super structure upto five storey in cement mortar 1 : 3 including centering and shuttering complete span upto 6 meter with Brick of class designation 75.	Cum.	7286.00
5.6	Add extra for additional cost of centering for arches exceeding 6m span including all shuttering , bolting wedging and removal.(Area of the soffit to be measured)	Cum	285.00

5.7	Half brick masonry in foundation & plinth using bricks of designation 75		
5.7.1	Cement mortar 1 : 3 (1 cement : 3 coarse sand)	Sqm.	477.00
5.7.2	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm.	452.00
5.7.3	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Sqm.	427.00
5.8	Half brick masonry in Superstructure , above plinth level upto floor V level using bricks of designation 75		
5.8.1	Cement mortar 1 : 3 (1 cement : 3 coarse sand)	Sqm.	530.00
5.8.2	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm.	503.00
5.8.3	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Sqm.	478.00
5.9	Add extra providing and placing in position 2 Nos. ,6mm Ø M.S. bar at every third course of half brick masonry .	Sqm	58.00
5.10	Brick work in partition in super structure upto five storey 7cm. thick (brick on edges) using bricks of class designation 75 in :		
5.10.1	Cement mortar 1 : 3 (1 cement : 3 coarse sand)	Sqm.	360.00
5.10.2	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm.	343.00
5.11	Honey comb brick work half brick thick with bricks of class designation 75 upto five storey in:		
5.11.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm.	327.00
5.11.2	Cement mortar1 : 6(1 cement : 6 coarse sand	Sqm.	311.00
5.12	Add extra over for laying brick work in or under water and or liquid mud including cost of pumping of or bailing out water and removal of slush etc. complete.	P.Cum. P.-Mtr. depth	266.00
Note :	The quantity will be calculated by multiplying the depth measured from the sub-soil water level up to centre of gravity of concrete under sub-soil water level with quantity of concrete in cum executed under sub-soil water. The depth of centre of gravity shall be reckoned correct to 0.1 m; 0.05 m or more shall be taken as 0.1 m and less than 0.05 m ignored.		
5.13	Add extra for laying bricks work in or under foul condition.	Cum.	108.00
5.14	Brick work in super structure upto five storey with selected bricks of class designation 75 in exposed brick work including making horizontal and vertical grooves 10mm. wide 12mm. deep complete in cement mortar 1:6.	Cum	4595.00
5.15	Brick work with clay flyash F.P.S. brick (IS 13757- 1993) of class designation 75 in superstructure above plinth level upto floor five level in :		
5.15.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Cum	4580.00
5.15.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Cum	4393.00

5.16	Brick work with mechanised autoclaved flyash lime bricks conforming to IS: 12894 :2002 of class designation 100 in superstructure above plinth level upto floor V level in.		
5.16.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Cum	4465.00
5.16.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Cum	4223.00
5.17	Add extra for brick work in superstructure above floor five level, for each four floors or part thereof.		
5.17.1	Quantity in Cum.	Cum.	236.00
5.17.2	Quantity in Sqm.	Sqm.	21.00
5.18	Add extra over item of brick work with class designation 100 in substitution of class designation 75 :		
5.18.1	Quantity in Cum.	Cum.	148.00
5.18.2	Quantity in Sqm.	Sqm.	17.00
5.19	Half Brick work with mechanised autoclaved flyash lime bricks conforming to IS: 12894 :2002 of class designation 100 in superstructure above plinth level upto floor V level in.		
5.19.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm	533.00
5.19.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Sqm	506.00
5.20	Half Brick work with mechanised autoclaved flyash lime bricks conforming to IS: 12894 :2002 of class designation 75 in superstructure above plinth level upto floor V level in.		
5.20.1	Cement mortar 1 : 4 (1 cement : 4 coarse sand)	Sqm	513.00
5.20.2	Cement mortar 1 : 6 (1 cement : 6 coarse sand)	Sqm	487.00
5.21.1	**Providing & Laying autoclave aerated blocks masonry with 100mm thick AAC Blocks of Grade-I Conforming to IS:2185 Part:III in superstructure above plinth level with cement mortar 1:4 (1 cement: 4 coarse sand) .The rates includes providing and placing in position 2 Nos. 6 mm dia M.S. bars at evry third course of masonary.**	**Cum.**	**4913**
5.21.2	**Providing & Laying autoclave aerated blocks masonry with 150mm/230mm/300mm thick AAC Blocks of Grade-I Conforming to IS:2185 Part:III in superstructure above plinth level with polymer modified adhesive mortar as per direction of Engineer in Charge.**	**Cum.**	**4320**

CHAPTER B-6
STONE MASONRY

Note :

1. Rates provided are for the complete finished items inclusive of cost of water and materials with all leads labour, all type of taxes, royalty, scaffolding, curing, laying, fixing finishing etc., hire charges for tools and plants, templates and other appliances, required for the purpose of execution of the items work as per specification.

2. In case of non-availability of stone header of proper size cement concrete header of mix 1:3 : 6 (M-100) with 25mm graded stone aggregate with smooth finish may be provided by the contractor at no extra cost.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
6.1	Random Rubble stone masonry for with hard stone in foundation and plinth in Cement Sand mortar above 30 CM thick wall in:		
6.1.1	Mud Mortar.	Cum.	1682.00
6.1.2	Dry Masonry.	Cum.	1585.00
6.1.3	Cement Mortar 1:3 (1-Cement : 3-Sand).	Cum.	3207.00
6.1.4	Cement Mortar 1:4 (1-Cement : 4-Sand).	Cum.	2893.00
6.1.5	Cement Mortar 1:5 (1-Cement : 5-Sand).	Cum.	2725.00
6.1.6	Cement Mortar 1:6 (1-Cement : 6-Sand).	Cum.	2580.00
6.2	Random Rubble stone masonry with hard stone in superstructure above plinth level and upto five level above 30cm. thick walls in :		
6.2.1	Mud Mortar.	Cum.	2264.00
6.2.2	Dry Masonry.	Cum.	2168.00
6.2.3	Cement Mortar 1:3 (1-Cement : 3-Sand).	Cum.	3790.00
6.2.4	Cement Mortar 1:4 (1-Cement : 4-Sand).	Cum.	3476.00
6.2.5	Cement Mortar 1:5 (1-Cement : 5-Sand).	Cum.	3308.00
6.2.6	Cement Mortar 1:6 (1-Cement : 6-Sand).	Cum.	3163.00
6.3	Add extra for Random Rubble stone masonry with hard stone in		
6.3.1	Square or Rectangular pillars	Cum.	157.00
6.3.2	Circular pillars	Cum.	472.00
6.3.3	Wall curved in plan to mean radius not exceeding 6 Mtr.	Cum	290.00
6.3.4	Wall upto 30cm. and less thickness.	Cum	169.00
6.4	Add extra for facing as per design and detailed specifications including additional cost of stones		
6.4.1	Course rubble facing First sort.	Sqm	562.00
6.4.2	Course rubble facing Second sort.	Sqm	446.00

6.4.3	Ashlar fine tooled facing.	Sqm	605.00
6.4.4	Ashlar rough tooled facing.	Sqm.	346.00
6.4.5	Ashlar pitched facing.	Sqm.	305.00
6.5	Extra for stone masory work,above floor V level for every four floors or part thereof.	Cum	495.00
6.6	Extra for laying stone work in or under water and/or liquid mud including cost of pumping or bailing out water and removing slush etc. complete.	Cum/ P Mt depth	279.00
6.7	Extra for laying stone work in or under foul position.	Cum	113.00
Note :-	Recover cost of useful stone @ 2/3rd of masonry work dismantaled quantity.	Cum	523.00
6.8	Stone work (machine cut edges) for Wall cladding/ Veneering work up to 10m height with 20 to 30 mm thick Red Sand Stone (Karauli) and backing filled with a grout of 12mm thick cement mortar 1:3 (1 cement : 3 coarse sand) including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment matching the stone shade :(To be secured to the backing by means of cramps which shall be paid separately)		
6.8.1	Exposed face rough dressed	Sqm.	1649.00
6.8.2	Exposed face machine cut	Sqm.	2273.00
6.9	Stone work (machine cut edges) for Wall cladding/ Veneering work up to 10 m height with 20 to 30 mm thick Pink Sand stone (Bansi Paharpur) and backing filled with a grout of 12mm thick cement mortar 1:3 (1 cement : 3 coarse sand) including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment matching the stone shade : (To be secured to the backing by means of cramps which shall be paid separately)		
6.9.1	Exposed face rough dressed	Sqm.	1683.00
6.9.2	Exposed face Machine cut	Sqm.	2305.00
6.10	Wall lining butch work upto 10m height with Red/ Pink Sand stone 20 to 30 mm thick (machine cut edges) stone strips of minimum length 300 mm and required width including embedding every tenth layer and bottom most layer in masonry or concrte after making necessary chases of size 75x75 mm and by providing layer of 75 mm thick strips i/c 12 mm thick bed of cement mortar 1 : 3 (1 cement : 3 Coarse sand) i/c ruled pointing in cement mortar 1 :2 (1 white cement : 2 stone dust) with an admixture of pigment to match the shade of stone complete as per direction of Engineer in Charge		

6.10.1	Exposed face rough dressed	Sqm.	1230.00
6.10.2	Exposed face machine cut	Sqm.	1323.00
6.11	Providing and fixing stainless steel cramps of required size and shape for anchoring stone wall lining to the backing or securing adjacent stones in stone wall lining in cement mortar 1:2 (1 cement : 2 coarse sand) including making the necessary chases in stone and holes in walls wherever required	Kg	643.00
6.12	Providing and fixing stone dowels 10x5x2.50 cm cut to double wedge shape as per design in cement mortar 1:2 (1 cement : 2 coarse sand) including making the necessary chases.	Each	26.00
6.13	Providing and fixing copper pins 7.5 cm long 6 mm diameter for securing adjacent stones in stone wall lining in cement mortar 1:2 (1 cement : 2 coarse sand) including making the necessary chases.	Each	24.00
6.14	Extra for stone work for wall lining on exterior walls of height more than 10 m from ground level for every additional height of 3 m or part there of.	Sqm.	58.00
6.15	Providing and fixing horizontal chajja of Red/ White sand stone 40 mm thick and upto 80 cm projection in cement mortar 1:4 (1 cement : 4 coarse sand) including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment matching the stone shade.	Sqm.	803.00
6.16	Providing dab stone over Chajjas duly fixed in cement sand mortar 1:6 complete :		
6.16.1	80mm thick.	Sqm.	837.00
6.16.2	50mm thick.	Sqm.	750.00
6.17	Supplying and fixing in walls machine cut and polished stone shelves, tands and in CM 1:3 with machine cut edges :		
6.17.1	Sand or other approved stone 25mm thick.	Sqm.	458.00
6.17.2	Sand or other approved stone 50mm thick.	Sqm.	612.00
6.17.3	Kota stone **minimum 30mm thick.**	Sqm.	554.00
6.18	Supplying and fixing stone lintels/bed plates of approved quarry rough dressed in cement mortar 1:4 :		
6.18.1	Upto 15 cm. thick.	Cum	8746.00
6.18.2	Above 15 cm. thick.	Cum	8329.00
6.19	Supplying and fixing machine cut fine dressed Red/Pink sand stone dasa or coping, with full moulding if required laid on cement mortar 1:4 including pointing with admixture of pigment matching with the stone shade.		

6.19.1	25 mm thick	Sqm.	1130.00
6.19.2	50 mm thick	Sqm.	1325.00
6.19.3	75 mm thick	Sqm.	1513.00
6.20	Supplying and fixing machine cut dressed 75 mm Pink sand stone (Bansi Paharpur) dasa/ coping wth ornamental shape having moulding and "PAN Shape" engraving as per required pattern with engraving depth of 5 mm (average) in single piece of length 90 cm to be laid on cement mortar 1:4 (12 to 20mm thick) including pointing with admixture of pigment matching with the stone shade.	Sqm.	4826.00
6.21	Providing and fixing Stone jali 40mm thick Red/Pink sand stone throughout in cement mortar 1:3 (1 cement : 3 coarse sand) including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment, matching the stone shade, jali patterns to be cut square to jali slab without any chamfers etc:	Sqm.	7137.00
6.22	Providing and fixing Stone jali 50mm thick Red/Pink sand stone as per ornamental design to fixed with white cement/epoxy mortar including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment, matching the stone shade.	Sqm.	8012.00
6.23	Providing and fixing Red/Pink sand stone post (Mataga) having size 100x100mm heights 600 to 750 mm as per ornamental design such as like flower octonogal head border etc. to fixed with white cement/epoxy mortar including pointing in white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment, matching the stone shade.	Each	727.00
6.24	Providing and fixing ornamental/aesthetic look cement concerte claddings/ veneers up to 100 mm with, over a base of cement Plaster 1:4, 20 mm thick including cement pointing 1:2 with an admixture of pigment to match the shades of veneer upto 4:5 mtr. Height above plinth level.	Sqm	918.00

CHAPTER : B - 7
MARBLE, GRANITE AND TILE WORKS

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
7.1	Providing and fixing Marble stone flooring table rubbed, 15-18 mm thick over 20mm (Av.) thick base of CM 1:4 (1 cement : 4 coarse sand) jointing with white cement mortar 1:2 (1 white cement : 2 marble dust) with pigment to match the shade of the marble slab including grinding, rubbing and polishing complete. OR Providing and fixing stone slabs table rubbed, 15-18 mm thick in Walls, Steps, Pillars, Counters, Jambs, Shelves, Sills etc. laid on 12mm (Av.) thick base of CM 1:3 (1 cement : 3 coarse sand) jointing with white cement mortar 1:2 (1white cement : 2 marble dust) with pigment to match the shade of the marble slab including grinding, rubbing and polishing complete.		
7.1	Makrana 'Adanga'.		
7.1.1	Upto to 1500 Cm^2 Tiles	Sqm.	1058.00
7.1.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	1227.00
7.1.3	Above 3601 Cm^2 Slabs	Sqm.	1522.00
7.2	Rajnagar 1st quality/Ageria with light spots.		
7.2.1	Up to 1500 Cm^2 Tiles	Sqm.	949.00
7.2.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	1052.00
7.2.3	Above 3601 Cm^2 Slabs	Sqm.	1186.00
7.3	Keseriaji/ Abu Ambaji (Green Marble)		
7.3.1	Up to 1500 Cm^2 Tiles	Sqm.	985.00
7.3.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	1153.00
7.3.3	Above 3601 Cm^2 Slabs	Sqm.	1522.00
7.4	Bhaislana,Abu,Kishangarh Zebra (Black Marble)		
7.4.1	up to 1500 Cm^2 Tiles	Sqm.	935.00
7.4.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	1018.00
7.4.3	Above 3601 Cm^2 Slabs	Sqm.	1237.00

7.5	Providing and fixing Granite stone slab mirror polished and machine edge cut in walls, pillars, steps, Shelves, Sills Counters, Floors etc. laid on 12mm (Av.) thick base of cement mortar 1:3 (1 cement : 3 coarse sand) jointing with white cement mortar 1:2 (1white cement : 2 marble dust) with pigment to match the shade of the marble slab including grinding, rubbing and polishing complete.		
7.5	Jhunjhunu / Jalore (Red/Choclate/**Black/Pink Colour**)		
7.5.1	Up to 1500 Cm^2 Tiles	Sqm.	1812.00
7.5.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	2562.00
7.5.3	Above 3601 Cm^2 Slabs	Sqm.	2753.00
7.6	South (Red)		
7.6.1	Up to 1500 Cm^2 Tiles	Sqm.	2230.00
7.6.2	1501 Cm^2 to 3600 Cm^2 Tiles	Sqm.	2676.00
7.6.3	Above 3601 Cm^2 Slabs	Sqm.	3122.00
7.7	**South Zed Black**		
7.7.1	**Up to 1500 Cm^2 Tiles**	Sqm.	2122.00
7.7.2	**1501 Cm^2 to 3600 Cm^2 Tiles**	Sqm.	2887.00
7.7.3	**Above 3601 Cm^2 Slabs**	Sqm.	3705.00
7.7.4	**Good quality South Black**		
7.7.4.1	**Up to 1500 Cm^2 Tiles**	**Sqm.**	2100.00
7.7.4.2	**1501 Cm^2 to 3600 Cm^2 Tiles**	**Sqm.**	2684.00
7.7.4.3	**Above 3601 Cm^2 Slabs**	**Sqm.**	3500.00
7.8	Extra for providing edge moulding to 15-18mm thick marble/ Granite/Kota stone counters, Vanities etc. including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge.		
7.8.1	Marble work		
7.8.1.1	**Full Edge moulding**	Mtr.	133.00
7.8.1.2	**Half Edge moulding**	**Mtr.**	80.00
7.8.2	Granite**/Kota Stone** Work		
7.8.2.1	**Full Edge moulding**	Mtr.	212.00
7.8.2.2	**Half Edge moulding**	**Mtr.**	130.00

7.9	Extra for fixing marble /granite stone over and above corresponding basic item, in facia and drops of width upto 150 mm with epoxy resin based adhesive including cleaning etc. complete	Mtr.	22.00
7.10	Extra for providing opening of required size & shape for wash basins/ kitchen sink in kitchen platform, vanity counters and similar location in marble/Granite/stone work including necessary holes for pillar taps etc. including rubbing and polishing of cut edges etc. complete	Each Opening	233.00
7.11	Extra for Mirror polishing on marble work where ever required to give high gloss finish complete.	Sqm.	93.00
7.12	Extra in marble/other flooring work for upto 50 mm wide strip of Green/White/ Jaisalmer stone in flooring work as per direction of Engineer in Charge	Rm	47.00
7.13	Providing and fixing cramps of required size & shape in RCC/ CC backing with cement mortar 1:2 (1 cement :2 coarse sand) including drilling necessary hole in stones and embedding the cramp in the hole .		
7.13.1	Gun Metal cramp	Kg	584.00
7.13.2	Stainless steel cramps	Kg	555.00
7.14	Providing and fixing 1st quality standard white, grey, ivory, fume red brown, light green, light blue and other light shades glazed tiles confirming to IS : 13753 & IS :15622 of size 200mm x 300mm in walls, floors, steps, pillars etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (excluding the cost of cement plaster on walls and pillar).	Sqm.	613.00
7.15	Providing and fixing 1st quality MAT finished ceremic tile size 300x300mm confirming to IS : 13755 and IS : 15622 colour such as white, grey, ivory, fume red brown, light green, light blue and other light shades in floors, steps, pillars etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (including the cost of cement mortar bed 1:4).	Sqm.	680.00
7.16	Extra for using Marble printed / Granite shade or dark shade tiles instead of white, grey, ivory, fume red brown, light green, light blue and other light shades in glazed tiles and MAT finished tiles.	Sqm	0.10
7.16.1	Extra for using size above 200x300mm in glazed tiles and for using 400mm x 400mm MAT finished tiles in place of 300mm x 300mm size.	Sqm	0.10

7.17	Providing and fixing 75 mm wide Decorative Border in different shades and in all coloures in glazed tiles laid over 12 mm thick bed of cement mortar 1:3 (1 cement : 3 coarse sand) and jointing with grey cement slurry @ 3.3 kg per sqm including pointing in white mixed with pigment of matching shade complete.	Rmt.	299.00
7.18	Providing and fixing Motive pieces all **colours** for glazed tiles laid over 12 mm thick bed of cement mortar 1:3 (1 cement : 3 coarse sand) and jointing with grey cement slurry @ 3.3 kg per sqm including pointing in white mixed with pigment of matching shade complete..		
7.18.1	Size 200 x 200mm	Each	118.00
7.18.2	Size 200 x 300mm	Each	141.00
7.18.3	**Size 300 x 300mm**	**Each**	199.00
7.18.4	**Size 300 x 450mm**	**Each**	227.00
7.18.5	**Size 300 x 600mm**	**Each**	255.00
7.19	P & F 1st qualityVitrified Porcelain Polished tiles on floor, skirting and steps etc.in different sizes (thickness to be specified by manufactuer) with water absorption less than 0.08% and conforming to IS 15622 of approved make in all colour and shade, laid with 20 mm thick CM 1 : 4 including grounting the joints with white cement and matching pigment etc complete.		
7.19.1	size 450 mm X 450 mm	Sqm	717.00
7.19.2	size 600 mm X 600 mm	Sqm	828.00
7.19.3	size 800mm X 800 mm	Sqm	1000.00
7.20	Add extra for Fixing glazed/ Ceramic/ Vitrified floor tiles with cement based high polymer modified quick-set tile adhesive (Water based) conforming to IS: 15477 , using 5kg. adhesive per **sqm** of tile area, in average 3mm thickness instead of cement mortar.	Sqm	116.00
7.21	Providing and fixing 1st quality standard white, grey, ivory, fume red brown, light green, light blue and other light shades ceramic glazed vitrified tiles with water absorption less than or equal to 0.08% confirming to IS:13753 & IS : 15622 in floors, steps etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (excluding the cost of cement plaster on walls and pillar).		
7.21.1	Size 600mm x 1200mm	Sqm	1330.00
7.21.2	Size 800mm x 1200mm	Sqm	1659.00
7.21.3	Size 196mm x 1215mm	Sqm	1658.00
7.22	Providing and fixing 1st quality MAT & GLOSSY finished ceramic tile confirming to IS : 13755 and IS : 15622 colour such as white, grey, ivory, fume red brown, light green, light blue and other light shades in floors, steps, pillars etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (including the cost of cement mortar bed 1:4).		

7.22.1	Size 250mm x 375mm	Sqm	661.00
7.22.2	Size 300mm x 450mm	Sqm	735.00
7.22.3	Size 300mm x 600mm	Sqm	813.00
7.22.4	Size 250mm x 1000mm	Sqm	981.00
7.23	P & F 1st qualityHeavy Duty Vitrified Polished Digital tiles on floor, skirting and steps etc.in different sizes (thickness minimum 10mm) with water absortion less than or equal 0.08% and conforming to IS 15622 of approved make in all colour and shade, laid with 20 mm thick CM 1: 4 including grounting the joints with white cement and matching pigment etc complete.		
7.23.1	Size 298mm x 298mm	Sqm	641.00
7.23.2	**Size 300mm x 450mm**	**Sqm**	791.00
7.23.3	**Size 600mm x 600mm**	**Sqm**	1024.00
7.24	**Providing non slip grooves to granite/ kota stone flooring, counters, vanities etc. using groove cutter V type including machine polishing to groove to give high gloss finish etc. complete as per design approved by engineer in charge**	**Mtr.**	95.00
7.25	**P & F 1st qualityHeavy Duty Vitrified Polished Digital tiles on floor, skirting and steps etc.in different sizes (thickness minimum 10mm) with water absortion less than or equal 0.08% and conforming to IS 15622 of approved make in all colour and shade, laid with 20 mm thick CM 1: 4 including grounting the joints with white cement and matching pigment etc complete.**		
7.25.2	**size 600 mm X 1200 mm**	**Sqm**	1410.00
7.26	**P & F 1st quality Heavy Duty Vitrified glazed MAT tiles on floor, skirting and steps etc.in different sizes (thickness minimum 10mm) with water absortion less than or equal 0.08% and conforming to IS 15622 of approved make in all colour and shade, laid with 20 mm thick CM 1: 4 including grounting the joints with white cement and matching pigment etc complete.**		
7.26.1	**size 600 mm X 600 mm**	**Sqm**	1178.00
7.26.2	**size 600 mm X 1200mm**	**Sqm**	1640.00
7.27	**P & F 1st quality Heavy Duty Vitrified Double Charged tiles on floor, skirting and steps etc.in different sizes (thickness minimum 10mm) with water absortion less than or equal 0.08% and conforming to IS 15622 of approved make in all colour and shade, laid with 20 mm thick CM 1: 4 including grounting the joints with white cement and matching pigment etc complete.**		
7.27.1	**size 600 mm X 600mm**	**Sqm**	946.00
7.27.2	**size 800 mm X 800mm**	**Sqm**	1178.00
7.27.3	**size 1000 mm X 1000mm**	**Sqm**	1331.00

CHAPTER : B-8
WOOD WORK

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
8.1	Providing & fixing door, window, chowkhats and other frames, including antitermite treatment, M.S. flat, hold fasts of size 250 x 40 x 3mm fixed with M.S. bolt 5mm dia 50mm long duly embedded in cement concrete M-15 grade :		
8.1.1	Sal wood Grade - I.	Cum	44406.00
8.1.2	Kiln seasoned and chemically treated Hollock wood. Grac	Cum	37719.00
8.1.3	Sheesham wood Grade - I.	Cum	92428.00
8.1.4	Nigeria teak wood Grade - I.	Cum	100125.00
8.1.5	Ghana teak wood Grade - I.	Cum	107776.00
8.1.6	Ivory coast wood Grade I	Cum	123212.00
8.2	Supplying and fixing wood work in frames of wall panelling, false ceiling partition wall etc. as per approved design and drawing including antitermite treatment.		
8.2.1	Hollock wood Grade - I.	Cum	42064.00
8.2.2	Chir wood Grade - I.	Cum	39443.00
8.3	Add extra in frames for circular members such as in fan lights, staircase railing etc.	Cum	10%
8.4	Providing and fixing fully panelled/partly panelled/ double leaf shutter frame for doors as per approved design and drawings with approved ordinary C.P./ oxidised steel fittings as per Annexure - 'A' and width of styles & top rails of 100mm and lock rails width of 150mm and bottom rails of 200mm width including teak wood beading of size 19mm x 12mm on both faces complete :		
8.4.1	35 mm thick		
8.4.1.1	Nigeria wood grade I	Sqm	2525.00
8.4.1.2	Ghana wood grade I	Sqm	2698.00
8.4.1.3	Shesham wood grade I	Sqm	2349.00
8.4.1.4	Ivory coast wood grade I	Sqm	3041.00
8.4.2	30 mm thick		

8.4.2.1	Nigeria wood grade I	Sqm	2241.00
8.4.2.2	Ghana wood grade I	Sqm	2389.00
8.4.2.3	Shesham wood grade I	Sqm	2076.00
8.4.2.4	Ivory coast wood grade I	Sqm	2688.00
8.5	Providing and fixing fully panelled /partly panelled double leaf shutters frame for windows and ventilators as per approved design and drawings with approved steel fittings as per Annexure 'A' and width of styles, bottom and top rails 75mm including intermediate rails 50mm wide, beading of 25mm x 15mm size on both faces :		
8.5.1	35 mm thick		
8.5.1.1	Nigeria wood grade I	Sqm	2262.00
8.5.1.2	Ghana wood grade I	Sqm	2409.00
8.5.1.3	Shesham wood grade I	Sqm	2102.00
8.5.1.4	Ivory coast wood grade I	Sqm	2705.00
8.5.2	30 mm thick		
8.5.2.1	Nigeria wood grade I	Sqm	2007.00
8.5.2.2	Ghana wood grade I	Sqm	2134.00
8.5.2.3	Shesham wood grade I	Sqm	1859.00
8.5.2.4	Ivory coast wood grade I	Sqm	2388.00
8.5.3	25 mm thick		
8.5.3.1	Nigeria wood grade I	Sqm	1754.00
8.5.3.2	Ghana wood grade I	Sqm	1854.00
8.5.3.3	Shesham wood grade I	Sqm	1617.00
8.5.3.4	Ivory coast wood grade I	Sqm	2070.00
8.6	Providing and fixing wood panelling or panelling and glazing in panelled or panelled and glazed shutters for doors, windows and clerestory windows (Area of opening for panel inserts excluding portion inside grooves or rebates to be measured). Panelling for panelled and glazed shutter 25 mm to 40 mm thick		
8.6.1	Wood panel grade I		
8.6.1.1	Nigeria wood Grade I	Sqm	2040.00
8.6.1.2	Ghana wood grade I	Sqm	2180.00
8.6.1.3	Shesham wood grade I	Sqm	1890.00
8.6.1.4	Ivory coast wood grade I	Sqm	2462.00
8.6.2	IS 710- 1976 marked 9 mm thick commerical ply	Sqm	824.00

8.6.3	IS 710- 1976 marked 9 mm thick Teak ply BWR one side	Sqm	1425.00
8.6.4	IS 710- 1976 marked 9 mm thick Teak BWR ply both side	Sqm	1655.00
8.6.5	IS 710- 1976 marked 12 mm thick commerical ply	Sqm	983.00
8.6.5A	**IS 710- 1976 marked 19 mm thick commerical ply**	**Sqm**	**1130.00**
8.6.6	IS 710- 1976 marked 12 mm thick Teak ply BWR one side	Sqm	1655.00
8.6.7	IS 710- 1976 marked 12 mm thick Teak ply BWR both side	Sqm	1885.00
8.6.8	Wire gauge 24 gauge 14 mesh of IS 1568-1970 marked 85 gauge of dia 0.56 mm	Sqm	632.00
8.6.9	SS wire gauge 14 mesh x 24 gauge	Sqm	927.00
8.6.10	4 mm plain float glass	Sqm	763.00
8.6.11	5 mm float glass	Sqm	927.00
8.6.12	8 mm thick glass	Sqm	1314.00
8.7	Extra for providing frosted glass panes instead of ordinary float glass panes in doors, windows and clerestory window shutters. (Area of opening for glass panes excluding portion inside rebate shall be measured).	Sqm	11.00
8.8	Deduct for providing pin headed glass panes instead of ordinary float glass panes weighing 4 mm thick in doors, windows and clerestory windows, shutters (Area of opening for glass panes excluding portion inside rebate shall be measured).	Sqm	87.00
8.9	Extra for providing IS 1341-1992 marked Stainless Steel butt hinges instead of black enamelled M.S. butt hinges with necessary screws. (Shutter area to be measured).	Sqm	115.00

8.10	Extra for providing movable/ fixed Nigeria / Ghana /Ivory Teak wood grade - I louvers instead of wooden panels in column No.4 for Nigeria / Ghana/Ivory Teak wood door shutters.	Sqm	10 %
8.11	Providing Single leaf shutter instead of double leaf.	Sqm	20 %
8.12	Add extra for etching work in glass work as per direction of engineer incharge (Area of opening for glass panes excluding portion inside rebate shall be measured)	Sqm	20%
8.13	Providing and fixing external grade board solid core single leaf flush door shutters ISI 2202-67 marked using Phenol formal dehyderesin in glue both sides with approved steel fittings complete as per annexure 'A' :		
8.13.1	25 mm thick .		
8.13.1.1	Commercial Veneer both side	Sqm	1533.00
8.13.1.2	Decorative teak veneer One side	Sqm	1722.00
8.13.1.3	Decorative teak veneer both side	Sqm	2160.00
8.13.2	30 mm thick .		
8.13.2.1	Commercial Veneer both side	Sqm	1674.00
8.13.2.2	Decorative teak veneer One side	Sqm	1848.00
8.13.2.3	Decorative teak veneer both side	Sqm	2218.00
8.13.3	35 mm thick .		
8.13.3.1	Commercial Veneer both side	Sqm	1810.00
8.13.3.2	Decorative teak veneer One side	Sqm	2044.00
8.13.3.3	Decorative teak veneer both side	Sqm	2277.00
8.14	Extra for providing external lipping with 2nd class teak wood battens 6 mm minimum depth on all edges of shutters (over all area of door shutter to be measured) Over item no. 8.13.	Sqm	128.00
8.15	Extra for providing vision panel not exceeding 0.1 Sqm in all type of flush doors (cost of glass excluded) (overall area of door shutter to be measured) :	Each	116.00
8.16	Extra if louvers (not exceeding 0.2 **sqm**) are provided in flush door shutters (overall area of door shutters to be measured). .	Sqm.	280.00

8.17	Extra for cutting rebate in flush door shutters for making double leaf shutter(Total area of the shutter to be measured).	Sqm.	81.00
8.18	Extra for providing other type of fittings as mention in annexure A (as approved by EI) :		
8.18.1	Brass fittings	Sqm.	20 %
8.18.2	Stainless steel fittings	Sqm.	10%
8.19	Providing and fixing Hollow core partition wall using 50x30 mm Chir wood frame 600 mm c/c on both directions,door opening as per drawing with steel fittings antitermite treatment complete as per specifications using :		
8.19.1	ISI 4 mm thick commercial ply with 0.80 mm mica with adhesive on both sides.	Sqm	1858.00
8.19.2	ISI 4 mm teak ply one side BWR	Sqm	1753.00
8.20	Providing and fixing wall panelling using soft wood frame of size 50x25mm @600 mm c/c in both directions duly treated with anti-termite treatment making grooves painting of grooves specification and drawing using :		
8.20.1	ISI 6 mm teak ply one side BWR	Sqm	1349.00
8.20.2	ISI 9 mm teak ply one side BWR	Sqm	1624.00
8.20.3	12mm thick Acoustical Board	Sqm	1296.00
8.20.4	12 mm thick one side laminated board Practical board (Phenol bonded)	Sqm	1589.00
8.21.1	Providing and fixing 50 mm thick resin bounded glass **wool** 32 kg/cum density wrapped in julities cloth bag (box type) and held in position using chicken mesh	Sqm	509.00
8.21.2	Providing and fixing Hassian cloth of approved colour and texture over soft wood frame glass wool & fibre glass **tissue** paper	Sqm	284.00
8.22	Providing and fixing expandable fastners of specified size with necessary plastic sleeves and galvanised M.S. screws including drilling holes in masonry work /CC/R.C.C. and making good etc. complete		
8.22.1	25 mm long	Each	12.00
8.22.2	32 mm long	Each	15.00
8.22.3	40 mm long	Each	16.00

8.22.4	50 mm long	Each	17.00
8.23	Providing and fixing Nigeria / Ghana /Ivory Coast Teak wood grade-I mouldings and beadings in **doors** and windows size :-		
8.23.1	50x12 mm	Mtr	81.00
8.23.2	50x20 mm	Mtr	115.00
8.23.3	25x12 mm	Mtr	65.00
8.23.4	25x20 mm	Mtr	80.00
8.24	Providing and fixing 18 mm thick, 150 mm wide pelmet of flat pressed 3 layer or graded wood particle board medium density grade I, IS : 3087 marked including top cover of 6 mm commercial ply wood conforming to IS: 303 BWR grade, nickel plate MS 20 mm dia curtain rod with nickel plated **bracket** including fixing with 25x3 mm MS flat 10 cm long all complete	Mtr	310.00
8.25	Providing and fixing curtain rods of 1.25 mm thick chromium plated brass plate, with two chromium plated brass brackets fixed with C.P. brass screws and wooden plugs, etc., wherever necessary complete:		
8.25.1	12 mm	Mtr..	210.00
8.25.2	20 mm	Mtr..	242.00
8.26	Providing and fixing IS : 3564 marked Aluminium die cast body tubular type universal hydraulic door closer with necessary accessories and screws etc. complete.	Each	825.00
8.27	Providing and fixing IS : 3564 marked aluminium extruded section body tubular type universal hydraulic door closer with double speed adjustment with necessary accessories and screws etc. complete.	Each	1344.00
8.28	Providing and fixing Bamboo jaffery/ fencing consisting of superior quality 25mm dia (Average) half cut bamboo placed vertically and fixed together with three numbers horizontal running members of wood in scantling of section 50X25mm fixed with nails and GI wire to existing surface complete as per direction of EI	Sqm	375.00

8.29	Providing and fixing wooden moulded corner beading of triangular shape to the junction of panelling etc. with iron screws, plugs and priming coat on unexposed surface etc. complete 2nd class teak wood. 50x50mm (base and height)	Rm	165.00
8.30	Providing and fixing bright finished Mortice lock of approved make Godrej or equivalent with pair of CP handles for doors with necessary screws etc complete (Best make of approved quality) as per direction of Engineer-in-charge.	Each	572.00
8.31	Extra for providing Brass handle instead of CP handle	Each	10%
8.32	Providing and fixing Almirah lock (50 to 70 mm size)of approved make without pair of handles for door with necessary screws etc complete as per direction of Engineer-in-charge.	Each	136.00
8.33	**Providing & Fixing of ward robe shutters made out of 19mm thick BWP block board ISI of superior quality make. Front side finishing with 4mm thick teak veneer approved make and 0.8mm thick approved shade mica in the inner side. Covering with teak wood beading 25x12mm all the sides of shutter & 35x12mm on outer frame for making good the wall joint including heavy brass S.S. fittings (Handle 200mm 2 Nos., Tower Bolt 150mm 2 Nos., Brass Lock 1 Nos., Magnetic Catcher 2 Nos., Brass Hinges 75x40x2.5 mm 6 Nos., SS Hanger Rod 20mm dia) complete with melamine polishing as per drawing and design approved by Engineer in Charge.**	**Sqm**	**6737.00**
8.34	**Providing and Fixing Bamboo Zafari/Fencing consisting of superior quality 25mm dia Solid Bamboo placed vertically & fixed together with 3 Nos Horizontal running members of wood in scantling of section size 50x40mm and fixed with nut and bolt to existing surface/angle post as per direction given by Engineer-in-charge**	**Sqm**	**855.00**

8.35	**Providing & Fixing mica of approved make for inner/outer side of shutters with fevicol & nails complete as per approved by engineer in charge.**		
8.35.1	**0.6mm thick mica**	**Sqm**	**570.00**
8.35.2	**0.8mm thick mica**	**Sqm**	**693.00**
8.36	**Providing & Fixing decorative veneer of approved make for inner/outer side of shutters with fevicol & nails complete as per approved by engineer in charge.**		
8.36.1	**4.0mm thick veneer**	**Sqm**	**1304.00**
8.36.2	**6.0mm thick veneer**	**Sqm**	**1426.00**
8.37	**Providing and fixing S.S. Rod 30mm dia (304 grade ,16SWG) including brackets fixed with C.P. brass screws and rawl plugs, etc., wherever necessary complete two side central all fittings.**	**Rmt**	**1034.00**

CHAPTER : B-9
STEEL & FENCING WORKS

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
9.1	Structural steel work in single section fixed with or without connecting plate including cutting, hoisting (height upto 10 m), fixing in position and applying a priming coat of approved steel primer all complete.	Kg.	63.00
9.1.1	**Add Extra if I/rail section is used**	**Kg.**	**10%**
9.2	Structural steel work riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting (height upto 10 m) , fixing in position and applying a priming coat of approved steel primer all complete:	Kg.	70.00
9.3	Steel work in built up tubular trusses including cutting, hoisting (height upto 10 m)fixing in position and applying a priming coat of approved steel primer, welded and bolted including special shaped washers etc. complete.		
9.3.1	Hot finished welded type tubes.	Kg.	91.00
9.3.2	Hot finished seamless type tubes.	Kg	99.00
9.3.3	Electric resistance or induction butt welded tubes.	Kg.	119.00
9.4	Extra for hoisting and fixing for every additional one metre or part thereof beyond 10 mtr height.	Kg.	5.00
9.5	Providing and fixing T-iron frames for doors, windows and ventilators of mild steel Tee-sections, joints mitred and welded with 15x3 mm lugs 10cm long embedded in cement concrete blocks 15x10x10 cm of 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominal size) or with wooden plugs and screws or rawl plugs and screws or with fixing clips or with bolts and nuts as require including fixing of necessary butt hinges and screws and applying a priming coat of approved steel primer.	Kg.	72.00

9.6	Providing and fixing pressed steel door frames confirming to IS 4351 manufactured from commercial mild steel sheet of 1.25 mm thickness including hinges jamb, lock jamb, bead and if required angle threshold of mild steel angle of section 50x25mm, or base ties of 1.25mm pressed mild steel welded or rigidly fixed together by mechanical means, adjustable lugs with split end tail to each jamb including steel butt hinges 2.5mm thick with mortar guards, lock strike-plate and shock absorbers as specified and applying a coat of approved steel primer after pre-treatment of the surface including filling with CC 1:2:4(M15 grade) complete as directed by Engineer-in-charge :		
9.6.1	Profile B	Mtr.	390.00
9.6.2	Profile C	Mtr.	426.00
9.6.3	Profile E	Mtr.	460.00
9.7.1	Supply and fixing in cement mortar welded hand railing made out of MS round or square bars, flats etc. for staircase or verandah as per design complete in all respect (wooden or PVC hand railing to be paid extra) with priming coat of red oxide.	Kg.	74.00
9.7.2	Extra for providing D type pipe head as top of hand railing	Kg.	10.00%
9.8.1	Providing and fixing steel gate, grating , and grills made of angles, tees, square bars, flats,or black pipe with holdfast and fittings complete as per design and drawing including cutting welding and fabrication with priming coat of red oxide	Kg.	87.00
9.8.2	Extra if square,rectangular hollow tubelar sections are used or grill made by flats only	Kg.	10.00%
9.9	Providing and fixing cow catcher made of R.S. joist angles channels MS rounds including cutting welding and fabrication with priming coat of red oxide (as per design).	Kg.	67.00
9.10	Providing and fixing in position collapsible steel shutters with vertical M.S. Channels 20 x 10 x 2mm and bracket with flat iron diagonals 20 x 5mm. size with top and bottom rail of T-iron 40 x 40 x 6mm. with 40mm dia steel pulleys/ball bearing complete with bolts, nuts locking arrangements inside and outside stoppers, handles etc. as per specification including applying a priming coat of approved steel primer. (To be measured and paid as per outer dimension).	Sqm.	4704.00
9.11	Providing and fixing 1mm thick M.S. sheet sliding-shutters with frame & diagonals braces of 40 x 40 x 6mm. angle iron 3mm M.S. gusset plates at the junction and corners, 25mm dia pulley 40x40x6mm angle and T-iron guide at the top and bottom respectively including applying a priming coat of approved steel primer.	Sqm	3254.00

9.12	Providing and fixing 1mm thick M.S. sheet garrage door shutters with frame of 40 x 40 x 6mm. angle iron 3mm M.S. gusset plates at the junction and corners including all fittings and applying a coat of approved steel primer : (Excluding cost of frames)		
9.12.1	Using M.S. angles 40 x 40 x 6mm for diagonal braces.	Sqm.	2855.00
9.12.2	Using flats 30 x 6mm for diagonal braces and central cross piece.	Sqm.	2721.00
9.13	Supplying and fixing rolling shutters of approved make, made of 80 x 1.25mm. M.S. laths interlocked together through their entire length and jointed together at the end by end locks mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete including cost of spring hooks, providing and fixing necessary 25.3cm. long wire springs grade No. 2 and M.S. top cover 1.25mm thick for rolling shutters as per design & IS 6248- 1979. (Clear openable area to be measured for payment).	Sqm.	1885.00
9.14	Providing & fixing ball bearing for rolling shutters	Each	469.00
9.15	Extra for providing mechanical device chain and crank operation for operating rolling shutters.		
9.15.1	Exceeding 10.00 Sqm and upto 16.80 Sqm in the area.	Sqm.	748.00
9.15.2	Exceeding 16.80 Sqm in the area.	Sqm.	806.00
9.16	Extra for providing grilled rolling shutters manufactured out of 8 mm dia. M.S. bar instead of laths as per design approved by Engineer-in-charge. (area of grill to be measured).	Sqm.	297.00
9.17.1	Providing and fixing M.S. sheet 1mm thick single leaf door shutter in angle iron frame 35x35x5mm suitabilly diagonally braced with 25x3mm flat iron above and below lock rail of size 50x5mm beading extra including all fittings, as per direction of Engg. incharge but excluding cost of chowkhats: including two coats of anit-corrosive red oxide primer paint	Sqm.	2608.00
9.17.2	Extra for double leaf shutters.	Sqm.	20%
9.18	Fixing standard steel glazed doors, windows and ventilators in walls with 15x3mm lugs 10cm long embedded in cement concrete blocks *15x10* x10 cm of 1:3:6 (1 cement: 3 coarse sand: 6 graded stone aggregate 20 mm nominal size) or with wooden plugs and screws or rawl plugs and screws or with fixing clips or with bolts and nuts as required, including fixing of glass panes with glazing clips and special metal-sash putty of approved make, or metal beading with screws (only steel windows with lugs, glass pane cut to size and glazing clips or metal beading with screws, shall be supplied by department free of cost.)	Sqm.	253.00

9.19	Providing and fixing steel glazed doors windows and ventilator shutters of standard rolled steel section (IS 1038-1983) joints mitred and welded with steel lugs 13x3mm, 10cm. long embedded in cement concrete block 15x10x10cm. of 1:3:6 (1cement : 3 coarse sand : 6 graded stone aggregate 20mm nominal size) or with wooden plugs and screws or rawl plugs and screws with fixing clips or with bolts and nuts as required including providing and fixing of plain glass panes 4mm thick with cooper glazing clips and special metal sash putty of approved make or metal beading with screws complete including priming coat of approval steel primer, excluding the cost of metal beading and other fitting except necessary hinges of pivots steel handles peg stay etc. as required :		
9.19.1	Doors	Sqm.	2817.00
9.19.2	Windows fixed	Sqm.	2809.00
9.19.3	Windows side hung /Ventilators top or centre hung (openable)	Sqm.	2850.00
9.19.4	Partly fixed and partly openable [fixed area not to exceed 33%]	Sqm.	2809.00
9.20	Providing brass handles/peg stays and fittings in place of oxidized fitting	Sqm.	5.00%
9.21	Aluminium handles/peg stays fitting instead of oxidized fitting.	Sqm.	3.00%
9.22	5mm thick plain glass instead of 4mm thick plain glass	Sqm.	86.00
9.23	Deduct for Not providing glass panels	Sqm.	283.00
9.24	Deduct for Pin headed 4mm thick glass panels is used instead of 4mm thick plain glass	Sqm.	5.00%
9.25	Deduct for Not provided oxidised fittings.	Sqm.	5.00%
9.26	Supplying and fixing fixed wire gauge of 14 mesh x 24 gauge to the metal frame of rolled section by metal beading 20x3mm with suitable screw at not exceeding 150mm distance.	Sqm.	561.00
9.27	Providing and fixing Square bars or other flat welded to window, ventilatiors etc.	Kg.	59.00

9.28	Providing and fixing steel glazed window frame made out of 80x40 mm hollow sheet section of 16 gauge thickness, joint mitred welded and grinded including hold fast of steel lugs 13mm x 3mm and 15 Cm long embedded in C C block 15 x 10 x 10 Cm of 1:3:6 nominal concrete and including fixing of pivoted hinges of superior quality, window shutters made out of 50 x 25.0 mm hollow steel section 15 mm paitam of 18 gauge thickness, joint mitred and grinded including 10mm x 10mm square bars welded to frame for paitam fixing float glass 4mm thick panes with glazing clips and metal sash putty and fixing of shutters frames peg stay, U shape handle 100 mm long, tower bolts 100 mm long of steel powder coated superior quality including fixing and jointing with frame hinges priming coat with steel primer complete in all respect as per direction of Engineer-in –charge		
9.28.1	Window openable.	Sqm	3457.00
9.28.2	Window fixed.	Sqm	3183.00
9.28.3	Extra for additional shutter in pipe section windows with wire gauge 14 mesh x 24 gauge	Sqm	1504.00
	FENCING WORK		
9.29	180cm high fencing with angle iron post 55x55x6mm. placed at every 3Mtr. apart 45cm. in ground embedded in cement concrete 1:3:6 (30x30x60cm) corner and every tenth post to be strutted with 55x55x6mm. angle iron provided with 6 horizontal lines and two diagonals of black barbed wire between two posts fitted and fixed with G.I. staples including earth work in excavation etc. complete.	Mtr.	533.00
9.30	150cm high fencing with angle iron 50x50x6mm. placed at every 3Mtr. apart 30cm. in ground embedded in cement concrete 1:3:6 (30x30x45cm) corner and every tenth post to be strutted with 50x50x6mm. angle iron provided with **5 horizontal lines** and two diagonals of black barbed wire between two posts fitted and fixed with G.I. staples including earth work in excavation etc. complete.	Mtr.	465.00
9.31	120cm. high fencing with angle iron posts 50x50x6mm. placed at every 3 Mtr. apart 30cm. in ground embedded in cement concrete 1:3:6 (30x30x45cm) corner & every tenth posts to be strutted with 50x50x6mm angle iron provided with **four horizontal lines** and two diagonals of black barbed wire between two posts fitted and fixed with G.I. staple including earth work in excavation etc. complete.	Mtr.	427.00

9.32	150 cm high fencing of precast R.C. posts of 15cmx15cm tapered to 10x10cm at top placed at every 3m apart 30cm in ground embedded in cement concrete 1:3:6 (30x30x45cm) corner and every tenth posts to be strutted with same R.C. posts provided with 6 horizontal lines and two diagonals of black barbed wire between the two posts fitted and fixed with G.I. staples including earth work in excavation etc. complete	Mtr.	457.00
9.33	Add or deduct for each wire line over basic item	Mtr.	11.00
9.34	Add or deduct for additional or less higher per 10 Cm. over basic item	Mtr	5.00%
9.35.1	Extra for using galvanized barbed wire instead of black barbed wire.	Mtr. Each **wire**	5.00%
9.35.2	**Add Extra if barbed wire fencing is done on wall upto height of 3.00 Mtr.**	**Mtr.**	**5.00%**
9.36	Supplying and fixing of chain link fencing with angle iron posts 50x50x6mm placed at every 3 Mtr. apart 30cm in ground embedded in cement concrete 1:3:6 (30x30x45cm) corner and every tenth post to be strutted with (50 x 50 x 6cm) angle iron provided and fixed and fitted with posts including earth work in excavation etc. complete with chain link size.		
9.36.1	50mm x 50mm x 3.15mm	Sqm.	572.00
9.36.2	75mm x 15mm x 3.15mm	Sqm.	514.00
9.36.3	100mm x 100mm x 3.15mm	Sqm.	474.00
9.36.4	150mm x 150mm x 3.15mm	Sqm.	436.00
9.37	Providing and fixing concertina coil fencing with required dia 610 mm (having 15 nos. round per 6 metre length) up to 3m height of wall with existing angle iron 'Y' shaped placed 2.4 m or 3.00 m apart and with 9 horizontal R.B.T. stud tied with GI staples and GI clips to retain horizontal including necessary bolts or GI barbed wire tied to angle iron all complete as per direction of Engineer In Charge with reinforced barbed tape (R.B.T.)/Spring core(2.5 mm thick) wire of high tensile strength of 165 kg/sq.mm with tape (0.52 mm thick) and weight 43.478 gm/metre (Cost of MS angle, CC block shall be paid extra)	Mtr.	242.00
9.38	Providing and fixing welded mesh/expanded metal mesh in frame work, flat iron beading 20x3mm including top cross laps inside and out sides welding, iron bolts, crews, clips etc. complete (excluding frame work) of size.		
9.38.1	Welded mesh 50 mmx 50 mm x 2.1 mm	Sqm.	630.00
9.38.2	Welded mesh 50 mmx 25 mm x 2.1 mm	Sqm.	656.00

9.38.3	Welded mesh 25 mmx 25 mm x 2.1 mm	Sqm.	682.00
9.38.4	Expended metal size 20-25 mm size 18SWG	Sqm.	708.00
9.39	Providing and fixing of double leaf steel shutter for cupboard, the frame is made of pressed steel ISI section 80x25 mmx 1.25 mm thick with 1.00 mm MS sheet to spot welded including holdfasts of 15x3mm, MS oxidized fittings such as butt hinges, sliding doors bolts, handles, tower bolts and twin peg etc complete including applying priming coat of approved steel primer of red oxide zinc chromate and spray painting in approved shade with deco paint of approved quality complete as per direction of Engineer In Charge	Sqm	3541.00

CHAPTER : B-10
ROOFING

Note :

1. Rates are for complete item of work and are inclusive of all centering, shuttering, curing and finishing including all leads and delivery of material at site of work with all lead and lift.
2. The rates for G.I. sheet roofing are inclusive of all necessary, overlaps and wastage in cutting and all standard screws nuts washers bolts and nuts 'J' or 'L' hooks etc. required as per specifications.
3. All precautions shall be taken to ensure that the finished roof is water tight and without any leakage under all weather conditions. Particular care shall be taken in providing good workmanship specially at vulnerable points.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	GALVANISED IRON SHEET ROOFING		
10.1	Providing corrugated G.S. sheet roofing including vertical / curved surface fixed with polymer coated J or L hooks, bolts and nuts 8 mm diameter with bitumen and G.I. limpet washers or with G.I. limpet washers filled with white lead and including a coat of approved steel primer and two coats of approved paint on overlapping of sheets complete (upto any pitch in horizontal/ vertical or curved surfaces) excluding the cost of purlins, rafters and trusses and including cutting to size and shape whenever required.		
10.1.1	1.00 mm thick with zinc coating not less than 275 gm/m^2	Sqm	822.00
10.1.2	0.80 mm thick with zinc coating not less than 275 gm/m^2	Sqm	695.00
10.1.3	0.63 mm thick with zinc coating not less than 275 gm/m^2	Sqm	586.00
10.2	Extra for straight cutting in C.G.S. sheet roofing for making opening of area exceeding 40 sq. decimetre for chimney stacks, sky light etc. :	P.Mtr. Peri- meter	28.00
10.3	Extra for circular cutting in C.G.S. sheet roofing for making opening of area exceeding 40 square decimetre :	P.Mtr. Peri- meter	130.00
10.4	Providing ridges or hips of width 60 cm over all width plain G.S. sheet fixed with polymer coated J or L hooks, bolts and nuts 8 mm dia. G.I. limpet and bitumen washers complete.		

10.4.1	0.80mm thick with zinc coating not less than 275	Mtr.	457.00
10.4.2	0.63mm thick with zinc coating not less than 275	Mtr.	401.00
10.5	Providing valleys of 90cm wide overall in plain G.S. sheet fixed with polymer coated J or L hooks, bolts and nuts 8mm dia. G.I. limpet and bitumen washers complete : 1.60mm thick with zinc coating not less than 350gm/m²	Mtr.	852.00
10.6	Providing flashing of 40 cm over all width in plain, G.S. sheet fixed with polymer coated J, or L hooks, bolts and nuts, G.I. limpet and bitumen washer complete, bent to shape and fixed in wall with cement mortar 1:3 (1cement : 3 coarse sand) 1.00mm thick sheet	Mtr.	325.00
10.7	Providing and fixing 15 cm wide 45 cm over all semi circular plain G.S. sheet gutter with iron brackets 40x3mm size, bolts, nuts and washers etc. including making necessary connections with rain water pipes complete.		
10.7.1	0.80mm thick	Mtr.	421.00
10.7.2	0.63mm thick	Mtr.	368.00
10.8	Providing non-asbestos reinforced by organic and/or inorganic reinforced cement 6 mm thick corrugated sheets (As per IS: 14871) roofing upto any pitch and fixing with polymer coated J, or L hooks, bolts and nuts 8mm dia. G.I. plain and bitumen washers or with self drilling fastener and EPDM washers etc. complete excluding the cost of purlins, rafters and trusses corrugated sheets and including cutting to size and shape wherever required	Sqm	348.00
10.9	Extra for straight cutting in non-asbestos reinforced by organic and/or inorganic reinforced cement 6 mm thicksheet roofing for making openings of area exceeding 40 square decimetre for chimney stacks, skylights etc.	Mtr.	25.00
10.10	Extra for circular cutting in non-asbestos reinforced by organic and/or inorganic reinforced cement 6 mm thick sheet roofing for making openings of area exceeding 40 square decimetre.	Mtr.	68.00
10.11	Extra for providing and fixing wind ties of 40x 6mm flat iron section.	Mtr.	103.00
10.12	Providing and fixing ridges and hips in non-asbestos reinforced by organic and/or inorganic reinforced roofing with suitable fixing accessories or self drilling fastener and EPDM washer etc. complete.		
10.12.1	One piece corrugated serrated adjustable ridges	Mtr.	352.00
10.12.2	Plain wing adjustable ridges	Mtr.	376.00

10.12.3	Close fitting adjustable ridges	Mtr.	464.00
10.12.4	Unserrated adjustable hips	Mtr.	396.00
10.13	Providing and fixing non-asbestos reinforced by organic and/or inorganic reinforced roofing accessories in all colours with polymer coated J or L hooks, bolts and nuts and or G.I. seam bolts and nuts, G.I. plain and bitumen washers or with self drilling fastener and EPDM washer etc. complete :		
10.13.1	Corrugated appron pieces	Mtr.	238.00
10.13.2	Eaves filler pieces	Mtr.	163.00
10.13.3	North light curve	Mtr.	353.00
10.13.4	Ventilator curve	Mtr.	442.00
10.13.5	Barge boards	Mtr.	361.00
10.13.6	Ridge final	pair	148.00
10.13.7	Special north light ventilator curve	Each	464.00
10.13.8	S type louvers	Mtr.	252.00
10.14	Providing flat iron brackets 50x3mm size with necessary bolts, nuts and washers etc. for fixing asbestos cement/G.S. sheets gutters with purlins.	Mtr.	38.00
10.15	Painting top of roofs with bitumen of approved quality at 17kg per 10 **sqm** impregnated with a coat of coarse sand at 60 cudm per 10sqm including cleaning the slab surface with brushes and finally with a piece of cloth lightly soaked in kerosene oil complete: With residual type petroleum bitumen of penetration 80/100	Sqm	130.00
10.16	Stone slab roofing on ground floor with fine grained stone slab from approved quarry including filling of joints of parapet and slab on both sides in cement sand mortar 1:4, with ceiling pointing in cement sand mortar 1 : 3 complete as per specification and instruction of Engineer In Charge	Sqm	1498.00
10.16.1	Add extra for subsequent story	Sqm	20%
10.17	Grading roof for water proofing treatment with water proffing compound		
10.17.1	Cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size)	Cum	4075.00
10.17.2	Cement mortar 1:3 (1 cement : 3 coarse sand)	Cum	5310.00
10.17.3	Cement mortar 1:4 (1cement : 4 coarse sand)	Cum	4420.00
10.18	Providing and laying brick kharanja on roofs with cement mortar 1:4 (1 cement : 4 coarse sand) mixed with 2% integral water proofing compound in square pattern		
10.18.1	75mm to 100mm thick average	Sqm	252.00
10.18.2	55mm to 60mm thick average	Sqm	228.00

10.19	Providing and laying brick tiles of class designation 100 over roofs grouted with cement mortar 1:3 (1 cement : 3 fine sand) mixed with 2% of integral water proofing compound by weight of cement, over a 12 mm layer of cement mortar 1:3 (1 cement : 3 fine sand) and finished neat .	Sqm	325.00
10.20	Providing and fixing 20 mm thick precast terrazo tiles of approved make, with marble chips of size upto 6mm laid over roof with neat cement slurry mixed with pigment to match the shade of the tiles, including cutting ,grinding rubbing with machine complete on 20 mm thick bed of cement sand mortar 1 : 4.	Sqm	318.00
10.21	Providing and laying integral cement based water proofing treatment including preparation of surface as required for treatment of roofs, balconies, terraces etc consisting of following operations. (a) Applying and grouting a slurry coat of neat cement using 2.75 kg/sqm. of cement admixed with proprietary water proofing compound conforming to IS. 2645 over the RCC slab including cleaning the surface before treatment. (b) Laying cement concrete using broken bricks/brick bats 25 mm to 100mm size with 50% of cement mortar 1:4 (1 cement : 4 coarse sand) admixed with proprietary water proofing compound conforming to IS : 2645 over 20 mm thick layer of cement mortar of mix 1:4 (1 cement :4 coarse sand) admixed with proprietary water proofing compound conforming to IS : 2645 to required slope and treating similarly the adjoining walls upto 300 mm height including rounding of junctions of walls and slabs. (c) After two days of proper curing applying a second coat of cement slurry admixed with proprietary water proofing compound conforming to IS : 2645. (d) Finishing the surface with 20 mm thick jointless cement mortar of mix 1:4 (1 cement :4 coarse sand) admixed with proprietary water proofing compound conforming to IS : 2645 and finally finishing the surface with trowel with neat cement slurry and making of curing and for final test. All above operations to be done in order and as directed and specified by the Engineer-in-Charge. With average thickness of 120mm and minimum thickness at khurra as 65 mm.	Sqm	430.00
10.22	Providing gola 75x75 mm in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 stone aggregate 10mm and down gauge) including finishing with cement mortar 1:3 (1 cement : 3 fine sand) as per standard design :	Mtr.	101.00

10.23	Making khurras 450 x 450mm with average minimum thickness 50mm cement concrete 1 : 2:4 over P.V.C. Sheet 1 Mtr. x 1 Mtr. x 40micron finished with 12mm cement plaster 1 : 3 and one coat of neat cement, rounding edges and making and finishing the out let complete.	Each	140.00
10.24	Providing & laying broken glazed titles on roof (broken glazed tiles not less than 9 Kg/Sqm) on top of hot bitumen @ 1.7 Kg/Sqm. (1.40 Kg of VG-40 (30/40)grade and 0.30 Kg/Sqm. Of VG-10 (80/100) grade) and joints filled with cement mortar 1 :2 (1 cement:2 marble dust mixed with water proofing compund all complete as per direction of Engineer-in-charge.	Sqm	288.00
10.25	Providing & fixing UV stabilised fiberglass reinforced plastic sheet roofing upto any pitch including fixing with polymer coated 'J' or 'L' hooks, bolts & nuts 8mm dia. G.I plain/bitumen washers complete but excluding the cost of purlins, rafters, trusses etc. The sheets shall be manufactured out of 2400 TEX panel rovigs incorporating minimum 0.3% Ultra-voilet stabiliser in resin system under approximately 2400 psi and hot cured. They shall be of uniform pigmentation and thickness without air pockets and shall confoim to IS 10192 and IS 12866.The sheets shall be opaque or translucent, clear or pigmented, textured or smooth as specified.		
10.25.1	2.0mm thick (Wt. 3.3 Kg./Mtr. Sq.)	Sqm	911.00
10.25.2	3.0mm thick (Wt. 5.0 Kg./Mtr. Sq.)	Sqm	1231.00
10.25.3	5.0mm thick (Wt. 8.4 Kg./Mtr. Sq.)	Sqm	1872.00
10.26	Extra for corrugated Fiberglass Reinforced Plastic (FRP) (2.5" /4.2"/6") or step down (2.5" /4.2"/6") as specified.	Sqm	10%
	FALSE CEILING		
10.27	Providing and fixing 12.5 mm thick tapered edge gypsum board conforming to IS 2095- Part I at all height false ceiling including	Sqm	750.00
10.28	Providing and fixing 14 mm thick Acoustical with fine fissured tiles/ Mineral fibre high density tiles with butt edges false ceiling tiles of	Sqm	966.00

10.29	Providing & fixing 12 mm thick tegular edge Glass Fibre Reinforced Gypsum False ceiling tiles of size 595 x 595 mm in true horizontal level suspended on inter locking metal grid of hot dipped galvanised steel sections (galvanized @ 170 gsm/sqm.) consisting of main “T” runner with suitably spaced joints to get required length and of size 24x38mm made from 0.30mm thick (minimum) sheet spaced at 1200mm center to center and cross “T” of size 24x25mm made of 0.30mm thick (minimum) sheet, 1200mm long spaced between main “T” at 600mm center to center to form a grid of 1200x600 mm and secondary cross “T” of length 600mm and size 24x25mm made of 0.30 mm thick (minimum) sheet to be interlocked at middle of the 1200x600mm panel to form grids of 600x600mm and wall angle of size 21x21x0.30 mm and laying false ceiling tiles of approved texture in the grid including, wherever, required, cutting/making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc. Main “T” runners to be suspended from ceiling using GI slotted cleats fixed to ceiling with 6 mm dia and 50mm long dash fasteners, 4mm GI adjustable rods with galvanised level clips spaced at	Sqm	1241.00
10.30	Providing and fixing calcium silicate board false ceiling at all heights including providing and fixing of frame work made of special sections		
10.30.1	6 mm thick	Sqm	830.00
10.30.2	8 mm thick	Sqm	932.00
10.31	Providing & fixing calcium silicate ceiling tiles of size 595 x 595 mm in true horizontal level suspended on inter locking metal grid of hot dipped galvanised steel sections (galvanized @ 170 gsm/sqm.) consisting of main “T” runner with suitably spaced joints to get required length and of size 24x38mm made from 0.30mm thick (minimum) sheet spaced at 1200mm center to center and cross “T” of size 24x25mm made of 0.30mm thick (minimum) sheet, 1200mm long spaced between main “T” at 600mm center to center to form a grid of 1200x600 mm and secondary cross “T” of length 600mm and size 24x25mm made of 0.30 mm thick (minimum) sheet to be interlocked at middle of the 1200x600mm panel to form grids of 600x600mm and wall angle of size 21x21x0.30 mm and laying false ceiling tiles of approved texture in the grid including, wherever, required, cutting/making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc. Main “T” runners to be suspended from ceiling using GI slotted cleats fixed to ceiling with 6 mm dia and 50mm long dash fasteners, 4mm GI adjustable rods with galvanised level clips spaced at 1200mm center to center along main		
10.31.1	6 mm thick	Sqm	590.00
10.31.2	8 mm thick	Sqm	677.00
10.32	Providing and fixing calcium silicate perforated board false ceiling at all heights including providing and fixing of frame work made of		
10.32.1	6 mm thick	Sqm	1071.00

10.32.2	8 mm thick	Sqm	1197.00
10.33	Providing & fixing calcium silicate perforated ceiling tiles of size 595 x 595 mm in true horizontal level suspended on inter locking metal grid of hot dipped galvanised steel sections (galvanized @ 170 gsm/sqm.) consisting of main “T” runner with suitably spaced joints to get required length and of size 24x38mm made from 0.30mm thick (minimum) sheet spaced at 1200mm center to center and cross “T” of size 24x25mm made of 0.30mm thick (minimum) sheet, 1200mm long spaced between main “T” at 600mm center to center to form a grid of 1200x600 mm and secondary cross “T” of length 600mm and size 24x25mm made of 0.30 mm thick (minimum) sheet to be interlocked at middle of the 1200x600mm panel to form grids of 600x600mm and wall angle of size 21x21x0.30 mm and laying false ceiling tiles of approved texture in the grid including, wherever, required, cutting/making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc. Main “T” runners to be suspended from ceiling using GI slotted cleats fixed to ceiling with 6 mm dia and 50mm long dash fasteners, 4mm GI adjustable rods with galvanised level clips spaced at 1200mm center to center along main		
10.33.1	6 mm thick	Sqm	777.00
10.33.2	8 mm thick	Sqm	899.00
10.34	Providing and fixing 15 mm thick densified tegular edged eco friendly light weight calcium silicate false ceiling tiles of approved texture as	Sqm	1500.00
10.35	**Providing and fixing stainless steel (Grade 304) railing made of Hollow tubes, channels, plates etc., including welding, grinding,**	**Kg.**	457.00
10.36	**Providing and fixing tiled false ceiling of approved materials of size 595X595 mm in true horizontal level suspended on inter**		
10.36.1	**GI Metal Ceiling Lay in Plain Tegular edge Global white colour tiles of size 595X595 mm, and 0.5 mm thick with 8 mm drop, made of G I Sheet having galvanizing of 100 gms/sqm (boths ides inclusive) and electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending.**	**Sqm**	1420.00
10.36.2	**GI Metal Ceiling Lay in perforated Tegular edge global white colour tiles of size 595x595mm and 0.5 mm thick with 8 mm drop made of GI Sheet having galvanizing of 100 gms/ sqm (both sides inclusive) and 20% perforation area with 1.8 mm dia holes and having NRC (Noise Reduction Coefficient) of 0.5, electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending and perforation and backed with a black Glass fiber acoustical fleece.**	**Sqm**	1536.00

10.37	**Providing & Fixing of Mineral Fibre Anti Bacterial Suspended Ceiling System with Bioguard (Bevelled Tegular) EDGE TILES**	**Mtr**	1561.00
10.38	**Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with (Bevelled Tegular) EDGE TILES WITH 15 mm**	**Mtr**	1479.00
10.39	**Providing and fixing 300C Aluminium Panel with perforation of 2mm diameter and 5mm centre to centre to give 15% open area with Non woven textile False Ceiling system of approved colour consisting of Panel 300 mm wide X 30 mm deep X 0.7 mm thick with bevelled edges having panel length upto 6 mtrs, Coil Coated on a Continuous Paint Line, Double baked and roll formed for higher strength and good roll forming characteristics. The panel ends will be raised upto 29mm. Panel shall be clipped to a baked enamelled aluminium panel carrier of 41.5mm wide x 62mm deep x 0.95 mm thick in standard length of 5 mtrs. made of double baked stove enamelled aluminium magnesium alloy AA3005 black with cut outs to hold the panels. The Carrier shall be suspended by means of G.I. threaded rod of 6-mm diameter directly fixed to the Carrier. The Carrier tabs to be bent down to secure the panels at all Panel Carrier intersections. The panels will be installed in a module of 300MM. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	4314.00

10.40	**Providing and fixing of perforated Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.7mm thick aluminium alloy 3105 with beveled edge of 4.15mm width height of 32mm. The tiles shall be perforated to 1.5mm dia with open area of 23%. The Tile will be manufactured on advanced includes several leveling stages in the manufacturing process by approved manufacturers. Tile ends will be raised with pips and slops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be bevel edged. The Tile shall be epoxy polyester powder coating in white colour. The Tile shall be cllpped into clip in rail of 0.5mm thick GI. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G .I rod, Hold on Clamp with Clip. POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	3432.00

10.41	**Providing and fixing of Un-perforated Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.7mm thick aluminium alloy 3105 with beveled edge of 4.15mm width height of 32mm. The Tile will be manufactured on advanced equipment that includes several leveling stages in the manufacturing process by approved manufacturers . Tile ends will be raised with pips and slops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be bevel edged. The Tile shall be epoxy polyester powder coating with in white colour. The Tile shall be cllpped into clip in rail of 0.5mm thick GI. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G .I rod, Hold on Clamp with Clip. POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in combination of Infrared & hot air oven in a certified plant as Akzo International approved applicator .** **The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8.** **(Tensile strength of suspension-344/413 mpa and fastner strength-110kg)** **All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	2896.00

10.42	**Providing and fixing of Un perforated Antibacterial Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.7mm thick aluminium alloy 3105 with beveled edge of 4.15mm width height of 32mm. The Tile will be manufactured on advanced equipment that includes several leveling stages in the manufacturing process by approved manufacturers. Tile ends will be raised with pips and slops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be bevel edged. The Tile shall be epoxy polyester powder coating with ANTIBAC in white colour which is in compliant to ROHS directive. The Tile shall be cllpped into clip in rail of 0.5mm thick GI. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G .I rod, Hold on Clamp with Clip. POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in** **The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications.** **The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8.** **(Tensile strength of suspension-344/413 mpa and fastner strength-110kg)** **All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	3743.00

10.43	**Providing and fixing of Perforated Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.5mm thick Galvanised Iron with beveled edge of 4.15mm width height of 32mm. The tiles shall be perforated to 1.5mm dia with open area of 23%. The Tile will be manufactured on advanced equipment that includes several leveling stages in the manufacturing process by approved manufacturers. Tile ends will be raised with pips and slops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be bevel edged. The Tile shall be epoxy polyester powder coating in white colour. The Tile shall be cllpped into clip in rail of 0.5mm thick GI. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G .I rod, Hold on Clamp with Clip. POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in** **The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8.** **(Tensile strength of suspension-344/413 mpa and fastner strength-110kg)** **All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	3068.00

10.44	**Providing and fixing of Unperforated Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.5mm thick Galvanised Iron with beveled edge of 4.15mm width height of 32mm. The tiles shall be plain. The Tile will be manufactured on advanced equipment that includes several leveling stages in the manufacturing process by approved manufacturers. Tile ends will be raised with pips and slops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be bevel edged. The Tile shall be epoxy polyester powder coating in white colour. The Tile shall be cllpped into clip in rail of 0.5mm thick GI. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G .I rod, Hold on Clamp with Clip. POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in combination of Infrared & hot air oven in a certified plant as The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	2313.00

10.45	**Supply & Fixing of torsion spring Plank Ceiling System, comprising of Plank of 600mm wide and 1200mm long manufactured out of 0.7mm thick Aluminium Alloy 3105 perforated 1.5mm dia With 22% open area. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel . The Plank will be manufactured on advanced CAD/CAM equipment that includes several levelling stages in the manufacturing process. Torsion Spring panel with two side legs die formed and two end legs die formed and punched to receive torsion springs (min two springs each end or side) for secure engagement into Tee Grid main runners which are factory punched to receive torsion springs. Planks will be square edged. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel. The Plank shall be Polyester powder coated in desired colour. Main Runners: 24mm deep, inverted "Tee" sections, 3.6m long, with factory punched flanges to receive torsion spring assembly. Main Tee on center spacing to match panel length. Cross Runners: 24 mm deep, inverted "Tee" sections designed to interlock in to web of main tee section on designated spacing. As per manufacturer standard considering type of plenum and its height. Paint finish – POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in combination of Infrared & hot air oven in a certified plant as Akzo International approved applicator . The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8 Acoustic Felt (Optional): Non-woven felt made of glass-reinforced fibre glued over the perforation for sound absorption. NRC- 0.7. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	4775.00

10.46	**Supply & Fixing of torsion spring Plank Ceiling System, comprising of Plank of 600mm wide and 600mm long manufactured out of 0.7mm thick Aluminium Alloy 3105 perforated 1.5mm dia With 22% open area. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel . The Plank will be manufactured on advanced CAD/CAM equipment that includes several levelling stages in the manufacturing process. Torsion Spring panel with two side legs die formed and two end legs die formed and punched to receive torsion springs (min two springs each end or side) for secure engagement into Tee Grid main runners which are factory punched to receive torsion springs. Planks will be square edged. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel. The Plank shall be Polyester powder coated in desired colour. Main Runners: 24mm deep, inverted "Tee" sections, 3.6m long, with factory punched flanges to receive torsion spring assembly. Main Tee on center spacing to match panel length. Cross Runners: 24 mm deep, inverted "Tee" sections designed to interlock in to web of main tee section on designated spacing. Suspension System: As per manufacturer standard considering type of plenum and its height. Paint finish – POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in combination of Infrared & hot air oven in a certified plant as Akzo International approved applicator . The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8. Acoustic Felt (Optional): Non-woven felt made of glass-reinforced fibre glued over the perforation for sound absorption. NRC- 0.7. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	4697.00

10.47	**Supply & Fixing of torsion spring Plank Ceiling System, comprising of Plank of 600mm wide and 1200mm long manufactured out of 0.6mm thick GALVANISED IRON perforated 1.5mm dia With 22% open area. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel . The Plank will be manufactured on advanced CAD/CAM equipment that includes several levelling stages in the manufacturing process. Torsion Spring panel with two side legs die formed and two end legs die formed and punched to receive torsion springs (min two springs each end or side) for secure engagement into Tee Grid main runners which are factory punched to receive torsion springs. Planks will be square edged. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel. The Plank shall be Polyester powder coated in desired colour. Main Runners: 24mm deep, inverted "Tee" sections, 3.6m long, with factory punched flanges to receive torsion spring assembly. Main Tee on center spacing to match panel length. Cross Runners: 24 mm deep, inverted "Tee" sections designed to interlock in to web of main tee section on designated spacing. Suspension System: As per manufacturer standard considering type of plenum and its height. Paint finish – POWDER COATING SPECIFICATION: Aluminium PANELS, cleaned through eco friendly cleaning ,process and powder applied thru electrostatic charged Automatic Carona technology spray guns and cured in combination of Infrared & hot air oven in a certified plant as Akzo International approved applicator . The powder is conforming with the performance requirements of the AAMA2603.02 & Qualicoat Claa 1 specifications. The coating thickness will be minimum 60 microns with properties of Salt Spray of 1500 hrs in exposure in SST Cabinet with Minimum blister rating 8. Acoustic Felt (Optional): Non-woven felt made of glass-reinforced fibre glued over the perforation for sound absorption. NRC- 0.7. (Tensile strength of suspension-344/413 mpa and fastner strength-110kg) All Ceiling Panels shall adhere to GreenPro Certifications.**	**Sqm**	4047.00

CHAPTER : B-11
FLOORING WORKS

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	BRICK FLOORING		
11.1	Brick on edge flooring with bricks of class designation 75 including cement slurry & pointing in CM (1 : 3) etc. complete laid on 20 mm thick bed of :		
11.1.1	Cement Sand Mortar 1 : 4	Sqm	464.00
11.1.2	Cement Sand Mortar 1 : 6	Sqm	443.00
11.2	Dry brick on edge flooring in required pattern with bricks of class designation 75 on a bed of 12 mm mud mortar including filling joints with River sand complete.	Sqm	409.00
	CEMENT CONCRETE FLOORING		
11.3	Cement concrete flooring 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate) finished with a floating coat of neat cement including cement slurry, making of lines or groove etc complete but excluding the cost of nosing of steps etc. complete.		
11.3.1	40mm thick with 20mm thick nominal size aggregate.	Sqm	252.00
11.3.2	75mm thick with 20mm thick nominal size aggregate.	Sqm	416.00
11.4	Providing and fixing 50mm thick cement concrete flooring with Metallic concrete hardener topping, under layer of 38mm thick cement concrete 1:2:4 (1-cement : 2-coarse sand : 4-graded stone aggregate 20mm thick nominal size) and top layer of 12mm thick metallic concrete hardener consisting of mix 1:2 (1 cement : 2 stone aggregate, 6mm nominal size) by volume & mixed with metallic hardening compound of approved quality @ 2Kg./Sqm including cement slurry, rounding off edges etc. but excluding the cost of nosing of step etc. complete.	Sqm	412.00
11.5	Cement plaster in skirting and/dado 18 mm thick with cement mortar 1:3 (1 cement : 3 course sand) finished with a floating coat of neat cement including rounding of junctions with floor	Sqm	258.00

TERRAZZO FLOORING

11.6	40mm thick marble chips flooring rubbed and polished to granolithic finish under layer 31mm thick cement concrete 1:2:4 (1 cement : 2 course sand : 4 graded stone aggregate 12.5mm nominal size) including fixing of dividing strip (excluding the cost of strips) and top layer 9mm thick with marble chips of approved colour & 4mm to 7mm nominal size laid in cement marble chips 2 : 3 (2 cement : 3 marble chips) by volume including cement slurry etc. complete		
11.6.1	Dark shade pigment with ordinary cement.	Sqm	441.00
11.6.2	Light shade pigment with ordinary cement.	Sqm	643.00
11.6.3	Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	490.00
11.6.4	White cement without any pigment.	Sqm	473.00
11.6.5	Light shade pigment with white cement.	Sqm	433.00
11.6.6	Ordinary cement without any pigment.	Sqm	409.00
11.7	40mm thick marble chips flooring rubbed and polished to granolithic finish under layer 28mm thick cement concrete 1:2:4 (1 cement : 2 course sand : 4 graded stone aggregate 12.5mm nominal size) including fixing of dividing strip (excluding the cost of strips) and top layer 12mm thick with marble chips of approved colour 7mm to 10mm nominal size laid in cement marble chips mix by weight in proportion of 2 : 3 (2 cement : 3 marble chips) by volume including cement slurry etc. complete :		
11.7.1	Dark shade pigment with ordinary cement.	Sqm	472.00
11.7.2	Light shade pigment with ordinary cement.	Sqm	543.00
11.7.3	Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	517.00
11.7.4	White cement without any pigment.	Sqm	511.00
11.7.5	Light shade pigment with white cement.	Sqm	533.00
11.7.6	Ordinary cement without any pigment.	Sqm	426.00
11.8	Marble chips skirting/dado rubbed and polished to granolithic finish 6mm thick with white marble chips of approved colour and size from smallest to 4mm nominal size laid in cement marble chips mix in proportion of 4:7(4 cement : 7 marble chips) by volume including fixing of dividing strips (excluding the cost of strips) complete in all respects : 18mm thick with under layer of 12mm thick cement plaster 1:3 (1 cement : 3 coarse sand) :		

11.8.1	Dark shade pigment with ordinary cement.	Sqm	568.00
11.8.2	Light shade pigment with ordinary cement.	Sqm	606.00
11.8.3	Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	590.00
11.8.4	White cement without any pigment.	Sqm	591.00
11.8.5	Light shade pigment with white cement.	Sqm	564.00
11.8.6	Ordinary cement without any pigment.	Sqm	548.00
11.9	Providing and fixing glass strips in joints of terrazo/ cement concrete floors.		
11.9.1	40mm wide and 4mm thick	Mtr.	28.00
11.9.2	20mm wide and 3mm thick for skirting.	Mtr.	15.00
11.10	Extra for laying terrazo flooring on staircase treads not exceeding 30 cm in width including cost of forming, nosing etc.	Sqm	24.00
11.11	Extra for terrazzo flooring in narrow band bands not exceeding 7.5cm in width.	Sqm	13.00
11.12	Crazy marble stone flooring 20mm thick marble stone with black or of specified colour topping including filling the gaps with cement marble chips mix in proportion of 4 : 7 (4cement : 7 white black or white or black marble chips) size from 1mm to 4mm by volume and under layer of 25mm thick cement concrete [1:2:4, graded stone aggregate 12.5mm nominal size] rubbing and polishing and cement slurry etc. complete : In ordinary cement.	Sqm	526.00
	TILE FLOORING		
11.13	Precast terrazo tiles of approved make, 20mm thick with marble chips of size upto 6mm laid in floors, and landing, jointed with neat cement slurry mixed with pigment to match the shade of the tiles, including rubbing and polishing complete on 20mm thick bed of cement sand mortar 1 : 4.		
11.13.1	Light shade using white cement.	Sqm	493.00
11.13.2	Medium shade using approximately 50% white cement and 50% ordinary cement.	Sqm	499.00
11.13.3	Dark shade using ordinary cement.	Sqm	430.00
11.13.4	Ordinary cement without any pigment.	Sqm	485.00
11.14	Extra if terrazo tiles are laid in treads of steps not exceeding 30 cm in width.	Sqm	29.00

11.15	Precast terrazzo tiles of approved make 20mm thick with marble chips of size upto 6mm in skirting & risers, treads of steps and dados 12mm thick cement plaster 1:3 (1 cement : 3 coarse sand) jointed with neat cement slurry mixed with pigment to match the shade of the tiles, including rubbing and polishing complete with tiles of :		
11.15.1	Light shade using white cement.	Sqm	627.00
11.15.2	Medium shade using approximately 50% white cement and 50% ordinary cement.	Sqm	619.00
11.15.3	Dark shade using ordinary cement.	Sqm	611.00
11.15.4	Ordinary cement without any pigment.	Sqm	620.00
11.16	Providing & laying Chequered terrazzo tiles of approved make 22mm thick with marble chips of size upto 6mm, in floors, jointed with neat cement slurry mixed wit pigment to match the shade of tiles including rubbing and polishing complete in all respect as per specification, on 28mm thick bed of C.M. 1:4 :		
11.16.1	Light shade using shade cement.	Sqm	622.00
11.16.2	Medium shade using approximately 50% white cement and 50% ordinary cement.	Sqm	621.00
11.16.3	Dark shade using ordinary cement.	Sqm	562.00
11.16.4	Ordinary cement without any pigment.	Sqm	517.00
11.16.5	Add extra for non slipping 9-12 mm ironite topping tiles	Sqm	10%
11.17	Chequerred precast cement concrete tiles 22 mm thick in footpath & courtyard jointed with neat cement slurry mixed with pigment to match the shade of tiles including rubbing and cleaning etc. complete on 20 mm thick bed of cement mortar 1:4 (1 cement : 4 coarse sand)		
11.17.1	Light shade using shade cement.	Sqm	519.00
11.17.2	Medium shade using approximately 50% white cement and 50% ordinary cement.	Sqm	612.00
11.17.3	Dark shade using ordinary cement.	Sqm	475.00
11.17.4	Ordinary cement without any pigment.	Sqm	430.00
11.17.5	**Add Extra for polished chequered precast cement concrete tiles over item No. 11.17.1, 11.17.2, 11.17.3, 11.17.4**		10%

KOTA STONE FLOORING

11.18	Kota stone slab flooring 25 mm thick over 20 mm (average) thick base laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab including rubbing and polishing complete with base of cement mortar 1 : 4 (1 cement : 4 coarse sand)		

11.18.1	For area of each slab from 901 to 2000 Sq.Cm :	Sqm	864.00
11.18.2	For area of each slab from 2001 to 5000 Sq.Cm	Sqm	930.00
11.19	Kota stone slabs 25 mm thick in risers of steps, skirting, dado and pillars laid on 12 mm (average) thick cement mortar 1:3 (1 cement 3 coarse sand) and jointed with grey cement slurry mixed with pigment to match the shade of the slabs, including rubbing and polishing complete		
11.19.1	For area of each slab upto 2001 Sqcm	Sqm	829.00
11.19.2	For area of each slab from 2001 to 5000 Sqcm	Sqm	911.00
11.20	Rough Chisel dressed Kota stone 35 mm thick flooring of approved shade set in pattern over 20mm thick base of cement mortar 1:6 (1 cement : 6 coarse sand) and pointing with cement sand mortar 1:3 with pigment to match the shade of stone with joint thickness up to 15 mm complete in all respects :	Sqm	570.00
	SAND STONE FLOORING		
11.21	Rough Chisel dressed Sand stone 35 mm thick flooring of approved shade set over 20mm thick base of cement mortar 1:6 (1 cement : 6 coarse sand) and pointing with cement sand mortar 1:3 with pigment to match the shade of stone with joint thickness up to 15 mm complete in all respects :		
11.21.1	Red sand stone (Karauli stone)	Sqm	563.00
11.21.2	Pink sand stone (Bansi Paharpur stone)	Sqm	594.00
11.21.3	Mandana stone	Sqm	594.00
11.22	Fine dressed & machine cut edges 30-35 mm thicksand stone flooring over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) jointed grey cement slurry mixed with pigment to match the shade of stone complete as per design drawing and instruction of EI.		
11.22.1	Red sand stone (Karauli stone)	Sqm	703.00
11.22.2	Pink sand stone (Bansi Paharpur stone)	Sqm	803.00
11.22.3	Mandana stone	Sqm	803.00
11.23	Fine grinded Gangsaw cut 30-35 mm thick stone flooring over 20 mm (average) thick base of cement mortar 1:4 (1 cement :4 coarse sand) jointed grey cement slurry mixed with pigment to match the shade of stone complete as per design drawing and instruction of EI.		
11.23.1	Red sand stone (Karauli stone)	Sqm	750.00
11.23.2	Pink sand stone (Bansi Paharpur stone)	Sqm	852.00

11.23.3	Mandana stone	Sqm	852.00
11.24	Extra for pre finished nosing in treads of steps of Kota stone/ sand stone slab.	Sqm	65.00
11.25	Extra for Mirror polishing on terrazzo/Kota stone work where ever required to give high gloss finish complete.	Sqm	110.00
11.26	Random rubble dry stone Kharanja under floor.	Cum	847.00
11.27	Filling of sunk portion of roof with earthen pots of required height including filling voids with 1:3:6 light concrete, using brick bats complete levelling and dressing the surface by 50mm thick cement concrete 1:2:4 as per specification.	Cum	1614.00
11.28	Providing and fixing designer cement concrete polished tiles over 20mm average thick base of CM 1 :4 (1 Cement : 4 sand) jointing with cement mortar 1 :2 (1cement : 2 sand) with pigment to match the shade of the tiles complete with all respect. **(as per IS 1237:2012)**	Sqm	1178.00
11.29	Providing and laying at or near ground level factory made kerb stone of M-25 grade cement in position to the required line, level and curvature jointed with cement mortar 1:3 (1 cement : 3 coarse sand) including making joints with without grooves (thickness of joints except at sharp curve shall not to more than 5mm) including making drainage opening wherever required complete including painting etc. as per direction of Engineer-in-charge (length of finished kerb edging shall be measured for payment). (Precast C.C. kerb stone shall be approved by Engineerin- charge).	Cum	5686.00
11.30	Providing and laying M30 grade controlled cement concrete pavements including mixing and vibrations concrete with necessary needle vibrators and surface with screed board vibrating complete in all respect with 20/25 mm stone aggregate (crusher broken) including anti-skid textured finish to required camber/ super **elevation** and grade including curing etc. as per specification with cement concrete M-30 grade including cost of Steel frame work for sides of C.C. pavement consisting of M. S. channels flats and angles with required steel pegs (some of them may left embedded in concrete/ including providing frame work)	Cum	5013.00

11.31	Extra for vacuum processed concrete including cost of equipments labour etc.	Sqm..	92.00
11.32	Cutting of construction joint/ longitudinal joint 4 to 6 mm. wide using mechanical concrete cutter including cost of diamond bit cutting wheel and filling of bitumen sealing compound in groove including cost of sealing compound.		
11.32.1	25 mm depth	P Mtr.	42.00
11.32.2	50 mm depth	P Mtr.	85.00
11.32.3	75 mm depth	P Mtr.	127.00
11.32.4	100 mm depth	P Mtr.	170.00
	WOODEN FLOORING		
11.33	25mm thick wooden planking, tongued and grooved in flooring including fixing with iron screws complete (Lower frame to be paid extra) with :First class Teak wood.	Sqm	2703.00
11.34	Providing and laying wooden flooring with a direct laminated surface treated with aluminum oxide on top of a high-density fiber board with a density of 850 Kg./per Mtr. having a click system tongue and groove joint to secure a long lasting joint. The planks are or termite resistant and water protected with both side lamination Wooden flooring Planks of size 1208 x 194 x 8 mm / 1215x294x8 mm (8mm thick laminated floor class of use 23/31 having a wear resistance level of AC-3 , impact resistance > 9mm, moisture resistance 3-10% as per (EN-424, EN – 438 EN – 425 EN – 417 – 2)or PREN 3329 complete in all respect as per direction of EI	Sqm	2101.00
11.35	Providing and laying of skirting 89 mm wide having both side laminations with **moulding** in upper side with matching of floor fixing with nails and screws & fevicol complete in all respect as per direction of EI.	Rm	890.00
11.36	Providing and laying transaction/adaptation profile with fixing of nails, screws & fevicol complete in all respect as per direction of EI.	Rm	824.00
	PVC FLOORING		
11.37	Providing and Fixing Cushion Vinyl/Fully flexible Vinyl Flooring confirming to I.S.-3462 Flooring of approved make with fibre base using rubber base adhesives including rolling with light wooden roller complete :		

11.37.1	1.0 mm thick.	Sqm	691.00
11.37.2	1.5 mm thick.	Sqm	788.00
11.37.3	2.0 mm thick.	Sqm	900.00
	Paver Block		
11.38	Providing and laying of paver block as per IS 15658 : 2006 (Indian standard for precast concrete block for paving-specification) and IRC : SP:63-2004 (guidelines for the use of interlocking concrete block pavement)		
11.38.1	M-30, **60** mm thick to be used in non traffic areas like building premises, monument premises, land scapes, public gardens/ park, domestic drives, paths and patios, Embankment slopes, sand stabilization Area etc.(As per table 1 of IS 15658:2006		
(A)	Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	Sqm	781.00
(B)	Category 'B' Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	Sqm	751.00
(C)	Category 'C'not denated on any it's faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	Sqm	688.00
11.38.2	M-35,60mm thick to be used in light traffic (Light traffic is defined as a daily traffic up to 150 commercial vehicles exceeding 30 KN laden weight, or an equivalent up to 0.5 million standard axles (MSA) for a design life of 20 years (A standard axle is defined as a single axle load of 81.6 KN) like pedestrian plazas, shopping complexes ramps, car parks, office driveways , housing colonies, office complexes, rural roads with low volume traffic, farm houses , beach sites, tourist resorts, local authority footways, residential roads, etc.		
(A)	Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	Sqm	921.00
(B)	Category 'B' Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	Sqm	893.00
(C)	Category 'C'not denated on any it's faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	Sqm	803.00
11.38.3	M-40,80mm thick to be used in medium- traffic (Medium traffic is defined as a daily traffic of 150-450 commercial vehicles exceeding 30 KN laden weight, or an equivalent of 0.5 to 2.0 MSA for a design life of 20 years.) like city streets, small and medium market roads, low volume roads, utility cuts on arterial roads, etc		
(A)	Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	Sqm	1051.00

(B)	Category 'B' Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	Sqm	1021.00
(C)	Category 'C'not denated on any it's faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	Sqm	962.00
11.38.4	M-50 100mm thick to be used in heavy-traffic (Heavy traffic is defined as a daily traffic of 450-1500 commercial vehicles exceeding 30 KN laden weight, or an equivalent of 2.0 to 5.0 MSA for a design life of 20 years.) like Bus terminals, industrial complexes, mandi houses roads on expansive soils, factory floor , service station, industrial pavements.etc.		
(A)	Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	Sqm	1272.00
(B)	Category 'B' Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	Sqm	1243.00
(C)	Category 'C'not denated on any it's faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	Sqm	1184.00
11.38.5	M-55 120mm thick to be used in very heavy-traffic (very heavy-traffic is defined as a daily traffic of more than 1500 commercial vehicle exceeding 30 KN laden weight, or an equivalent of more than 5.0 MSA for a design life of 20 years.) like container terminals, ports, block yards, mine access roads, bulk cargo handling areas, airport pavements, etc.		
(A)	Category A Denated units to key into each other on four faces zigzag shape as per IRC SP 63:2004	Sqm	1482.00
(B)	Category 'B' Denated only two side like I,Z,T shape etc. as per IRC SP 63:2004	Sqm	1453.00
(C)	Category 'C'not denated on any it's faces like Hexagon, Rectangular, square shape as per IRC SP 63:2004	Sqm	1394.00
11.39	Add extra over item No. **11.38, A to C** if reflective paver block, manufactured with wet cast vibration system in two layers is used.	Sqm	10%
11.40	Add extra over rates of item No. **11.38, A to C** if colored paver blocks are used	Sqm	20%
11.41	Providing and fixing 25 mm thick Non slippery reflective type designer paving tile with metallic hardening using 2 Kg/sq.mtr made out of cement concrete of required grade, sizes of tile upto 300X300 mm Manufactured with wet cast Vibration system in two layers and laid over base of 20 mm thick cement morter1:4 using neat cement slurry in cluding finishing of joints etc. complete.		
(A)	Using M-40 grade cement Concrete	Sqm	1372.00
(B)	Using M-30 grade cement Concrete	Sqm	1244.00
11.42	Add extra over **11.41, A & B** if **coloured** wearing course (as per IS 1237:1980) is used.	Sqm	20%

CHAPTER : B-12
FINISHING WORK

Note :

1. The rates are for complete work including racking of joints, curing, grinding, mixing and finishing T & P and scaffolding material and cost of material with all lead and lift
2. The rates cover protection of all places and things, requiring protection and cleaning such places and thing of all droppings and splashes of mortar etc.
3. The Rate for external plaster are for height up to 10 m from ground level unless otherwise stated.
4. Cement mortar is to be made by mechnical mixer only.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	CEMENT PLASTER		
12.1	Plaster on new surface on wall in cement sand mortar 1:3 including racking of joints etc. complete fine finish :		
12.1.1	25 mm thick	Sqm.	202.00
12.1.2	20mm thick	Sqm	176.00
12.1.3	12mm thick	Sqm	137.00
12.2	Plaster on new surface on walls in cement sand mortar 1:4 including racking of joints etc. complete fine finish :		
12.2.1	25 mm thick.	Sqm	189.00
12.2.2	20mm thick .	Sqm	168.00
12.2.3	12mm thick .	Sqm	136.00
12.3	Plaster on new surface on walls in cement sand mortar 1:6 including racking of joint etc. complete fine finish :		
12.3.1	25 mm thick	Sqm	176.00
12.3.2	20mm thick.	Sqm	158.00
12.3.3	12mm thick.	Sqm	133.00
12.4	Plaster on new surface on walls in cement sand mortar 1:8 including racking of joint etc. complete fine finish :		
12.4.1	25mm thick.	Sqm	147.00
12.4.2	20mm thick.	Sqm	137.00

12.4.3	12mm thick.	Sqm	117.00
12.5	6 mm thick cement plaster to ceiling of mix 1:3 (1cement : 3-fine sand)	Sqm.	116.00
12.6	6mm thick cement plaster 1:3 (1cement : 3 fine sand) finished with a floating coat of neat cement and thick coat of Lime wash on top of walls when dry for bearing of R.C.C slabs and beams.	Sqm.	136.00
12.7	Finishing with neat cement (punning).	Sqm	33.00
12.8	Extra for providing and mixing water proofing material in cement plaster work in proportion recommended by the manufacturers.	Kg.	38.00
	ROUGH CAST PLASTER		
12.9	Rough cast plaster upto 10m height above ground level with a mixture of sand and gravel or crushed stone from 6mm to 10mm nominal size dashed over and including the fresh plaster in two layers, under layer 12mm cement plaster 1:4 (1 cement : 4 coarse sand) and top layer 10mm cement plaster 1:3 (1 cement : 3 fine sand) mixed with 10% finely grounded hydrated lime by volume of cement. Ordinary cement finish using ordinary cement	Sqm	309.00
12.10	Pebble dash plaster upto 10m height above ground level with a mixture of washed pebble or crushed stone 6mm to 12.5mm nominal size dashed over and including fresh plaster in two layers under layer 12mm cement plaster 1:4 (1 cement : 4 coarse sand) and top layer 10mm cement plaster with cement mortar 1:3 (1 cement : 3 fine sand) with 10% finely grounded hydrated lime by volume of cement	Sqm	294.00

12.11	Washed stone grit plaster on exterior walls of height upto 10 M. above level in two layers, under layer 12mm cement plaster 1:4 (1 cement : 4 coarse sand) furrowing the under layer with scratching tool, applying cement slurry on the under layer @ 2kg of cement per sqm, top layer 15 mm cement plaster 1:1/2:2 (1 cement : 1/2 coarse sand : 2 stone chipping size 10 mm nominal size) in panel with groove all around as per approved pattern including scrubbing and washing, the top layer with brushes and water to expose the stone chipping, complete as per specification and direction of Engineer in charge. (Payment for providing grooves shall be made separately)	Sqm	464.00
12.12	Forming groove of uniform size 15x15mm and upto 25x15mm in the top layer of washed stone grit plaster as per approved pattern using wooden battens, nailed to the under layer including removal of wooden battens, repair to the edges of panels and finishing the groove complete as per specification and direction of Engineer in charge	Rm	23.00
12.13	Extra for using white cement in place of ordinary cement in the top layer of the item of washed stone grit plaster.	Sqm	33.00
12.14	Extra for plastering exterior walls of height more than 10 m from ground level for every additional height of 3 m or part thereof.	Sqm	31.00
12.15	Extra for plastering on circular work not exceeding 6 m in radius:	Sqm	14.00
12.16	Extra for plastering done on moulding cornices or architraves including neat finish to line and level:		
12.16.1	In one coat	Sqm	191.00
12.16.2	In two coats	Sqm	316.00
12.17	Extra for plastering :		
12.17.1	Spherical ceiling	Sqm	58.00
12.17.2	Groined ceiling	Sqm	64.00
12.17.3	Flewing soffits	Sqm	38.00
12.18	Extra for lining out plaster to imitate stone or concrete blocks walling	Sqm	31.00

12.19	Making grooves as per design in plaster 10mm to 20mm wide.	Rm	6.00
12.20	Making of drip course (Tapka) in cement sand mortar 1:6 width upto 40 mm and 10 mm thick as per approved design	Rm	10.00
12.21	12 mm thick plain cement mortar bands in cement mortar 1:4 (1 cement : 4 fine sand) :		
12.21.1	Flush band.	Cm P Rm	2.30
12.21.2	Sunk band.	Cm P Rm	2.40
12.21.3	Raised Band	Cm P Rm	2.80
12.21.4	Moulded band.	Cm P Rm	4.70
12.22	Providing and applying white cement based putty over plastered surface to prepare the surface even and smooth complete		
12.22.1	**New Plastered Surface (three or more coats)**	Sqm	76.00
12.22.2	**Old Plastered Surface (two or more coats)**	**Sqm**	42.00
12.23	Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete	Sqm	63.00
12.24	Providing & Laying POP moulding & beading in ceiling including nailing & scafollding etc. complete of size :		
12.24.1	25mm x 12mm	Rm	55.00
12.24.2	25mm x 25 mm	Rm	65.00
12.25	Providing & Laying POP Cornice of required design & pattern in ceiling including nailing & scafollding etc. complete with fine finishing of size:		
12.25.1	50mm x 50mm	Rm	102.00
12.25.2	65mm x 65mm	Rm	113.00
12.25.3	75mm x 75mm	Rm	133.00
12.26	Lime plaster (up to 40 mm thick in 3 Coats) on new surface on walls with lime sand mortar 1:2 (1 lime putty : 2 sand) using admixture like Gur, Methi & Gugal @ 1.0 Kg./each for every 10 Sqm of area including racking of joints curing etc. complete	Sqm	321.00

12.27	Coloured Araish work 1 : 2 (1-lime putty : 2-Zikki) over lime plaster or plain back ground including preparation of lime for 6 months by slaking of lime with curd and changing the water every week. The kara plaster of not more than 6 mm thick is to be left for maximum of 3 months to appear the shrinkage & temperature cracks over the kara plaster 2 mm layer of grinded lime putty is to be done and is to be rubbed gently by Akik stone then inserting the design of colour border/ flowers/pictures etc including desired colour stone pigment to match the existing design/pattern after this, applying Khopra paste & rubbing with cloth to give a uniform picture. The work to be executed strict as per direction of Engineer Incharge	Sqm	3346.00
12.28	Yellow/pink colour wash with Khameera mixed with pigment including making gola garden, Kangoora and ornamental lining work as per aprroved Jaipur style and pattern complete in all respest as per direction of Engineer in Charge		
12.28.1	On New Surface (3 coats)	Sqm	80.00
12.28.2	On old Surface (3 coats)	Sqm	44.00
12.29	Extra for making Ornamental design like Kangooras Gola etc. as per Jaipur practice in cement plaster	Sqm	50%
	POINTING		
12.30	Pointing on brick work or brick flooring with cement mortar 1:3 (1 cement : 3 fine sand) :		
12.30.1	Flush / Ruled/ Struck or weathered pointing.	Sqm	73.00
12.30.2	Raised and cut pointing	Sqm	121.00
12.31	Pointing on stone masonry in cement sand mortar 1:3 (1 cement : 3 sand) :		
12.31.1	Flush / Ruled/ Struck or weathered pointing.	Sqm	113.00
12.32.2	Raised and cut pointing.	Sqm	212.00
12.32	Pointing on stone slab ceiling with cement mortar 1:2 (1 cement : 2 fine sand): Flush/ Ruled pointing	Sqm	71.00
12.33	Extra for pointing on walls on the outside at height more than 10 m from ground level for every additional height of 3 m or part there of.	Sqm	4.00

INTERIOR FINISHING

12.34	White washing with lime to give an even shade including all scaffolding:		
12.34.1	New work (three or more coats).	Sqm	10.00
12.34.2	Old work (two or more coats) including scrapping old surface and repairing with whiting where ever necessary.	Sqm	5.00
12.34.3	Old work (one or more coats) including scrapping old surface and repairing with putty where ever necessary.	Sqm	2.75
12.35	Colour washing of all shades to give an even shade including all scaffolding :		
12.35.1	New work (two or more coats) with a base coat of white washing with lime.	Sqm	13.00
12.35.2	New work (two or more coats) with a base coat of whiting.	Sqm	14.00
12.35.3	Old work (one or more coats) with whiting with a base coat of white washing including scrapping old surface.	Sqm	6.50
12.35.4	Old work (two or more coats) with whiting with a base coat of white washing including scrapping surface.	Sqm	5.50
12.35.5	Old work (one or more coats) with lime including scrapping surface.	Sqm	3.70
12.35.6	Old work (one or more coats) with whiting including scrapping surface.	Sqm	3.70
12.35.7	Removing white or colour wash by scrapping and sand papering and preparing the surface smooth including necessary repairs to scratches by sandla/loi. (only for colour changing).	Sqm	3.25
12.36	Distempering with dry distemper of approved brand and shade (two or more coats) and of required shade on new work, over and including, priming coat of whiting to give an even shade including all scaffolding.	Sqm	48.00
12.37	Distempering with oil bound washable distemper of approved brand and manufacture to give an even shade including all scaffolding:		
12.37.1	New work (two or more coats) over and including scrapping and priming coat with cement primer.	Sqm	68.00
12.37.2	Old work (one or more coats) including scrapping surface and necessary repairs.		
a	Colour Change	Sqm	14.00
b	Same colour	Sqm	10.00

12.37.3	Removing dry or oil bound distemper by scrapping sand papering and preparing surface smooth including necessary repairs to scratches by sandla/loi and all scaffolding (for colour changing only)	Sqm	3.90
12.38	Distempering with 1st quality acrylic washable distemper (ready mixed) of approved manufacturer and of required shade and colour completeincluding all scaffolding as per manufacturer's specification. Two or more coats on new work including prepartion of base with primer, putty, lippy etc complete in all respect.	Sqm	39.00
12.39	Applying one coat of Cement primer of approved brand and manufacture on wall surface including all scaffolding:	Sqm	13.00
12.40	Wall painting with plastic emulsion paint of approved brand and manufacture to give an even shade including all scaffolding:		
12.40.1	Two or more coats on new work including prepartion of base with primer, putty, lippy etc complete in all respect.	Sqm	73.00
12.40.2	Old work (One or more coats)	Sqm	36.00
	EXTERIOR FINISHING		
12.41	Finishing wall with water proofing cement paint of approved brand and manufacture and or required shade to give an even shade including all scaffolding:		
12.41.1	New work (Two or more coats applied @ 3.84 kg/10 sqm).	Sqm	48.00
12.41.2	Old work (One or more coats).	Sqm	18.00
12.42	Finishing walls with Acrylic Smooth exterior paint of required shade including all scaffolding. New work (Two or more coat applied @ 1.67 ltr/10 sqm over and including base coat of water proofing cement paint applied @ 2.20 kg/ 10 sqm).	Sqm	78.00
12.43	Painting exterior surface of Wall with 100% acrylic exterior paint of approved brand and manufacture to give an even shade with two or more coats including prepartion of base with sand papering, primer, putty, etc complete in all respect **including scaffolding and safety provision**		
12.43.1	New Work	Sqm	112.00

12.43.2	Old Work	Sqm	45.00
12.44	Finishing walls with textured exterior paint of required shade as per approved colour complete as per manufacturers specifications including primer coat and protecting coat **including scaffolding and safety provision**	Sqm	
12.44.1	Roller Finish av. thickness 400 to 500 microns.	Sqm	155.00
12.44.2	Trowel Finish av. thickness 1000 to 1200 microns.	Sqm	315.00
12.44.3	Trowel Finish av. thickness 1500 to 2000 microns.	Sqm	367.00
12.44.4	Trowel Finish av. thickness 2000 to 2500 microns.	Sqm	430.00
	PAINTING		
12.45	Applying priming coat :		
12.45.1	With ready mix pink or gray primer of approved brand and manufacture on wood work hard and soft wood.	Sqm	26.00
12.45.2	With ready mix Aluminium primer of approved brand and manufacture on resinous wood and plywood.	Sqm	26.00
12.45.3	With ready mixed red oxide zinc chromate primer of approved brand and manufacture on steel galvanised iron/steel works	Sqm	21.00
12.45.4	With ready mixed red oxide zinc chromate primer of approved brand and manufacture on steel work (second coat)	Sqm	11.00
12.46	Painting with synthetic enamel paint of approved brand and manufacture to give an even shade :		
12.46.1	Two or more coats on new work	Sqm	63.00
12.46.2	One or more coats on old work.	Sqm	31.00
12.47	Varnishing with varnish of approved brand and manufacture :		
12.47.1	Two or more coats glue sizing with copal varnish over an under coat of flatting varnish.	Sqm	81.00
12.47.2	Two or more coats glue sizing with spar varnish or an under coat of flatting varnish.	Sqm	85.00
12.48	French sprit polishing :		
12.48.1	Two or more coats on new work including a coat of wood filler.	Sqm	152.00
12.48.2	One or more coats on old work.	Sqm	52.00
12.49	Polishing on wood work with ready mixed wax polish of approved brand and manufacture :		

12.49.1	New Work	Sqm	63.00
12.49.2	Old Work.	Sqm	26.00
12.50	Providing and applying Melamine polish on teak wood work of approved brand and manufacture to give an even shade with two or more coats by compressor including sand papering, wood filler coat etc complete in all respect:		
12.50.1	New Surface	Sqm	806.00
12.50.2	Old Surface	Sqm	551.00
12.51	Painting one thin coat with white lead of approved brand and manufacture on wet or patchy portion of plastered surfaces.	Sqm	33.00
12.52	Painting (two or more coats) on rain water soil, waste and vent pipes and fitting with black anticorrosive bitumastic paint of approved brand and manufacture over and including a priming coat of ready mixed zinc chromate yellow primer on new work :		
12.52.1	75mm dia. Pipes.	Mtr.	15.00
12.52.2	100mm dia. Pipes	Mtr.	24.00
12.52.3	150mm dia. Pipes.	Mtr.	36.00
12.53	Painting (one or more coats) on rain water soil, waste and vent pipes and fitting with black anticorrosive bitumastic paint of approved brand and manufacture on old work :		
12.53.1	50mm dia. Pipes.	Mtr.	3.00
12.53.2	75mm dia. Pipes.	Mtr.	4.85
12.53.3	100mm dia. Pipes	Mtr.	6.00
12.53.4	150mm dia. Pipes.	Mtr.	9.00
12.54	Painting (two or more coats) on rain water, soil, waste and vent pipes and fittings with synthetic enamel paint of approved brand and manufacture and required colour over a priming coat of approved steel primer on new work.		
12.54.1	75mm dia. Pipes.	Mtr.	15.00
12.54.2	100mm dia. Pipes	Mtr.	26.00
12.54.3	150mm dia. Pipes.	Mtr.	36.00

12.55	Painting (two or more coats) on rain water, soil, waste and vent pipes and fittings with synthetic enamel paint of approved brand and manufacture and required colour over a priming coat of approved steel primer on old work.		
12.55.1	50mm dia. Pipes.	Mtr.	3.00
12.55.2	75mm dia. Pipes.	Mtr.	4.80
12.55.3	100mm dia. Pipes	Mtr.	6.00
12.55.4	150mm dia. Pipes.	Mtr.	7.00
12.56	Painting with oil type wood preservative of approved brand and manufacture :		
12.56.1	New work (Two or more coats).	Sqm	19.00
12.56.2	Old work (One or more coats).	Sqm	10.00
12.57	Providing and applying two coats of fire retardant paint unthinned on cleaned wood/ply surface @ 3.5 **sqm** per litre per coat including preparation of base surface as per recommendations of manufacturer to make the surface fire retardant.	Sqm	326.00
12.58	Coal-tarring coats on new work using 0.16 and 0.12 litre of coal tar per Sqm in the first and second coat respectively.	Sqm	21.00
12.59	Painting with Aluminium paint of approved brand and manufacture to give an even shade :		
12.59.1	Two or more coats on new work.	Sqm	48.00
12.59.2	One or more coats on old work.	Sqm	15.00
12.60	Painting with acid proof paint of approved brand and manufacture of required colour to give an even shade :		
12.60.1	Two or more coats on new work .	Sqm	49.00
12.60.2	One or more coat on old work .	Sqm	24.00
12.61	Painting with black anti-corrosive bitumastic paint of approved brand and manufacture to give an even shade :		
12.61.1	Two or more coats on new work.	Sqm	43.00
12.61.2	One or more coat on old work.	Sqm	17.00
12.62	Lettering with black Japan paint of approved brand and manufacture.	P. Letter Per cm. height	2.00

12.63	Re-lettering with black Japan paint of approved brand and manufacture.	P. Letter Per cm. height	0.97
12.64	**Providing and applying superior quality wall painting with Velvet touch emulsion paint of approved brand and manufacture to give an even shade with putty and preparation of surface applying with the help of paint roller the work complete in all respect as per direction of Engineer-in-charge.**		
12.64.1	**New Work Three or More coat**	**Sqm**	**224.00**
12.64.2	**Old Work Analysis**	**Sqm**	**147.00**
12.65	**Spray painting in approved shade and approved quality with Duco paint of required colour to give even shade with two or more coats by compressor including sand papering, NC putty filler coat etc complete in all respect as per design given by Engineeer incharge on new work.**	**Sqm**	**1690.00**

CHAPTER : B-13
REPAIRS TO BUILDING

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
13.1	Repairs to plaster of thickness 12mm to 20mm in patches of area 2.5 sq. metres and under including cutting the patch in proper shape, raking out joints and preparing and plastering the surface of the walls complete including disposal of rubbish to the dumping ground within 50 m lead : With cement mortar 1 :4 (1 cement : 4 coarse sand)	Sqm	214.00
13.2	Fixing chowkhats in existing opening including embedding chowkhats in floors or walls cutting masonry for holdfasts embedding hold fasts in cement concrete blocks with cement concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominaal size)painting two coats of coal tar to sides of chaukhats and making good the damages to walls and floors as required complete including disposal of rubbish to the dumping ground within 50 metres lead		
13.2.1	Door chowkhats	Each	559.00
13.2.2	Window chowkhats	Each	348.00
13.2.3	Clerestory window chowkhats	Each	246.00
13.3	Making the opening in masonry including dismantling in floor or walls by cutting masonry and making good the damages to walls, flooring and jambs complete to match existing surface i/c disposal of mulba/ rubbish to the nearest municipal dumping ground. For door/windows/clerestory windows	Sqm	406.00
13.4	Renewing glass panes, with putty and nails wherever necessary:		
13.4.1	Float glass panes of thickness 4 mm	Sqm	561.00
13.4.2	Float glass panes of thickness 5 mm	Sqm	694.00
13.5	Renewing glass panes with wooden fillets wherever necessary :		

13.5.1	4mm thick plain glass panes.	Sqm	695.00
13.5.2	5mm thick plain glass panes.	Sqm	829.00
13.6	Renewing glass panes and refixing existing wooden fillets :		
13.6.1	4mm thick plain glass panes.	Sqm	578.00
13.6.2	5mm thick plain glass panes.	Sqm	709.00
13.7	Supplying and fixing new wooden fillets wherever necessary: 2nd class teak wood fillets	Rm	26.00
13.8	Renewal of old putty of glass panes (length)	Rm	16.00
13.9	Refixing old glass panes with putty and nails	Sqm	194.00
13.10	Fixing old glass panes with wooden fillets (Excluding cost of fillets)	Sqm	160.00
13.11	Providing and fixing 16 mm M.S. Fan clamps of standard shape and size in existing R.C.C. slab including cutting chase and making good and painting exposed portion of the clamps complete.	Each	195.00
13.12	Repair to stone masonary in cement sand mortar 1:6 with old and new stone.	Sqm	289.00
13.13	Repair to brick masonary in cement sand mortar 1:6 with new brick.	Sqm	322.00
13.14	Cement plaster 1:3 under old stone slabs with admixture (gur,nails,binding wire to mesh and scraping scales & dressing joints of slabs complete.		
13.14.1	Proofing upto 25 mm thickness.	Sqm	262.00
13.14.2	Proofing above 25 mm thickness& upto 60 mm thickness	Sqm	417.00
13.15	Patch repair to cement concrete floor including digging out old floor in regular shape and removal of rubbish curing etc. complete in all respects.	Sqm	288.00
13.16	Washing floor with soda, soap or other cleaning material.	Sqm	1.00
13.17	Removal and refixing stone chhajja 50 to 75 mm thick including finishing, complete in all respects.	Sqm	168.00

13.18	Replacing broken roofing slab in patches including lime terrace, ceiling plaster etc. including removal of broken slabs.	Sqm	1676.00
13.19	Making holes in stone masonry wall upto 450 mm x 450 mm size and making good in CM 1:6.	Each	165.00
13.20	Grinding of existing floor with		
13.20.1	60, 80, 120 and 360 No. Carborundum stone including cost of wax polishing complete :	Sqm	63.00
13.20.2	120 and 360 No. Carborundum stone including cost of wax polishing complete :	Sqm	39.00
13.21	Add extra for mirror polishing on existing marble work/ Kota stone flooring work where ever required to give high gloss finish complete.	Sqm	110.00
13.22	Providing and fixing double scaffolding system on the exterior side, upto five story made with 4" **wooden** member 1.5 m c/c horizontal and vertical direction joining with rassi, clamps, challies. The scaffolding system shall be stiffened with bracing, runner, connection of building if necessary with all essential safty features for the workmen etc complete as per direction of EI.The lavation area of the scaffolding shall be **measured** for payment. The **payment** will be made once irrespective of duration of scaffolding. Note : This item to be used for maintence work judically	Sqm.	147.00
13.23	Repair of existing steel glazed doors, windows and ventilator, MS sheet door, garge doors, compound wall gates etc. by welding drilling complete include. Cost of welding rods and hire charge of welding meachine and TP etc complete	Sqm.	302.00
13.24	Labour charges for repair to door / window without using any material such as applying randha (finishing) tightening of bolts & screw etc.	Sqm.	200.00
13.25	L/C for fixing of GI/AC sheet include. Cutting jhiri and repair , cutting of sheet , fixing of sheet with J or L hook and bitumen washer as per direction of EI	Sqm.	233.00

13.26	Repair of rolling shutter with replacement of shutter spring incl. removal of old spring in the existing rolling shutter incld. Cost of new spring	Each	787.00
13.27	Repair of rolling shutter with using any material such as bend repair , tightening of spring,boltand screw etc complete	Each	551.00
13.28	Refebrication work with existing steel work including cutting, welding & refebrication as per requirement with fixing at site complete as per direction of Engineer In charge	Kg	31.00
13.29	Repair to damaged RCC slabs, beams and columns etc for strengtheing with cement mortar 1 : 2 in following steps i) Choping losse and cracked concrete to required depth, cleaning the exposed surface of steel and concrete with wire brush and sand papering , cleaning of concrete surface with water as necessary and getting dried. ii) Applying Epoxy primer on the exposed steel bars. iii) Applying Epoxy hardener and resin mix (as per manufectured's specification) on concrete and steel bars with brush. iv)Speading dry cement mortar 1 : 2 mix over epoxy treated wet surface and allow it to dry for one day. v) Plastering the surface with cement sand mortar 1 : 3 upto depth of 30mm complete. vi) Cure the plastered surface for 15 days **All work including scaffolding etc. complete in all respect.**	Sqm	1425.00
13.30	Replacement of stone Chajja with sriwan in damaged portion as per existing thickness and width in CM 1 : 3 with removal of old damaged pieces of chajja including top plaster of chajja and bottom pointing in CM 1 : 3 with scoffolding upto **Ground floor** level as per direction of Engineer in charge	Sqm	2564.00
13.30.1	**Add Extra for each additional floor**	Sqm	240.00

13.31	Repair to damaged cement plaster of stone chajja with CM 1 : 3 of required thickness on top & bottom as per existing pattern and making patta etc complete up to three floor level as per direction of Engineer in charge including all necessary scoffolding etc	Sqm	575.00
13.32	Mechanised cleaning of storage tanks comprises of following 6 stage as mentioned below : 1. Mechanical dewatering 2. Desalting removal of leftover dirty water and sludge with sludge pump 3. Cleaning of wall, ceiling and floor by high pressure water jet with help of equipments which creates a pressure of 100-150 bar 4. Removal of remaing sludge from floor with the help of industrial vacuum cleaner. 5. Spraying of non toxic, bio degradable ecofriendly and bacterial agent certified by a govt approved laborary to disinfect the tank from all harmfull pathogens. The dose should be safe as per the UECD guidelines & 23 adopted on 17 th december 2001. 6. Treatment of inside tank by exposing UV radiation to kill further suspended or floating bacteria if any		
13.32.1	**Upto 500 ltr.**	**Ltr.**	0.40
13.32.2	**Above 500 ltr. to 1000 Ltr.**	**Ltr.**	0.35
13.32.3	**Above 1000 ltr.**	**Ltr.**	0.30
13.33	Repair of door, window shutters including cost of material and labour for change of damaged :		
(a)	Style / rail of teak wood 30mm thick/ Aluminium Section		
(i)	4"wide	Rmt.	242.00
(ii)	6" wide	Rmt.	334.00
(b)	Panel by		
(i)	Shuttering Ply - 9mm thick (Commercial)	Sqm.	544.00
(ii)	Wire Gauge 14 mesh x 24 gauge	Sqm.	485.00
(iii)	Plain Glass Panes 4mm thick	Sqm.	392.00

(c)	Aldrop of mild Steel		
(i)	12" long x12mm dia	Each	134.00
(ii)	8" long x 10mm dia	Each	96.00
(d)	Stopper of Aluminium with		
(i)	Single Rubber Gutka	Each	30.00
(ii)	Double Rubber Gutka	Each	43.00
(e)	Door Springs of mild Steel	Each	145.00
(f)	Steel Hinges		
(i)	For Wooden Chowkhat 3" size.	Each	22.00
(ii)	For Wooden Chowkhat 4" size	Each	31.00
(iii)	For Stone Chowkhat ,4"size	Each	26.00
(g)	Mild Steel Tower Bolt 6" long	Each	35.00
(h)	Steel Locking Bolt	Each	48.00
(i)	Steel Handle 6" size	Each	22.00
13.34	**Making of hole in R.C.C. slab/wall/beams and masonary wall (width up to 450 mm) by core cutting machine complete as per direction of EI**		
13.34.1	**From diameter 100mm to 150mm**	**Nos**	**1027.00**
13.34.2	**Diameter above 150mm**	**Nos**	**1544.00**

CHAPTER : B-14

DISMANTLING & DEMOLISHING WORK

Note : All items of demolition & dismantling of building shall be conform to IS 4130 for handling , IS 7969 for storage and IS 3696 for scaffolding

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
14.1	Demolishing lime concrete manually/ by mechanical means and disposal of material within 50 metres lead as per direction of Engineer-in-charge.	Cum	231.00
14.2	Demolishing cement concrete manually/ by mechanical means including disposal of material within 50 metres lead as per direction of Engineer-in-charge.		
14.2.1	1: 3: 6 or richer mix.	Cum	579.00
14.2.2	1: 4: 8 or leaner mix.	Cum	357.00
14.3	Demolishing R.C.C. work manually/ by mechanical means including stacking of steel bars and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge.	Cum	845.00
14.4	Demolishing R.B. work manually/ by mechanical means including stacking of steel bars and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge.	Cum	756.00
14.5	Extra for cutting reinforcement bars manually/ by mechanical means in R.C.C. or R.B. work (Payment shall be made on the cross sectional area of R.C.C. or R.B. work) as per direction of Engineer-in-charge	Sqm	288.00
14.6	Extra for scrapping, cleaning and straightening reinforcement from R.C.C. or R.B. work	Kg	2.20
14.7	Demolishing brick work manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge.		
14.7.1	In mud mortar.	Cum	168.00
14.7.2	In lime mortar.	Cum	201.00
14.7.3	In cement mortar.	Cum	489.00

14.8	Removing mortar from bricks and cleaning bricks including stacking within a lead of 50 meters (stacks of cleaned bricks shall be measured):		
14.8.1	From bricks work in mud mortar.	1000 Nos.	1134.00
14.8.2	From brick work in lime mortar	1000 Nos.	1334.00
14.8.3	From brick work in cement mortar.	1000 Nos.	1685.00
14.9	Demolishing stone rubble masonry manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50 metres lead as per direction of Engineer-in-charge:		
14.9.1	In lime mortar.	Cum	275.00
14.9.2	In cement mortar.	Cum	583.00
14.9.3	In mud mortar	Cum	163.00
14.9.4	dry masonry	Cum	139.00
14.10	Dismantling dressed stone work ashlar face stone work, marble work or precast concrete work manually/ by mechanical means including stacking of serviceable and disposal of unserviceable material within 50 metres lead as per direction of Engineer in charge :		
14.10.1	In lime mortar.	Cum	348.00
14.10.2	In cement mortar.	Cum	683.00
14.11	Removing mortar from stones and cleaning stones and concrete articles including stacking (net quantity of stacks of cleaned materials will be measured):		
14.11.1	Lime mortar.	Cum	116.00
14.11.2	Cement mortar.	Cum	168.00
14.12	Dismantling doors, windows and clearstory windows steel or wood shutter including chowkhats and holdfasts etc. complete and stacking within 50 meters lead:		
14.12.1	Of area 3 square meter and below.	Each	95.00
14.12.2	Of area exceeding 3 square meter.	Each	131.00
14.13	Taking out doors, windows and clearstory windows shutters (steel or wood) including stacking within 50 meters lead:		
14.13.1	Of area 3 square meter and below.	Each	38.00
14.13.2	Of area exceeding 3 square meter.	Each	50.00
14.14	Dismantling wood work in frames, trusses, purlins and rafters up to 10 meter span and 5m height including, levelling stacking the material within 50m lead:		
14.14.1	Of sectional area 40 Sq.cm and above	Cum	1244.00
14.14.2	Of sectional area below 40 Sq.cm.	Mtr.	5.00

14.15	Extra for dismantling trusses, rafters, purlins etc. of wood work for every additional span of one metre or part thereof beyond 10 metres :		
14.15.1	of sectional area 40 Sq.cm and above	Cum per Meter span	152.00
14.15.2	of sectional area below 40 Sq.cm.	Mtr. .Per meter span	0.38
14.16	Extra for dismantling trusses, rafters, purlins etc. of wood work for every additional height of one metre or part thereof beyond 5 metres :		
14.16.1	of sectional area 40 Sq.cm and above	Cum per Meter span	216.00
14.16.2	of sectional area below 40 Sq.cm.	Mtr. per Meter span	0.78
14.17	Dismantling steel work in single sections including dismembering and stacking within 50 meters lead in:		
14.17.1	R.S. joists.	Kg	0.85
14.17.2	Channels, angles, tees & flats.	Kg	0.60
14.18	Dismantling steel work in built up sections in angles, tees, flats and channels including all gusset plates, bolts, nuts, cutting rivets, welding etc. including dismembering and stacking within 50metres lead.	Kg	1.55
14.19	Dismantling steel work manually/ by mechanical means in built up sections without dismembering and stacking within 50 metres lead as per direction of Engineer-in-charge.	Kg	1.00
14.20	Extra for dismantling trusses, rafters, purlins etc. of steel work for every additional span of one metre or part thereof beyond 10 metres	Kg per Mtr.	0.30
14.21	Extra for dismantling trusses, rafters, purlins etc. of steel work for every additional height of one metre or part thereof beyond 5 metres.	Kg per Mtr.	0.30
14.22	Extra for making of structural steel work required to be re-erected.	Kg	1.55

14.23	Dismantling tile work in floors and roofs laid in cement mortar including stacking of serviceable material and disposal of unserviceable material within 50 meter lead:		
14.23.1	For thickness of tiles 10mm to 25mm.	Sqm	18.00
14.23.2	For thickness of tiles 25mm to 40mm.	Sqm	31.00
14.24	Dismantling dry brick pitching in floors drains etc. including stacking of serviceable material and disposal of unserviceable material within 50 meters lead.	Cum	312.00
14.25	Dismantling stone slab/ terrazo chip flooring laid in cement mortar including stacking of serviceable material and disposal of unserviceable material within 50 meters lead.	Sqm	73.00
14.26	Dismantling brick tiles covering in terracing including stacking of serviceable material and disposal of unserviceable material within 50 meters lead.	Sqm	31.00
14.27	Demolishing mud phuska/ Lime dhar/ Kharanja in terracing and disposal of unserviceable material within 50 meters lead.	Cum	259.00
14.28	Dismantling roofing including ridges, hips, valleys and rafters etc. and stacking the within 50 meter lead of:		
14.28.1	G.I. Sheet.	Sqm	44.00
14.28.2	Asbestos Sheet.	Sqm	21.00
14.29	Dismantling stone slab roofing over wooden karries and battens (dismantling karries and battens to be paid for separately) including stacking of serviceable material and disposal of unserviceable material within 50 meter lead:	Cum	636.00
14.30	Dismantling Jack arch roofing and floors including stacking of serviceable material and disposal of unserviceable material within 50-meter lead.	Sqm	59.00
14.31	Dismantling tiled roofing with battens boarding etc. complete including stacking of serviceable material and disposal of unserviceable material within 50-meter lead.	Sqm	49.00
14.32	Dismantling thatch roofing including mats bamboos, jafries etc. complete including stacking of serviceable material and disposal of unserviceable material within 50 meter lead	Sqm	13.00
14.33	Dismantling wooden ballies in posts and struts including stacking within 50 meters lead.	Mtr.	5.00
14.34	Dismantling and stacking within 50 meters lead fencing post, struts, including earth work dismantling of concrete etc. in the case of:		
14.34.1	T or Angle Iron posts	Each	65.00

14.34.2	R.C.C. posts.	Each	73.00
14.35	Cutting ballies or wooden posts of fencing at the point of projection above the concrete at ground and stacking the same within 50 meter lead	Each	6.00
14.36	Dismantling barbed wire or chain link fencing including bending making rolls and stacking within 50-meter lead.	Kg	9.00
14.37	Dismantling wooden trellis work excluding frames but including bending, making rolls and stacking within 50 meter lead.	Sqm	15.00
14.38	Dismantling expanded metal or I.R.C. fabric with necessary battens and beading including stacking the serviceable material within 50-meter lead.	Sqm	19.00
14.39	Dismantling wooden boards in lining of walls and partition excluding supporting members but including stacking within 50 meter lead:		
14.39.1	Up to 10mm thick	Sqm	15.00
14.39.2	Thickness above 10mm up to 25mm	Sqm	23.00
14.39.3	Above 25mm thick.	Sqm	33.00
14.40	Dismantling precast concrete or stone slabs in wall partitions etc. including stacking within 50 meter lead:		
14.40.1	Thickness up to 40mm.	Sqm	70.00
14.40.2	Thickness 40mm up to 75mm	Sqm	106.00
14.41	Dismantling CI or asbestos rain water pipe with fitting and clamps including stacking within 50 meter lead:		
14.41.1	Up to 80mm dia pipe.	Mtr.	17.00
14.41.2	100mm dia pipe.	Mtr.	17.00
14.41.3	150mm dia pipe.	Mtr.	18.00
14.42	Dismantling tar felt over roofing complete including stacking of serviceable material and disposal of unserviceable material within 50-meter lead.	Sqm	21.00
14.43	Dismantling GI pipes (external work) including excavation and refilling trenches after taking out the pipes and including stacking of pipes within 50 meter lead:		
14.43.1	15mm to 40mm nominal bore.	Mtr.	37.00
14.43.2	Above 40mm nominal bore.	Mtr.	43.00
14.44	Dismantling CI pipes including excavation and refilling trenches after taking out the pipes, breaking lead caulked joints, melting of lead and making into block, including stacking of pipes lead at site within 50 meter, of dia:		
14.44.1	Up to 150mm dia.	Mtr.	99.00
14.44.2	Above 150mm dia. up to 300mm dia.	Mtr.	134.00

14.44.3	Above 300mm dia.	Mtr.	179.00
14.45	Dismantling steel cylinder, R.C. pipes etc. including excavation and refilling trenches after taking out the pipes, breaking lead caulked joints, melting of lead and making into blocks including stacking of pipes, within a lead of 50 meter, of dia:		
14.45.1	Up to 600mm dia.	Mtr.	179.00
14.45.2	Above 600 mm dia.	Mtr.	452.00
14.46	Dismantling asbestos cement pressure pipes including excavation and refilling trenches after taking out the pipes, and stacking the same within a lead of 50 meter, of dia:		
14.46.1	Up to 150mm.	Mtr.	78.00
14.46.2	Above 150mm.	Mtr.	95.00
14.47	Taking out CI cover with frame from stone slab / RCC slab of inspection chambers of various sizes including dismantling of stone slab / RCC slab roofing and stacking of useful materials near the site and disposal of unserviceable materials within 50-meter lead.	Each	170.00
14.48	Taking out C.I. cover with frame from stone slab / RCC slab of manholes of various sizes including dismantling of stone slab / RCC slab roofing and stacking of useful materials near the site and disposal of unserviceable materials within 50-meter lead.	Each	100.00
14.49	Dismantling of RCC spun vent shaft including excavating the cement concrete pit completely, taking out the shaft, refilling the excavated gap stacking the useful material near the site and disposal of unserviceable materials within 50 meter lead.	Each	1155.00
14.50	Dismantling of gully chamber of various sizes including CI grating with frame including stacking of useful materials near the site and disposal of unserviceable material within 50 meter lead including refilling the gap.	Each	232.00
14.51	Dismantling of flushing cistern of any size including stacking of useful materials near the site and disposal of unserviceable material within 50 meter lead.	Each	252.00
14.52	Dismantling of CI sluice valve including stacking of useful materials within a lead of 50 meter:		
14.52.1	Up to 150mm dia.	Each	90.00
14.52.2	Above 150mm dia.	Each	321.00
14.53	Dismantling of spindle fire hydrant including stacking of useful material within 50 meter lead.	Each	194.00

14.54	Dismantling of cement concrete platform along with curtain walls and base concrete etc. including stacking of useful materials near the site and disposal of unserviceable materials within 50 metres lead :		
14.54.1	**Upto 1.44 Sqm**	Sqm	283.00
14.54.2	**Above 1.44 Sqm to 2.52 Sqm**	Sqm	420.00
14.54.3	**Above 2.52 Sqm to 3.84 Sqm**	Sqm	630.00
14.55	Dismantling old plaster or skirting, raking out joints and cleaning the surface for plaster including disposal of rubbish to the dumping ground within 50 meter lead.	Sqm	14.00
14.56	Dismantling CC jali including cost of making grooves in plaster and making good and including disposal of unserviceable surplus material and stacking of serviceable material with in 50 meters lead as directed by Engineer-in-charge. .	Sqm	73.00
14.57	**Removing** aluminium/Gypsum partitions, doors, windows, fixed glazing and false ceiling including disposal of unserviceable material and stacking of serviceable material with in 50 meters lead as directed by Engineer-in-charge.	Sqm	15.00
14.58	Making of hole (width up to 150 mm) in R.C.C. slab/wall/beams by power driven drilling machine complete as per direction of EI		
14.58.1	upto 150mm thick	Each	160.00
14.58.2	above 150 mm thick	Each	245.00
14.59	Dismantling manually/ by mechanical means including stacking of serviceable material and disposal of unserviceable material within 50metres lead as per direction of Engineer-in-charge :		
14.59.1	Water bound macadam road (average thickness 250 mm)	Sqm	63.00
14.59.2	bituminous road (average thickness 300 mm)	Sqm	122.00

CHAPTER : B-15
WATER PROOFING AND EXPANSION JOINT TREATMENT

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	WATER PROOFING TREATMENT		
15.1	Providing and laying water proofing treatment to vertical and horizontal surfaces of depressed portions of W.C., kitchen and the like consisting of : i) Ist course of applying cement slurry @ 4.4 Kg/sum mixed with water proofing compound conforming to IS 2645 in recommended proportions including rounding off junction of vertical and horizontal surface. ii) IInd course of 20mm cement plaster 1:3 (1 cement: 3 coarse sand) mixed with water. proofing compound in recommended proportion including rounding off junction of vertical and horizontal surface.. iii) IIIrd course of applying blown or residual bitumen applied hot at 1.7 Kg per sqm. of area. iv) IVth course of 400 micron thick PVC sheet .(Overlaps at joints of PVC sheet should be 100 mm wide and pasted to each other with bitumen @ 1.7 Kg/sqm.)	Sqm.	466.00
15.2	Providing and Placing in position suitable PVC water stops conforming to IS:12200 for construction/ expansion joints between two RCC members and fixed to the reinforcement with binding wire before pouring concrete etc. complete :		
15.2.1	Serrated with central bulb (225 mm wide, 8-11 mm wide)	Rm	451.00
15.2.2	Dumb bell with central bulb (180 mm wide, 8 mm wide)	Rm	418.00
15.2.3	Kickers (320 mm wide, 5 mm thick)	Rm	429.00
15.3	Providing and laying water proofing treatment in sunken portion of WCs, bathroom etc., by applying cement slurry mixed with water proofing cement compound consisting of applying : a) first layer of slurry of cement @ 0.488 kg/sqm mixed with water proffing cement compound @0.253 kg//sqm .This layer will be allowed to air cure for 4 hours. b) Second layer of slurry of cement @ 0.242 kg/sqm mixed with water proffing cement compound @ 0.126 kg/sqm This layer will be allowed to air cure for 4 hours followed by water curing for 48 hours. The rates includes preparation of surface treatment & sealing of all joints ,corners, junction with polymer mixed slurry.	Sqm.	250.00
15.4	Providing and laying water proofing treatment on roofs of slabs by applying cement slurry mixed with water proofing cement compound consisting of applying : a) after surface preparation, first layer of slurry of cement @ 0.488 kg/sqm mixed with water proffing cement compound @ 0.253 kg/sqm. b) laying second layer of Fiber glass cloth when the first layer is still green . Overlaps of joints of fiber cloth should not be less than 10 cm . c) Third layer of 1.5 mm thickness consisting of slurry of cement @ 1.289 kg/sqm mixed with water proffing cement compound @ 0.670 kg/sqm and coarse sand @ 1.289 kg/sqm. This will be allowed to air cure for 4 hours followed by water curing for 48 hours. The entire treatment will be taken upto 30cm on parapet wall and tucked into groove in parapet all around d) fourth and final layer of brick tiling ‘with cement mortar or any other approved protective coarse. (which will be paid for separately) For the purpose of measurement the entire treated surface will be measured	Sqm.	338.00

15.5	Providing and laying four courses water proofing treatment with bitumen felt over roofs consisting of first and third courses of blown bitumen 85/25 or 90/15 conforming to IS : 702 applied hot 1.45Kg. per square meter of area for each course, second course of roofing felt type 3 grade –I (Hessian based self finished bitumen felt)and fourth and final course of stone grit 6mm and down size of pea sized gravel spread at 6cubes diameter per square meter including preparation of surface, but excluding grading, complete with Bitumen felt (Hessian base) type 3 grade - I conforming to : 1322.	Sqm.	322.00
15.6	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifth course of blown bitumen 85/25 or 90/15 conforming to IS : 702 applied hot @ @ 1.45 ,1.20 and 1.45 Kg per sqm. of area respectively second and fourth courses of roofing felt type 3 grade-I conforming to IS 1322 (Hessian based self finished bitumen felt) conforming to IS 1322 and sixth and final course of stone grit 6mm and down size of pea sized gravel spread at 6 cubic dm. per square metre including preparation of surface but excluding grading complete .	Sqm.	498.00
15.7	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifth courses of blown or/and residual bitumen applied hot at 1.45, 1.20 and 1.70 Kg. per Sqm. of area respectively second and fourth courses of roofing felt type 2 grade-I (fiber base self finished bitumen felt) and sixth and final course of stone grit 6mm and down size or pea sized gravel spread at 8cu.dm per sqm including preparation of surface excluding grading complete confirming to IS : 1322 and 1346.	Sqm.	541.00
15.8	Providing and laying six courses water proofing treatment with bitumen felt over roofs consisting of first, third and fifth courses of blown or/and residual bitumen applied hot at 1.45, 1.20 and 1.70Kg. per Sqm of area respectively second and fourth courses of roofing felt type 2 grade 2 (fiber base self finished bitumen felt) and sixth and final course of stone grit 6mm and down size of pea sized gravel spread at 8cu.dm per sqm including preparation of surface excluding grading complete confirming to IS : 1322 and 1346.	Sqm.	541.00
15.9	Supplying and applying bituminous solution primer on roofs or on wall surface at 0.24 liter per sqm.	Sqm.	21.00
15.10	Deduct for omitting in water proofing treatment final course of spreading stone grit 6mm. and down size of pea sized gravel :		
15.10.1	At 6dm^3 per Sqm.	Sqm.	8.80
15.10.2	At 8dm^3 per Sqm.	Sqm.	11.00
15.11	Providing and laying APP (Atactic Polypropylene Polymer) modified prefabricated five layer 3mm thick water proofing membrane, black finished reinforced with glass fibre matt consisting of a coat of bitumen primer for bitumen membrane @ 0.40 ltr/sq.mt. by the same membrane manufacture of density at 25°C, 0.87-0.89 kg/ltr and viscosity 70-160 cps. Over the primer coat the layer of membrane shall be laid using Butane Torch and sealing all joints etc., and preparing the surface complete. The vital physical and chemical parameters of the membrane shall be as under Joint strength in longitudinal and transverse direction at 23°C as 650/450N/5cm. Tear strength in longitudinal and transverse direction as 300/250N. Softening point of membrane not less than 150°C. Cold flexibility shall be upto -2°C when tested in accordance with ASTM, D-5147. The laying of membrane shall be got done through the authorized applicator of the manufacturer of membrane. 3 mm thick	Sqm.	443.00

15.12	Providing and laying APP (Atactic Polypropylene Polymer) modified prefabricated five layer 3mm thick water proofing membrane, black finished reinforced with non-woven polyester matt consisting of a coat of bitumen primer for bitumen membran @ 0.40 Itr/sq.mt. by the same membrane manufacture of density at 25°C, 0.87-0.89 kg/ltr and viscosity 70-160 cps. Over the primer coat the layer of membrane shall be laid using Butane Torch and sealing all joints etc., and preparing the surface complete. The vital physical and chemical parameters of the membrane shall be as under Joint strength in longitudinal and transverse direction at 23°C as 650/450N/5cm. Tear strength in longitudinal and transverse direction as 300/250N. Softening point of membrane not less than 150°C. Cold flexibility shall be upto -2°C when tested in accordance with ASTM, D-5147. The laying of membrane shall be got done through the authorized applicator of the manufacturer of membrane. 3 mm thick	Sqm.	512.00
15.13	Extra for covering top of membrane with Geotextile, 120gsm non woven, 100% polyester of thickness 1 to 1.25mm bonded to the membrane with intermittent touch by heating the membrane by Butane Torch as per manufactures recommendation.	Sqm.	51.00
15.14	Extra over items of Cement Plaster / Cement Concrete flooring / Plain or RCC work Providing and mixing admixture of synthetic Fiber 6mm / 12mm. @ rate of 125gm. pack per 50 Kg. of Cement or in the ratio of as specified by manufacture specification and direction of Engineer In-charges with all leads and lifts.	P Pack 125 gm	52.00
	EXPANSION JOINT TREATMENT:		
15.15	Supplying and providing of two components poly sulphide sealant containing solid contents, confirming to standard specification and approved compatible primer, for priming vertical faces prior to application of poly sulphide sealant including all labour charges for sealing of expansion joints after drill mixing of two components sealing compound. Cleaning and preparation of expansion joint using wire brush sand papering with emery paper and filing with suitable mechanical means. The work includes racking and cleaning of joints, sticking, debonding tape on bottom of joint, applying primer on edges, using masking tape on edges, providing of sealant and removal of masking tape immediately after providing sealant in joint.		
15.15.1	To vertical joint between RCC columns, bricks & stone masonry in size 40 mm width and 12 mm depth, including providing back up material (Thermocoal foam) if required.	Mtr.	753.00
15.15.2	To joints in flooring, as per drawing with first layer of poly sulphide sealant in size 20 mm width and 10 mm depth over PE Foam and then providing polyethylene foam back up rod 25 mm dia size and finally providing another layer of poly sulphide sealant in size 20 mm width and 10 mm depth over P.E. back up rod as per application procedure and direction of Engineer-in-charge.	Mtr.	786.00
15.16	Providing and fixing in position bitumen impregnated fiberboard confirming to IS: 1838 in expansion joints including the cost of primer sealing compound.		
15.16.1	12 mm. thick	Sqm.	550.00
15.16.2	18 mm. thick	Sqm.	720.00
15.16.3	25 mm. thick	Sqm.	940.00
15.17	Providing and fixing expansion joint board for building RCC column beams, walls, slab of Polyethylene foam of density 30 to 35 Kg/m^3 of Armour Board.		
15.17.1	20mm Thick	Sqm.	378.00
15.17.2	25 mm Thick	Sqm.	448.00
15.17.3	30 mm Thick	Sqm.	512.00
15.17.4	40 mm Thick	Sqm.	652.00
15.18	Providing and fixing 150 mm wide sheet covering over expansion joints with iron screws as per design to match the colour / shade of wall treatment.		

15.18.1	Non-asbestos fibre cement board 6 mm thick as per IS:14862. 1	Rm.	105.00
15.18.2	Aluminum fluted strips 2 mm thick	Rm.	372.00
15.18.3	Acrylic sheet 5 mm thick	Rm.	100.00
15.19	Providing and filling in position bitumen sealing compound grade A (IS 1834) in expansion joints.	P.Cm. depth P. Cm. Width for 100mtr. length	709.00
15.20	Providing and filling in position bitumen mix filler of proportion 80 Kg. of hot bitumen, 1 Kg. of cement and 0.25 cubic meter of coarse sand for expansion joints.	Per cm width per cm depth per 100 mtr. length	208.00
	R.C.C. FRAMES		
15.21	Providing and fixing in cement sand mortar 1 : 3 single/double rebated R..C. Door Window and ventilator frames (Chowkhats) including 1% reinforcement as per design and concrete mix 1 : 2 : 4 cutting etc. complete incl. horns. and hold fasts :		
15.21.1	Size 75 x 50mm	Mtr.	117.00
15.21.2	Size 75 x 75mm	Mtr.	133.00
15.21.3	Size 100 x 75mm	Mtr.	140.00
15.21.4	Size 125 x 75mm	Mtr.	150.00
15.22	Extra for RC frames (Chowkhats) Curved in plan	Mtr.	10%
15.23	Extra for RCC frame finishing the exposed face with terrazo (Mosaic) finish.	Mtr.	20%
	CUT STONE DOOR AND WINDOW FRAMES		
15.24	Providing and fixing in cement sand mortar 1 : 4 single paitam (rebated) stone doors windows and ventilator frames of approved quarry :		
15.24.1	Size 75 x 60mm	Mtr.	125.00
15.24.2	Size 75 x 75mm	Mtr.	142.00
15.24.3	Size 100 x 60mm	Mtr.	160.00
15.24.4	Size 100x 75mm	Mtr.	170.00
15.24.5	Size 125 x 100mm	Mtr.	180.00
15.25	Providing and fixing in CM 1 : 4 double paitam (rebated) stone door window and ventilator frames of approved quarry :		
15.25.1	Size 100 x 75mm.	Mtr.	180.00
15.25.2	Size 125 x 100mm	Mtr.	190.00
15.26	Providing & mixing admixture in Cement Concrete flooring/Cement Plaster/Plain or RCC work, Synthetic Polyster Triangular Construction Fiber of length 6mm/12mm/18mm with specific gravity 1.34 to 1.40 and diameter 10-40 microns and melting point>220 degrres centigrade by using 125 gm fiber for 50 kgs cement used as per direction of Engineer-in-charge.	Kg	444.00

15.27	Providing & mixing admixture in Cement Mass Concrete flooring CC Road, RCC work, Canal Lining and Shotcrete Non Circular with ribbed surface Structural Synthetic Macro Fiber 42-50mm length, specific gravity of > 1.30 and diameter 0.8 - 1.20mm and melting poing > 220 degree centigrade @ 0.75% -100% by cement weight in mix as per direction of Engineer-in-charge.	Kg	581.00
15.28	**Nano technology water proofing, thermal insulation protection coating**		
15.28.1	**Nano technology water proofing, thermal insulation protection coating 12 layer process for increasing life of building and damage control for roof.-- before coating scaling, cleaning, and preparing surface for treatment. Opening all joint and cracks and filling them with nano technology crack filling hydrophobic chemical with 2/3 coats. then after drying doing a PCC coat on cracks. One coat of nano technology hydrophobic/oiliophobic primer. Nano technology water proofing treatment in3 coats non silicon based applied with fiber mesh sheets in medium. Nano technology thermal insulation coating highly elasticity in nature with 3 coats with 1000 micron width.2top coats of anti stain, anti slippery, natural transparent high shine chemical for better life of the treatment.**	**Sqm**	1162.00
15.28.2	**Nano technology water proofing, thermal insulation protection coating 9 layer process for increasing life of building and damage control protection for walls.-- before coating scaling, cleaning, and preparing surface for treatment. Opening all joint and cracks and filling them with nano technology crack filling hydrophobic chemical with 2 coats. then after drying doing a PCC coat on cracks. One coat of nano technology hydrophobic/oiliophobic primer. Nano technology water proofing treatment in 2 coats non silicon based thermal insulation coating highly elasticity in nature with 2 coats with 1000 micron thickness 2 top coats of anti slippery, natural transparent high shine chemical for better life of the treatment.**	**Sqm**	829.00
15.28.3	**Nano technology water proofing or anti stain protection coating 5-7 layer process for increasing life of all stone fascia/facades/flooring or stone building and damage control protection, life 7-9 years, warranty 5 years-- before coating washing of stone and degreasing of that area is done 1 coat. (Removing of algae/moss/blackness/stains/rust etc. is extra) then after drying doing 1 coat of Nono technology hydrophobic prime. After this on dry area Nano technology water proofing treatment in 3 coats wet on wet. In case of anti stain treatment 2 top coats of anti stain, anti slippery, natural transparent high shine or matt chemical treatment is done**	**Sqm**	893.00
15.29	**Providing and Applying ACRYLIC emulsion polymer waterproofing chemical coating, on over roof on over roof slabs/terrace/balconies/parapet wall etc.with following method/specification:** **Paint the surface with regular paint brush so as to obtain approximately 0.3mm thickness coating. apply the 2nd coat next day at right angles to the 1st coat so that total thickness of the coating should be approximately 0.6mm coverage by using above formulation will be 40 ft2/ litre of chemical. painted surface does not require any external water curing since chemical will hold sufficient water for curing.After the primer coat Apply 2 coats of modified acrylic emulsion polymer water proofing compound having solar heat reduction properties coating @ 0.330 KG/SQM on the surface of the interval of 3-4 hours. Undulation on the surface is repair before the waterproofing work.**	**Sqm**	526.00

15.30	**Providing & applying one coat of UV resistant polymer cement based waterproofing slurry coating of approved make and quality on concrete or plastered surface as per specifications including preparation of surface by cleaning brushing and dust removing, filling of cracks and making the surface leak proof by mixing one part of chemical with 2 parts of cement and 2 parts of water (covering area 1.86 Sq.M/Kg of chemical) including cost of cement, labour, tools & plant, scaffolding if necessary etc., complete as directed by Engineer in Charge.** **Note: Undulations/cracks found on the surface should be rectified with Cement Motor before application and the same will be paid under relavent items.**	**Sqm**	244.00
15.31	**Providing & applying Three coat of Acrylic Water Proof membrane having sunlight and UV resistant properties over waterproofing slurry coat , initial coat of 50-75 micron thickness with diluting 1 part of chemical with 2 parts of clean water, over which applying 2 coats at 2 Hrs .intervals each with diluting 1 part of chemical with 20% of clean water for total thickness of 400 to 450 microns (covering area 0.9 Sq.M/Kg of chemical including preparation of surface by cleaning, brushing and dust removing, scaffolding if necessary etc., with contractor's labour, tools, machinery etc, complete as directed by Engineer in Charge.**	**Sqm**	486.00
15.32	**Providing & applying Three coat of pure Acrylic Water based elastomeric brush applied insulating coating on exterior surface overwaterproofing slurry coat , initial coat of 50-75 micron thickness with diluting 1 part of chemical with 2 parts of clean water, over which applying 2 coats at 2 Hrs, intervals each with diluting 1 part of chemical with 10% of clean water for total thickness of 400 to 450 microns (covering area 0.9 Sq.M/Kg of chemical) including preparation of surface by cleaning, brushing and dust removing, scaffolding if necessary etc., with contractor's labour, tools, machinery etc, complete as directed by Engineer in Charge.**	**Sqm**	522.00

CHAPTER : B-16
ALUMINIUM & P.V.C. WORK

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
16.1	Providing and fixing aluminum work for doors ,windows, ventilators and partition with extruded built up standard tubular / appropriate Z sections and other sections of approved make conforming to IS :733 and IS :1285, fixed with rawl plugs and screws or with fixing clips ,or with expansion hold fasteners including necessary filling up of gap. at junctions , at top ,bottom and sides with required PVC/neoprene felt etc. Aluminium section shall be smooth ,rust free, straight ,mitered and jointed mechanically wherever required including cleat angle Aluminium snap beading for glazing /paneling , C.P. brass/ stainless steel screws Al. Tower bolt & Al. handle & Al. Aldrop etc.,all complete as per architectural drawings and the directions of Engineer- in – charge .(Glazing and paneling to be paid for separately).		
16.1.1	For fixed portion		
16.1.1.1	Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer)	Kg.	296.00
16.1.1.2	Powder coating aluminum (minimum thickness of powder coating 50 micron)	Kg.	319.00
16.1.1.3	Polyester Powder coating aluminum (minimum thickness of powder coating 50 micron)	Kg.	331.00
16.1.2	For shutters builtup standard tubular sections of openable doors ,windows & ventilators including providing & fixing hinges / rollers etc. and making provision for fixing of fittings wherever required (lockes shall be paid for separately).		
16.1.2.1	Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer)	Kg.	323.00
16.1.2.2	Powder coating aluminum (minimum thickness of powder coating 50 micron)	Kg.	346.00
16.1.2.3	Polyester Powder coating aluminum (minimum thickness of powder coating 50 micron)	Kg.	358.00
16.2	Extra for shutters of sliding aluminum windows		0.10
16.3	Extra for work done in curved / circular shape.		0.10
16.4	Providing and fixing 12 mm thick prelaminated three layer medium density (exterior grade) particle board Grade I, Type II conforming to IS : 12823 bonded formaldehyde synthetic resin ,of approved brand and manufacture in paneling fixed in aluminium doors ,windows shutters and partition frames with C.P. brass / stainless steel screws etc. complete as per architectural drawings and direction of engineer-in-charge.		
16.4.1	Pre-laminated particle board with decorative lamination on one side and balancing lamination on other side.	Sqm	936.00
16.4.2	Prelaminated particle board with decorative lamination on both sides.	Sqm	993.00
16.5	Providing and fixing glazing in aluminium door, window ,ventilator shutters and partitions etc. with PVS / neoprene gasket etc. complete as per the architectural drawings and the directions of engineer-in-charge.(Cost of aluminium snap beading shall be paid in basic item.)		
16.5.1	With glass panes of 4.0 mm thickness (weight not less than 10.0 kg/sqm) as per IS : 2835.	Sqm	719.00
16.5.2	With float glass panes of 5.0 mm thickness (weight not less than 13.50 kg/sqm)	Sqm	820.00
16.5.3	With float glass panes of 8 mm thickness	Sqm	1123.00

16.6	Providing and fixing double action hydraulic floor spring of approved brand and manufacture IS : 6315 marked, for doors including cost of cutting floors as required, embedding in floors and cover plates with brass pivot and single piece M.S.		
16.6.1	With stainless steel cover plate	Each	1296.00
16.6.2	With brass cover plate	Each	1796.00
16.7	Filling the gap upto 5 mm depth and 5 mm width in between aluminium frame & adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.	Rm	71.00
16.8	Providing and fixing 100mm brass locks (best make of approved quality) for aluminium doors including necessary cutting and making good etc. complete.	Each	298.00
16.9	Providing and fixing aluminium **U shape handle** of **size** 100mm with SS screws etc. Complete as per direction of Engineer-in-charge		
16.9.1	Anodized (AC 15) aluminium	Each	61.00
16.9.2	Powder coated minimum thickness 50 micron aluminium	Each	66.00
16.9.3	Polyester powder coated minimum thickness 50 micron aluminium .	Each	68.00
16.10	Providing and fixing expanded grill made of aluminum as per design and drawing having members section of size 7.5mm x 6.0mm and opening of size 102mm x 99mm to aluminum window/vent with required screws, Y and H aluminium sec (as per drawing) at the ends and middle joints respectively complete in all respect as per direction of engineer in charge.		
16.10.1	Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer)	Sqm.	1614.00
16.10.2	Powder coating aluminum (minimum thickness of powder coating 50 micron)	Sqm.	1675.00
16.10.3	Polyester Powder coating aluminum (minimum thickness of powder coating 50 micron)	Sqm.	1797.00
16.11	S&f fixed wire gauge of stainless steel of 14 mesh x 24 gauge to the aluminium window by Aluminium beading 20x3mm with suitable screws at not exceeding 150mm distance at all heights with all lead and lift with scafolding.	Sqm	724.00
16.12	Providing and fixing vertical Blinds , headrail shall be oriented aluminum alloy 50 mm **wide** x 18 mm high the thickness shall be 1.2 mm. Control unit shall be made of high grade polymer and shall be housed in side the headrail, which shall transfer motion from chain tilter to tilt rod. Chain tilter shall be made of plastic bead of diameter 4.5 mm through which 2 mm thick polyester code passes. Tilt rod shall be extruded aluminum having 3 key ways. Runner shall be moulded plastic and shall have to plastic wheels mounted on plastic exles to enable runner to slide unhindered in headrail. Luver shall be made of fabric 100 mm colour fast and shall have protective scotchgard coating to resist stains and dust. Bottom mechanism consist bottom weight made of powder coated galvanized steel for maximum corrosion resistance. The bottom weight shall be attached with bottom.	Sqm	1475.00
16.13	Providing and fixing Horizontal blinds /Venation blind shall be cold rolled formed U shape 23 mm high x 25 mm wide with flanged adages. It shall be made of heavy gaze 0.56 mm galvanized steel, Slates shall be made of virgin Al. alloy (having 3.5 – 4.5% of Mg. Content of width 25 mm. The slat shall have dust guard paint coating, the cord lock shall be of thick tomized steel and shall be securely attach in the head rail. It shall provide locking arrangement of the cord. Tilt rod shall be made of galvanized steel it shall have a D shaped with an average cross section with diameter of 6 mm to achieve minimum torsional deflection.	Sqm	1475.00

16.14	Providing and fixing Drapery rod shall be made of 20 gauge cold rolled steel strip. Which is electrical resistance welded, The section will be available 25.4 mm diameter the section shall be coated with acid resistance polymer finial shall made from ABS material in nearest matching colour of rod. The ring shall be made from ABS material in nearest matching colour to rod. Brackets of made of galvanized steel.	Rm	431.00
16.15	Providing and fixing sun control film, thickness of film 1 Mil Visible Light Transmittance 4-24% & Solar Energy Transmittance 42-51% & Total solar Energy rejection 37-43%of approved colour and shade , complete in all respect , at all level , as directed by engineer-in -charge.	Sqm	581.00
	PVC ITEMS		
16.16	Providing and fixing factory made of uPVC Extruded section having an overall dimension as below (tolerance+/-1mm) with wall thickness 2.0mm+/-0.2mm,corners of the door frame to be mitred and welded of plastic ,galvanized brackets and stain less steel screws.The hinge side vertical of the frames reinforced by galvanized M.S.tube of size 19x19mm and 1mm+/-0.1mm wall thickness and 3 nos.stainless steel hinges fixed to the frame complete as per manufacturers specification and direction of Engineer-in-charge.		
16.16.1	Extruded section profile size 48x40 mm	Rm	178.00
16.16.2	Extruded section profile size 42x50 mm	Rm	198.00
16.17	Providing and fixing to existing door frames 24mm thick factory made PVC door shutters made of stiles and rails of a uPVC hollow section of size 59x24 mm and wall thickness 2mm +/- 0.2 mm with inbuilt edging on both sides.The stiles and rails mitred and joined at the corners by means of M.S.galvanized/plastic brackets of size 75x220mm having wall thickness 1.0mm+/- 0.1mm and stainless steel screws. The stiles of the shutter reinforced by inserting galvanized M.S.tube of size 20x20mm and 1mm +/-0.1mm wall thickness. The lock rail made up of 'H' section,a uPVC hollow section of size 100x24mm and 2mm +/- 0.2mm wall thickness fixed to the shutter stiles by means of plastic/ galvanized M.S.'U' cleats.The shutter frame fitted with a uPVC multi-shambered single panel of size not less than 610mm, having overall thickness of 20mm and 1mm +/- 0.1mm wall thickness. The panels fitted vertically and tie bar at two places by inserting horizontally 6mm galvanized M.S.rod and fastened with nuts and washers, complete as per manufacturer's specification and direction of Engineer-in-charge (For W.C.and bathroom door shutter).	Sqm	1601.00

16.18	Providing and fixing to existing door frames 30mm thick factory made polyvinyl chloride (PVC) door shutter made of stiles and rails of a uPVC hollow section of size 60x30 mm and wall thickness 2mm +/- 0.2mm with inbuilt decorative moulding edging on one side. The stiles and rails mitred and joined and the corners by means of M.S.galvanized/plastic brackets of size 75x220 mm having wall thickness 1.0mmand stainless steel screws. The stiles of the shutter reinforced by inserting galvanized M.S.tube of size 25x20 mm and 1mm +/- 0.1mm wall thickness. The lock rail made up of 'H' section, a uPVC hollow section of size 100x30 mm and 2mm +/- 0.2mm wall thickness fixed to the shutter stiles by means of plastic/galvanized M.S.tube 'U' cleats. The shutter frame filled with a uPVC multi-chambered single panel of size not less than 610mm,having overall thickness of 20mm and 1mm+/- 0.1mm wall thickness. The panels filled vertically and tie bar at two places by inserting horizontally 6mm galvanized M.S.rod and fastened with nuts and washers.complete as per manufacturer's specification and direction of Engineer-in-charge.(For W.C.and bathroom door shutter)	Sqm	1663.00
16.19	Providing and fixing factory made P.V.C. door frame of size 50x47mm with a wall thickness of 5mm, made out of extruded 5mm rigid PVC foam sheet mitred at corners and joined with 2 Nos. of 150mm long brackets of 15x15mm M.S. square tube, the vertical door profiles to be reinforced with 19x19mm M.S. square tube of 19 gauge, EPDM rubber gasket weather seal to be provided through out the frame. The door frame to be fixed to the wall using M.S. screws of 65/100mm size complete as per manufacturers specification and direction of Engineer-in-Charge.	Rm	252.00
16.20	Providing and fixing to existing door frames.		
16.20.1	30mm thick factory made solid panel PVC door shutter consisting of frame made out of M.S. tubes of 19 gauge thickness and size of 19mm x 19mm for stiles and 15 mm x15mm , top & bottom rails. M.S .frame shall have a coat of steel primers of approved make and manufacture M.S. frame covered with 5mm thick heat moulded PVC 'C channel of size 30mm thickness, 70mm width out of which 50mm shall be flat and 20mm shall be tapered in 45 degree angle on either side forming stiles; and 5mm thick, 95mm wide PVC sheet out of which 75 mm shall be flat and 20mm shall be tapered in 45 degree on the inner side to form top and bottom rail and 115mm wide PVC sheet out of which 75 mm shall be flat and 20mm shall be tapered on both sides to form lock rail. Top bottom and lock rails shall be provided either side of the panel. 10mm (5mm x 2) thick, 20mm wide cross PVC sheet shall be provided as gap insert for top rail & bottom rail. Panelling of 5mm thick PVC sheet to be fitted in the M.S. frame welded/ sealed to the stiles & rails with 7mm (5mm +2mm) thick x 15mm wide PVC sheet bending on inner side, and joined together with solvent cement adhesive. An additional 5mm thick PVC strip of 20mm width is to be	Sqm	2919.00
16.20.2	30mm thick factory made solid both side Pre laminated panel PVC door shutter consisting of frame made out of M.S. tubes of 19 gauge thickness and size of 19mm x 19mm for stiles and 15x15mm top & bottom rails. M.S .frame shall have a coat of steel primers of approved make and manufacture M.S. frame covered with 5mm thick heat moulded PVC 'C channel of size 30mm thickness, 70mm width out of which 50mm shall be flat and 20mm shall be tapered in 45 degree angle on either side forming stiles; and 5mm thick, 95mm wide PVC sheet out of which 75mm shall be flat and 20mm shall be tapered on the inner side to form top and bottom rail and 115mm wide PVC sheet out of which 75 mm shall be flat and 20mm shall be tapered on both sides to form lock rail. Top bottom and lock rails shall be provided either side of the panel. 10mm (5mm x 2) thick, 20mm wide cross PVC sheet shall be provided as gap insert for top rail & bottom rail. Panelling of 5mm thick PVC sheet to be fitted in the M.S. frame welded/ sealed to the stiles & rails with 7mm (5mm +2mm) thick x 15mm wide PVC sheet bending on inner side, and joined together with solvent cement adhesive. An additional 5mm thick PVC strip of 20mm width is to be stuck on the interior side of the 'C Channel using PVC solvent adhesive etc.	Sqm	3043.00

16.21	Providing and fixing of false ceiling with grid of M.S.tube section of size 19mm X 19mm and with a wall thickness of 1mm+/- 0.2mm.The side pipe and the intermediate pipes is fixed to the roof from the top with the help of ceiling angles/wire grid shall be covered by the uPVC profile section of size 6mm X 250mm with wall thickness of 0.80mm +/- 0.2mm,with the help of self tapping screw of 6mm X 13mm, 6mm X 19mm.Necessary cutouts for electric connections, lighting, air conditioning etc, shall be provided at required place. The perimeter edge shall be covered by extruded PVC corner beading section of size 9mm X 28mm or 25mm X30mm with a wall thickness of 1mm+/-0.2mm fixed by applying cynoacrylic adhesive complete as per manufacturer's specification and direction of Engineer-in-charge.	Sqm	905.00
16.22	Providing and fixing of wall paneling on grid made out of PVC profile section of size 21x17mm with a wall thickness of 1.5mm+/-0.2mm,fixed on the existing wall with wood screw of size 50mm X 8mm with rowel plug at a spacing of 900mm center.uPVC profile section of size 150mm X 10mm with a wall thickness of 1.0mm+/-0.2mm to be fixed on the grid by self tapping screw of size 19mm X 6mm.Necessary cutouts for electrical connection to be provided at required places. The edge and periphery, finally to be covered by extruded PVC beading section of size 28mm X 12mm with a wall thickness of 1.0mm+/-0.2mm complete as per manufacture's specification and direction of Engineer-in-charge.	Sqm	1267.00
16.23	Supply of " Computer Table" of Modular nature having all the necessary arrangement for keyboard, CPU, UPS, Printer and wire manager Table Top : The table top of size 900 x 600mm(WxD) would be made out of post formed on 25 mm thick particle board with 0.7mm thick Carvica sheet in half round post formed finish on the front and rear side, matching color compensating laminate fixed on the bottom side. Other two sides are edge banded and duly sealed with 1.5 mm thick PVC edge banding of matching color with high pressure through feed machine. Keyboard Below the table top the keyboard pull out tray having sliding arrangement fixed on side supporting legs with 450mm G.I. telescopic channel. Keyboard of size 750 x 450 made out of 18 mm thick Particle board laminate with 0.7mm thick laminate sheet having all four sides Flat edge banded and duly sealed with 1.5 mm thick PVC edge banding of matching color with high pressure through feed machine. Side Support :Two supporting side leg of size 600 x 750mm(WxH) made out of post formed on 18 mm thick particle board with 0.7mm thick Carvica sheet in half round post formed finish on two sides and other two sides are edge banded and sealed with 1.5mm thick PVC edge banding of matching color with high pressure through feed machine.Modesty Panel : Made out of 18 mm thick Particle board laminate with 0.7mm thick laminate sheet of size 900 x 750mm(WxH) having all four sides Flat edge banded and duly sealed with 1.5 mm CPU, UPS & Printer Places Table having the provision for CPU of size 300 x 600 x 600mm(WxDxH) below the table top on the right side of table adjoining to this on left side below the table the provision for UPS of size 450 x 600 x 250mm (WxDxH) and above this the provision for the Printer and stationary having overall dimension of 450 x 600 x 300mm(WxDxH). All this is made out of 18 mm thick Particle board laminate with 0.7mm thick laminate sheet having all four sides flat edge banded and duly sealed with 1.5 mm thick PVC edge banding of matching color with high pressure through feed machine. All the parts of the table are duly fixed with mini fix dowel system with each other having matching drilled holes to be made on CNC multi hole boring machine for proper matching and alignment of the different part of the table, 4nos. of Self Adjusting level screw of size 45 x 6mm are provide at the bottom of the both legs for the proper leveling of the table.	Each	7392.00

16.24	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with post forming three drawer pedestal unit one KBT or one CPU trolley. Size 4800 x 1500 x 1200mm single seat in combination	Each	14132.00
16.25	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with four side edage banding one drawer and one open able shutter pedestal unit one KBT or one CPU trolley. Size 6000 x 3000 x 1200mm single seat in combination	Each	22410.00
16.26	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick MDF board with Membrane three drawer pedestal unit one KBT or one CPU trolley. Size 4600 x 2100 x 1200mm single seat in combination	Each	22346.00
16.27	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particaleboard with post formed three drawer pedestal unit one KBT or one CPU trolley. (Rate of this item is effective from 01.12.2016) Size 4500 x 2400 x 1200mm single seat in combination	Each	20418.00
16.28	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick MDF board with Membrane three drawer pedestal unit one KBT or one CPU trolley. Size 6600 x 3000 x 1200mm single seat in combination	Each	27209.00
19.29	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with post formed one drawer or one open able shutter pedestal unit one KBT or one CPU trolley. Size 4500 x 3000 x 1200mm single seat in combination	Each	21252.00
16.30	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with post formed three drawer pedestal unit one KBT or one CPU trolley. Size 3600 x 2400 x 1200mm single seat in combination	Each	15773.00
16.31	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with four side edage banding 1200mm height storage unit with two drawer one KBT or one CPU trolley. Size 1500 x 3000 x 1650mm single seat in combination	Each	45590.00
16.32	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with post formed three drawer one KBT or one CPU trolley. Size 2100 x 2100 x 1200mm single seat in combination	Each	7943.00
16.33	Supply and fixing of Modular work Station with Partition 32mm thick Aluminum top 25 mm thick particale board with post formed legas 18mm particle board with four side edage banding one KBT or one CPU trolley. Size 2100 x 2100 x 1200mm single seat in combination	Each	12980.00

16.34	Supply and fixing of Modular work Station with Partition 60mm thick Aluminum top 25 mm thick particale board with post formed legas 18mm particle board with four side edage banding with three drawer one KBT or one CPU trolley Two 1200mm height storage unit. Size 6000 x 3000 x 1200mm single seat in combination	Each	142900.00
16.35	Supply and fixing of Modular work Station with Partition 22mm thick Aluminum top 25 mm thick particale board with post formed one drawer or one open able shutter one KBT or one CPU trolley Two 1200mm height storage unit. Size 3000 x 2700 x 1200mm single seat in combination	Each	16514.00
16.36	Supply and fixing of Modular work Station with Partition 22mm thick Aluminum top 25mm thick particale board with post formed one drawer or one open able shutter one KBT or one CPU trolley Two 1200mm height storage unit. Size 2400 x 1200 x 1200mm single seat in combination	Each	13554.00
16.37	Supply and fixing of Modular work Station with Partition 60mm thick Aluminum top 25mm thick particale board with post formed one drawer or one open able shutter one KBT or one CPU trolley Two 1200mm height storage unit. Size 1200 x 1200 x 1200mm single seat in combination	Each	15965.00
16.38	Supply and fixing of Modular work Station with Partition 22mm thick Aluminum top 25mm thick particale board with post formed one drawer or one open able shutter one KBT or one CPU trolley Two 1200mm height storage unit. Size 2400 x 600 x 1200mm single seat in combination	Each	16268.00
16.39	Supply and fixing of Modular work Station with Partition 60mm thick Aluminum top 25mm thick particale board with post formed one drawer or one open able shutter one KBT or one CPU trolley Two 1200mm height storage unit. Size 1800 x 1800 x 1950mm single seat in combination	Each	56971.00
16.40	**Providing and fixing Structural glazing work with toughened glass of 6mm thick and reflective colour as approved in dark shade like Blue/Dark Blue etc. with frame work of curtain section 65x60mm (Weight 1.64 kg/m min.) made of aluminium with chrome plating. Internal frame work will also be of curtain section 65x60mm of chrome plated aluminium section with20x20mm aluminium tube section and structural spacer made of rubber. The aluminium frame work size as per instruction of Engineer-in-charge and including openable panels with all necesary sections and fitting.The work include silicon structural adhesive for complete gripping of glass with framing members and silicon weather seal sealant to prevent moisture/water through joints. Glass will be fixed in plumb and with uniform spacing of 10-12mm wide duly filled with silicon weather seal sealant. The work includes all labour,scaffolding , structural aluminium work, toughened glasses,silicon weather seal sealant, silicon structural adhesive structrucal spacer, all accessoreis like handles, peg stays, M-15 grade cement concrete for fixing holdfast .(Face area of glass shall be measured)**	**Sqm**	4892.00

16.41	**Supply and fixing of polycrete wall fencing of existing wall, fencing of 0.9 mtr. Height above the existing wall, main member PVC profile section of size 125mmx125mmx900mm (c/c 1.67 mtr. length) with a wall thickness of 2.00mm+0.20mm and Horizontal panel PVC profile section of size 305mmx20mm (3 nos. 1.6 mtr. length) with wall thickness 1.00mm + 0.1mm and side edge covered with PVC C-Section of size 24mm x18mm with a wall thickness of 2.00mm+0.20mm Horizontal panel reinforced with G.I. section (C-Type) with a wall thickness of 1.20+0.10mm and horizontal panel having side in main member groove (vertical pillar 125x125mm) all complete as per design specification and as direction by Engineer in charge.**	**Sqm**	2336.00
16.42	**Providing and fixing of uPVC casement fixed windows or ventilators manufactured in ISO 9001:2000 & 14001:2004 certified company. Frame : Made from the Extruded uPVC Window Profile Section of size 60 x 60mm having outer wall thickness of 2.25mm (+/- 0.2mm) and 3 box multi-chamber construction, White in finish, duly reinforced with 1.2mm thick G/J/U/O TYPE GI section. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Fix Mullion made of 76 x 60mm uPVC Profile Section with steel reinforcement shall be provided in windows as per the requirement. Frame shall have 'O' type EPDM gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Glazing bead of size 34 x 20mm with 'K' & 'O' type inner and outer EPDM weather seal gaskets alongwith 5mm thick ISI make plain float glass.All welding joints of frame and shutter shall be cleaned and milled with the CNC mechanism to provide uniform grooved finish on all visible joints. Hardware : Installation at Site : Complete window is to be installed on site with 10 x 100mm fastners with white cap in**	**Sqm**	4884.00
16.43	**Providing and fixing expanded grill made of aluminum as per design and drawing having members section of size 7.5mm x 6.0mm and opening of size 75mmx72mm to aluminum window/vent with required screws, Y and H aluminium sec (as per drawing) at the ends and middle joints respectively complete in all respect as per direction of engineer in charge.**		
16.43.1	**Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer) Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer)**	**Sqm**	1675.00
16.43.2	**Powder coating aluminum (minimum thickness of powder coating 50 micron)**	**Sqm**	1736.00
16.43.3	**Polyester Powder coating aluminum (minimum thickness of powder coating 50 micron)**	**Sqm**	1858.00
16.44	**Providing and fixing expanded grill made of aluminum as per design and drawing having members section of size 7.5mm x 6.0mm and opening of size 50mmx48mm to aluminum window/vent with required screws, Y and H aluminium sec (as per drawing) at the ends and middle joints respectively complete in all respect as per direction of engineer in charge.**		
16.44.1	**Anodised aluminum (anodised transparent 81 dyed to required shade according to IS : 1868. (Minimum anodic coating of grade AC 15, Anodizing to be got done from approved Anodizer)**	**Sqm**	1736.00

16.44.2	**Powder coating aluminum (minimum thickness of powder coating 50 micron)**	**Sqm**	1797.00
16.44.3	**Polyester Powder coating aluminum (minimum thickness of powder coating 50 micron)**	**Sqm**	1919.00
16.45	**Providing and fixing of uPVC Sliding windows: Frame Made from the Extruded uPVC Window Profile Section of size 106w x 50h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and 3 box multi-chamber construction, White in finish, duly reinforced with 1.2 mm thick GI section. Frame shall have three track configuration, two for sliding of window shutter and one for mosquito mesh shutter. Vertical member of frame which bears the sliding shutter load shall have aluminium rail/track for smooth sliding of shutter rollers. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Shutter : The shutter of sliding window shall be made of 39w x 69h mm Extruded 3 box multi- chamber uPVC Window Profile Section of white colour having outer wall thickness of 2.5mm (+/- 0.2mm) provided with reinforcement of 1.5mm thick GI section duly mitered cut & thermal welded at all corners and fitted with uPVC glazing bead of size 22 x 20 mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass. Mesh shutter**	**Sqm**	8443.00
16.46	**Providing and fixing of uPVC Sliding windows: Frame Made from the Extruded uPVC Window Profile Section of size 106w x 50h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and 3 box multi-chamber construction, Laminated (colour) finish, duly reinforced with 1.2 mm thick GI section. Frame shall have three track configuration, two for sliding of window shutter and one for mosquito mesh shutter. Vertical member of frame which bears the sliding shutter load shall have aluminium rail/track for smooth sliding of shutter rollers. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Shutter : The shutter of sliding window shall be made of 39w x 69h mm Extruded 3 box multi- chamber uPVC Window Profile Section of laminated (colour) having outer wall thickness of 2.5mm (+/- 0.2mm) provided with reinforcement of 1.5mm thick GI section duly mitered cut & thermal welded at all corners and fitted with uPVC glazing bead of size 22 x 20 mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass. Mesh**	**Sqm**	12965.00
16.47	**Providing and fixing of uPVC Sliding windows: Frame Made from the Extruded uPVC Window Profile Section of size 106w x 50h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and 3 box multi-chamber construction, White in finish, duly reinforced with 1.2 mm thick GI section. Frame shall have three track configuration, Vertical member of frame which bears the sliding shutter load shall have aluminium rail/track for smooth sliding of shutter rollers. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Shutter : The shutter of sliding window shall be made of 39w x 69h mm Extruded 3 box multi- chamber uPVC Window Profile Section of white colour having outer wall thickness of 2.5mm (+/- 0.2mm) provided with reinforcement of 1.5mm thick GI section duly mitered cut & thermal welded at all corners and fitted with uPVC glazing bead of size 22 x 20 mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass.**	**Sqm**	6343.00

16.48	**Providing and fixing of uPVC Sliding windows: Frame Made from the Extruded uPVC Window Profile Section of size 106w x 50h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and 3 box multi-chamber construction, Laminated (colour) finish, duly reinforced with 1.2 mm thick GI section. Frame shall have three track configuration, Vertical member of frame which bears the sliding shutter load shall have aluminium rail/track for smooth sliding of shutter rollers. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Shutter : The shutter of sliding window shall be made of 39w x 69h mm Extruded 3 box multi- chamber uPVC Window Profile Section of laminated (colour) having outer wall thickness of 2.5mm (+/- 0.2mm) provided with reinforcement of 1.5mm thick GI section duly mitered cut & thermal welded at all corners and fitted with uPVC glazing bead of size 22 x 20 mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass.**	**Sqm**	10187.00
16.49	**Providing and fixing of UPVC casement openable windows (outward / inward): Outer Frame made from the Extruded UPVC Windows profile section of size 60w X 60h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and three chamber constuction, White in finish, duly reinforced with 1.5mm thick GI section. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Fix Mullion made of 60w X 80h mm UPVC profile section with steel reinforcement shall be provided in windows having 2 or more openable shutters, as per the requirement. Frame shall have co-extruded with EPDM/TPE-E gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter: The shutter of outward openable window shall be made of size 60w X 74h mm and shutter of inward openable window shall be made of size 60w x 77h mm**	**Sqm**	9118.00
16.50	**Providing and fixing of UPVC casement openable windows (outward / inward): Outer Frame made from the Extruded UPVC Windows profile section of size 60w X 60h mm having outer wall thickness of 2.5mm (+/- 0.2mm) and three chamber constuction in Laminated (colour) finish, duly reinforced with 1.5mm thick GI section. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Fix Mullion made of 60w X 80h mm UPVC profile section with steel reinforcement shall be provided in windows having 2 or more openable shutters, as per the requirement. Frame shall have co-extruded with EPDM/TPE-E gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter: The shutter of outward openable window shall be made of size 60w X 74h mm and shutter of inward openable window shall be made of size 60w x 77h mm Extruded 3 box multi-chamber UPVC Window profile section of laminated(colour) having outer wall thickness of 2.25mm (+/- 0.2mm) provided with reinforcement of 1.5mm thick GI section duly mitered cut & thermal welded at all corners and fitted with UPVC glazing bead of size 36 X 20mm with inner and outer co-extruded EPDM/TPE-E weather seal gasket alongwith 6mm thick ISI make plain float glass. All welding joints of frame and shutter shall be cleaned and milled with the CNC mechainism to provide uniform grooved finish on all visible jaints. Hardware: Window shutter are fixed with frame on 2.5mm thick SS301 Grade friction hinge system to keep the window opened at desird angle. Friction Hinge also enables easy cleaning of glass on both side from the inside of building.**	**Sqm**	13474.00

16.51	**Providing and fixing of UPVC casement fully (french) open-able windows (outward / inward) : Frame made from the Extruded UPVC Windows profile section of size 60w X 60h mm having outer wall thickness of 2.5 mm (+/- 0.2mm) and 3 box multi-chamber constuction, White in finish, duly reinforced with 1.5 mm thick G/J/U/O TYPE GI section. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Fix Mullion made of 60w X 80h mm UPVC profile section with steel reinforcement shall be provided for the fix portion of the window and for the window area having 2 or more openable shutters / fix part as per the requirement. Frame shall have co-extruded EPDM/TPE-E gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter: The shutter of outward openable window shall be made of size 60w X 74h mm and shutter of inward openable window shall be made of size 60w X 77h mm**	**Sqm**	11027.00
16.52	**Providing and fixing of UPVC casement fully (french)open-able windows Laminated Profile (Colour) (outward / inward) : Frame made from the Extruded UPVC Windows profile section of size 60w X 60h mm having outer wall thickness of 2.5 mm (+/- 0.2mm) and 3 box multi-chamber constuction, Laminated(colour) finish, duly reinforced with 1.5 mm thick GI section. All the four corners shall be mitered cut & thermal welded so as to form window frame. Frame shall be milled with drain and air equalizer hole in order to be water tight and for drainage of accumulated water, if any, to outer side. Fix Mullion made of 60w X 80h mm UPVC profile section with steel reinforcement shall be provided for the fix portion of the window and for the window area having 2 or more openable shutters / fix part as per the requirement. Frame shall have co-extruded EPDM/TPE-E gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter: The shutter of outward openable window shall be made of size 60w X 74h mm and shutter of inward openable window shall be made of size 60w X 77h mm**	**Sqm**	14272.00
16.53	**Providing and fixing of openable uPVC Doors with or without fix top: Frame Made of Extruded uPVC Window Profile Section of size 60w x 70h mm having outer wall thickness of 2.75mm (+/- 0.2mm) and 3 box multi-chamber construction, White in finish, duly reinforced with 1.2mm thick GI section. All the corners shall be mitered cut & thermal welded so as to form door frame. Fix Mullion made of 60w x 80h mm uPVC Profile Section with steel reinforcement shall be provided for the top fix portion of the doors or in case of the doors having more than two openable shutters, as per the requirement / drawing. Frame of openable door shall have „O" type EPDM gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter : The shutters of openable door shall be made of 60w x 95h mm Extruded 3 box multi-chamber uPVC Window Profile Section of white colour having outer wall thickness of 2.75mm (+/- 0.2mm) provided with reinforcement of 2.0 mm thick GI section duly mitered cut & thermal welded at all corners, provided with center mullion (if specified in drawing) of size 60w x 80h mm uPVC Profile Section and fitted with uPVC glazing bead of size 36 x 20mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass.**	**Sqm**	9930.00

All welding joints of frame and shutter shall be cleaned and milled with the CNC mechanism to provide uniform grooved finish on all visible joints. Hardware: Each openable door shutters is to be fixed with three load beading butt hinges and locking of doors is to be provided with multi-point transmission gear (ESPAG) with white finish powder- coated lockable handle. Installation at site: Complete door is to be installed on site with 10 X 100mm fastners with white cap in existing pre-finished wall cut-out cemented to glass level plane at all height & width and silicon glue is applied to fill up the crevices between wall and door frame. Complete in all respect as per the drawing and specification and direction of engineer-in-charge.

16.54	**Providing and fixing of openable uPVC Doors with or without fix top: Frame Made of Extruded uPVC Window Profile Section of size 60w x 70h mm having outer wall thickness of 2.75mm (+/- 0.2mm) and 3 box multi-chamber construction, Laminated (colour) finish, duly reinforced with 1.2mm thick GI section. All the corners shall be mitered cut & thermal welded so as to form door frame. Fix Mullion made of 60w x 80h mm uPVC Profile Section with steel reinforcement shall be provided for the top fix portion of the doors or in case of the doors having more than two openable shutters, as per the requirement / drawing. Frame of openable door shall have co-extruded EPDM/TPE-E gasket fitted in in-built groove of frame profile for proper air & sound insulation of the shutter. Shutter : The shutters of openable door shall be made of 60w x 95h mm Extruded 3 box multi-chamber uPVC Window Profile Section of laminated(colour) having outer wall thickness of 2.75mm (+/- 0.2mm) provided with reinforcement of 2.0 mm thick GI section duly mitered cut & thermal welded at all corners, provided with center mullion (if specified in drawing) of size 60w x 80h mm uPVC Profile Section and fitted with uPVC glazing bead of size 36 x 20mm with inner and outer co-extruded EPDM/TPE-E weather seal gaskets alongwith 6mm thick ISI make plain float glass.** **All welding joints of frame and shutter shall be cleaned and milled with the CNC mechanism to provide uniform grooved finish on all visible joints. Hardware: Each openable door shutters is to be fixed with three load beading butt hinges and locking of doors is to be provided with multi-point transmission gear (ESPAG) with white finish powder- coated lockable handle. Installation at site: Complete door is to be installed on site with 10 X 100mm fastners with white cap in existing pre-finished wall cut-out cemented to glass level plane at all height & width and silicon glue is applied to fill up the crevices between wall and door frame. Complete in all respect as per the drawing and specification and direction of engineer-in-charge.**	**Sqm**	13558.00
16.55	**Providing and Fixing Aluminium composite panel of approved make 3mm thick with 0.5mm thick skin & PVDF coated approved shade for wall panelling. Main framing work to be done using 50mmx25mmx1.6mm tubular section of approved make. Stainless steel screw to be used to fix the ACP panel to be fixed to the main frame with industrial adhesive tape and panel groove to be sealed with weather proof silicon sealant suitable for glass/aluminium etc. complete in all respect as approved by Engineer-in-charge.**	**Sqm**	2358.00

CHAPTER : B-17
ADDITIONAL MISCELLANEOUS ITEMS

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
17.1	Supply and installation of Multi Rib Proofing/cladding Sheet Manufacture out of 0.50mm TCT (Total Coated Thickness) high tensile Zinc aluminum alloy coated galvaiumeseel (as per 150gsm zinc aluminum coated 550Mpa tuekd strength) confirming to IS: 1397/astm A-792 sheet to have wide pens 30mm high NB at 250 centre and centre width of 1020mm shets be casted with regular modified plyster system on a continuous with line on centre for face and with a polyster coating respectively sheet shall have proportionality with siphoned plate at made at to prevent leakage sheet shall be fixed by more of sell drilling lift tapping hot dip zinc coated hox head fasterner of size 12x14x55mm long. The sheet shall be supplied in cost one length and foe a minimum up to 12mts with all scaffolding.	Sqm	626.00
17.1.1	**Add extra for 0.60mm TCT (Total Coated Thickness) high tensile Zinc aluminum alloy coated galvaiumeseel sheet in place of 0.50mm thick**	**Sqm**	**15%**
17.1.2	**Add extra for 0.80mm TCT (Total Coated Thickness) high tensile Zinc aluminum alloy coated galvaiumeseel sheet in place of 0.50mm thick**	**Sqm**	**45%**
17.2	<u>Supply and Installation of ROOF – PREFABRICATED INSULATED PPGL FACE PANEL:</u> Roof: Insulated sandwich panels for roof of total thickness mentioned below exclusive of crest height of top corrugation. Panel shall be made of 0.50mm (+/- 0.04mm) thick pre coated metal sheet on both sides of rigid polyurethane insulation. The top external facing metal sheet shall have corrugated shape with crest. The trapezoidal shape crowns/crest height should not be less than 30mm. The pre-coated metal sheet shall be PPGL (Pre Painted Galvalume Sheet with AZ coating) sheet having SDP (Super Durable Polyester) colour coat of 18 micron (min). The bottom sheet must have tongue and groove type flange (U shape) of 8mm x 8mm on groove side and 8mm x 12mm on tongue side. Flange must be on lengthwise sides for joining adjacent panels. Panel's bottom side PPGL sheet must also be pre-strengthened by full lengthwise stiffening with semi-circular beading of 12mm width & 3 to 4mm depth at 240 (+/-5) mm pitch. The top corrugated sheet must have at-least one open crown/crest for overlapping adjacent panel widthwise. Similarly the top corrugated sheet must have suitable open extension of min 6" on bottom widthwise side of panel for overlapping further panels joined lengthwise also. The insulation		
	Thickness 65mm (+/- 2mm)	Sqm.	2292.00
	Thickness 105mm (+/- 2mm)	Sqm.	2593.00
17.3	<u>Supply and Installation of WALL – PREFABRICATED INSULATED PPGL PANEL:</u>		

Walls: Insulated sandwich panels for wall of total thickness mentioned below. Panel shall have 0.50mm (+/-0.04mm) thick pre coated metal sheet on both sides of rigid PUR insulation. The pre-coated metal sheet shall be PPGL (Pre Painted Galvalume Sheet with AZ coating) sheet having SDP (Super Durable Polyester) colour coat of 18 micron (min). The panels shall be vertically joined together with adjacent panel by tongue and groove joints. For joining, wall panel's metal sheet facings must have lengthwise flange (U shape) of 8mm x 8mm on groove side and 8mmx12mm on tongue side of panel. The pre-coated metal sheet (PPGL) must also be pre-strengthened by full lengthwise stiffening with semi-circular beading of 12mm width & 3 to 4mm depth at 240 (+/-5) mm pitch. Panels may have optional provision for inbuilt electrical conduit running lengthwise atvertical center of specified panels. The insulation core shall be self-extinguishing DIN 4102,Class B3 fire retardant class rigid PUR with thermal conductivity of 0.025 W/mK and with density of 40kg/m3 (+/-2kg) suitable for temperature range of -30 degree C to +80 degree C. The panel must be made using glue-free high pressure pressing self-bonding method. PPGL sheet on

Thickness 55mm (+/- 2mm)	Sqm.	2109.00
Thickness 65mm (+/- 2mm)	Sqm.	2292.00

17.4 Supply and Installation of WALL – PREFABRICATED INSULATED CEMENT FACE PANEL:

Factory-readymade structural reinforced insulated wall panel of thickness mentioned below. Panel shall be made of 6mm (+/-0.6mm) thick asbestos-free fiber cement sheet/ Cement bonded particle board as per IS:14276:1995 on both sides of rigid PUR insulation. The wall panel shall be vertically joined together with adjacent panel by biscuit joints lengthwise with suitable construction adhesive. Joining member (biscuit) shall be made of approx 12mm thick cement sheet /Cement bonded particle board as per IS:14276:1995. Panels may have optional provision for inbuilt electrical conduit running lengthwise at vertical center of specified panels. Both vertical lengthwise sides of panels must have full length MS tubes of min 1.0mm thickness for better structural strength.The final panel surface must be free from undulations. The Insulating core shall be self-extinguishing DIN 4102,Class B3 fire retardant class rigid PUR with thermal conductivity of 0.025 W/mK and with density of 40kg/m3 (+/-2kg) suitable for temperature range of -30 degree C to +80 degree C. The panel must be made using glue-free high pressure hot pressing self-bonding method and shall be free from major surface undulations.

Thickness 65mm (+/- 2mm)	Sqm.	2292.00
Thickness 75mm (+/- 2mm)	Sqm.	2410.00

17.5 Supply and Installation of WALL – PREFABRICATED INSULATED LAMINATED FACE PANEL:

Factory-readymade interior wall panel of thickness mentioned below. The panels shall have suitable provision for joining adjacent panels. Interior panels may have optional provision for inbuilt electrical conduit running lengthwise at vertical center of specified panels. The insulation core shall be self-extinguishing DIN 4102,Class B3 fire retardant class rigid PUR with thermal conductivity of 0.025 W/mK and with density of 40kg/m3 (+/-2kg) suitable for temperature range of -30 degree C to +80 degree C. The panel must be made using glue-free high pressure hot pressing self-bonding method and shall be free from major surface undulations. both sides of panel shall have a min15 micron protective plastic guard film to avoid scratches during transportation.

1. Panel shall be made of min 4mm (+/- 0.5mm) thick laminated wood based medium density fiber board sheet on both sides of rigid PUR insulation.

Thickness 65mm (+/- 2mm)	Sqm.	2410.00
Thickness 75mm (+/- 2mm)	Sqm.	2528.00

17.6 Supply and Installation of SMALL ROOM (TOILET & GUARD ROOM) – PREFABRICATED INSULATED PPGL FACE PANEL:

Insulated closed lip panels (for door, wall and roof) of thickness mentioned below. Panel shall have 0.50mm (+/-0.04mm) thick pre-coated metal sheet on both sides of rigid PUR insulation. The pre-coated metal sheet shall be PPGL (Pre Painted Galvalume Sheet with AZ coating) sheet having SDP (Super Durable Polyester) colour coat of 18 micron(min). The Pre coated metal sheet on both sides must be pre-strengthened by full lengthwise stiffening with semi-circular beading of 12mm wide & 3mm depth at 240 (+/-5) mm pitch. Panels shall have closed lengthwise sides made by right angle flange on both side of Pre coated metal sheets, free from cap type covers. The top and bottom part on widthwise sides of panels must have full width MS Square tube insert of (min 25mm x 1.0mm) for better screw retention and strength. The insulation core shall be self-extinguishing DIN 4102,Class B3 fire retardant class rigid PUR with thermal conductivity of 0.025 W/mK and with density of 40kg/m3 (+/-2kg) suitable for temperature range of -30 degree C to +80 degree C. The panel must be made using glue-free high pressure hot pressing self-bonding method free from major undulations. PPGL sheet on both sides of panel shall have a

	Unit	Rate
Thickness of 32mm (+/-2mm)	Sqm.	2055.00
Thickness of 50mm (+/-2mm)	Sqm.	2206.00

17.7 AccessoriesSupply and Installation of (Labour rate included in main item)

Bottom and Top U track :

U-track for mounting panels with floor. The channel shall be made from minimum 1.15 mm thick PPGL sheet with flange height of minimum 35 mm and width as per thickness of panel used. RMT 200.00

17.7.1 Flashings (inner and outer):

Corner angle flashings of equal shape for covering various corners and open sides of panels. Flashings shall be made from 0.5mm thick pre-coated PPGL sheet in following sizes:

	Unit	Rate
50mm x 50mm	RMT	150.00
100mm x 100mm		200.00

17.7.2 Roof Ridge Cap:

Ridge flashing for roof panels shall be made from 0.5mm thick pre-coated PPGL sheet of

200mm x200 mm. RMT 200.00

17.7.3 Roof Front Corrugated Cap:

Front side flashing for roof panels (corrugated trapazodial shape) shall be made from 0.5mm thick pre-coated PPGL sheet. RMT 200.00

17.7.4 Roof End Cap:

End cap for closing ends of roof panels made from 0.5mm thick PPGL sheet. RMT 150.00

17.7.5 Frame for Door/Window:

Frame shall be made out of min 1.15mm thick press steel sheet. RMT 200.00

17.7.6 Panel End Caps:

U or L shape caps suitable for thickness of panel used, with 30mm flange. It should be made frem 0.5mm Pre coated PPGL sheet. RMT 100.00

17.7.7 Self Drilling Screws:

	5”	Each	12.00
	3.5”	Each	9.00
	1”	Each	5.00
17.7.8	Adhesives	KG.	180.00
17.8	Supply and fixing of Cement Bonded Particle Board (as per IS 14276) false ceiling using 6MM thick Cement Board with galvanized and pre painted steel T section of size 24mm x 27mm x 0.4mm for main T duty pre punched to accept cross T section of size 24mm x 25mm x 0.4mm duly punched at both ends for insertion into main T. The grid size will be 610mm x 610mm. The frame work (grid) is suspended to the roof by using G.I flat of 0.60mm or 14-gauge G.I wire with necessary fixers to the roof and frame work. Cement Board panel is laded on the grid. The cost inclusive of necessary hardware, labor, one coat of primer (Both side) and two coats of smooth finish on visible side.	Sqm	594.00
17.9	Supply and fixing of Cement Bonded Particle Board (as per 14276) wall paneling using 10 MM thick Cement Board with frame work made of GI stud section of size 48mm x 35mm x 0.55mm thick placed at every 610 mm C/C intervals vertically and at 1200mm internals horizontally fixed to the wall by means of self expansion screws and caps. Cement Board panel is to be fixed to frame by means of self taping screws placed at every 300 mm intervals leaving 3mm gap between two panels. The cost inclusive of one coat of Altek primer and two coats of Altek smooth finish and cost & conveyance of all materials, labor charges etc. complete as per the direction of Engineer in charge.	Sqm	1526.00
17.10	Supply and fixing of Cement Bonded Particle Board (as per 14276) double skin partition using with 10 MM thick Cement Board made out of G.I. track section of size 50mmx 35mm x 0.55 mm for fixing to the roof and floor and stud section of size 48mm x 35mm x 0.55mm placed in track section vertically at 610mm intervals and at 1200mm intervals horizontally fixed to the wall by means of self expansion screws and caps to the wall and roof. Cement Board panel is to be fixed both sides of frame work by using self taping screws fixed at every 300mm intervals to the frame work leaving 3 mm gap between two panels. The cost inclusive with one coat of Alltek primer and two coats of Alltek smooth finish and cost and conveyance of all materials to site, labor charges etc., complete as per the direction of Engineer in charge.	Sqm	2208.00
17.11	Supply and fixing of Factory Laminated Cement Bonded Particle Board double skin partition using with 10MM Factory Laminated Cement Bonded Board made out of G.I track section of size 50 mm x 35mm x 0.55mm for fixing to the roof and floor and stud section of size 48mm x 35mm x 0.55mm placed in track section vertically at 610mm intervals and at 1200mm internals horizontally. The frame is fixed by means of self expansion screws and caps to the wall / roof. Laminated Cement Bonded Particle Board is to be fixed both sides of the frame work by using 2mm thick electro plated CR flat section leaving 3mm gap between two panels. The cost inclusive of cost and conveyance of all materials to sites, other materials incidentals and labor charges as per the direction of Engineer in charges.	Sqm	3092.00

17.12	Supply and fixing of Bath room / Toilet door shutter with 16mm Factory Laminated Cement Bonded Particle Board BISON Lam (IS 14276) door shutter with all-round 'U' lipping of PPS section of size 12mm X 18mmX 12mmX0.6mm thickness with a hardware of 12mm dia X200mm long aluminum aldrop – 2 nos2 No of 125mm long handles and also 3 no Of IS304 grade Patee Hinges 3mmX12mmX180mm long with pole receiver of 10mm dia pole X40mm long welded on 2mm X40mm(ss304) plates works as receiver for RT patte hinges. The price inclusive of all material at site.	Sqm	2188.00
17.13	<u>Solid Rectangular Block (400x200x200, 400x200x100, 400x200x150, 225x200x100)</u>		
	Providing and fixing up to floor five level of cement concrete solid blocks (M-20 Grade) of different sizes (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand), cost of required centering, shuttering complete.	Cum	9370.00
17.14	<u>Hollow Rectangular Block (400x200x200, 400x200x100)</u>		
	Providing and fixing up to floor five level of cement concrete Hollow Rectangular blocks of different sizes (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand), cost of required centering, shuttering complete.	Cum	6675.00
17.15	<u>Insulated Rectangular Block (400x200x200 mm)</u>		
	Providing and fixing up to floor five level of cement concrete insulated blocks of different sizes having a layer of polysterene as insulating material (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand) including cost of required centering, shuttering complete.	Cum	9920
17.16	<u>Interlocking Paver Blocks having 80 mm thickness in M-40 grade concrete</u>		

	Providing and Laying 80 mm thick cement concrete blocks of M-40 Grade (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds(Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) spreading 25mm thick sand underneath and filling joints with sand on existing W.B.M.base, as per IS15658:2006 and all materials shall conform to MoRTH Specification Clause 602.	Sqm.	1100.00
17.17	<u>Interlocking Paver Blocks having 100 mm thickness in M-50 grade concrete</u>		
	Providing and Laying 100 mm thick cement concrete blocks of M-50 Grade(Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) spreading 25mm thick sand underneath and filling joints with sand on existing W.B.M.base, as per IS15658:2006 and all materials shall conform to MoRTH Specification Clause 602.	Sqm.	1289.00
17.18	<u>Interlocking Paver Blocks having 120 mm thickness in M-50 grade concrete</u>		
	Providing and Laying 120 mm thick cement concrete blocks of M-50 Grade(Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy)on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) spreading 25mm thick sand underneath and filling joints with sand on existing W.B.M.base, as per IS15658:2006 and all materials shall conform to MoRTH Specification Clause 602.	Sqm	1424.00
17.19	<u>Interlocking Paver Blocks having 100 mm thickness in M-55 grade concrete</u>		

	Providing and Laying 100 mm thick cement concrete blocks of M-55 Grade (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy)on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) spreading 25mm thick sand underneath and filling joints with sand on existing W.B.M.base, as per IS15658:2006 and all materials shall conform to MoRTH Specification Clause 602.	Sqm	1389.00
17.20	<u>Interlocking Paver Blocks having 120 mm thickness in M-55 grade concrete</u>		
	Providing and Laying 120 mm thick cement concrete blocks of M-55 Grade (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) spreading 25mm thick sand underneath and filling joints with sand on existing W.B.M.base, as per IS15658:2006 and all materials shall conform to MoRTH Specification Clause 602.	Sqm.	1489.00
17.21	<u>Kerb Stones of 1200x300x150, 1000x450x150, 1000x600x150, 600x300x150 size (M-30)</u>		
	Providing and Laying cement concrete Kerb Stones of different sizes in M-30 grade (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) in position to the required line, level and curvature jointed with cement mortar 1:3 (1 Cement : 3 Coarse sand) including making of joints, drainage opening wherever required.	Cum	8572.00
17.22	<u>Kerb Stones of 1200x300x150, 1000x400x150, 1000x600x150, 600x300x150 size (M-40)</u>		

Providing and Laying cement concrete Kerb Stones of different sizes in M-40 grade (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) in position to the required line, level and curvature jointed with cement mortar 1:3 (1 Cement : 3 Coarse sand) including making of joints, drainage opening wherever required. Cum 9067.00

17.23 Cellular Concrete Block for Pitching and garden paths (600x400x80) in M-30

Providing and fixing Cellular concrete blocks (M-30 Grade), 600x400x80 mm sizes (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand), cost of required centering, shuttering complete. Sqm. 739.00

17.24 Pre Cast Concrete Flooring (Slab 600x600x100, 1000x650x100)

Providing and fixing Precast concrete flooring slabs (M-30 Grade) of different sizes(Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand), cost of required centering, shuttering complete and spreading 25mm thick sand underneath. Sqm. 894.00

17.25 Pre Cast U/Semi circular Drains (Outer size 600x400 mm, wall thickness 80 mm)

Providing and fixing Precast concreteUdrains (M-30 Grade) of different sizes (Manufactured with fully mechanised dry cast process with forced action mixer whose arms revolve along the main axis and rotate about its own axis also, having humidity sensor for moisture control of w/c ratio in concrete mix. Manufactured with Steel moulds (Milled and manufactured in CSI diamond or CSI Nitro process for accuracy) on steel base plate of min 16 mm thick; Having Vibration and pressing action both together with the help of mutiplesynchronised vibrators. Power of hydraulic station should be minimum 50KW for compaction and cured in controlled chambers having humidity control and automated air circulation system) including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 Coarse Sand), cost of required centering, shuttering complete and spreading 25mm thick sand underneath. R.M. 417.00

17.26	**Dry Cleaning of sofa sets using shampooing machine with all materials & drying it do the level using machinery at designated place by engineer-in-charge one seat of sofa (payment for multi seater sofa will be done as the number of seats of that sofa.)**		
17.26.1	**Single Seater Sofa**	**Each**	**140.00**
17.26.2	**Two Seater Sofa**	**Each**	**211.00**
17.26.3	**Three Seater Sofa**	**Each**	**302.00**
17.27	**Dry Cleaning of Woolen Carpet of any size using shampooing machine with all material & drying it using machinery at the designated place as told by engineer in charge.**	**Sqm**	**93.00**
17.28	**Complete replacement of Tapestry cloth of Sofa set costing @ Rs. 300/- per Mtr. Approx of superior quality including stitching charges (As approved by the Engineer in Charge) as per existing pattern/design.**		
17.28.1	**Single seater sofa**	**Each**	**1317.00**
17.28.2	**Three seater sofa**	**Each**	**3583.00**
17.29	**Complete replacement of damaged foam of Sofa set and using 100mm thick Joint less P.U. foam in seat of 32 density feather foam of superior quality of approved make (as approved by the Engineer-in-charge) pasted with Adhesive (Rubber Solution) of superior quality.**		
17.29.1	**Single seater sofa**	**Each**	**685.00**
17.29.2	**Three seater sofa**	**Each**	**1921.00**
17.30	**Complete replacement of damaged foam of Sofa set and using 50mm thick joint less P.U. foam of 32 density feather foam superior quality of approved make (as approved by the Engineer-in-charge) in back duly pasted with Adhesive (Rubber Solution) of superior quality.**		
17.30.1	**Single seater sofa**	**Each**	**412.00**
17.30.2	**Three seater sofa**	**Each**	**1172.00**
17.31	**Complete replacement of damaged foam of Sofa set and using 25mm thick joint less P.U. foam of 32 density feather foam superior quality of approved make (as approved by the Engineer-in-charge) in Hand rest, in seat and back and design work as required of superior quality.**		
17.31.1	**Single seater sofa**	**Each**	**779.00**
17.31.2	**Three seater sofa**	**Each**	**1652.00**
17.32	**Repair of wooden frame of Sofa set using Tat, Nail, Screw, Bolt (if required) and re-assembling sofa set (Cost Including all Consumble Item) complete in all respect as per direction of the Engineer-in-charge**		
17.32.1	**Single seater sofa**	**Each**	**625.00**
17.32.2	**Three seater sofa**	**Each**	**1903.00**
17.33	**Repair using replacement of Jata, Dori, wooden batten & legs everything in all respect as per approval of Engineer-in-charge.**		
17.33.1	**Single seater sofa**	**Each**	**456.00**
17.33.2	**Three seater sofa**	**Each**	**686.00**

17.34	**Providing and Fixing machine made Synthetic Carpet having total Carpet weight 850 gm Psqm. and 5mm thickness superior quality complete in all respect as per direction of Engineer-in-charge.**	**Each**	**440.00**
17.35	**Providing and Fixing Chiken wire mesh at the joint of masonry members/ R.C.C. Members with Nails complete in all respect.**	**Sqm**	**70.00**

CHAPTER : S-1
SANITARY INSTALLATION WORK

NOTE: All items which are manufactured ISI marked, shall only be used. this condition shall be always incorporated in special condition of contract.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	WATER CLOSETS		
1.1	P. & F Squatting Pan (Indian type) white glazed vitreous china 1st quality W.C. Pan (IS : 2556 Mark) with 100 mm vitreous china P or S trap including cutting and making good the wall and floor (excluding the pair of foot rest.)		
1.1.1	Size 450mm.	Each	891.00
1.1.2	Size 510 mm	Each	972.00
1.1.3	Size 580mm	Each	1012.00
1.2	P & F *Indian type* white glazed vitreous china 1st quality W.C. orissa pan (IS :2556 Mark) with 100 mm vitreous china P or S trap including cutting and making good the wall and floor:		
1.2.1	Size 530x410mm.	Each	2400.00
1.2.2	Size 580x440mm.	Each	2800.00
1.3	P & F European type white glazed vitreous china 1st quality W.C pan (IS : 2556 Mark) with P or S trap including cutting and making good the wall and floor	Each	2200.00
1.4	P & F European type white glazed vitreous china 1st quality Double syphonic W.C (IS :2556 Mark) with P or S trap including cutting and making good the wall and floor	Each	4725.00
1.5	P & F Universal (anglo-Indian type) white glazed vitreous china W.C. pan 1st quality (IS 2556 Marked) with P or Strap including cutting and making good the wall and floor	Each	3698.00
1.6	P & F *European type* white glazed vitreous china 1st quality W.C. (IS : 2556 Mark) including cutting and making good the wall and floor		
1.6.1	Wall Mounting	Each	3143.00
1.6.2	Wall Mounting (extended)	Each	3526.00

1.6.3	Wall Mounting (syphonic)	Each	4037.00

Note :-

(A)	Add extra over item No. 1.1 to 1.7, 1.14, 1.15, 1.19, 1.22 ,1.23,1.38 & 1.42 for using approved light coloured vitreous china ware i.e.ivory champagne , perl , soft green , soft blue ,pink, ,light grey etc. instead of white vitreous chinawares.	Each	50%
(B)	Add extra over item No.1.1 to 1.7, 1.14, 1.15, 1.19, 1.22 ,1.23,1.38 & 1.42 for using approved dark colored vitreous china ware i.e.Cherry Red , Dark blue, Jet black, Magenta, Burgandy instead of white vitreous chinawares.	Each	65%
1.7	P & F water closet Seat Covers with brass hinges complete :		
1.7.1	Solid PVC (IS 2548 marked) grade-I Black for EWC.	Each	**365.00**
1.7.2	-do- White.	Each	**441.00**
1.7.3	-do- Coloured.	Each	**473.00**
1.7.4	-do- White.	Each	**320.00**
1.7.5	-do- Coloured.	Each	**365.00**
1.8	P & F C.P. lugs for toilet seat cover.	Pair	72.00
1.9	Labour charges for removing W.C. pan (any type) of all sizes with care including all necessary fittings P or S trap.	Each	**260.00**
1.10	P & F White glazed Conversion Bend for W.C. from P trap to S trap. .	Each	133.00
1.11	P & F vitreous china P trap of approved quality	Each	195.00
1.12	Add extra if C.I.P trap is used in place of Vitreous china P trap for W.C.	Each	**542.00**
1.13	P & F White Vitreous China *Double Syphonic European W.C.* (I.S:2556Mark) with mounted W.V.C. flushing cistern of (IS : 2556 Mark) of 10 litre capacity complete with all necessary internal fittings including cutting and making good the wall and floor.	Each	**11013.00**

URINALS:

1.14	P & F Ist quality *WVC* Urinal (IS:2556 mark) with 25mm dia G.I. waste pipe, dome waste couplings, concealed iron brackets or screws etc complete.		
1.14.1	Flat Back (small) size 456x355x265mm.	Each	**1665.00**
1.14.2	Flat Back (large) or half stall size 610x400x80mm.	Each	**4215.00**
1.14.3	Mini stall size 455x385x350mm.	Each	**2079.00**
1.15	P & F W.V.C. *Urinal Partition Plate* of approved standard make of size 630x315mm including making good the wall.	Each	904.00
1.16	P&F Marble *Urinal Partition slab* both sides polished of approved design cut from marble size 900x600mm.		
1.16.1	Makrana II quality marble having light spots	Each	839.00
1.16.2	Makrana \`Albeta\` Marble having light streaks.	Each	784.00
1.16.3	Makrana \`Adanga\` or Raj Nagar 1St Quality White or Andhi Indo-Italian marble.	Each	728.00
1.17	P & F W.V.C. *Half-Round Channel* of approved make in C.M. 1:3, joints finished in white cement of:		
1.17.1	Size 100mm dia.	Mtr.	**541.00**
1.17.2	Size 150mm dia.	Mtr.	**688.00**
1.18	P & F W.V.C. (IS: 2556 Mark) *Squatting Urinal* of standard make complete. Including cutting & making good the floor:		
1.18.1	Size 450x350mm	Each	775.00
1.19	Labour charges for removing and refixing of urinals (all type & sizes) including all necessary fittings.	Each	308.00
1.20	P & F *Low level Flushing Cistern* of 10 litres capacity (IS : 2556mark). Of approved make with complete fittings C.I. brackets duly painted, brass ball cock with ball, (IS: 1703 mark) complete including cutting and making good the wall:		
1.20.1	WVC with C.P. brass bend.	Each	**2292.00**
1.20.2	PVC with PVC bend as per IS : 7231 .	Each	**1174.00**
1.20.3	PVC with PVC bend **and superior internal fittings**	Each	**1495.00**
1.20.4	WVC for symphonic EWC.	Each	**2363.00**
1.20.5	PVC with CP brass long band.	Each	**1355.00**
1.21	P & F *Automatic Flushing Cistern* of first quality (IS : 774 mark) of approved make fitting, siphon with copper tube, C.I. brackets complete including cutting and making good the wall.		

	WVC (as per IS : 2556) 10 litres.	Each	988.00
Note :-	Deduct if plastic syphon is used.	Each	90.00
1.22	P & F *Flush Pipe* of approved make with complete fittings for high level flushing cistern:		
1.22.1	G.I. \`B\` class (Concealed) 25mm.	Each	399.00
1.22.2	G.I. \`B\` class (Concealed) 32 mm.	Each	466.00
1.22.3	PVC exposed (heavy) 32mm.	Each	153.00
1.22.4	GI (Sheet) hot galvanized exposed (10 litres).	Each	231.00
1.23	P & F WVC (10 litres) low-level flushing cistern with cover.	Each	753.00
1.24	P & F WVC (10 litres) low-level flushing cistern cover only	Each	278.00
1.25	P & F WVC syphon with plunger plate only.	Each	222.00
1.26	P & F WVC Auto cistern with cover only.	Each	420.00
1.27	P & F WVC Auto cistern cover only.	Each	132.00
1.28	P & F C.P. brass *Urinal Flush Pipe* set complete with flanged brackets of approved make.		
1.28.1	For 4 Urinals	Each	1210.00
1.28.2	For 3 Urinals	Each	971.00
1.28.3	For 2 Urinals	Each	669.00
1.28.4	For 1 Urinal	Each	458.00
1.29.1	P & F C.P. brass *Urinal Spreader* for stall urinal of approved make.	Each	**330.00**
1.29.2	P & F Synthetic (PTMT) Urinal Spreader for stall urinal of approved make.	Each	**224.00**
1.30.1	P & F C.P. brass *Doom Waste Coupling* 32 mm dia.	Each	138.00
1.30.2	P & F C.P. brass *Doom Waste Coupling* 40 mm dia.	Each	187.00
1.31	P & F *Plastic Syphon* of approved make for automatic flush tank.	Each	143.00
1.32	P & F Copper tube Syphon for auto flush tank.	Each	341.00
1.33	Removing and re-fixing of wash hand basin including all necessary fittings.	Each	217.00

1.34	P & F metal brackets duly painted including cutting & making good the wall.	Pair	150.00
1.35	P & F *C.P. Brass flush Valve* with C.P. flush bend of approved make.		
1.35.1	Lever type with Elbow 32 mm dia	Each	**2792.40**
1.35.2	Push type with Elbow 32 mm dia	Each	**2917.00**
1.35.3	Concealed type for 115 mm thick wall	Each	**3340.00**
1.35.4	Concealed type for 230 mm thick wall	Each	**3448.00**
1.35.5	Economy model 25 mm dia.	Each	**1996.00**
1.35.6	Economy model 32 mm dia.	Each	**2185.00**
	WASH BASINS:		
1.36	P & F WVC Wash basin (Ist quality IS:2556 Mark) of approved make with C.I. brackets duly painted 1 No. 15 mm C.P. Pillar cock (IS:8934 Mark) & 32 mm C.P. brass waste coupling of approved make, **P.V.C Waste pipe with PVC nut 32 mm** complete including cutting & making good the wall :		
1.36.1	Size 450 mm x 300 mm	Each	**2062.00**
1.36.2	Size 510 mm x 400 mm	Each	**2189.00**
1.36.3	Size 550 mm x 400 mm	Each	**2362.00**
1.36.4	Size 580 mm x 450 mm	Each	**2495.00**
1.36.5	Size 610 mm x 510 mm	Each	**3161.00**
1.36.6	Size 400 mm x 400 mm or 450 mm x 390 mm Corner	Each	**2029.00**
1.36.7	460 dia for counter top	Each	**2688.00**
1.36.8	Size 550 mm x 400 mm dia for counter top	Each	**2722.00**
1.36.9	Size 630 mm x 500 mm dia for counter top	Each	**2968.00**
1.36.10	**Add extra for using 25mm GI waste pipe instead of PVC waste pipe**	**Each**	
1.37	P & F Terrazzo (Mosaic) wash hand basin with supporting brackets waste coupling and PVC waste pipe complete, including painting of brackets making good the walls:		
1.37.1	450 mm x 280 mm	Each	962.00
1.37.2	550 mm x 400 mm	Each	1038.00
	KITCHEN & LAB. SINKS:		
1.38	P & *F Kitchen & Lab. Sink* of approved make with C.I. brackets duly painted, 40 mm C.P. waste coupling, C.P. Brass chain with rubber plug, 40 mm G.I. waste pipe up-to floor level complete including cutting and making good the wall & floor :		
1.38.1	Fire clay glazed kitchen sink (I.S 771-1985 Marked) size 600 x 450 x 250mm.	Each	**3536.00**

1.38.2	Fire clay glazed kitchen sink (I.S 771-1985 Marked) size 600 x 450 x 200 mm.	Each	**3403.00**
1.38.3	WVC Lab sink (IS:2556 mark) size 530 x 430 x 180 mm	Each	**2504.00**
1.38.4	WVC Lab sink (IS:2556 mark) size 500 x 350 x 150 mm	Each	**2071.00**
1.38.5	WVC Lab sink (IS:2556 mark) size 450 x 300 x 150 mm	Each	**1872.00**
1.38.6	Mosaic fine polished kitchen sink size 600 x 450 x 200mm.	Each	**1472.00**
1.38.7	Mosaic fine polished kitchen sink size 450 x 450 x 200mm.	Each	**1300.00**
1.38.8	Mosaic fine polished kitchen sink size 600 x 450 x 200 mm with 600 x 450mm drain board	Each	**1672.00**
1.38.9	1.0 mm thick stainless steel AISI -304 & IS 13983-1994 kitchen sink of approved make as per Engineer-in-charge with large waste coupling.		
	Overall size (inches) / <u>Bowl size</u> (inches)		
1.38.9.1	18 x 16 x 6 / 16x14x6	Each	**3470.00**
1.38.9.2	22 x 18 x 7 / 20x16x7	Each	**3936.00**
1.38.9.3	24 x 18 x 8 / 20x16x8	Each	**4602.00**
1.38.9.4	37 x 18 x 7 / 17x15x7 (with drain board)	Each	**5069.00**
1.38.9.5	45 x 20 x 8 / 20x16x8 (with drain board)	Each	**7932.00**
Note :-	Deduct for item No 1.40.9 for using any other make than specified above.	Each	35%
1.39	P & F *Drain Board* of approved make with C.I. brackets duly painted compolete including cutting & making good the wall:		
1.39.1	WVC (I.S.: 2556 Mark) size 530 x 450mm	Each	1091.00
1.39.2	Mosaic fine polished size 600 x 450	Each	370.00
1.40	P & F 1st quality WVC pedestal for wash basin.	Each	1313.00
	MISCELLANEOUS		
1.41	P & F Bottle Trap of approved make :		
1.41.1	C.P Brass 32 mm	Each	**531.00**
1.41.2	C.P Brass 40 mm	Each	**575.00**
1.41.3	Synthetic Material (PTMT)32 mm	Each	**465.00**
1.41.4	Synthetic Material (PTMT)42 mm	Each	**520.00**
1.41.5	P.V.C	Each	**118.00**
1.42	P& F waste Coulping with fittings of approved quality /make:		
1.42.1	C.P Brass 32 mm dia	Each	**130.00**
1.42.2	C.P Brass 40 mm dia	Each	**198.00**
1.42.3	C.P. Brass 50 mm dia	Each	**238.00**

1.42.4	SS 125 mm	Each	**277.00**
1.43	P& F waste Pipe with all fitting		
1.43.1	P.V.C with C.P nut 32 mm	Each	132.00
1.43.2	P.V.C with PVC nut 32 mm	Each	55.00
1.43.3	P.V.C. with C.P. nut 40 mm dia	Each	143.00
1.43.4	G.I(B class) 32mm	Each	204.00
1.43.5	G.I(B class) 40 mm	Each	270.00
1.44	P & F *Bevelled edge Mirror/mirror with teak wood lipping around* of special glass of approved make as per direction of Engineer-in-charge complete with 6mm thick commercial ply base fixed to wooden screws & washers.		
1.44.1	Size 600 x 450mm x 4 mm thick	Each	**523.00**
1.44.2	Other sizes	Sqm.	**1398.00**
Note :-	For other quality mirrors from approved make mirrors deduct 30%		
1.45	P & F *Looking Mirrors* with P.V.C. frame of approved make as per direction of Engineer-in-charge		
1.45.1	Size 500x400mm	Each	**319.00**
1.45.2	Oval shape 450x350 mm	Each	**467.00**
1.45.3	Round 500mm dia	Each	**499.00**
Note :-	For other quality mirrors from approved make mirrors deduct 30%		
1.46	P & F Toilet Shelf of approved quality/make:		
1.46.1	W.V.C. (I.S.: 2556 Mark) Size 300mm	Each	**400.00**
1.46.2	W.V.C. (I.S.: 2556 Mark) Size 550 mm	Each	**473.00**
1.46.3	Synthetic Material with tumbler of approved quality/make size 470 x 114 x 36mm	Each	**392.00**
1.46.4	Hard Plastic of approved make as per direction of EI of size 590x140mm	Each	**473.00**
1.46.5	Acrylic shelf on C.P. brass casted brackets & guard rail of approved make as per direction of EI of size 550x125mm	Each	**906.00**
1.46.6	Glass Shelf with edges rounded off Anodised Aluminium angle frame & C.P. brass brackets & guard rail size 600x120mm	Each	**328.00**
1.47	P & F *Towel Rail or Ring* of approved quality/make:		
1.47.1	C.P. brass Towel Rail elbow type with concealed screws size 450mm (Heavy duty).	Each	**425.00**

1.47.2	C.P. brass Towel Rail elbow type with concealed screws size 600 mm (Heavy duty).	Each	**469.00**
1.47.3	C.P. brass towel rail with brackets 450 x 20mm.	Each	**269.00**
1.47.4	C.P. brass towel rail with brackets 600 x 20mm.	Each	**306.00**
1.47.5	Synthetic Material(PTMT) of approved quality/make size 525 x90 x 75mm	Each	**138.00**
1.47.6	Synthetic Material(PTMT) of approved quality/make size 675 x90 x 75mm	Each	**150.00**
1.47.7	WVC (1st quality, IS2556 Marked) of approved make. Recessed towel Hanger size 108 x 108mm.	Each	**131.00**
1.47.8	C.P. Brass Towel Ring revolving type	Each	**231.00**
1.47.9	Synthetic Material (PTMT) Towel Ring of approved quality/make.	Each	**150.00**
1.48	P & F Grating of approved quality/make:		
1.48.1	Stainless Steel Sheet size 125mm dia.	Each	**56.00**
1.48.2	Stainless Steel Sheet size 125mm dia. Heavy Quality of approved make	Each	**88.00**
1.48.3	Synthetic Material(PTMT) of approved quality/make size 125mm dia.	Each	**81.00**
1.48.4	(-do-) 150mm dia.	Each	**88.00**
1.48.5	(-do-) Square 150 x 150 x 8mm. (Anti Cockroach)	Each	**325.00**
1.48.6	C.P. brass with frame (Heavy) & superior quality size 125mm dia.	Each	**119.00**
1.48.7	(-do-) Light & superior quality size 125mm dia	Each	**50.00**
1.48.8	Cast Iron.	Each	**63.00**
1.49	P & F *Liquid Soap Container* with brackets complete of approved make:		
1.49.1	C.P. brass	Each	**216.00**
1.49.2	Synthetic Material of approved quality/make:	Each	**149.00**
1.50	P & F *Tooth Paste & Brush Holder* of approved quality/make:		
1.50.1	C.P. brass	Each	**115.00**
1.50.2	C.P. brass (Heavy and Superior quality)	Each	**473.00**
1.50.3	Stainless Steel	Each	**290.00**
1.51	P & F *Toilet Paper Holder* with rod of approved quality/make:		
1.51.1	C.P. brass	Each	**203.00**
1.51.2	C.P. brass heavy & Superior quality.	Each	**547.00**
1.51.3	WVC (1st quality, IS 2556 Marked) recessed size 150x150mm.	Each	**277.00**
1.51.4	(-do-) exposed (-do-)	Each	**216.00**

1.52	P & F *Soap Dish or Tray* of approved quality/make		
1.52.1	C.P. brass	Each	**115.00**
1.52.2	C.P. brass heavy and superior quality.	Each	**142.00**
1.52.3	WVC (1st Quality, IS 2556 Marked) recessed size 150 x 150 mm.	Each	**257.00**
1.52.4	(-do-) 200 x100mm.	Each	**297.00**
1.52.5	(-do-) Soap tray 170mm.	Each	**223.00**
1.52.6	(-do-) Soap Dish 380mm.	Each	**297.00**
1.53	P & F Twin- peg of approved quality / make		
1.53.1	C.P. brass	Each	**72.00**
1.53.2	C.P. brass (heavy)Superior quality	Each	**135.00**
1.53.3	Acrylic	Each	**69.00**
1.54	P & F Shower arm 15mm dia of approved quality/make.		
1.54.1	C.P.brass heavy & superior quality 150mm Projection.	Each	**180.00**
1.54.2	(-do-) size 220mm(-do-)	Each	**279.00**
1.54.3	(-do-) size 300mm.(-do-)	Each	**306.00**
1.54.4	(-do-) size 370mm.(-do-)	Each	**342.00**
1.55	P & F Bath Shower of approved quality/make.		
1.55.1	C.P. brass fixed.	Each	**135.00**
1.55.2	C.P. brass of Heavy & superior quality 150mm.	Each	**342.00**
1.55.3	C.P. brass of Heavy & superior quality, revolving with adjustable key 150mm.	Each	**521.00**
1.56	P & F Extension Pipe for bib cocks of approved quality/make.		
1.56.1	C.P. brass of Heavy & Superior quality 50x15mm	Each	**77.00**
1.56.2	(-do-) 100 X 15mm.	Each	**108.00**
1.56.3	(-do-) 150 X 15mm.	Each	**153.00**
1.57	P & F *Acrylic Toilet accessories set* consisting of Towel rail, shelf, towel ring, soap dish, paste & brush holder & twin peg of approved quality/make.	Each	**1072.00**
1.58	Labour Charges for removing WVC pan:		
1.58.1	Indian/Orissa pan	Each	**206.00**
1.58.2	European W.C.	Each	**206.00**
1.58.3	Flushing cistern	Each	**96.00**
1.59	P & F *Jet spray* for water closet with C.P. Copper Tube flange of approved make.	Each	**346.00**

1.60	Providing & fixing C.P.flange for 15mm dia taps.	Each	**16.00**
1.61	Providing and fixing 1st quality standard white, grey, ivory, fume red brown, light green, light blue and other light shades glazed tiles confirming to IS : 13753 & IS :15622 of size 200mm x 300mm in walls, floors, steps, pillars etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (excluding the cost of cement plaster on walls and pillar).	Sqm.	613.00
1.62	Providing and fixing 1st quality MAT finished ceremic tile size 300x300mm confirming to IS : 13755 and IS : 15622 colour such as white, grey, ivory, fume red brown, light green, light blue and other light shades in floors, steps, pillars etc. laid on a bed of neat cement slurry finished with flush pointing in the white cement mixed with pigment to match the shade of the tile complete (including the cost of cement mortar bed 1:4).	Sqm.	680.00
1.63	Extra for using Marble printed / Granite shade or dark shade tiles instead of white, grey, ivory, fume red brown, light green, light blue and other light shades in glazed tiles and MAT finished tiles.	Sqm	10%
1.64	Extra for using size above 200x300mm in glazed tiles and for using 400mm x 400mm MAT finished tiles in place of 300mm x 300mm size.	Sqm	10%
1.65	P & F C.P. Health Faucet with 1Mtr. Long Tube & Hook of approved make and heavy as per direction of Enginner-in-Charge	Each	550.00
1.66	P & F C.P. brass Urinal Spreader for Large Urinal of heavy duty of approved make as per direction of Enginner-in-Charge	Each	561.00
1.67	Removing & Re fixing of P.V.C. Cistern fittings (Syphon, Ball cock & Handle Set) complete as per direction of Enginner-in-Charge	Each	**425.00**

1.68	**Providing & Fixing Premium Quality white glazed vitreous china Sanitary Wares (IS:2556 Mark) of approved make as per direction of Engineer-in-charge including cutting and making good the wall and floor.**		
1.68.1	**W.C. orissa pan with 100 mm vitreous china P or S trap (Size 580x440 mm.)**	**Each**	**2786.00**
1.68.2	**Universal (anglo-Indian type) with P or S trap**	**Each**	**5805.00**
1.68.3	**Wall Mounting European type W.C.**	**Each**	**8247.00**
1.68.4	**Double Syphonic European W.C. with mounted W.V.C. flushing cistern of 10 litre capacity complete with all necessary internal fittings with Hydraulic seat cover.**	**Each**	**16217.00**
1.69	**Providing & Fixing premium qulaity WVC Urinal with PVC waste pipe, dome couplings, concealed iron brackets or screws etc. complete.**		
1.69.1	**Flat Back (small) size 440x265x315 mm.**	**Each**	**2381.00**
1..69.2	**Flat Back (large) or half stall size 590x375x390 mm.**	**Each**	**5678.00**
1.70	**Providing & Fixing premium qulaity W.V.C. Urinal Partition plate of size 630x315 mm**	**Each**	**2396.00**
1.71	**Providing & Fixing premium qulaity WVC Wash basin with C.I. brackets duly painted & 32 mm C.P. brass waste coupling of approved make with PVC waste pipe complete**		
1.71.1	**Size 550 mm x 400 mm**	**Each**	**2122.00**
1.71.2	**Size 550 mm x 400 mm dia for counter top.**	**Each**	**3065.00**
1.72	**P & F Premiumquality water closet Seat Covers with brass hinges complete Solid PVC (ISI marked) grade-I White for EWC.**	**Each**	**649.00**
1.73	**P & F Premium quality Low level Flushing Cistern of 10 litres capacity (IS:2556 mark) of approved make as per Engineer-in-charge with complete fittings C.I. brackets duly painted, brass ball cock with ball, (IS:1703 mark) complete including cutting and making good the wall :**	**Each**	**1657.00**

1.74	P&F Looking Mirror 5mm thick of approved make with ornamental frame size 50x12 with 6mm Ply Board on Back Side Complete as per approved sample	Sqm	2313.00
1.75	Supply of Hydraulic Seat Cover for Europeon WC of superior quality approved make in white Colour	Each	3688.00
1.76	Providing & Fixing chair for wall hung W.C.	Each	1458.00
1.77	Providing & Fixing Concealed cistern with Floor mounting frame and flush plate of superior quality and approved sample by Engineer-in-charge.	Each	7560.00
1.78	Providing and fixing new spindle for Bib cock/pillar cock for repairing purpose to solve leakage and proper functioning of tap.	Each	104.00
1.79	Providing & Fixing premium quality Toilet corner Glass Shelf size 225x225mm with C.P. brass brackets & guard rail edges rounded of SS frame	Each	1420.00
1.80	Providing & Fixing premium quality waste coupling 32mm half thread as per approved by Engineer in Charge.	Each	472.00
1.81	Providing & Fixing Premium quality Bottle trap internal portion 32mm, size 250mm long wall connection pipe & wall flange as per approved by Engineer in Charge.	Each	1454.00

CHAPTER : S-2
WATER SUPPLY INSTALLATION

Chapter Code No.	Description	Unit	Final BSR Rate 2019
	2	3	4
2.1	P & F *G.I. pipes (Internal Work)* with G.I. Fittings excluding union (IS:1239 Mark) & MS clamps including cutting and making good the walls and floors:		
(a)	Exposed on wall		
2.1.1	15 mm dia nominal bore		
	'A' Class	Mtr.	**184.00**
	'B' Class	Mtr.	**209.00**
2.1.2	20mm dia nominal bore		
	'A' Class	Mtr.	**240.00**
	'B' Class	Mtr.	**256.00**
2.1.3	25mm dia nominal bore		
	'A' Class	Mtr.	**296.00**
	'B' Class	Mtr.	**327.00**
2.1.4	32mm dia nominal bore		
	'A' Class	Mtr.	**325.00**
	'B' Class	Mtr.	**396.00**
2.1.5	40mm dia nominal bore		
	'A' Class	Mtr.	**445.00**
	'B' Class	Mtr.	**483.00**
2.1.6	50mm dia nominal bore		
	'A' Class	Mtr.	**551.00**
	'B' Class	Mtr.	**612.00**
2.1.7	65mm dia nominal bore		
	'A' Class	Mtr.	**667.00**
	'B' Class	Mtr.	**721.00**
2.1.8	80mm dia nominal bore		
	'A' Class	Mtr.	**762.00**
	'B' Class	Mtr.	**875.00**

(b)	Concealed pipe in wall including painting with anticorrosive bitumen paint		
2.1.9	15 mm dia nominal bore		
	'A' Class	Mtr.	**197.00**
	'B' Class	Mtr.	**220.00**
2.1.10	20mm dia nominal bore		
	'A' Class	Mtr	**237.00**
	'B' Class	Mtr	**253.00**
2.1.11	**25mm dia nominal bore**		
2.1.11.1	**A' Class**	**Mtr**	**311.00**
2.1.11.2	**B' Class**	**Mtr**	**390.00**
2.2	P & F *G.I. Pipes (External Work)* with G.I. fittings excluding union (IS : 1239 Mark) including trenching & refilling earth etc.		
2.2.1	15mm dia nominal bore		
	'A' Class	Mtr.	**148.00**
	'B' Class	Mtr.	**173.00**
2.2.2	20mm dia nominal bore		
	'A' Class	Mtr.	**187.00**
	'B' Class	Mtr.	**202.00**
2.2.3	25mm dia nominal bore		
	'A' Class	Mtr.	**223.00**
	'B' Class	Mtr.	**253.00**
2.2.4	32mm dia nominal bore		
	'A' Class	Mtr.	**275.00**
	'B' Class	Mtr.	**311.00**
2.2.5	40mm dia nominal bore		
	'A' Class	Mtr.	**329.00**
	'B' Class	Mtr.	**359.00**
2.2.6	50mm dia nominal bore		
	'A' Class	Mtr.	**396.00**
	'B' Class	Mtr.	**446.00**
2.2.7	65mm dia nominal bore		

	'A' Class	Mtr.	**524.00**
	'B' Class	Mtr.	**565.00**
2.2.8	80mm dia nominal bore		
	'A' Class	Mtr.	**605.00**
	'B' Class	Mtr.	**702.00**
2.2.9	100mm dia nominal bore		
	'A' Class	Mtr.	**812.00**
	'B' Class	Mtr.	**967.00**
2.2.10	125mm dia nominal bore		
	'A' Class	Mtr.	**1179.00**
	'B' Class	Mtr.	**1323.00**
2.3	Making connection of G.I. distribution branch with G.I. main including all fitting.		
2.3.1	Upto 25mm dia	Each	**241.00**
2.3.2	Beyond 25mm & upto 100mm dia	Each	**354.00**
2.4	Fixing water meter (IS: 779 mark) and stop cock in G.I. pipe line including making chamber of approved size excluding cost of water meter & stop cock.	Each	**280.00**
2.5	P & F water meter (IS : 779 mark) of approved make.		
2.5.1	15mm dia	Each	**863.00**
2.5.2	20mm dia	Each	**1576.00**
2.5.3	25 mm dia	Each	**2162.00**
	COCKS, MIXERS, DIVERTORS & VALVES		
Note:			
(a)	The rates are for ISI marked item.		
(b)	If the items are Non- ISI marked, reduce the rate by 25%.		
2.6	P & F Pilllar Cocks (IS :8934 Mark) of superior quality and approved make:		
2.6.1	C.P. Pillar cock, 15 mm dia nominal bore	Each	**708.00**
2.6.2	C.P. Pillar cock Elbow action, 15 mm nominal bore.	Each	**745.00**
2.6.3	C.P. Pillar cock fancy type ,15mm nominal bore.	Each	**831.00**

2.6.4	C.P. Pillar cock, Swan-neck with casted spout, 15mm nominal bore.	Each	**1133.00**
2.6.5	C.P. Pillar cock, high-neck, 15 mm nominal bore.	Each	**708.00**
2.6.6	Synthetic Material (PTMT) Pillar cock of approved quality / make 15mm nominal bore.	Each	**353.00**
2.7	P & F Bib Cock (IS : 8931 Mark), Superior quality of approved make:		
2.7.1	Brass 400 **gm**,15mm nominal bore.	Each	**271.00**
2.7.2	P.V.C.heavy duty, 15mm nominal bore.	Each	**96.00**
2.7.3	C.P. Brass bib cock,15mm nominal bore.	Each	**536.00**
	C. P. Brass swan - neck, 15mm nominal bore.	Each	**Deleted**
2.7.4	(-do-) Long body, 15mm nominal bore weight not less than 690 gm	Each	**712.00**
2.7.5	(-do-) Long nose, 15mm nominal bore weight not less than 810 gm	Each	**916.00**
2.7.6	(-do-) Elbow-action, 15mm nominal bore.	Each	**866.00**
2.7.7	Synthetic Material(PTMT) bib cock of approved quality/ make, nominal bore, 15mm, 86 mm long.	Each	**170.00**
2.7.8	(-do-) Fancy Type, 122 mm long.	Each	**211.00**
2.7.9	(-do-) Long Body, 175 mm long	Each	**268.00**
2.8	P & F *Stop Cock* (IS :8931 Mark), superior quality & of approved make:		
2.8.1	Brass 400 gm. 15mm nominal bore.	Each	**290.00**
2.8.2	(-do-) 750 gm. 20 mm nominal bore.	Each	**418.00**
2.8.3	(-do-) 1350gm. 25 mm nominal bore.	Each	**663.00**
2.8.4	C.P. Brass 15mm nominal bore.	Each	**434.00**
2.8.5	(-do-) Concealed Stop Cock 15mm.	Each	**917.00**
2.8.6	Synthetic material (PTMT) stop cock of approved quality/make 15mm.	Each	**210.00**
2.8.7	(-do-) Concealed stop cock 15mm	Each	**235.00**
2.9	P & F Sink Cocks of superior quality & approved made :		
2.9.1	C.P. Sink cock swinging spout 15mm.	Each	**888.00**
2.9.2	(-do-) Casted Spout 15 mm.	Each	**888.00**
2.10	P & F *Push Cock* superior quality, of approved make		
2..10.1	C.P. Brass 15 mm.	Each	**336.00**

2.10.2	Synthetic Material (PTMT) push cock of approved quality/make 15mm.	Each	**186.00**
2.11	P & F *Ball Cock* (IS :1703 Mark) with Rod & P.V.C. Ball complete :		
2.11.1	Brass wt. **Not less than 300 gm**,15mm.	Each	**285.00**
2.11.2	Brass wt.**Not less than 400 gm**, 20mm.	Each	**338.00**
2.11.3	(-do-) wt**.Not less than 600 gm**, 25mm.	Each	**400.00**
2.11.4	(-do-) wt.**Not less than 800 gm**, 32mm.	Each	**436.00**
2.11.5	(-do-) wt.**Not less than 1000 gm** 40mm.:	Each	**662.00**
2.11.6	(-do-) wt.**Not less than 1200 gm** 50mm.	Each	**717.00**
2.11.7	Synthetic material (PTMT) of approved make 15mm nominal size.	Each	**252.00**
2.11.8	(-do-) 20 mm nominal size.	Each	**348.00**
2.11.9	(-do-) 25 mm nominal size.	Each	**645.00**
2.11.10	(-do-) 50 mm nominal size.	Each	**1674.00**
2.12	P *& F Inlet Connection* (Angle Valves) Superior quality, of approved make, for Wash basin, Gyser etc.		
2.12.1	C.P. Inlet connection 15mm.Brass (IS : 8931 marked)	Each	**455.00**
2.12.2	---do--- with 37 cm long pipe and nut.	Each	**536.00**
2.12.3	---do--- with 60 cm long thread and nut	Each	**569.00**
2.12.4	Synthetic Material (PTMT) Inlet connection of approved quality/make 15 mm nominal size.	Each	**218.00**
2.13	P & F 15 mm. Dia *Connection Pipe* of approved quality/make :		
2.13.1	PVC pipe with C.P. Brass nuts upto length, 300mm	Each	**66.00**
2.13.2	(-do-) upto length 370mm	Each	**72.00**
2.13.3	(-do-) 450mm.	Each	**92.00**
2.13.4	(-do-) 600mm.	Each	**105.00**
2.13.5	CP Brass Pipe with CP Brass nuts upto length 300mm.	Each	**88.00**
2.13.6	(-do-) 370mm.	Each	**112.00**
2.13.7	(-do-) 450mm.	Each	**124.00**
2.13.8	(-do-) 600mm.	Each	**160.00**
2.14	P & F *Flush Cock / Flush Valve* (IS : 9758 Mark) for WC of approved quality make :		
2.14.1	CP brass push type 15mm nominal bore.	Each	**325.00**

2.14.2	(-do-) 20mm nominal bore.	Each	**492.00**
2.14.3	(-do-)exposed 25mm nominal bore.	Each	**874.00**
2.14.4	(-do-) 32mm nominal bore.	Each	**1183.00**
2.14.5	(-do-) concealed 25mm nominal bore.	Each	**886.00**
2.14.6	(-do-) concealed 32mm nominal bore.	Each	**1293.00**
2.14.7	(-do-) Half-turn exposed 25mm nominal bore.	Each	**712.00**
2.14.8	(-do-) Concealed 25mm nominal bore.	Each	**820.00**
2.14.9	Brass / Gun metal Half-turn wt. 1 Kg. of approved quality / make, 25mm nominal bore.	Each	**501.00**
2.14.10	CP Brass flush valve 32mm nominal bore	Each	**1812.00**
2.15	P & F *Full-way Valve* (IS:778 Mark) or wheel valve of approved make :		
2.15.1	Gun-metal 15mm nominal bore.	Each	**206.00**
2.15.2	(-do-) 20mm nominal bore.	Each	**325.00**
2.15.3	(-do-) 25mm nominal bore.	Each	**481.00**
2.15.4	(-do-) 32mm nominal bore.	Each	**639.00**
2.15.5	(-do-) 40mm nominal bore.	Each	**976.00**
2.15.6	(-do-) 50mm nominal bore.	Each	**1331.00**
2.15.7	(-do-) 65mm nominal bore.	Each	**1621.00**
2.15.8	(-do-) 80mm nominal bore.	Each	**2427.00**
2.16	P & F *Gun-metal Non-return Valve or Check Valve* (IS :Make) of approved make, superior quality:		
2.16.1	15mm nominal bore.		
(a)	Horizontal	Each	**247.00**
(b)	Vertical	Each	**216.00**
2.16.2	20mm nominal bore.		
(a)	Horizontal	Each	**312.00**
(b)	Vertical	Each	**276.00**
2.16.3	25mm nominal bore.		
(a)	Horizontal	Each	**451.00**
(b)	Vertical	Each	**366.00**
2.16.4	32mm nominal bore.		
(a)	Horizontal	Each	**634.00**
(b)	Vertical	Each	**530.00**
2.16.5	40mm nominal bore.		
(a)	Horizontal	Each	**882.00**
(b)	Vertical	Each	**674.00**

2.16.6	50mm nominal bore.		
(a)	Horizontal	Each	**1287.00**
(b)	Vertical	Each	**876.00**
2.17	P & F *Flush Valve* (Hydraulic) with level complete (IS : 9758 Mark) of approved make :		
2.17.1	32mm nominal bore.	Each	**2409.00**
	25mm nominal bore.	Each	**Deleted**
2.18	P & F Oxidised Gas Taps, Bench-type of superior quality and approved make:		
2.18.1	One way	Each	**209.00**
2.18.2	Two way	Each	**281.00**
2.18.3	Three way	Each	**354.00**
2.18.4	Four way	Each	**415.00**
2.19	P & F Chromium plated *Swan-neck Laboratory Cock* of approved quality / make:		
2.19.1	One way	Each	**349.00**
2.19.2	Two way	Each	**477.00**
2.19.3	Three way	Each	**574.00**
2.19.4	Four way	Each	**628.00**
2.20	P & F *Spouts* of superior quality and approved make:		
2.20.1	C.P. brass spout	Each	**602.00**
2.20.2	(-do-) Extra heavy with flange.	Each	**736.00**
2.21	P & F *Mixing Range* (15mm)		
2.21.1	C.P. One hole/close hole basin mixer with casted spout.	Each	**1582.00**
2.21.2	(-do-) with swinging spout casted.	Each	**1819.00**
2.21.3	C.P. sink mixer with swinging spout J-pipe.	Each	**1705.00**
2.21.4	(-do-) With casted spout.	Each	**1760.00**
2.21.5	C.P. valve mixer non-telephonic type.	Each	**1636.00**
2.21.6	C.P. wall mixer telephonic type crutch & telephonic shower.	Each	**2890.00**
2.21.7	(-do-) without crutch & telephonic shower.	Each	**2310.00**
2.21.8	C.P. Bath-tub mixer without telephonic shower but with crutch.	Each	**2860.00**
2.21.9	C.P. Surgical basin mixer elbow action.	Each	**2310.00**
2.21.10	(-do-) with shower (wall type)	Each	**4164.00**

2.22	P & F *Divertors* of superior quality and approved make:		
2.22.1	C.P. brass Tip-tone divertor bath spout with mixing valve.	Each	**2324.00**
	(-do-) Concealed Five-way divertors.	Each	**Deleted**
	(-do-) Four way (-do-)	Each	**Deleted**
2.22.2	(-do-) Three way (-do-)	Each	**2233.00**
2.22.3	(-do-) Two way with 22Cm. Pipe flange	Each	**1282.00**
2.23	P & F foot valve of superior quality and approved make with fittings :		
2.23.1	Brass 25 mm.	Each	**156.00**
2.23.2	CI 50 mm	Each	**392.00**
2.23.3	CI 65 mm	Each	**494.00**
2.23.4	CI 80 mm	Each	**755.00**
2.24	P & F *Precast R.C.C. Water Storage Tank* with R.C.C. cover 25mm thick and 15 Cm. Long G.I. over-flow pipe including making connections etc. & hoisting upto 1M. ht. From ground level having:		
2.24.1	340 litres water capacity size 90 x 75 x 60 Cm.	Each	**847.00**
2.24.2	270 litres water capacity size 90 x 60 x 60 Cm.	Each	**687.00**
2.25	Hoisting Charges of precast R.C.C. water storage tank on each floor :		
2.25.1	340 litres water capacity.	Each	**198.00**
2.25.2	270 litres water capacity.	Each	**147.00**
2.26	P & F *PVC Storage Tank* ISI Marked (IS : 12701) indicating the BIS license No), of approved make with cover, 25mm dia 1M long G.I. over-flow pipe & 25 Cm. long wash out pipe with plug & socket, including making connection etc., complete of approved design:		
2.26.1	200 litres capacity.	Each	**1426.00**
2.26.2	300 (-do-)	Each	**2138.00**
2.26.3	500 (-do-)	Each	**3564.00**
2.26.4	750 (-do-)	Each	**5346.00**
2.26.5	1000 (-do-)	Each	**7128.00**

2.26.6	2000 (-do-)	Each	**14256.00**
2.26.7	3000(-do-)	Each	**21384.00**
2.26.8	5000(-do-)	Each	**35640.00**
2.27	Hoisting Charges for each floor :		
2.27.1	For items 2.26.1 to 2.26.4	Each	**100.00**
2.27.2	For items 2.26.5 to 2.26.8	Each	**143.00**
2.28	Providing and fixing Bitumen Paint Coating over G.I. Pipes Complete as per Specifications. –		
2.28.1	15 mm dia	Rmt.	**1.35**
2.28.2	20 mm dia	Rmt.	**1.90**
2.28.3	25 mm dia	Rmt.	**2.20**
2.28.4	40 mm dia	Rmt.	**3.60**
2.29	Providing and fixing Polyethylene - Aluminium - Polyethylene (PE-AL-PE) Composite Pressure Pipes as per 15450 : 2004 U.V. stablised with carbon black having thermal stabillity for hot and cold water supply, capable to withstand temperature upto 80 oC including all special f withittings of composite material (engineering plastic blend and brass inserts whwnever required) e.g. elbows, tees, reducers, couplers connectors etc.with clamps at 1.00 metre spacing.This include the cost of cutting chases and including testing of joints complete as per direction of Engineer In charge Internal work - Exposed on Wall		
2.29.1	1216 (16mm OD) pipe	Metre	**214.00**
2.29.2	1620 (20mm OD) pipe	Metre	**245.00**
2.29.3	2025 (25mm OD) pipe	Metre	**298.00**
2.29.4	2532 (32mm OD) pipe	Metre	**395.00**
2.29.5	3240 (40mm OD) pipe	Metre	**508.00**

2.30 Providing and fixing Polyethylene - Aluminium - Polyethylene (PE-AL-PE) Composite Pressure Pipes as per 15450 : 2004 U.V. stablised with carbon black having thermal stabillity for hot and cold water supply, capable to withstand temperature upto 80 oC including all special f withittings of composite material (engineering plastic blend and brass inserts whwnever required) e.g. elbows, tees, reducers, couplers connectors etc.with clamps at 1.00 metre spacing.This include the cost of cutting chases and including testing of joints complete as per direction of Engineer In charge
Concealed installation including cutting chases and making good the walls etc. complete.

2.30.1	1216 (16mm OD) pipe	Metre	**182.00**
2.30.2	1620 (20mm OD) pipe	Metre	**201.00**
2.30.3	2025 (25mm OD) pipe	Metre	**237.00**
2.30.4	2532 (32mm OD) pipe	Metre	**308.00**
2.30.5	3240 (40mm OD) pipe	Metre	**376.00**

2.31 Providing and fixing Polyethylene - Aluminium - Polyethylene (PE-AL-PE) Composite Pressure Pipes as per 15450 : 2004 U.V. stablised with carbon black having thermal stabillity for hot and cold water supply, capable to withstand temperature upto 80 oC including all special f withittings of composite material (engineering plastic blend and brass inserts whwnever required) e.g. elbows, tees, reducers, couplers connectors etc.with trenching & refilling and testing of joints complete as per direction of Engineer In charge
External work

2.31.1	1216 (16mm OD) pipe	Metre	**162.00**
2.31..2	1620 (20mm OD) pipe	Metre	**181.00**
2.31.3	2025 (25mm OD) pipe	Metre	**217.00**
2.31..4	2532 (32mm OD) pipe	Metre	**288.00**
2.31.5	3240 (40mm OD) pipe	Metre	**356.00**

2.32 Supply & fixing G.I. Union ISI marked in G.I. pipe line as required complete in all respect of size :

2.32.1	15mm dia nominal bore	Each	**92.00**
2.32.2	20mm dia nominal bore	Each	**127.00**
2.32.3	25mm dia nominal bore	Each	**149.00**
2.32.4	32mm dia nominal bore	Each	**220.00**
2.32.5	40mm dia nominal bore	Each	**278.00**
2.32.6	50mm dia nominal bore	Each	**421.00**
2.32.7	65mm dia nominal bore	Each	**820.00**
2.32.8	80mm dia nominal bore	Each	**946.00**
2.33	Providing and fixing Superior quality CP Brass fittings of approved make as per direction of Engineer-in-charge		
2.33.1	Pillar Cock 15 mm nominal size	Each	**748.00**
2.33.2	Swan neck pillar cock 15 mm nominal size	Each	**1436.00**
2.33.3	Bib Cock with long thread 15 mm nominal size	Each	**748.00**
2.33.4	Long body Bib Cock 15 mm nominal size	Each	**985.00**
2.33.5	Two way Bib Cock 15 mm nominal size	Each	**1067.00**
2.33.6	Angular stop cock with long thread 15 mm nominal size	Each	**693.00**
2.33.7	Stop cock long thread	Each	**726.00**
2.33.8	Stop cock with reduced body	Each	**924.00**
2.33.9	Concealed stop cock 15 mm nominal size	Each	**1056.00**
2.33.10	Sink cock with swinging casted spout (table mounted) 15 mm nominal size	Each	**1474.00**
2.33.11	Single lever wash basin mixer with 450 mm long copper connection pipe	Each	**4235.00**
2.33.12	Central hole basin mixer	Each	**2453.00**
2.33.13	Wall mixer 3 in 1 system with provision of tetephonic bath shower with bend pipe	Each	**3403.00**
2.33.14	600 mm long towel rail	Each	**1637.00**
2.33.15	Revolving type towel ring	Each	**1001.00**
2.33.16	Soap dish	Each	**974.00**
2.33.17	Double coat hook	Each	**748.00**
2.33.18	190 mm rain shower	Each	**5075.00**
2.33.19	Shower arm	Each	**1100.00**
2.33.20	Paper holder	Each	**1023.00**
2.33.21	Hand shower with 8 mm dia 1 mt long PVC tube & wall hook complete	Each	**1665.00**

2.34	Providing and fixing 3-layer PP-R (Polypropylene random coplymer) pipes SDR 7.4 UV stabilized and antimicrobial fusion welded, having thermal stability for hot & cold water supply including all PP-R plain & brass threaded Polypropylene random fittings and fixing the pipe with clamps at 1.00 spacing. This includes testing of joint complete as per direction of Engineer - in charge.(INTERNAL WORK- EXPOSED ON WALL)		
2.34.1	PN-16 Pipe, 16 mm OD	Rm	**157.00**
2.34.2	PN-16 Pipe, 20 mm OD	Rm	**180.00**
2.34.3	PN-16 Pipe, 25 mm OD	Rm	**258.00**
2.34.4	PN-16 Pipe, 32 mm OD	Rm	**361.00**
2.34.5	PN-16 Pipe, 40 mm OD	Rm	**500.00**
2.34.6	PN-16 Pipe, 50 mm OD	Rm	**726.00**
2.35	Providing and fixing 3-layer PP-R (Polypropylene random coplymer) pipes SDR 7.4 UV stabilized and antimicrobial fusion welded, having thermal stability for hot & cold water supply including all PP-R plain & brass threaded Polypropylene random fittings and fixing the pipe with clamps at 1.00 spacing. This includes cost of cutting chases and making good the same including testing of joint complete as per direction of Engineer - in charge.(CONCELED WORK INCLUDING CUTTING CHASES AND MAKING GOOD THE WALL ETC.)		
2.35.1	PN-16 Pipe, 16 mm OD	Rm	**215.00**
2.35.2	PN-16 Pipe, 20 mm OD	Rm	**241.00**
2.35.3	PN-16 Pipe, 25 mm OD	Rm	**311.00**
2.35.4	PN-16 Pipe, 32 mm OD	Rm	**414.00**
2.36	Providing and fixing 3-layer PP-R (Polypropylene random coplymer) pipes SDR 7.4 UV stabilized and antimicrobial fusion welded, having thermal stability for hot & cold water supply including all PP-R plain & brass threaded Polypropylene random fittings including trenching, refilling and testing of joint complete as per direction of Engineer - in charge. (EXTERNAL WORKS)		
2.36.1	PN-16 Pipe, 16 mm OD	Rm	**142.00**
2.36.2	PN-16 Pipe, 20 mm OD	Rm	**168.00**

2.36.3	PN-16 Pipe, 25 mm OD	Rm	**238.00**
2.36.4	PN-16 Pipe, 32 mm OD	Rm	**342.00**
2.36.5	PN-16 Pipe, 40 mm OD	Rm	**480.00**
2.36.6	PN-16 Pipe, 50 mm OD	Rm	**619.00**
2.36.7	PN-16 Pipe, 63 mm OD	Rm	**1042.00**
2.37	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply including all CPVC plain & brass threaded fittings i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge. (Internal work Exposed on wall)		
2.37.1	15 mm nominal outer dia pipes	Rm	**174.00**
2.37.2	20mm nominal outer dia pipes	Rm	**202.00**
2.37.3	25 mm nominal outer dia pipes	Rm	**256.00**
2.37.4	32mm nominal outer dia pipes	Rm	**335.00**
2.37.5	40 mm nominal outer dia pipes	Rm	**472.00**
2.37.6	50 mm nominal outer dia pipes	Rm	**729.00**
2.38	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply including all CPVC plain & brass threaded fittings i/c fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cost of cutting chases and making good the same including testing of joints complete as per direction of Engineer in Charge. (Concealed work including cutting chases and making good the walls etc.,)		
2.38.1	15 mm nominal outer dia pipes	Rm	**254.00**
2.38.2	20mm nominal outer dia pipes	Rm	**263.00**
2.38.3	25 mm nominal outer dia pipes	Rm	**342.00**
2.38.4	32mm nominal outer dia pipes	Rm	**433.00**

2.39	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply including all CPVC plain & brass threaded fittings, this included jointing of pipes & fittings with one step CPVC solvent cement , trenching refilling & testing of joints complete as per direction of Engineer in Charge. (External work).		
2.39.1	15 mm nominal outer dia pipes	Rm	**152.00**
2.39.2	20mm nominal outer dia pipes	Rm	**178.00**
2.39.3	25 mm nominal outer dia pipes	Rm	**238.00**
2.39.4	32mm nominal outer dia pipes	Rm	**312.00**
2.39.5	40 mm nominal outer dia pipes	Rm	**428.00**
2.39.6	50 mm nominal outer dia pipes	Rm	**688.00**
2.40	**Providing and fixing Premium quality Towel rack 600mm long 260mm wide with lower hangers CP Brass fittings of approved make as per direction of Engineer-in-charge**	**Each**	**2808.00**
2.41	**Labour Charges for Removing & Refixing the water Storage Tank of all sizes and including all Necessary fitting complete in all respect as per approved sample and direction of engineer in charge.**	**Each**	**335.00**
2.42	**Providing and fixing premium quality Angular stop cock 15mm with wall flange complete as per sample approved by Engineer in charge**	**Each**	**1429.00**
2.43	**Providing and fixing premium quality Two way Bib cock 15mm with wall flange complete as per sample approved by Engineer in charge**	**Each**	**2123.00**
2.44	**Providing and fixing superior quality PVC Tank Cover complete as per approved by Engineer In Charge**		
2.44.1	**300 Lt. capacity**	**Each**	**169.00**

2.44.2	500 Lt. capacity	Each	183.00
2.44.3	Above 500 Lt. capacity	Each	196.00
2.45	Providing & fixing C.I. Saddle pieces 100mm dia with nut & bolt & washer & Rubber Gasket complete as per direction of engineer in charge	Each	178.00
2.46	Supply and installation of Premium quality sluice valve 100mm dia including all fittings as per approved by Engineer in charge.	Each	4584.00
2.47	Providing Laying Jointing DI Pipe (Ductile iron pipe) of 100 mm dia ISI marked K-7 grade as per IS : 5329-2000 with internal cement mortar lining for potable water with rubber ring (EPDM/ SBR) Joints per IS : 5382-1985 complete including Hydraulic testing & commissioning as per Technical Specification.	Mtr	989.00
2.47	Providing Laying Jointing DI Pipe (Ductile iron pipe) of 100 mm dia ISI marked K-7 grade as per IS : 5329-2000 with internal cement mortar lining for potable water with rubber ring (EPDM/ SBR) Joints per IS : 5382-1985 complete including Hydraulic testing & commissioning as per Technical Specification.	Mtr	989.00

CHAPTER : S-3
DRAINAGE / DISPOSAL WORK

Note :-

1 For laying of S.W. pipes beyond 1.2m depth, earthwork & close timbering or open timbering (as per site conditions) is to be paid extra.

2 The rates for S.W. pipes are applicable to work executed in soils above sub-soil water level. Extra allowance is to be made for working under sub-soil water level.

3 Rock cutting wherever required will be paid extra.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
	STONEWARE PIPES & FITTING		
3.1	Providing Laying & jointing glazed S.W. pipes grade `A` (IS;651 marked) of approved make with spun yarn & stiff mixture of cement mortar 1:1, excavation & refilling of earth in all types of soil upto 1.2m depth, S.W. fittings, including testing of joints etc, complete.		
3.1.1	100mm dia	Mtr.	**229.00**
3.1.2	150mm dia	Mtr.	**369.00**
3.1.3	200mm dia	Mtr.	**514.00**
3.1.4	230mm dia	Mtr.	**581.00**
3.1.5	250mm dia	Mtr.	**677.00**
3.1.6	300mm dia	Mtr.	**949.00**
3.1.7	400mm dia	Mtr.	**1387.00**
Note:-			
1	If grade `AA S.W. pipes (IS:651 mark) are used then add 5%		
2	Deduct 5% if jointing is done in CM 1:3.		
3.2	Providing & Laying C.C. 1:5:10 (Stone aggregate 40mm nominal size) all around S.W. pipe including 15 Cm. thick bed concrete (Width W= Diameter of pipe in Cm+30 Cm.)(Excluding cost of pipe)		
3.2.1	100mm dia S.W. pipe	Mtr.	**415.00**
3.2.2	150mm dia S.W. pipe	Mtr.	**500.00**
3.2.3	200mm dia S.W. pipe	Mtr.	**509.00**
3.2.4	230mm dia S.W. pipe	Mtr.	**648.00**
3.2.5	250mm dia S.W. pipe	Mtr.	**685.00**
3.2.6	300mm dia S.W. pipe	Mtr.	**788.00**

3.2.7	350mm dia S.W. pipe	Mtr.	**892.00**
3.2.8	400mm dia S.W. pipe	Mtr.	**995.00**
3.3	Providing & Laying C.C. 1:5:10 (Stone aggregate 40mm nominal size) upto haunches of S.W. pipes including 150 mm thick bed concrete (Width W= Dia of pipe in Cm+ 30Cm.) concrete width to be reduced from width W at center of pipe to \`0\` at top of pipe. (Excluding cost of pipe)		
3.3.1	100mm dia S.W. pipe	Mtr.	**197.00**
3.3.2	150mm dia S.W. pipe	Mtr.	**321.00**
3.3.3	200mm dia S.W. pipe	Mtr.	**375.00**
3.3.4	230mm dia S.W. pipe	Mtr.	**412.00**
3.3.5	250mm dia S.W. pipe	Mtr.	**438.00**
3.3.6	300mm dia S.W. pipe	Mtr.	**506.00**
3.3.7	350mm dia S.W. pipe	Mtr.	**577.00**
3.3.8	400mm dia S.W. pipe	Mtr.	**648.00**
3.4	Providing & fixing S.W. *square mouth Gully trap* \`A\` grade (IS 651-1992 marked) of approved make & design in existing man-hole.		
3.4.1	Size 100 X 100mm	Each	**165.00**
3.4.2	Size 150 X 100mm	Each	**177.00**
3.4.3	Size 225 X 150mm	Each	**230.00**
3.5	P & F *Square-mouth* gully trap \`A\` grade (IS 651-1992 :marked) of approved make complete with C.I. grating, frame & Cover masonary chamber in C.M. 1:6, 100mm thick C.C. 1:5:10 base concrete, 40mm thick C.C.M-15 grade coping, 12mm thick cement plaster in CM 1:3 complete as per approved design & type.		
3.5.1	Trap size 150 X 100mm, ferrow cement cover with frame, Internal Chamber size 300 X 300mm.	Each	**1045.00**
3.5.2	Trap size 225 X 150 mm, ferrow cement cover with frame Internal Chamber size 450 X 450mm.	Each	**1390.00**
3.6	P & F S.W. *Master Trap* (Inter-cepting) \`A\` grade (I.S.: 651 marked) of approved quality/make		
3.6.1	100mm dia	Each	**313.00**
3.6.2	150mm dia	Each	**442.00**
3.7	Labour charges for Removing S.W. pipes including breaking of joints & bed concrete, stacking of useful material at site within 50 mtr. Lead and disposal of unserviceable material within 0.5 Km. lead:		
3.7.1	100 & 150mm dia pipe	Mtr.	**21.00**

3.7.2	200, 230 & 250mm dia pipe	Mtr.	**23.00**
3.7.3	300mm dia pipe	Mtr.	**24.00**
3.7.4	350mm dia pipe	Mtr.	**27.00**
3.7.5	400mm dia pipe	Mtr.	**28.00**
	R.C.C. NON PRESSURE PIPES		
3.8	Providing, Laying & Jointing R.C.C. class NP-2 Non-Pressure pipes (IS : 458mark) of approved make with collars, jointed with C.M. 1:2 or having Spigot and socket ends with flexible rubber rings joint including testing of joints etc. complete :		
3.8.1	100mm dia Internal	Mtr.	**444.00**
3.8.2	150mm dia Internal	Mtr.	**545.00**
3.8.3	200mm dia Internal	Mtr.	**613.00**
3.8.4	250mm dia Internal	Mtr.	**810.00**
3.8.5	300mm dia Internal	Mtr.	**901.00**
3.8.6	450mm dia Internal	Mtr.	**1409.00**
3.8.7	500mm dia Internal	Mtr.	**1578.00**
3.8.8	600mm dia Internal	Mtr.	**2026.00**
3.8.9	700 mm dia Internal	Mtr.	**2474.00**
3.8.10	800 mm dia Internal	Mtr.	**3404.00**
3.8.11	900 mm dia Internal	Mtr.	**4022.00**
Note :	Deduct 20 % for Non ISI marked pipes of all size		
	SAND-CAST IRON SOIL, WATER & RAIN WATER PIPES.		
3.9	P & F *Sand-cast Iron* (S.C.I.) Pipe (IS : 1729mark) of approved make in wall or in floor with M.S. holder bat clamps in 10 X 10 X 10 Cm. M-15 grade concrete blocks, joints filled with CM 1:4 with spun yarn including cutting holes and making good the wall:	.	
3.9.1	50mm dia	Mtr.	**690.00**
3.9.2	75mm dia	Mtr.	**865.00**
3.9.3	100mm dia	Mtr.	**1116.00**
3.9.4	150mm dia	Mtr.	**1851.00**
3.10	Add extra for *Lead caulked joints* (IS:782-1978) in sand cast iron pipes/spun-		
3.10.1	75mm dia, 0.88 Kg. Pig lead	Each	**263.00**
3.10.2	100mm dia, 0.98 Kg. Pig lead	Each	**317.00**
3.10.3	150mm dia, 1.48 Kg. Pig lead	Each	**411.00**
3.11	P &F *S.C.I. Fittings* (IS : 1729 mark) in CM 1:4 with spun yarn.		
3.11.1	50mm w/o door		

(i)	Socket	Each	**290.00**
(ii)	Cowel	Each	**368.00**
(iii)	Tee	Each	**534.00**
(iv)	Bend	Each	**433.00**
3.11.2	75 mm w/o door		
(i)	Offset	Each	**486.00**
(ii)	Socket	Each	**343.20**
(iii)	D/door Y-Tee	Each	**802.80**
(iv)	Cowel	Each	**432.00**
(v)	Tee	Each	**607.20**
(vi)	Bend	Each	**499.20**
(vii)	Door Cross	Each	**589.20**
3.11.3	100mm-do-		
(i)	Offset	Each	**804.00**
(ii)	Socket	Each	**409.00**
(iii)	D/door Y-Tee	Each	**1118.00**
(iv)	Cowel	Each	**510.00**
(v)	Tee	Each	**832.00**
(vi)	Bend	Each	**602.00**
(vii)	Door Cross	Each	**642.00**
3.11.4	150mm w/o door		
(i)	Socket	Each	**916.00**
(ii)	Cowel	Each	**1030.00**
(iii)	Bend	Each	**1284.00**
3.11.5	50mm with door		
(i)	Tee	Each	**622.00**
(ii)	Bend	Each	**497.00**
3.11.6	75mm-do-		
(i)	Tee	Each	**646.00**
(ii)	Bend	Each	**550.00**
3.11.7	100mm-do-		
(i)	Tee	Each	**863.00**
(ii)	Bend	Each	**695.00**
3.11.8	150mm-do-		
(i)	Tee	Each	**1799.00**
(ii)	Bend	Each	**1493.00**

3.12	P & F *Sand –cast Floor Trap (Nahni)* of (IS:1729 mark) of approved quality/make with socket down or hinged grating including cutting and making good the floor.		
3.12.1	Size 100 x 100mm	Each	**756.00**
3.12.2	Size 100 x 75mm	Each	**720.00**
3.12.3	Size 100 x 50mm	Each	**396.00**
3.13	P & F *Sand-cast* Iron `P` or `S` Trap (IS:1729 make) of approved make.	Each	**843.00**
3.14	Labour charges for making connection of sewer line through main hole.	Each	**590.00**
	Making connection with existing G.I. pipe lines of all sizes.	Each	**Deleted**
3.15	Labour charges for *removing C.I./A.C*. pipes of all sizes including accessories with care and stacking of useful material.	Mtr.	**30.00**
	RIGID PVC PIPE		
3.16	P&F rigid *PVC Pipe* (IS:4985 mark) class II/ (4 Kg. /Cm^2 . approved quality /make including joining the pipe with solvent cement rubber ring and lubricant.		
3.16.1	63 mm dia	Mtr	**134.00**
3.16.2	75 mm dia	Mtr	**161.00**
3.16.3	110 mm dia	Mtr	**256.00**
3.16.4	160 mm dia	Mtr	**421.00**
3.16.5	**Providing & Laying C.C. 1:3:6 (Stone Agreegate 20mm nominal size) all around PVC Pipe including 7.5 cm thick bed concerte [Width & Height = Dia of Pipe in CM+15CM] excluding cost of Pipe & Excavation of trenches etc.**		
3.16.5.1	**PVC pipe 110 mm dia**	**Mtr.**	295.00
3.16.5.2	**PVC pipe 160 mm dia**	**Mtr.**	386.00
3.17	P&F rigid *PVC pipe fittings* (IS: 4985 mark) of approved quality /make including joining the pipe with solvent cement rubber ring and lubricant:		
3.17.1	Coupler (socket)		
(i)	75mm dia	Each	**79.00**
(ii)	110mm dia	Each	**98.00**
(iii)	160 mm dia	Each	**108.00**

3.17.2	Reducer 110 X 75 mm		
(i)	75mm dia	Each	**82.00**
(ii)	110mm dia	Each	**88.00**
(iii)	160 mm dia	Each	**104.00**
3.17.3	Plain Tee		
(i)	75mm dia	Each	**104.00**
(ii)	110mm dia	Each	**170.00**
(iii)	160 mm dia	Each	**186.00**
3.17.4	Door Tee		
(i)	75mm dia	Each	**161.00**
(ii)	110mm dia	Each	**194.00**
(iii)	160 mm dia	Each	**208.00**
3.17.5	Door 'Y'		
(i)	75mm dia	Each	**170.00**
(ii)	110mm dia	Each	**209.00**
(iii)	160 mm dia	Each	**224.00**
3.17.6	Double 'Y'		
(i)	75mm dia	Each	**177.00**
(ii)	110mm dia	Each	**282.00**
(iii)	160 mm dia	Each	**300.00**
3.17.7	Double Door 'Y'		
(i)	75mm dia	Each	**208.00**
(ii)	110mm dia	Each	**355.00**
(iii)	160 mm dia	Each	**371.00**
3.17.8	Bend 45		
(i)	75mm dia	Each	**73.00**
(ii)	110mm dia	Each	**116.00**
(iii)	160 mm dia	Each	**130.00**
3.17.9	Bend 87 .5		
(i)	75mm dia	Each	**88.00**
(ii)	110mm dia	Each	**146.00**
(iii)	160 mm dia	Each	**161.00**
3.17.10	Door Bend (Top Side)		
(i)	75mm dia	Each	**120.00**
(ii)	110mm dia	Each	**140.00**
(iii)	160 mm dia	Each	**156.00**

3.17.11	Door Bend (Left & Right)		
(i)	75mm dia	Each	**144.00**
(ii)	110mm dia	Each	**177.00**
(iii)	160 mm dia	Each	**192.00**
3.17.12	Vent Cowel		
(i)	75mm dia	Each	**39.00**
(ii)	110mm dia	Each	**51.00**
(iii)	160 mm dia	Each	**69.00**
3.17.13	Socket plug		
(i)	75mm dia	Each	**62.00**
(ii)	110mm dia	Each	**74.00**
(iii)	160 mm dia	Each	**92.00**
3.17.14	Door Cap		
(i)	75mm dia	Each	**40.00**
(ii)	110mm dia	Each	**56.00**
(iii)	160 mm dia	Each	**72.00**
3.17.15	Pipe Clip		
(i)	75mm dia	Each	**22.00**
(ii)	110mm dia	Each	**33.00**
(iii)	160 mm dia	Each	**48.00**
3.17.16	W.C. connector with lipring		
	110mm dia	Each	**194.00**
3.17.17	W.C. connector bent type		
	110mm dia	Each	**203.00**
3.17.18	Plain floor trap		
	110mm dia	Each	**209.00**
3.17.19	Multi floor trap		
	110mm dia	Each	**242.00**
3.17.20	height Raiser		
	110mm dia	Each	**194.00**
3.17.21	Top Tile& straine		
	110mm dia	Each	**73.00**

3.17.22	Top Tile only 110mm dia	Each	**56.00**
3.17.23	P- Trap 110mm dia	Each	**347.00**
3.17.24	Q- Trap 110mm dia	Each	**389.00**
3.17.25	S- trap 110mm dia	Each	**486.00**
3.17.26	Ring fitting 110mm dia	Each	**98.00**
3.17.274	Gully trap 110mm dia	Each	**510.00**
3.18	Raising manhole cover and frames slab to required level including dismantling existing slab and making good the damage as required (raising depth of manhole masonry to be paid separately).		
3.18.1	Rectangular manhole 90x60 cm	Each	**884.00**
3.18.2	Rectangular manhole 150x120 cm	Each	**1352.00**
3.19	Construction of *manhole* in all type of soil inner size 90 X 60 Cm. 300 mm thick masonry in CM 1:6, 10 Cm. thick cement concrete 1:5:10 in foundation, 20 mm thick inside plaster in CM 1:6, finished with floating neat cement, 50mm thick M-15 grade C.C. flooring, making channels, 80mm thick stone slab covering with 40mm thick M-15 grade C.C. flooring, Cement cover with frame of 450mm dia, earthwork etc. complete as per design including disposal of surplus earth within 50 mtr. lead.		
3.19.1	Depth up to 0.5 M	Each	**4990.00**
3.19.2	Add extra over item 3.19.1 for every additional 0.10m depth above 0.5 m depth.	Each	**187.00**
3.19.3	**Deduct in Item No 3.19.1 for manhole with brick work 230 mm thick in place of 300 mm thick stone masonry**	**Each**	**947.00**

3.20	Construction of Man hole, in all types of soil, inner size 1.5 x 1.2 M, with 450mm thick masonry in CM 1:6, 10 Cm. thick cement concrete 1:5:10 in foundation, 20mm thick inside plaster in CM 1:6, finished with floating neat cement, 50 mm thick M-15 grade C.C. flooring making channels, 80 mm thick stone slab covering with 40 mm thick M-15 grade C.C. flooring, Ferro Cement cover with frame of 450mm dia, earthwork etc. complete including disposal of surplus earth within 50 mtr. lead as per design :		
3.20.1	Depth up to 1.5 M	Each	**18196.00**
3.20.2	Add extra over item 3.20.1 for every additional 0.10m depth above 1.5 m depth .	Each	**639.00**
3.21	Construction of *chamber* in all type of soil with 300 mm thick masonry in CM 1:6 m,10 cm thick C.C. 1:5:10 in foundation, 20mm thick insider plaster in Cm 1:6, finished with floating neat cement, 50mm thick M-15 grade C.C. flooring , earthwork etc. complete as per design including disposal of surplus earth within a lead of 50 mtr.		
3.21.1	Inside size 300 X 300 mm depth upto 0.5 M Cement cover with frame.	Each	**954.00**
3.21.1.1	**Deduct for using brick work 230 mm thick in place of 300 mm thick stone masonry**		**414.00**
3.21.2	-do- size 450 x 450mm depth upto 0.5 M Cement cover with frame	Each	**2120.00**
3.21.2.1	**Deduct for using brick work 230 mm thick in place of 300 mm thick stone masonry**		**462.00**
3.21.3	-do- size 600 x 450mm depth upto 0.5 M Cement cover with frame.	Each	**2999.00**
3.21.3.1	**Deduct for using brick work 230 mm thick in place of 300 mm thick stone masonry**		**481.00**
3.22.4	**-do-** depth upto 0.5 M to 1.0 M –do-	Each	**3298.00**
3.21.4.1	**Deduct for using brick work 230 mm thick in place of 300 mm thick stone masonry**		**737.00**
3.22	Construction of soakage well in all type of soil with 300 mm thick dry masonry, top and bottom 300 mm course in CM 1:6, 80mm thick stone slab, jointing of slab in CM 1 : 3, Ralthal, Kharanja, 40 mm thick M-15 grade C.C flooring , earthwork complete as per approved drawing including disposal of earth within a lead of 50 mtr.:		
3.22.1	Size 300 Cm. dia outside & 300 Cm. depth .	Each	**21941.00**

3.22.2	-do- & 240 Cm. Depth.	Each	**20125.00**
3.22.3	Size 240 Cm. dia outside & 240 Cm. depth.	Each	**14636.00**
3.23	Construction of soakage Trench in all types of soil with 300 mm thick dry stone masonry, 80 mm thick stone slab covering, jointing of slab in CM 1:3 Ralthal, Kharanja, 40 mm thick M-15grade C.C. flooring earthwork etc. complete as per approved drawing including disposal of earth within a lead of 50 Mtr.:		
3.23.1	Top width 180Cm., bottom width 90Cm. & depth 240Cm.	Mtr.	**5590.00**
3.23.2	-do- 240 Cm., -do- 150Cm. & depth 300 Cm	Mtr.	**9452.00**
3.24	Construction of Soakage well in all types of soil of approved drawing, top 90 Cm .Portion in 450mm thick masonry with CM 1:6, 80 mm thick stone slab covering, jointing of slab in CM 1:3 ,Ralthal, kharanja 40mm thick M-15 grade C.C flooring, earth work etc . complete including disposal of surplus earth within a lead of 50 mtr .		
3.24.1	Inner dia 90 Cm & 10 to 12 Mtr deep.	Each	**4980.00**
3.24.2	Inner dia 60 Cm & 10 to 12 Mtr deep.	Each	**4691.00**
3.25	Labour charges for construction of Soakage well of 90 Cm. dia (Earth –work & its disposal ,with in a lead of 0.5 Km .only).	Mtr	**90.00**
3.26	Add extra if Soakage Well is filled with dry stones.	Cum	**363.00**
3.27	Construction of septic Tank in all types of soil with 40 Cm .thick masonry in CM 1:6, 15 Cm thick C.C bed of 1:5:10, M-15 grade C.C floor & **RCC slab covering with M15 grade c.c. floor,** 50 mm thick stone **slab** partition walls, 20 mm thick plaster in CM 1:6 finished with neat floating cement, 4 Nos **stone** foot rests of approved design ,two No. 450 mm dia each Ferro cement cover with frame, earth work etc. complete as per approved drawing including disposal of surplus earth within a lead of 50 mtr:-		
3.27.1	Size 200x100x130 cm.(for 10 users) with 115 mm thick RCC (M-20) slab with Tor steel reinforcement 8mm ϕ @15 cm c/c bothways including shuttering complete in all respect.	Each	**24838.00**

3.27.2	Size 230 x110 x150 cm.(for 20 users) with 115 mm thick RCC (M-20) slab with Tor steel reinforcement 10mm ϕ @15 cm c/c bothways including shuttering complete in all respect.	Each	**34999.00**
3.27.3	Size 400 x140x150 cm.(for 50 users) with 115 mm thick RCC (M-20) slab with Tor steel reinforcement 10mm ϕ @15 cm c/c bothways including shuttering complete in all respect.	Each	**54418.00**
Note :-	Add or subtract proportionately for every 0.5 M increase or decrease in Length -12.5% Breadth & Depth -18 % each		
3.28	Emptying of Septic tank ,Soakage well etc. by disposing or sludge upto 5 Km. lead & taking out sewage including cleaning of site.	Cum	**260.00**
3.29	Add extra if R.C.C slab of designed thickness & reinforcement is provided on manhole in place of stone slab(Actual area of R.C.C slab to be measured).	Sqm	**195.00**
3.30	P & F M.S. Foot Rests in with C.M.,1:3, standard design in 20mm square M.S. bar 50 mm embedded in wall & 10 cms. projected out complete.	Each	**203.00**
3.31	Supplying and fixing C.I. Cover with frame, C.I. gratings for chambers & man holes.	Kg.	**65.00**
3.32	Supplying and fixing Ferro- cement cover with C.I. frame for chambers & man holes of size.		
3.32.1	450mm dia, 5Kg. C.I. frame to with stand 3 MT center point load & minimum unfractured load of 1 MT. (Light duty).	Each	**1232.00**
3.32.2	500mm dia 12 Kg.-do-15 MT centre point load (medium duty).	Each	**2261.00**
3.32.3	500mm dia 22 Kg.-do-22 MT centre point load (Heavy duty).	Each	**3551.00**
3.32.4	500mm dia 25 Kg.-do-70 MT centre point load seal type (Heavy duty.)	Each	**3709.00**
3.32.5	Rectangular shape 600 x 450 mm, to withstand 3 MT center point load & minimum unfractured load of 1 MT. (Light duty).	Each	**1234.00**
3.33	Supplying & Fixing R.C.C. Manholes covers with frame of approved make (Light duty).		
3.33.1	Size 500mm dia (Double seal)	Each	**299.00**
3.33.2	Size 450mm dia	Each	**265.00**

3.33.3	Size 400mm dia	Each	**199.00**
3.33.4	Size 600 X 600mm (Double seal)	Each	**456.00**
3.33.5	Size 600 X 450mm	Each	**383.00**
3.33.6	Size 450 X 450mm	Each	**290.00**
3.33.7	Size 300 X 300mm	Each	**90.00**
3.33.8	Size 250 X 250mm	Each	**74.00**
3.33.9	Size 180 X 180mm	Each	**66.00**
3.34	Construction of Open Surface Drain with 112mm thick brick masonry in CM 1:4,110mm thick base concrete 1:5:10, 37mm thick M-15 grade C.C. flooring ,12mm 1:4 cement plaster on all exposed faces of walls including top surface excavation & disposal of earth complete as per approved design/drawing:		
3.34.1	112mm drain,225mm Av. depth	Mtr.	**246.00**
3.34.2	150mm drain,225mm Av. depth	Mtr.	**258.00**
3.34.3	250mm drain,300mm Av. depth	Mtr.	**345.00**
3.35	Construction of open Drain with 112mm thick brick masonry in CM 1:4 Semi circular S.W. glazed channel (confirming to IS Standards) fixed with cement mortar 1:1 over a bed of 100mm thick C.C. 1:5:10, filling of edged with M-15 grade C.C. flooring, 12mm cement plaster 1:4, over all exposed faces, excavation & disposal of earth with in a lead of 50 mtr complete as per approved drawings :		
3.35.1	112 mm drain,225mm Av. depth width 10 Cm S.W. glazed channel.	Mtr	**395.00**
3.35.2	150 mm drain,225mm Av. depth width 15 Cm S.W. glazed channel.	Mtr.	**456.00**
3.35.3	250 mm drain,225mm Av. depth width **25** Cm S.W. glazed channel.	Mtr.	**796.00**
3.36	Extra For additional 5 Cm. Average depth of drain of all sizes.	Mtr.	**32.00**

3.37	P & F precast dense cement concrete (Vibro-pressed) drain section Block of different size as per approved drawing and design and strength as per IS code 2185 part I of grade 'D' (5.00mm^2) proper quality of additive/admixture like plasticizer etc. added to produce high quality and durable drain section Blocks of 'L' shape 'U' shape etc. as per approved design and drawing complete with fixing and joining in C.M.1:4 in proper grade and level complete in all respect including earth work and disposal of surplus earth within 0.5 Km. lead. (cost of Steel/ welded mesh etc. shall be paid extra).		
3.37.1	40mm thick	Sqm.	**380.00**
3.37.2	Upto 60mm thick	Sqm.	**471.00**
3.37.3	Upto 80mm thick	Sqm.	**576.00**
3.38	Construction of Cement Concrete Chamber 125mm dia.		
3.38.1	C.P. Brass grating	Each	**152.00**
3.38.2	C.P. Brass doom grating	Each	**161.00**
3.38.3	C.I. Grating	Each	**138.00**
3.38.4	Stainless Steel Sheet Grating	Each	**155.00**
3.39	Construction of Cement Concrete Chamber 125mm dia on depressed floor upto 45cm height from floor with :		
3.39.1	C.P. Brass grating	Each	**215.00**
3.39.2	C.P. doom grating	Each	**239.00**
3.39.3	C.I. Grating	Each	**198.00**
3.39.4	Stainless Steel Sheet Grating	Each	**215.00**
	SPUN-CENTRIFUGALLY CAST IRON SOIL WATER PIPES:-		
3.40	Providing & fixing Spun –Centrifugally cast Iron Pipes (IS : 3989 Mark) of Approved make in wall or in floor with M.S. Holder bat clamps in 10 X 10 X10 Cm. M-15 Grade concrete blocks, joints filled with CM 1:4 with spun yarn including cutting holes and making good the wall:-		
3.40.1	75 mm dia	Mtr	**965.00**
3.40.2	100 mm dia	Mtr	**1181.00**
3.40.3	150 mm dia	Mtr	**1987.00**
3.41	Providing & fixing Spun –Centrifugally cast Iron Pipes fitting (IS : 3989 Mark) in CM 1:4 with spun Yarn.		

3.41.1	75mm w/o door		
(i)	Offset	Each	**540.00**
(ii)	Socket	Each	**329.00**
(iii)	D/Door Y-Tee	Each	**788.00**
(iv)	Bend	Each	**429.00**
(v)	Door Cross	Each	**1079.00**
(vi)	Cowel	Each	**393.00**
(vii)	Tee	Each	**609.00**
3.41.2	100 mm –do-		
(i)	Offset	Each	**735.00**
(ii)	Socket	Each	**472.00**
(iii)	D/Door Y-Tee	Each	**1040.00**
(iv)	Bend	Each	**595.00**
(v)	Door Cross	Each	**1156.00**
(vi)	Cowel	Each	**490.00**
(vii)	Tee	Each	**814.00**
3.41.3	150 mm –do-		
(i)	Socket	Each	**950.00**
(ii)	D/Door Y-Tee	Each	**2389.00**
(iii)	Bend	Each	**1195.00**
(iv)	Door Cross	Each	**2190.00**
(v)	Cowel	Each	**981.00**
(vi)	Tee	Each	**1785.00**
3.41.4	75mm with door		
(i)	Bend	Each	**502.00**
(ii)	Door Cross	Each	**706.00**
(iii)	Tee	Each	**656.00**
3.41.5	100mm with door		
(i)	Bend	Each	**662.00**
(ii)	Door Cross	Each	**1147.00**
(iii)	Tee	Each	**899.00**
3.41.6	150mm with door		
(i)	Bend	Each	**1372.00**
(ii)	Door Cross	Each	**2717.00**
(iii)	Tee	Each	**1967.00**
3.42	Providing and Fixing Spun –Cast Iron floor trap (Nahani) of (IS :3989 Mark) approved quality / make with socket down or hinged grating including cutting and making good the floor :-		

3.42.1	Size 100 x 100 mm	Each	**909.00**
3.42.2	Size 100 x 75 mm	Each	**711.00**
3.43	Add 10% extra Over Item No. 3.41 if the pipes are laid under suspended condition under the roof.		
	UPVC SOIL WASTE & RAIN WATER (SWR) PIPES		
3.44	Providing and Fixing Unplasticized Poly Vinyl Chloride (UPVC) SWR Pipes Type B for sciland waste discharge system (IS:13592 : 1992 Marked) of approved quality /make		
3.44.1	75 mm dia	Mtr.	**152.00**
3.44.2	110 mm dia	Mtr.	**305.00**
3.45	Providing and Fixing Unplasticized Poly Vinyl Chloride (UPVC) SWR Pipes fittings type B for sciland waste discharge system (IS:13592 : 1992 Marked) of approved quality /make		
3.45.1	Coupler		
(i)	75mm dia	Each	**65.00**
(ii)	110mm dia	Each	**98.00**
3.45.2	Repair Coupler		
(i)	75mm dia	Each	**69.00**
(ii)	110mm dia	Each	**113.00**
3.45.3	Single Push Fit Coupler		
(i)	75mm dia	Each	**66.00**
(ii)	110mm dia	Each	**115.00**
3.45.4	Bend 87.5 Dg.		
(i)	75mm dia	Each	**85.00**
(ii)	110mm dia	Each	**141.00**
3.45.5	Bend 45 Dg.		
(i)	75mm dia	Each	**71.00**
(ii)	110mm dia	Each	**120.00**
3.45.6	Door Bend 87.5 Dg.		
(i)	75mm dia	Each	**113.00**
(ii)	110mm dia	Each	**175.00**
3.45.7	Single 'Y'		
(i)	75mm dia	Each	**131.00**

(ii)	110mm dia	Each	**253.00**
3.45.8	Single 'Y' with Door		
(i)	75mm dia	Each	**170.00**
(ii)	110mm dia	Each	**316.00**
3.45.9	Double 'Y'		
(i)	75mm dia	Each	**178.00**
(ii)	110mm dia	Each	**321.00**
3.45.10	Double 'Y' with Door		
(i)	75mm dia	Each	**271.00**
(ii)	110mm dia	Each	**415.00**
3.45.11	Socket Plug		
(i)	75mm dia	Each	**56.00**
(ii)	110mm dia	Each	**69.00**
3.45.12	Pipe Clip		
(i)	75mm dia	Each	**16.00**
(ii)	110mm dia	Each	**28.00**
3.45.13	Double tee with Door		
(i)	75mm dia	Each	**304.00**
(ii)	110mm dia	Each	**520.00**
3.45.14	W.C. Connector		
	110mm dia	Each	**196.00**
3.45.15	W.C. Connector (Bend) with ring		
	110mm dia	Each	**205.00**
3.45.16	Single Tee		
(i)	75mm dia	Each	**109.00**
(ii)	110mm dia	Each	**191.00**
3.45.17	Single Tee with Door		
(i)	75mm dia	Each	**124.00**
(ii)	110mm dia	Each	**220.00**
3.45.18	Double Tee (4 Ways)		
(i)	75mm dia	Each	**193.00**
(ii)	110mm dia	Each	**334.00**

3.45.19	Multi floor trap		
	110mm dia	Each	**219.00**
3.45.20	Offset		
	110mm dia	Each	**256.00**
3.45.21	Extra Square Jali		
	110mm dia	Each	**34.00**
3.45.22	Extra Round Jali		
	110mm dia	Each	**24.00**
3.45.23	Extra 4 SWR Ring		
(i)	75mm dia	Each	**13.00**
(ii)	110mm dia	Each	**15.00**
3.45.24	Gully Trap		
	110mm dia	Each	**623.00**
3.45.25	Nahni Trap with Jali 4”		
	110mm dia	Each	**155.00**
3.45.26	Nahni Trap without Jali 4”		
	110mm dia	Each	**138.00**
3.45.27	Nahni Trap with Jali 3”		
	75 mm dia	Each	**139.00**
3.45.28	Nahni Trap without Jali 3”		
	75 mm dia	Each	**123.00**
3.45.29	Vent Cowel		
	75 mm	Each	**44.00**
3.45.30	Vent Cowel		
	110mm dia	Each	**54.00**
3.46	Providing & fixing Spun –Centrifugally cast Iron Pipes fitting (IS : 3989 Mark) in CM 1:4 with spun Yarn 100mm w/o door as per direction of Engineer-in-Charge.		
(a)	P Trap of standard size	Each	**851.00**
(b)	Long P Trap (900mm)	Each	**2450.00**
(c)	Long P Bend	Each	**839.00**
(d)	Long Floor Trap (600mm)	Each	**1505.00**

3.47	Removing & Re fixing of S.C.I. Pipe and Spun Pipe with C.I.clip as per direction of Engineer-in-Charge of following diameters :-		
(a)	100mm dia	R.Mtr	**165.00**
(b)	150mm dia	R.Mtr	**132.00**
3.48	Cleaning of sewer line of various sizes as per requirement at site of different sizes including 100mm,150mm & 225mm dia of Stoneware pipe as per direction of Engineer-in-Charge.	Rmt	**106.00**
3.49	Cleaning of C.I. Pipe Line of various sizse as per requirement at site of different sizes including 100mm,150mm & 225mm dia of C.I.Pipes as per direction of Engineer-in-Charge.	Rmt	**66.00**
3.50	Repairing of Cement Concrete Chambers of different size as per requirement at site and as per direction of Engineer-in-charge.	Each	**55.00**
3.51	**Providing & fixing Hubless Spun –Centrifugally cast Iron Pipes (IS : 15905 Mark) of Approved make in wall or in floor with M.S. Holder bat clamps in 10 X 10 X10 Cm. M-15 Grade concrete blocks, joints filled with CM 1:4 with spun yarn including cutting holes and making good the wall:-**		
3.51.1	**75 mm dia**	**Rmt**	**1025.00**
3.51.2	**100 mm dia**	**Rmt**	**1247.00**
3.51.3	**150 mm dia**	**Rmt**	**2036.00**
3.52	**Providing & fixing Hubless Spun –Centrifugally cast Iron Pipes fitting (IS : 15905 Mark) in CM 1:4 with spun Yarn.**		
3.52.1	**75mm w/o door**		
(i)	**Offset (130mm)**	**Each**	**511.00**
(ii)	**Shielded (SS) Coupling (With EPDM Rubber)**	**Each**	**418.00**
(iii)	**D/Door Y-Tee**	**Each**	**543.00**
(iv)	**Bend**	**Each**	**273.00**
(v)	**Cowel**	**Each**	**348.00**
(vi)	**Tee**	**Each**	**364.00**
3.52.2	**100mm W/o door**		
(i)	**Ofset (130mm)**	**Each**	**709.00**
(ii)	**Shielded (SS) Coupling (With EPDM Rubber)**	**Each**	**456.00**
(iii)	**D/Door Y-Tee**	**Each**	**1000.00**

(iv)	Bend	Each	388.00
(v)	Cowel	Each	449.00
(vi)	Tee	Each	639.00
3.52.3	150mm W/o door		
(i)	Shielded (SS) Coupling (With EPDM Rubber)	Each	639.00
(ii)	D/Door Y-Tee	Each	1861.00
(iii)	Bend	Each	820.00
(iv)	Cowel	Each	953.00
(v)	Tee	Each	1840.00
3.52.4	75mm with door, bend	Each	371.00
3.52.5	100mm with door		
(i)	Bend	Each	593.00
(ii)	Tee	Each	809.00
3.53	Mechanised Cleaning of sewer line of diameter upto 200mm with the help of trolley mounted electrical driven machine having flexible steel springs of length 20ft. Each which are rotated with the help of electric motor and clutch assembly.	mtr	140.00
3.54	Mechanised sewer line Cleaning upto diameter 100mm with the help of PU coated glass fibre rod mounted on a wheel with brush attachment.	mtr	97.00
3.55	Mechanised Cleaning and desilting of manhole/sewer line chambers With the help of mechanical bucket which gets open and close through levers.		
3.55.1	depth upto 0.5 mtr.	Each	89.00
3.55.2	Depth upto 1.0 mtr.	Each	165.00
3.55.3	Depth upto 2.0 mtr.	Each	308.00
3.55.4	Depth upto 3.0 mtr.	Each	446.00
3.55.5	Depth upto 5.0 mtr.	Each	710.00
3.56	Suplying and Fixing Ductile iron Man Hole Covers/storm water grating and gratingwith frame of various sizes weight and types and load bearing capacity as per EN -124 ,superior quality or equivalent, comlete as per Direction of Engineer In charge.	Kg.	165.00

3.57	**Providing and fixing circular air tight composite resin (FRP) manhole cover without frame of approved brand with grade designation C-250 HD as per EN-124 standard. Manhole Cover shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing in existing manhole frame etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal dia 0.56m**	**Each**	**10617.00**
3.58	**Providing and fixing circular air tight composite resin (FRP) manhole cover without frame of approved brand with grade designation D-400 EHD as per EN-124 standard. Manhole Cover shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing in existing manhole frame etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal dia 0.56m**	**Each**	**13271.00**

3.59	**Providing and fixing circular air tight composite resin (FRP) manhole cover with frame of approved brand with grade designation C-250 HD as per EN-124 standard. Manhole Cover & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal dia 0.56m**	**Each**	**12155.00**
b.	**Internal dia 0.60m**	**Each**	**13700.00**
3.60	**Providing and fixing circular air tight composite resin (FRP) manhole cover with frame of approved brand with grade designation D-400 EHD as per EN-124 standard. Manhole Cover & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal dia 0.56m**	**Each**	**14809.00**
b.	**Internal dia 0.60m**	**Each**	**19009.00**

3.61	**Providing and fixing rectangular air tight composite resin (FRP) Manhole Cover with frame of approved brand with grade designation C-250 HD as per EN-124 standard. Manhole Cover & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal Size 0.45 m x 0.60 m**	**Each**	**11347.00**
b.	**Internal Size 0.45 m x 0.90 m**	**Each**	**14319.00**
c.	**Internal Size 0.60 m x 0.90 m**	**Each**	**21186.00**
d.	**Internal Size 0.90 mx 1.20 m**	**Each**	**57584.00**
3.62	**Providing and fixing rectangular air tight composite resin (FRP) Manhole Cover with frame of approved brand with grade designation D-400 EHD as per EN-124 standard. Manhole Cover & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal Size 0.45 m x 0.60 m**	**Each**	**14839.00**
b.	**Internal Size 0.45 m x 0.90 m**	**Each**	**18650.00**
c.	**Internal Size 0.60 m x 0.90 m**	**Each**	**24678.00**
d.	**Internal Size 0.90 mx 1.20 m**	**Each**	**70157.00**

3.63	**Providing and fixing rectangular air tight composite resin (FRP) Grating with frame of approved brand with grade designation C-250 HD as per EN-124 standard. Grating with & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. completein all respect as directed by Engineer in Charge.**		
a.	**Internal Size 0.45 m x 0.45 m**	**Each**	**10590.00**
b.	**Internal Size 0.50 m x 0.60 m**	**Each**	**13591.00**
c.	**Internal Size 0.60 m x 0.75 m**	**Each**	**20885.00**
d.	**Internal Size 0.90 m x 0.60 m**	**Each**	**22303.00**
e.	**Internal Size 0.90 m x 0.90 m**	**Each**	**36730.00**
f.	**Internal Size 1.20 m x 0.60 m**	**Each**	**32331.00**
3.64	**Providing and fixing rectangular air tight composite resin (FRP) Grating with frame of approved brand with grade designation D-400 EHD as per EN-124 standard. Grating & Frame shall have top abrasion resistant layer conforming to IS-15658:2006 and permissible permanent set value of 1.9mm after the application of 2/3 test load in five continuous application as per EN-124, the testing shall be performed in fully equipped and NABL certified test lab with random batch test (2% of lot size) as per testing procedure stated in EN-124 standard including fixing with M-20 concrete etc. complete in all respect as directed by Engineer in Charge.**		
a.	**Internal Size 0.45 m x 0.45 m**	**Each**	**14403.00**
b.	**Internal Size 0.50 m x 0.60 m**	**Each**	**18759.00**
c.	**Internal Size 0.60 m x 0.75 m**	**Each**	**28149.00**
d.	**Internal Size 0.90 m x 0.60 m**	**Each**	**31970.00**
e.	**Internal Size 0.90 m x 0.90 m**	**Each**	**46300.00**
f.	**Internal Size 1.20 m x 0.60 m**	**Each**	**43227.00**

CHAPTER : S-4
TUBE WELL

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
4.1	Construction of tube-well from ground level and upto 100 Meter depth and above sizes in all types of soils in alluvium strata by “bailing” method and without gravel packing as per IS: 2800 (Part - I& II) 1979 specifications (The work includes formation of cavity well at bottom by development with appropriate air compressor or bailer pumping and also lowering of casing pipe but excluding cost of the casing pipe. The tube-well should have a throughout bore as per nominal bore of casing pipe. The work would be completed after obtaining sand free water.)		
4.1.1	100 mm dia Nominal bore.	R Mtr.	254.00
4.1.2	125mm-do-.	R Mtr.	290.00
4.1.3	150mm-do-	R Mtr.	315.00
4.2	Construction of Tube-well upto 100 Meter depth and above in all type of rocks by DTH system and over burden, to accommodate casing pipe of following sizes in all types of soils and over burden including lowering of casing pipes, but excluding cost of casing pipes as per IS : 2800 (Part I & II) 1979 specifications. The work would be completed after obtaining sand free water. The tube well should have a throughout bore as per nominal dia of casing pipe:		
4.2.1	100 mm dia Nominal bore.	R.Mtr.	345.00
4.2.2	125mm-do-.	R.Mtr.	363.00
4.2.3	150mm-do-	R.Mtr.	605.00
4.2.4	200 mm –do-	R.Mtr.	908.00
4.3	Construction of *tube-well* from ground levels and upto 100 Meter depth and above to accommodate housing and assembly pipe of following sizes in all types of alluvium strata by percussion/ rotary drilling method and with gravel as per IS:4097-1967 and packing as per IS:2800 (Part I -& II) 1979 as amended upto date (the work includes the cost of gravel & its primary packing and packing during development, lowering of housing & strainer assembly pipes, with supply and wrapping of coir-rope, wherever necessary, for arresting fine sand particles. The work will not include cost of housing pipe and strainer pipe assembly and development work, but work would be completed after obtaining sand free water).		
4.3.1	150 mm Nominal bore.	R Mtr.	968.00

4.3.2	200mm –do-.	R Mtr.	1331.00
4.3.3	250mm –do-.	R Mtr.	1694.00
4.4	Development of tube well as per IS specification using suitable compressor to give sand free water for required yield of the gravel packed tube well.	Hr.	545.00
4.5	Supply of *ERW M.S. black casing pipe* ISI marked (IS:4270/1992) of grade Fe410 of following sizes at site of work.		
4.5.1	100 mm Nominal bore of pipe	Mtr.	747.00
4.5.2	125 mm Nominal bore of pipe	Mtr.	930.00
4.5.3	150 mm Nominal bore of pipe	Mtr.	1113.00
4.5.4	200 mm Nominal bore of pipe	Mtr.	1571.00
4.5.5	250 mm Nominal bore of pipe	Mtr.	2567.00
4.6	Supply of *strainer pipes* made of ERW M.S. black pipe ISI mark of following sizes at the site of work including required size of slotting as per IS:8110-1985.		
4.6.1	150 mm Nominal Bore.	Mtr.	1388.00
4.6.2	200mm-do-.	Mtr.	1846.00
4.6.3	250mm-do-.	Mtr.	2842.00
4.7	Testing verticality of tube-well by plumbing system and yield test and draw down test by pumping system as per IS: 2800 (Part II) 1979.	Each	7623.00
4.8	S & F *tube well* cover of M.S.sheet (6mm thick) with nuts and bolts complete for casing size:		
4.8.1	100mm dia	Each	169.00
4.8.2	125mm dia	Each	206.00
4.8.3	150mm dia	Each	254.00
4.8.4	200mm dia	Each	339.00
4.8.5	250mm dia	Each	363.00
4.9	S & F M.S. *clamp,* set of 50 ×6mm flat iron with nuts and bolts etc. for holding the riser pipe assembly of submersible pump set.	Each	242.00
4.10	Supply of 50mm dia G.I. medium class pipe(each length 5.8 m to 6.0m) tripod as per drawing and design with chain pulley of. 3MT capacity	Each	10890.00
4.11	Installation of submersible motor pump set in Tube-well/open well complete (labour charges only) including transportation of tripod, chain pulley block & any other material required for lowering purpose.	Each	3388.00

4.12	P & F 60 cm. wide *M.S. ladder* in open well made of flat iron 32×5mm and 10mm dia bars @ 200mm c/c with angle iron hold fasts at suitable interval complete as per drawing.	Mtr.	206.00
4.13	Installation of India mark *II hand pump* set, complete on existing plate form.	Each	847.00
4.14	Construction of 185 cm. dia platform as per approved design and drawing of UNICEF.	Each	2662.00
4.15	Supplying of India mark *II hand pump* set complete marked with cylinder and ten connecting rods etc. IS 1239	Each	7865.00
4.16	Supplying and installation of 32 mm dia *G.I. pipe* medium class IS 1239 marked in 3m length including all accessories and fittings required for installation of hand pump.	Mtr.	303.00
4.17	Supplying and installation of *submersible pumping motor pump sets* I.S.I. Marked (IS : 8034-1989) of approved make, making connection suitable for T.W./D.C.B./Open well. The job includes screwing and welding of flanges on G.I. riser pipes, installation of complete fitting and accessories, with riser pipes, jointing of electrical cables and other connections making switch board electrical connections, all labour for testing of submersible pump set and supply of water to water mains, complete in all respect (Refer Electrical BSR)		
4.18	For item of earthing refer to Electrical BSR .		
4.19	For item of submersible cable refer to Electrical BSR.		
4.20	for Item of panel board refer to Electrical BSR		
4.21	Providing & lowering of G.I. Pipes, flange pipe including rubber washer and nuts of 8 mm dia complete in all respect I.S.1239 Marked.		
4.21.1	B Class 50 mm dia	R. Mtr.	453.00
4.21.2	B Class 80 mm dia	R. Mtr.	655.00
4.21.3	B Class 100 mm dia	R. Mtr.	865.00
4.22	Un-lowering of submersible pump set with all piping from a tube-well/ open-well complete (labour charge only) including transportation of tripod, chain, pulley-block and other materials required and lowering of repaired/ new pump in the tube-well / open-well complete in all respect.	Each Job	4235.00

RAIN WATER HARVESTING

4.23	Construction of tubewell up to 100m depth and above to accomodate housing and assembly pipe in all types of alluvium strata by rotay drilling method with "code of practice for construction and testing of tubewells (IS:2800 (Part-II) : 1991 and IS:2800 (Part-II) :1979 both amended up to date) with supply of bail plug if necessary.The work will not include cost of housing pipe and strainer pipe assembly cost, gravel and development work. Work would be deemed completed only after obtaining sound free water during pumping.		
4.23.1	150 mm nominal bore	Rm	726.00
4.23.2	200 mm nominal bore	Rm	1089.00
4.23.3	250 mm nominal bore	Rm	1452.00
4.24	Construction of Recharge Shaft in filter pit including Earth work Excavation , lifting excavted earth with all lift and 50 m lead and centering & shuttering evenever required.		
4.24.1	Diameter 900 mm	Rm	242.00
4.24.2	Diameter 750 mm	Rm	212.00
4.25	Supply & fixing Pre-cast RCC Rinngs at the mouth of recharge shaft of M20 mix with 40 mm thickness		
4.25.1	Diameter 900 mm	Rm	696.00
4.25.2	Diameter 750 mm	Rm	605.00
4.26	Supply & lowering air line of 25mm dia, G.I. pipe 'B' class having perforation 3 to 5mm size all arrow of G.I. pipe in Zig-Zag fashion & making air vent at top including the cost of Tee-Nipples, Elbow, Socket etc.	Rm	273.00
4.27	Supply and lowering vent air pipe 63mm dia rigid PVC pipe(IS 4985) mark class II (4kg per cm2) with required all necessary fittings including perforation of 3 to 5 mm size all around the pipe at a distance of 180 to 220mm in zig zag fashion and making air vent at top complete as per direction of Engineer incharge.	Rm	140.00
4.28	Supply & lying River-gravels of following sizes in recharge shaft, filter pit after screening & washing from kachha shift without deduction of voids		
4.28.1	Size 9 to 12 mm	Cum	1815.00
4.28.2	Size 3 to 5 mm	Cum	2662.00
4.29	Supply & laying of clean, clear sharp river sand of size 1 to 2mm coarse, without deduction of voids.	Cum	3388.00

CHAPTER H-1
SUPPLY OF MANURE, INSCTICIDES, PLANTS, POTS, T & P ETC.

Note : The rates are inclusive of all taxes etc. and for estimation purpose only.

Chapter Code No.	Description	Unit	Final BSR Rate 2019
1	2	3	4
1.1	Supply of dry manure including loading unloading,transportation & stacking at site.		
1.1.1	Farm yard manure(organic)	Cum.	720.00
1.1.2	Compost	Cum.	960.00
1.1.3	Goat dungmanure.	Cum.	1200.00
1.1.4	**Vermi compost**	**Per 50 kg Bag**	**250.00**
1.1.5	**Neemcack**	**Per 50 kg Bag**	**1250.00**
1.2	Supply of Chemical Fertilizers at store in bags weighing not less than 50 kg each including loading unloading & transportation		
1.2.1	Urea	Per 50 kg Bag	400.00
1.2.2	DAP	Per 50 kg Bag	1500.00
1.2.3	Super Phosphate	Per 50 kg Bag	350.00
1.2.4	Murrate of Photash	Per 50 kg Bag	450.00
1.2.5	Straw Meal	Per 50 kg Bag	1080.00
1.2.6	**Bonemeal**	**Per 50 kg Bag**	**1800.00**
1.2.7	**Calcium Nitrate**	**Per 50 kg Bag**	**2500.00**
1.3	Supply & Stacking good soil of earth at site complete including loading unloading & Transportation etc.		
1.3.1	Good soil of earth	Cum.	240.00
1.4	Supply of insecticides & Pesticides at store in dust/liquid form complete.		
1.4.1	Methyl Prathion 2%	Per 25 kg Bag	500.00
1.4.2	Karathion (Liquid)	Per 250 ml	600.00

1.4.3	Metacid(Liquid)	Per Litre	444.00
1.4.4	Rogor(Liquid)	Per Litre	325.00
1.4.5	Cholorophyriphos 20 Ec.	Per Litre	400.00
1.4.6	Bauiston 50%	Per 100 Grams	120.00
1.5	Supply of Garden Tools & implements at store complete.		
1.5.1	Spade weighing 1.5kg with Handle	Each	175.00
1.5.2	Khurpa weighing 500gm with Handle	Each	90.00
1.5.3	Gurvana weighing 800 gm with Handle(Kamani Steel)	Each	108.00
1.5.4	Hedge Shear(Kamani Steel)	Each	720.00
1.5.5	Secaetur	Each	360.00
1.5.6	Garden Rack with Handle	Each	192.00
1.5.7	Pick Axe	Each	180.00
1.6	Supply of different sized pots at site store including loading unloading & transportation etc. complete.		
1.6.1	Cement pot 300 mm size	Each	150.00
1.6.2	Cement pot 350 mm size	Each	216.00
1.6.3	**Cement pot 375 mm size**	**Each**	
1.6.4	**Cement pot 450 mm size**	**Each**	
1.6.5	**Cement pot 525 mm size**	**Each**	
1.6.6	Asbestos pot 350 mm size	Each	120.00
1.6.7	**Asbestos pot 400 mm size**	**Each**	
1.6.8	**Asbestos pot 450 mm size**	**Each**	
1.6.9	**Asbestos pot 600 mm size**	**Each**	
1.7	Supply of different sized earthen pots at site/store including loading unloading & transportation etc.complete		
1.7.1	Earthen pot 150 mm size	Each	18.00
1.7.2	Earthen pot 225 mm size	Each	30.00
1.7.3	Earthen pot 300 mm size	Each	43.00
1.7.4	**Earthen pot 350 mm size**	**Each**	**125.00**
1.7.5	**Earthen pot 225 mm size(Kundi shape)**	**Each**	**30.00**
1.7.6	**Earthen pot 300 mm size(Kundi shape)**	**Each**	**45.00**
1.7.7	**Earthen pot 350 mm size(Kundi shape)**	**Each**	**125.00**
1.8	Supply of different sized lawn mower including transportation etc. as store complete.		

1.8.1	300 mm size Hand Lawn Mower(Roller Type)	Each	4320.00
1.8.2	350 mm size Hand Lawn Mower(Roller Type)	Each	4800.00
1.8.3	**300 mm Electric Lawn Mower**	**Each**	**18000.00**
1.8.4	**350 mm Electric Lawn Mower**	**Each**	**20000.00**
1.8.5	**400 mm Electric Lawn Mower**	**Each**	**22000.00**
1.8.6	**450 mm Electric Lawn Mower**	**Each**	**25000.00**
1.8.7	**500 mm Electric Lawn Mower**	**Each**	**28000.00**
1.8.8	**Electric Lawn Mower sarfex 16 EL**	**Each**	**30000.00**
1.8.9	**Electric Lawn Mower sarfex 18 EL**	**Each**	**36000.00**
1.8.10	**Electric Lawn Mower sarfex 21 EL**	**Each**	**40000.00**
1.9	Supply of different size Tokaras/Tokaries etc at store complete		
1.9.1	Big size(600mm dia)	Each	150.00
1.9.2	Small size(450 mm dia)	Each	90.00
1.9.3	**Jute Band(Ssutli)**	**Per kg**	**140.00**
1.9.4	**Broom (Bamboo)**	**Per kg**	**50.00**
1.9.5	**Anti Bird Net**	**Running Mtr.**	**25.00**
1.9.6	**Shed Net 50%**	**Square Mtr.**	**25.00**
1.9.7	**Shed Net 75%**	**Square Mtr.**	**30.00**
1.9.8	**Shed Net 90%**	**Square Mtr.**	**35.00**
1.10	Supply of PVC hose pipe of different sizes at store including transportation etc. complete.		
1.10.1	50 mm dia	R/Mtr.	165.00
1.10.2	38 mm dia	R/Mtr.	140.00
1.10.3	25 mm dia	R/Mtr.	80.00
1.10.4	17 mm dia	R/Mtr.	65.00
1.10.5	12.5 mm dia	R/Mtr.	45.00
1.11	Supply of different varities of tree/shrubs/climbers according to height and age of the plant at side including loading/unloading & transportation etc.complete		
1.11.1	Different flowering plants 900mm to1500mm	Each	50 to 150

1.11.2	Different varieties of shrubs 600mm to 900 mm	Each	25 to 160
1.11.3	Different varieties of climbers 900mm to 1500 mm	Each	48 to 144
1.12	Supply of different varieties of bougainvillea in ply bags/earthen pot according to height growth and age of plant at site including loading unloading and transportation etc. complete		
1.12.1	Local variety of bougainvillea 600mm to 900mm	Each	25 to 60
1.12.2	Different varieties of bougainvillea 600mm to 900mm	Each	60 to 120
1.13	Supply of different varieties of rose plants in poly bag/ earthen pot according to height growth and age of the plant at site including loading unloading and transportation etc. complete		
1.13.1	Desi rose / root stock plant	Each	40 to 50
1.13.2	**Ganganagari rose plant in poly beg**	**Each**	**80 to 125**
1.13.3	Different varieties of grafted rose plants in poly bag	Each	84 to 102
1.13.4	**Different varieties of grafted rose plants in poly pot/earthen pot**	**Each**	**200 t0 250**
1.14	Supply of different varieties of palms in poly bags/earthen pot according to height growth and age of plant at site including loading unloading and transportation etc. complete		
1.14.1	**Raffish palm 2 to 2.2 feet**	**Each**	**350.00**
1.14.2	**Deshi palm 2 to 3 feet**	**Each**	**150.00**
1.14.3	**China palm 1 to 1.5 feet**	**Each**	**125.00**
1.14.4	**Arica palm many branches 5 to 6 feet**	**Each**	**400 to 500**
1.14.5	**Bottle palm 5 to 8 feet**	**Each**	**700 to 900**
1.14.6	**Fox tale palm 5 to 8 feet**	**Each**	**900 to 1200**
1.14.7	**Fish tale palm 5 to 8 feet**	**Each**	**900 to 1200**
1.14.8	**T cycuss 3 feet 5 feet**	**Each**	**900 to 1200**
1.15	Supply of different varieties of indoor plants in poly bags/earthen pot according to height growth and age of plant at site including loading unloading and transportation etc. complete		

1.15.1	Deshi crotons	Each	50 to 60
1.15.2	Selective varieties of crotons 2 to 3 feet	Each	72 to 180
1.15.3	**Crotons Bangalore 1.5 to 2 feet**	**Each**	**250 to 2300**
1.15.4	Rubber plants etc	Each	120 to 150
1.15.5	DAFFEN Bachia **1.5 to 2 feet**	Each	100 to 250
1.15.6	Song of India **1.5 to 2.5 feet**	Each	100 to 200
1.15.7	Aglonmia **1.5 to 2 feet**	Each	110 to 175
1.15.8	Drassina **1 to 2 feet**	Each	100 to 240
1.15.9	Sin gonium with moss stick **2 to 3 feet**	Each	125 to 250
1.15.10	Cycussevelta 50mm to 75 mm dia	Each	1200
1.15.11	Furcaria gignita 50mm to 75 mm	Each	900 to 1000
1.15.12	Different fycus plants tolairy 100mm to 125 cm height	Each	800 to 1000
1.15.13	Different champa 125 cm to 215 cm height	Each	400 to 900
1.16	**Supply of different varieties of blubs including loading unloading and transportation etc. complete**		
1.16.1	**Supply of different varieties of caladium blubs**	**Each**	**18.00**
1.16.2	**Supply of different varieties of amrilis lilly bulb**	**Each**	**25.00**
1.16.3	**Supply of tube rose bulbs**	**Each**	**12.00**
1.16.4	**Supply of hemaniths bulb**	**Each**	**28.00**
1.16.5	**Supply of Asiatic Lilly bulb**	**Each**	**30.00**
1.16.6	**Supply of oriental lilly bulb**	**Each**	**42.00**
1.16.7	**Supply of different variety of gladioli bulb**	**Each**	**18.00**
1.17	**Supply of flower Garlands & Flower petal's including transportation at site.(Complete)**		
1.17.1	**Merigold Garland A Class (Big flowers) 2 feet long**	**Each**	**75.00**
1.17.2	**Merigold Garland B Class (Small flowers) 1.5 feet long**	**Each**	**50.00**
1.17.3	**Green Garland (Ashok Leaf) 2 feet long**	**Each**	**15.00**

1.17.4	Rose Garland 1.5 feet long	Each	125.00
1.17.5	Jalusi Garland(VIP A Class) Double fold	Each	500.00
1.17.6	Jalusi Garland(VIP B Class) Single fold	Each	400.00
1.17.7	Mogra Garland 1.5 feet long	Each	125.00
1.17.8	Kunj Garland 1.0 feet long	Each	150.00
1.17.9	Rose petal's	Per Kg.	400.00
1.17.10	Merigold petal's	Per Kg.	250.00
1.17.11	Crisenthimum petal's	Per Kg.	300.00
1.17.12	Chandine flower	Per Kg.	300.00
1.17.13	Mogra Flower	Per Kg.	450.00
1.17.14	Kunj Flower	Per Kg.	475.00
1.18	Supply of Cut Flower at site including transportation Complete		
1.18.1	Tata Rose	Each	15.00
1.18.2	Carnation	Each	18.00
1.18.3	Gladioli	Each	20.00
1.18.4	Tube rose single	Each	12.00
1.18.5	Tube rose double	Each	15.00
1.18.6	Ilium Asiatic	Each	75.00
1.18.7	Ilium Oriental	Each	125.00
1.18.8	An thorium	Each	75.00
1.18.9	Arched Mix	Each	40.00
1.18.10	Zarbera	Each	20.00
1.18.11	Dasy	Each	25.00
1.18.12	Chrysanthemum	Each	20.00

1.18.13	Candytuft	Each	4.00
1.18.14	Aster	Each	6.00
1.18.15	Lotus	Each	40.00
1.19	Supply of Flower Bucket & flower Bunch at site including transportation (Complete)		
1.19.1	Flower Bunch / Bucket (20 Roses)	Each	500.00
1.19.2	Flower Bunch / Bucket (30 Roses, Carnation)	Each	700.00
1.19.3	Flower Bunch / Bucket (7 Lily & 15 Roses)	Each	950.00
1.19.4	Flower Bunch / Bucket (10 Lily, 20 Roses & Carnation Mix)	Each	1800.00
1.19.5	Flower Bunch / Bucket (15 Lily)	Each	1650.00
1.19.6	Flat Flower Bucket (20 Mix Flower's)	Each	490.00
1.19.7	Flat Flower Bunch (25 Mix Flower's) lily Carnation Rose	Each	1200.00
1.19.8	Flower Bunch / Bucket (Small 15 Flower Stic)	Each	400.00
1.19.9	Flower Bunch / Bucket (10 Roses, 5 Carnation)	Each	350.00
1.19.10	Flower Bunch / Bucket (15 Roses)	Each	280.00

CHAPTER H-2
MAINTENANCE OF GRADENS

The rates are inclusive of all taxes etc. and for estimation purpose only.

Chapter Code No.	Description	Unit	Final BSR Rate 2018
1	2	3	4
2.1	Maintenance of lawn	-	
2.1.1	Spreading of manure , good earth in required thickness (cost of manure and good earth to be separately)	cub	24.00
2.1.2	Irrigation of lawn	Per hect. Per day	1440.00
2.1.3	Uprooting weeds from the lawn plots and trenched area after 10 to 15 days of its flooding with including disposal of uprooted vegetation	100 sqm	240.00
2.1.4	Moving manually / with power machines		
(a)	Manually	Per hect	1560.00
(b)	with power machine	Per hect	1200.00
2.1.5	Maintenance of shrubbery or hedges cutting including disposal of rubbish with all leads and lifts	100 sqm	480.00
2.1.6	Sweeping / cleaning of gardens roads lawns paths and disposal of all rubbish	100sqm	3.50
2.1.7	Security of the garden per gate for 24 hours in 3 shift with the exilitary persons	Per day 24 hours	19440.00
2.1.8	Maintenance /fitting of permanent ornamental pots/ seasonal pots		
(a)	Mixing earth and manure in per portion as specified or directed for pots	cum	42.00
(b)	Filling and planting of sapling and watering of pots	Per 100 pots per day	180.00
(c)	Weeding hoeing stacking and coloring of pots material will be supplied by the department	Per 100 pots per day	180.00
2.1.9	Preparation of beds for budging and shrubbery by excavating 60cms. Deep and trenching the excavated base to a further depth of 30 cms . refilling the excavated earth after breaking clods and mixing with manure in ratio of 8:1 (8 part of stacked volume)	Each	24.00

2.2	Digging pits in ordinary soil and refilling the same with the excavated earth mixed with manure in the ratio of 2:1 by volume (2parts of stacked volume of earth after reduction by 20% 1 part of stacked volume of manure after reduction by 8%) flooding wit	each	24.00
2.3	Planting of annual saplings watering in the bed	sqm	2.40
2.4	watering of group plantation in bed	sqm	0.30
2.5	Preparation of **edging** lawn	Rmt	0.72
2.6	Group plantation in the bed at specified place	sqm	2.40

CHAPTER H-3
DEVELOPMENT OF GRADENS

The rates are inclusive of all taxes etc. and for estimation purpose only.

Chapter Code No.	Description	Unit	Final BSR Rate 2018
1	**2**	**3**	**4**
3.1	Laying of lawns including ploughing leveling breaking of clods manuring and removal of stones etc. (cost of manure or extra good earth to be paid separately)maintenance of lawn foe 30 days or more till the grass forms a thick lawns free from weeds and fit for moving in the duration of planting if lawn die some where the contractor will replant it at his own cost		
3.1.1	In rows 15 cm apart in either direction	100 sqm	3000.00
3.1.2	in rows 7.5 cm apart in either direction	100sqm	5500.00
3.1.3	in rows 5 cm apart in either direction	100 sqm	6600.00
3.2	Renovating lawns including weeding cheeping the grass forking the ground top dressing with manure . Mixing tthe same with forked soil watering maintaining the lawn for 30 days for till the grass grass forms a thick lawns free from weeds and fit for moving in the duration of planting if lawn die some where the contractor will replant it at his own cost	100sqm	4500.00
3.3	Planting of trees /shrubs/ hedge and climbers plants at desired distance in row and watering	100 plants	30.00
3.4	Preparation of different sized bed as directed	100 sqm	456.00

CHAPTER H-4
MAINTENANCE OF RODS SIDE PLANTATION

The rates are inclusive of all taxes etc. and for estimation purpose only.

Chapter Code No.	Description	Unit	Final BSR Rate 2018
1	2	3	4
4.1			
(a)	Preparation of soil including cleaning & removing of unwanted sharbs removeal of stones & Garbage.	Per 100 Sqm	150.00
(b)	Digging of pits size 60x60x60 cms. Including removal of stones Manuring applicable of insecticides & watering atleast 15 litre per plant after plan ting.	Per Pit	30.00
(c)	Loading & unloading of plant from Deptt.Nursery to site perkm.	Per 1000 Plants	150.00
(d)	Preparation of Thavala sizing 60 cms. Diametre & 10 cms. Deep once in a month.	Per Thawala	1.20
(e)	Inter cultural operations hoeing & weeding etc. in 90 cms. Dia & 15 cms. Deep.	Per Plants	2.00
(f)	Watering of plants with the own tanker i.e. at least 20 literwater for plant at one time	Per Plants	1.40
(g)	Transportation of tree guards from Depot to the plantation site including loading & unloading.	Per Tree Guard	18.00
(h)	Security of plants & fencing etc.	Per Plant Per month	0.90
(i)	Prunning & trumming and cutting of old big trees on the road side The contractor will deposit the cut wood in the Deptt.	Per tree	120.00
4.2	Mantencance of Plants by the contractor including of pits/bids watering preparation of Thavala Hoeing weeding etc. & application of insecticides etc. & security if the plant die during maintenance contractor has to replace same height plant at his own cost.	Per Plant Per month	18.00

CHAPTER H-5

DETAIL OF PLANTATIONS AND MAINTENANCE OF PLANTS ON ROAD SIDE FOR FIVE YEARS

The rates are inclusive of all taxes etc. and for estimation purpose only.

Chapter Code No.	Description	Unit	Final BSR Rate 2018
1	2	3	4
5.1	Supply & Planting of shady plants (Neem. Peepal, Goolar, Jamun, Imli or other specified) along road side including the following activities:- 1. Preparation of soil including cleaning & removing of unwanted shrubs removal of stones & garbage. 2. Supply of plant at site of two years of age & height more than 1.50 Mtr. 3. Supply of dry manure(Farm yard manure organic) 4. Supply of Insecticides. 5. Digging of pits size 60x60x60 cms. Including removal of stones. Manuring application of insecticides & watering at least 1.5 Liter per plant after planting 6. Half brick Circular tree guard in second class bricks, internal dia 1.25 meter and height 1.20 meter above ground and 0.20 meter above ground and 0.20 meterr below ground. Bottom two courses laid dry and top three couses in cement mortar 1:6 (1 Cement : 6 Sand) and the intermediate courseses being in dry honey comb masonry. As per design complete. 7. Watering to plants 8. Maintance of plants by the contractor including pits, preparation of Thavala Hoeing weeding etc & application of insecticiedes etc. & security. If the plant die during maintence contractor has to rplace same heingh plant at his own cost.	Per Plant	1886.00
5.2	Supply & Panting of ornamental plants (Kachnar, Amaltas, Gulmohar, Arjun, Kala, Siris, Jacranda etc. or other specified) along road side including the following activities :- 1. Preparation of soil including cleaning & removing of unwanted shrubs removal of stones & garbage 2. Supply of plant at site of two years of age & height more than 1.50 Mtr.	Per Plant	2198.00

3. Supply of dry manure(Farm yard manure organic)
4. Supply of Insecticides
5. Digging of pits size 60x60x60 cms. Including removal of stones. Manuring application of insecticides & watering at least 1.5 liter per plant after planting.
6. Half brick Circular tree guard in second class bricks. Internal dia 1.25 meter and height 1.20 meter above ground and 0.20 meter below ground. Bottom two courses laid dry and top three courses in cement morter 1:6(1 Cement : 6 Sand) and the intermediate courses being in dry honey comb masonry. As per design complete.
7. Waterring to Plants.
8. Maintenance of plants by the contractor including pits. Preparation of Thavala Hoeing weeding etc. & application of insecticides etc. & security. If the plant die during maintencance contractor has to replace same height plant at his own cost.

5.3	Planting & Maintenance of shrubs. Planting of shrubs in central verge 1. Preparation of soil including cleaning & removing of unwanted shrubs removal of stones & garbage. 2. Supply of plant at site like Bouganvillia & Kaner. 3. Supply of dry manure(Farm yard manure organic) 4. Supply of Insecticides 5. Digging of pits size 60x60x60 cms. Including removal of stones. Manuring application of insecticides & watering at least 1.5 liter per plant after planting. 6. Waterring to Plants. 7. Maintenance of plants by the contractor including pits. Preparation of Thavala Hoeing weeding etc. & application of insecticides etc. & security. If the plant die during maintencance contractor has to replace same height plant at his own cost.	Per Plant	1069.00

Detail of Plantation and Maintenance of Plants on Road sides

Ch. Code	Description	Unit	Qty.	Rate (Rs.)	Cost (Rs.)
1	Supply & Planting of shady plants (Neem, Peepal, Goolar, Jamun, Imli or other specified) along road side including the following activities :-				
	1. Preparation of soil including cleaning & removing of unwanted shrubs removal of stones & garbage.	P.Sqm	1.00	1.00	1.00
	2. Supply of plant at site of two years of age & height more than 1.50 Mtr.	P.Plant	1.00	100.00	100.00
	3. Supply of dry manure (Farm yard manure organic)	P.Cum	0.10	400.00	40.00
	4. Supply of Insecticides.	L.S.			25.00
	5. Digging of pits size 60x60x60 cms. Including removal of stones, Manuring application of insecticides & watering at least 15 liter per plant after planting.	P.Plant	1.00	20.00	20.00
	6. Half brick Circular tree guard in second class bricks, internal dia 1.25 meter and height 1.20 meter above ground and 0.20 meter below ground, bottom two courses laid dry and top three courses in cement mortar 1:6 (1 Cement : 6 Sand) and the intermediate courses being in dry honey comb masonry, as per design complete.	P.Plant	1.00	773.00	773.00
	7. Watering to plants.	P.Tanker	1.00	300.00	300.00
	8. Maintenance of plants by the contractor including pits, preparation of Thavala Hoeing weeding etc. & application of insecticides etc. & security. If the plant die during maintenance contractor has to replace same height plant at his own cost.	P.Plant Per Month	60.00	2.50	150.00
	Total	P.Plant			**1409.00**
	Less depreciated cost of tree guards @ 20% of cost of tree guards	P.Plant			-154.60
	Sub Total	P.Plant			**1254.40**
	Add Contractors profit @ 10%	P.Plant			125.00
	Total cost Per Plant	P.Plant			**1379**

Sr No	Description	Unit	Quantity	Rate Rs	Cost Rs
1	**Half Brick Circular Tree Guard, in 2nd Class Brick, internal diametre 1.25 metres, and height 1.2 metres, above ground and 0.20 metre below ground**				

	Half brick circular tree guard, in 2nd class brick, internal diametre 1.25 metres, and height 1.2 metres, above ground and 0.20 metre below ground, bottom two courses laid dry, and top three courses in cement mortar 1:6 (1 cement 6 sand) and the intermediate courses being in dry honey comb masonry, as per design complete				
	***Unit* = Each**				
	***Taking output* = one tree guard**				
	a) Labour				
	Mate	day	0.050	200.00	10.00
	Mason	day	0.250	250.00	62.50
	Mazdoor	day	0.250	180.00	45.00
	b) Material				
	Brick 2nd class including carriage	each	230.000	2.30	529.00
	Cement mortar 1:6	cum	0.025	2236.00	55.90
	c) Overhead charges @ 10 % on (a+b)				70.24
	Rate per tree Guard = a+b+c				772.64
				say	***773.00***

Sub-analysis (D)	**Cement mortar1:6 (1cement :6 sand)**				
	***Unit* = 1 cum**				
	***Taking output* = 1 cum**				
	a) Materials				
	Cement	MT	0.29	5300.00	1526.40
	Sand	cum	1.20	450.00	540.00
	b) Labour				
	Mate	day	0.04	200.00	8.00
	Mazdoor	day	0.90	180.00	162.00
	Total Material and Labour = (a+b)				2236.00

OFFICE OF THE CHIEF ENGINEER, PWD (B & R), RAJASTHAN, JAIPUR.

CIRCULAR No. 19 (Building 6)

Sub : General Instructions to be followed while preparing detailed estimates.

It has been observed that while submitting estimates for Technical sanction instructions contained in Section 5 of the Manual of Orders of PWD/B&R and those issued by previous Circular are not followed strictly, it is therefore, enjoined upon all the detailed estimates for works costing more than Rs. 3.00 lacs and requiring technical sanction by Addl. Chief Engineer and submitted to this office should contain the following schedules and details :

1. SCHEDULES :

1.1 **Schedule-I (Technical Report) :**
Proforma at Annexure 'A' may be used for submission of a detailed report which should be full & comprehensive and preferably written by the Executive Engineer.

1.2 **Schedule-II (General Abstract) :**
It should indicate abstract cost of the sub-work included in the sanctioned proposal and provide for the increase in the market rates as may be approved by Superintending Engineer.

1.3 **Schedule-III**
Detailed estimates of each sub-work with abstract of cost and details of quantities.

1.4 **Schedule-IV (Drawing) :**

(a) The drawings accompanying the detailed estimates should be as follows :-

(i) A site plan showing the situation of the proposed building with reference to others, the various structure in proximity to the intended site, the North point, the prevailing direction of the wind and other matters capable of graphic delineation which may have influenced the selection.

(ii) A ground floor plan or plans of the building and plans of the foundations and of various storeys as required.

(iii) Section through the buildings of such number and in such directions are necessary to exhibit the intended form and dimensions of every part.

(iv) Such elevations as are necessary.

(v) A drawing or drawings showing the general arrangement of the floor and roof and the distribution of timbers, iron work etc. and such working drawing as will enable the officer responsible for the project to judge the details.

(b) The drawings should bear the signatures of authority approving the plan and other details duly attested by Superintending Engineer.

(c) The drawings should clearly indicate the proposed works, existing portions, dismantling work (if required in the existing building) in different colours.

2. GENERAL CONDITIONS :

2.1 The total estimated cost of the proposed construction should not exceed the Administrative sanction. Only such work which can be built within the sanctioned amount should be included in the detailed estimate. Detailed estimate for the portion of work, which is although essential as per requirement of concerned department but can not be included on account of inadequate sanction may be submitted as part 'B' to facilitate to move the case for Revised/Supplementary Administrative sanction.

2.2 In the case of estimate for Additions and Alterations the drawings and specifications of existing buildings should also be submitted along with the estimate. If there is any deviation from existing specifications, proper justification for the changes may be reported.

2.3 Estimate should be submitted in quadruplicate so that a copy of sanctioned estimate may be retained in Chief Engineer's office and the remaining three could be made available for circle, divisional and sub divisional office.

2.4 In the detailed estimates, the dimensions assumed in working out the quantities should be susceptible of identification of the dimensions shown on the drawings. Cross references to the dimensions of the estimates and also drawings should be indicated.

2.5 The items of detailed estimate should conform to the specifications described in technical report and also the PWD specifications.

2.6 Provisional rates for non-BSR items must be supported invariably by detailed rate analysis approved by Superintending Engineer.

2.7 All building estimate must include detailed estimates for sanitary, electrification, water supply and development works and no lump-sum provision on percentage basis should be provided in the estimate.

2.8 Estimates for works incorporating RCC items must include complete RCC details and bar bending schedules so as to facilitate in arriving at the correct quantities of concrete and steel.

2.9 Wherever necessary, detailed estimates may be supported by sketches to supplement drawings.

2.10 It those cases where a portion of work or entire work has already been taken in hand/executed before issue of sanction the quantities of complete portion should be taken as per actual measurements and a certificate to this effect may be recorded by the Executive Engineer. The specifications already adopted should also be clearly indicated in the technical report.

2.11 Detailed design for foundations and other structural components should be sent alongwith the estimate.

2.12 The type of soil met within the foundation, the test carried out for assessing the SBC of the soil at the foundation level, the SBC of the soil and the justification for the type of foundation adopted should be given.

2.13 The Executive Engineer and Superintending Engineers should particularly examine the foundation, the specifications for floors, roof windows and doors, ornamental features, finishing item etc. and observe in the forwarding letter or in the report itself the need for adopting expensive specifications.

Annexure 'A' to Circular No. 19 (Building-6)

TECHNICAL REPORT

1. GENERLAL

1.1 Name of Work

1.2 Circle

1.3 Division

2. AUTHORITY

2.1 Reference of Sanction

2.2 Amount of Administrative/Financial sanction (including/excluding the prorata charges).

2.3 Reference and Abstract details of Forecast estimate (if any).

2.4 Reference of Technical sanction of work if any issued earlier.

3. LOCATION AND LAND

3.1 Location

3.2 Area

3.3 Possession taken over/not taken over.

4. REFERENCES OF APPROVED DRAWING

S. No.	Details	Drawing No.	Authority
1.	Site Plan		
2.	Plans		
3.	Sections		
4.	Elevations		
5.	Details of components.		
6.	Sanitary, Water Supply and Drawing		
7.	Electrifications		
8.	Other drawings.		

5. DISCUSS THE PLANS SECTION & ELEVATIONS GENERAL

6. ESTIMATED COST

Estimated cost of portion shown in 'Red' Drawing No…….. including/exclusive of prorata charges has been worked out as Rs………….

7. SOURCE OF MATERIAL

8. TYPE OF SOIL/S.B.C.

9. BRIEF SPECIFICATION (In case of A/A specification of existing may also be indicated)

9.1 Foundation concrete.
9.2 Masonry in Foundation and plinth.
9.3 Damp proof course.
9.4 Superstructure (Type and specifications)
9.5 Lintels, Sun-shades etc.
9.6 Roofing
9.7 Interior finish
9.8 Exterior finish
9.9 Flooring (a) Base (b) Finish
9.10 Skirting and Dados
9.11 Windows (Frame, Panels, Wire gauging, Safety bars)
9.12 Doors (Frame and Shutters)
9.13 White/Colour/Cement lime/Decorative finish
9.14 Painting of Doors, Windows and Walls
9.15 Electrification (Type of Wiring, Fittings and Fixtures)
9.16 Sanitary & Water Supply
9.17 Compound wall and it's gate
9.18 Other specifications.

10. SPECIAL FITTINGS AND FIXTURES (Fans, Tube-lights, Exhaust, Pumps, Elevators etc.)

11. AGENCY FOR EXECUTION

12. PRESENT STAGE OF EXECUTION

13. COST ANALYSIS

13.1 B.S.R. adopted and premium approved by Superintending Engineer.
13.2 Tender Premium (Received/expected).
13.3 Total plinth area.
13.4 Break up of cost.

S.No.	Component	Total Cost	Cost/Sqft.	% age of total cost
(a)	Foundation & plinth			
(b)	Superstructure work			
(c)	Roofing.			
(d)	Doors, Windows & Frames.			
(e)	Flooring.			
(f)	Finishing.			

(g) Sanitary & water supply.

(h) Electrification.

(i) Development.

(j) Other items.

Total-

Total plinth area cost of Rs……../Sft. is higher/lower than the plinth area rate Rs…….. adopted in forecast estimate.

14 OTHER SALIENT FEATURE THAT NEEDS SPECIFIC MENTION

Assistant Engineer Executive Engineer

Superintending Engineer

Name of Work : Forecast Estimate for ______________________________

S. No	Compo- nent	Cost of Construction					Land		Special Fixtures Fan etc.	Other Expenses	Total (G) To (K)	Contin-gencies & Super-vision Charges 1.5% Of (L)	Quality Control 1% of (L-H)	Total (M+N)	Prorata Charges 13% of (L)	Grand Total (I+ O+ P)	Remarks
		Plinth Area	Plinth Area Rate	Cost	Internal Service 24½% (for hospitals and hostels 26½%)	Total (E+F)	Cost	Development Cost 10% of Const. Cost									
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)

Junior Engineer **Assistant Engineer** **Executive Engineer**

FORECAST ESTIMATE OF ROAD WORK

(1) Name of the road

- (a) Length in Km.
- (b) Off-take point/Km
- (c) Obligatory point (3) with population.
- (d) Classification : UR / ODR / MDR / SHW / NHW

(2) Name of constituency with District.

(3) PWD Division

(4) PWD Circle

(5) PWD Region

(6) Type of terrain

(7) CBR value of Sub-grade
(tested/As per soil classification)

(8) Traffic intensity.

- (a) Observed on
- (b) Anticipated after…..........…years

(9) Crust thickness as per above CBR/anticipated traffic intensity.

(10)	Sub grade		Cost per Km
(a)	Earthwork including	(a)	Average height
	Compaction	(b)	Compaction

(11) WBM crust

- (a) Sub base
- (b) Base course
- (c) Wearing course

(12) Surfacing
(a) Single coat surface Painting /20 mm PMC/- ………..   ....
Cost per KM   Cost of ....…..KMs

(13)	CD works	Type	Span	Total Amt.

- (i) Approximate No. of Culverts/ Causeways
- (ii) Protection works, if any

(14) Land acquisition

(i) Land width proposed :

(15) Miscellaneous : Caution boards/ direction boards/KM stones/Ganghuts. ....………

Total

Add : extra for difficulties such as hilly area/difficult areas / shortage of material etc./special items. ....……………..

Total

Add : 4.5% contingencies and supervision charges

Add : 1% for quality control

....……………..

Total

Add: ……..% prorata charges

....……………..

Total

...……………..

Grant Total

Cost/KM including all .....……………

Add : for escalation………% per year (as per norms)

CIRCULAR No. 36

(6 Building)

Check-list for taking over newly constructed buildings from contractors

*****

It has been observed that at the time of handing over the completed buildings by this department to client department/users lot of deficiencies are pointed out/reported by them. This is due to the fact that while taking over these buildings from building contractors or other agencies, the departmental officers do not examine them carefully and meticulously. Also no uniform and proper procedure is adopted at present for the inspection of such completed buildings except for checking items executed as per nomenclature mentioned in the 'G' Schedule of agreements. To overcome this difficulty, a check-list for taking over/handing over of such buildings is enclosed to enable the departmental officials to examine/inspect the buildings at the time of their taking over/handing over in a systematic manner.

The check-list is in two parts (Part-I & II) and should be adopted for inspection at the initial stage and then after 3-4 months specially during rainy season. These be prepared by concerned Engineering Sub-ordinate incharge of work at the time taking over building/elect. Works/sanitary works from respective contractors and shall be countersigned by AEN Incharge of the work after necessary test-check of the list. One copy of such taking over would always be kept in the record of the concerned Ex. Engineer for information and reference. Superintending Engineers will inspect these records during their periodical inspections of the works of the divisions.

Sd/-

(**M.M. SWAROOP**)
Chief Engineer, PWD
Rajasthan : Jaipur

BUILDING DIGEST

CENTRAL BUILDING RESEARCH INSTITUTE, INDIA

CHECK-LIST FOR TAKING-OVER NEWLY CONSTRUCTED BUILDING

INTRODUCTION

A large number of construction activities are being carried out by various organizations such as CPWD, MES, State PWD, Housing Boards, Development Authorities, Improvement Trust and Private Sector. There is a greater demand for the construction of houses on outright sale basis. It has been observed that a number of complaints about the newly constructed buildings are received by the authorities from the buyers/user at the time of taking over the houses. Since there is mass construction going on, it has been observed that before taking over the building from the builder/contractor, the departmental authorities are not able the examine the buildings carefully, There are no proper guidelines laid down for the inspection of buildings at the time of taking over, except the general conditions as laid down in the specifications for the construction. There are several items involved in building and unless these are clearly defined and listed, it is not feasible to check them. To meet the demand, a check-list has been prepared to enable the authorities to examine/inspect the salient features before taking over the buildings. Such lists should be prepared for every building and in case of large buildings, the same may be divided in convenient parts and separate check-list prepared for each part.

APPROACH

An endeavor has, therefore, been made to give a check-list covering the different items so that the inspecting authorities, before taking over the buildings, can examine/inspect the buildings in systematic way. The list has been made in two parts. The Part-I gives the main building items and points to be observed just after the completion of the buildings and before taking over. In case there are any defects observed in the first taking-over inspection, the action for rectification is to be incorporated and a second inspection is to be made before actually taking over building. The Part-II gives the items and points that should be examined/inspected after about 3-4 months and specially during the monsoon period keeping the time well before the expiry of the maintenance guarantee period. This will provide the performance of the water supply and sanitary installation in use and also the leakage and dampness in the building, which may not be possible to be examined fully at the first stage of taking-over. For each building these check-list proforma will be filled in, separately, at the time of inspection.

These proforma will also give useful information about the performance of various specifications and construction techniques in use, at the initial stage, in different parts of the country, so that a judicious selection for proper type in the region can be achieved.

CHECK-LIST FOR INSPECTION OF NEW BUILDINGS

Name & number of the building……………………………………………………………...

Date of completion of the building……......……………Date of inspection…………………

Date of taking over the building…......……………Date of handing over.………………....

From the builder/contractor……………………to user……………………………………............

Item of Inspection		Type of Details		Follow up Action	Remarks
		Location	Nature		
PART – I (Inspection to be carried out before taking over newly constructed building)					
A - Inside Building					
1. Walls					
(i) Are Walls to plumb ? (If no, indicate the location, nature and action to be taken)	Yes/No				
(ii) Are Walls in proper alignment? (If no, indicate the location, nature and action to be taken)	Yes/No				
(iii) Are there any cracks in Walls? (If yes, indicate the location, nature i.e. vertical/ horizontal/ inclined and action to be taken)	Yes/No				
(iv) Are there any signs of dampness/leakage on Walls? (If yes, indicate the location, magnitude and action to be taken)	Yes/No				
(v) Are all Walls examined ?	Yes/No				
2. Floors					
(i) Are there any cracks in floor ? (If yes, indicate location and action to be taken)	Yes/No				
(ii) Is there any settlement in floor ? (If yes, indicate location and action to be taken)	Yes/No				
(iii) Is the floor laid to proper slope ? (If no, indicate location and action to be taken)	Yes/No				

Item of Inspection			Type of Details		Follow up Action	Remarks
			Location	Nature		
(iv)	Is the floor properly finished/polished? (If no, indicate location, nature and action to be taken)	Yes/No				
(v)	Are there any cracks in skirting/dado ? (If yes, indicate location, nature and action to be taken)	Yes/No				
(vi)	Are all floors / dado examined?	Yes/No				
3.	**Roofs.**					
(a)	**Flat Roofs**					
(i)	Are there any cracks in bottom/top? (If yes, indicate location, nature and action to be taken)	Yes/No				
(ii)	Are there any leaks in roof? (If yes, indicate location, nature and action to be taken)	Yes/No				
(iii)	Has the water-proofing, treatment been laid properly and to slope ? (If no, indicate location, nature and action to be taken)	Yes/No				
(iv)	Are rain water pipes properly fitted and free of choking ?	Yes/No				
	(If no, indicate location, nature and action to be taken)	Yes/No				
(b)	**Sloping roofs**					
(i)	Is the roof laid to proper pitch/slope ?	Yes/No				
(ii)	Are there any breakages in tiles/sheet ?	Yes/No				
(iii)	Is the gutter laid to slope with proper joints? (If no, indicate location, nature and action to be taken)	Yes/No				
(iv)	Has proper preservative treatment been given to the steel/timber truss ? (If no, indicate action to be taken)	Yes/No				

Item of Inspection			Type of Details		Follow up Action	Remarks
			Location	Nature		
(v)	Is there any sagging/cracking in ceiling ? (If yes, indicate location, nature and action to be taken)	Yes/No				
(vi)	Is there sign of dampness in ceiling ? (If yes, indicate location, nature and action to be taken)	Yes/No				
(vii)	Are all roofs and ceilings examined ?	Yes/No				
4.	**Doors, Windows and Ventilators**					
(i)	Are all doors, windows and ventilators opening and closing smoothly ? (If no, indicate location, nature and action to be taken)	Yes/No				
(ii)	Are all doors windows and ventilators properly painted/ polished ? (If no, indicate location, and number to be done)	Yes/No				
(iii)	Are all fittings viz. locking arrangement, tower bolts, door stoppers, hooks and hinges etc. working smoothly ? (If no, indicate location, number to be set right)	Yes/No				
(iv)	Are all glass panes properly fitted cleaned & crack-free ? (If no, indicate location, number to be attend)	Yes/No				
5.	**Finishing**					
(i)	Are all rooms properly white washed, colour washed or distempered. (If no, indicate location, number to be attended)	Yes/No				
6.	**Fixtures**					
(a)	**Cupboards/shelves**					
(i)	Are doors of cup boards opening and closing smoothly? (If no, indicate location and number to be set right)	Yes/No				
(ii)	Are all fittings viz. hinges, locking arrangement, tower bolts etc. working smoothly?	Yes/No				

Item of Inspection			Type of Details		Follow up Action	Remarks
			Location	Nature		
(iii)	Are all cup-boards/shelves examined?	Yes/No				
(b)	**Chimney**					
(i)	Is the hood and slope provided to, have proper function ? (If no, indicate action to set right)	Yes/No				
7.	**Water Supply and Sanitation**					
(i)	Are pipes properly fixed to wall at required spacing and are not loose ? (If no, indicate location and number to be set right)	Yes/No				
(ii)	Are there any leakages in the pipe joints ? (If yes, indicate location, number and action to be taken)	Yes/No				
(iii)	Are all taps, valves, showers, wash basins, sinks working properly and there are no chocking / leakage ?	Yes/No				
(iv)	Are flushing cisterns correctly fitted and work properly ? (If no, indicate location number and action to set right)	Yes/No				
(v)	Are water seals in water closets and floor taps properly working ? (If no, indicate location, number and action to be taken)	Yes/No				
(vi)	Are W.C. Pans, wash basins and sinks free of crack/breakage ? (If no indicate location & number to be replaced)	Yes/No				
8.	**Electrical Fittings**					
(i)	Are the following working properly ?					
(a)	Switches	Yes/No				
(b)	Plug points	Yes/No				
(c)	Fans	Yes/No				
(d)	Regulators	Yes/No				

Item of Inspection			Type of Details		Follow up Action	Remarks
			Location	Nature		
(e)	Meter (If no, indicate location & number to be set right)	Yes/No				
(ii)	Are all the fuses wired properly ? (If no, indicate location and number to be set right)	Yes/No				
(iii)	Is the earth-wire connected properly ? (If no, indicate action to be taken)	Yes/No				
	B- Outside Building					
(i)	Is the surrounding free of debris cleaned and Levelled. (If no, indicate action to be taken)	Yes/No				
(ii)	Is the compound wall and gate provided ?	Yes/No				
(iii)	Is the approach road properly laid ? (If no, indicate action to be taken)	Yes/No				
(iv)	Are the outside drains provided & clear ?	Yes/No				
(v)	Are the steps & staircase provided & to proper slope ?	Yes/No				
(vi)	Is the outside building finishing complete ?	Yes/No				

HANDED OVER
....
Signature of the
Builder/Contractor.

TAKEN OVER
....
Signature of the
Engineering Subordinate, PWD

...................Section
...................Sub Dn.
...................Division

COUNTER SIGNED
( )

Signature of the Asstt. Engineer
..................Sub. Dn
.................Division

Copy submitted to the following for record and reference :-
1. The Executive Engineer, PWD ....
2. The Asstt. Engineer, PWD ....

( )

Engg. Subordinate

Part - II

(Inspection to be carried out during/after monsoon and well before expiry of maintenance-guarantee period)

1. Building :

Are there any leakages/dampness in the following?

(i)	Walls	Yes/No
(ii)	Floors	Yes/No
(iii)	Roofs	Yes/No
(iv)	Ceiling	Yes/No
(v)	Parapet	Yes/No
(vi)	Sunshades (If yes, indicate location, nature number & action)	Yes/No

2. Service:

Are there any leakage in the following ?

(i)	Rain Water Pipe	Yes/No
(ii)	Sewer Pipe	Yes/No
(iii)	Water Pipe.	Yes/No
B.	Is the whole system working leak proof ?	Yes/No
C.	Are the following working satisfactorily ?	
(i)	Septic Tank	Yes/No
(ii)	Soak Pit	Yes/No
(iii)	Outside drain	Yes/No

Assistant Engineer, PWD,
Sub Dn.....................

Copy forwarded to the Executive Engineer, PWD ....for reference and necessary action.
Copy forwarded to the Engg. Subordinate, PWD .... for early action.

Assistant Engineer : PWD
Sub Dn.......................

CIRCULAR No. 3/76

(5 - Roads)

It has been observed that the property holders along our Highways, instead of providing proper soak pit etc. for sullage, let it out on our Highways with the result that the road surface gets damaged causing lot of hindrances to the traffic.

In this regard section 278 and 431 of the Indian Penal Code are reproduced below :

"278 -- Making atmosphere noxious to health : Whoever voluntarily vitiates the atmosphere in any place so as to make it noxious to the health of persons in general dwelling or carrying on business in the neighbourhood or passing along a public way, shall be punished 'with fine which may extend to rupees five hundred.'

"431 -- Mischief by injury to public road, bridge, river or channel.

Whoever commits mischief by doing an act which renders or which he knows to be likely to render any public road, bridge, navigable river or navigable channel or artificial impassable or less safe for traveling or conveying property, shall be punished with imprisonment of either description for a term which may be extended to five years or with fine or with both.

Thus, it would be evident that under these two sections of I.P.C. anybody who vitiates the atmosphere by not making proper arrangement for the disposal of sullage etc. and causes injury to our roads, can be prosecuted by lodging a report with the nearest police station.

It is hereby enjoined that the officers of the department would ensure that proper police case is made out against those persons who vitiate the atmosphere or cause damage to our highways or any other property.

Sd/-

CHIEF ENGINEER : (B & R)
Dated: 17-09-1976

No.F.59(I)Sec.II/76

Dated: 8th April, 1977

CIRCULAR No.5

(5 - Roads)

Preparing & Scrutinizing Estimate For Road Works

It has been observed that adequate attention is not being paid by the Executive/Superintending Engineers in preparing & scrutinizing estimates for Road works submitted to this office with the result that the estimates have either to be returned or concerned staff has to be called to this office for clarification modification etc. due to which the technical sanction is delayed.

In order to improve the standard of estimates the following points may be kept in view :-

1. The alignment of the road should clearly be marked in the Index Plan showing the village falling on the alignment.

2. **The estimate should be accompanied with proper "L" sections, Cross sections, Plans etc. so as to have a clear picture of the gradients to be adopted and various Nallahs and other important features, utmost economy should be Exercised in marking the Gradients and formation levels.**

3. **A detailed design for crust thickness adopted must be submitted giving all the necessary investigations carried out at site and the Laboratory to Justify a proper type of specification adopted in the estimate.**

4. **A certificate for availability of material at the various quarries required under the circular issued vide this office No. ACE/F, 16/Lab/D-547, dated 02.02.76 must be sent with the estimate.**
5. Proper quarry chart showing leads of various materials proposed to be utilized in the work must be submitted with the estimate.

6. **In case of improvement of existing work a detailed liner chart showing present position of the work as also be detailed plan showing existing features of the road such as width of pavement, radius of curves etc. must be submitted with full justification for improvement of crust thickness, gradients widening etc.**

7. **In case of C.D. works full justification for water way type of structure adopted should be finished along with a design based on IRC Special Publication No. 13 - "Guide for small bridges and culverts."**

8. The drawings of C.D. works must show all the details including details of Reinforcement of the super - structure masonry section adopted etc.

9. **While marking various reduced levels in the drawing of C.D. works it must be ascertained that this tallies with the reduced levels shown in the L-section and Cross sections of the road on which the C.D. works are proposed.**

10. Detailed specifications of C.D. works must be given in the estimate.

It has been noticed that the Technical Report accompanying the estimate are very much incomplete due to which the various items adopted in the estimates can not be understood property. The Technical report should be prepared in details under following heads :-

1. Authority :

It should be clearly mentioned here as to under which Govt. reference number the work has been administratively sanctioned and whether it is under State Plan, Repairs or under Deposit Scheme etc. In case of Deposit work sanction of the competent authority for accepting the deposit must also be clearly mentioned. The Job No. should also be quoted here.

2. Necessity :

The necessity of work undertaken should clearly be mentioned giving justification of the alignment by indicating the population of villages falling on the alignment.
In case of improvement work, the deficiency of the existing work should be clearly stated and justification given for improvement of crust thickness, geometrics, width of pavement, camber, Gradients and curves etc. should clearly by spelled out and separate statements should be enclosed in the form of Appendices where necessary. A reference to the Liner chart of existing features should also be quoted here.
In case of incomplete works, details of existing position of work should be submitted in a Liner chart, the reference of which should be mentioned here.

3. Design Data :
Details of laboratory and field investigations carried out and results there of should be mentioned here or attached as Appendices in case of detailed design of C.D. works, Design of curves etc.

4. Proposals :
Detailed proposals based on design data should be mentioned here.

5. Specifications :

Detailed specifications adopted in view of various design features should be mentioned giving justification of particular specifications adopted comparing it with alternate possible specifications. The specifications should be grouped under following heads :-

1. Earth work.
2. Metalling & material.
3. B.T. work.
4. C.D. work.
6. Source of materials :

The source of material adopted must be clearly mentioned here, for example the name of quarries and stores from where store material such as cement, steel, bitumen etc. will be obtained.

7. Machinery :

Details of machinery required in the execution of work be mentioned here. It should also be made clearly whether the required machinery is available or some of items may have to be procured from other divisions.

8. Rates :

(i) Instead of writing "rates based on current B.S.R." it should be clearly mentioned as to on which B.S.R. of year the estimate has been prepared.

(ii) Non B.S.R. items taken in the estimate must be accompanied with detailed rate analysis which should be signed by the Supdtg. Engineer in taken of his approval and no word like provisional rate should be mentioned in the rate adopted in the estimate.

(iii) A statement showing the rate of various materials in each KM of road based on the leads adopted in the quarry chart should be submitted so that in statement the rates for various Kms. as also the leads could be checked.

In future it will be the responsibility of the Executive Engineers/Superintending Engineer to see that the estimates are accompanied with details mentioned above. If any deficiency is noticed in the estimates, technical report etc. the estimate shall be returned by this office and responsibility for delay shall have to be fixed on the Executive Engineers/Superintending Engineers

Sd/-

CHIEF ENGINEER (B&R)
PUBLIC WORKS DEPTT.

SUB. : STANDING ORDER. No. X-R.-8

REF. : CE No. 5/I27/Sec.II dated 22.05.1974.

Attention is invited to this Office Standing Order No.56 and Circular Order No. Nil dated the 26th May, 1962 (Copy enclosed) requiring that all cross-drainage works should be inspected before the rainy season and necessary repairs and precautionary measures be undertaken well in time to prevent any damage or mishap during the floods. Without prejudice to what it contained in Standing Order No.56 and Circular Order refereed above following additional instructions are issued which have to be strictly followed. Non compliance of these instructions by any officer will account to dereliction of duty and the officer concerned will be responsible of all the mishaps that may happen due to his omission that he has not inspected, will be no excuse.

All cross-drainage works will be inspected twice in a year before and after rains by the authorities mentioned below :-

1. All bridges Vent-Causeways, Flush, Pakka Causeway of span 300 ft. or more will be inspected by the Superintending Engineer himself.
2. All bridges Vented Causeways, Flush, Pakka, Causeways of span 100 ft. and upto 300 ft. will be inspected by the Executive Engineers.
3. All bridges Vented Causeways, Flush, Pakka, Causeways below 100 ft. span will be inspected by the Assistant Engineers.
4. The form in which date will be furnishing is given at Appendix-I.

The Superintending Engineer will keep a record in his office of the inspections in Form No.1 of all the cross drainage works and will take such measures as necessary to safe-guard against the expected damage before rains or immediately after rains. Important cases where Superintending Engineers feels can be refereed to this office for guidance.

Superintending Engineer will submit a certificate in Form No.2 in the Chief Engineer Office every year twice on 15th June and 15th December that all the cross-drainage works in his areas has been inspected by the concerned officer and the inspection form have been received in his office and have been examined by him.

Encl: 'As above'.

FORM – 1
INSPECTION REPORT OF C.D. WORKS

Before Rain
After Rain

Name of Road : ……………………………………………………………………………
Category ;- NHW / SHW / MDR / ODR Km. ………… to …………..

Sub Dn.. :- ……………………………………………

I	II – Location & Name		III – Existing Data						
S. No.	Location (No. of C.D. Works)	Name if any	Water Ways				Type of Foundations	Type of Sub – Structure	Type of Super – Structure & Decking
			No. of spans	Size	Height	Area of water way			
1	2	3	4	5	6	7	8	9	10

IV – Inspection Comments					
Type of Training / Safety Works if any provided	Type of bearings	condition of foundation with special mention regarding scour compared with original bed levels and structural safety. Bed profile may have to be drawn if necessary by taking in the river bed with note regarding Approns etc. if provided	Condition of Sub – Structure	Condition of Super Structure & Decking	General Remarks of the Inspecting Authority
11	12	13	14	15	16

Name of Inspecting Authority :

Date of Inspection :

Date of Submission of Higher Authority :

Comments of Higher Authority

FORM -2

Certificate by Superintending Engineer, P .W .D ./ B&R, Rajasthan as per Standing Order No.56 and as modified vide No. 5/ Misc./ 127/ Sec.II/ dated 22nd May, 1974 of the Chief Engineer, P.W.D.(B&R), Rajasthan, Jaipur regarding Inspection of Cross-drainage works before/after rains (to be submitted by 15th June & 15th December) to the Chief Engineer, Rajasthan P.W.D., (B.& R.), Jaipur.

1. Certified that all the Cross-drainage works in my jurisdiction (Districts of…………) have been inspected by the concerned Officers as required.

2. The inspection forms have been received/prepared in my office in respect of all these C.D. works and have been examined by me.

3. Necessary action where required have been taken in respect of each Bridge/Culvert/Causeway separately.

SUPERINTENDING ENGINEER,
P .W .D. (B. & R.)

OFFICE OF THE CHIEF ENGINEER, P.W.D.,RAJASTHAN, JAIPUR

No.S&S/92/D-1383 Dated :- 30 May, 1978

CIRCULAR

Sub :- Inspection of Works

Towards the close of the preceding financial year, the Government had ordered to evaluate the performance of the Divisions/Circles under the various PWD Regions and, accordingly detailed instructions were issued, vide this office Circular No.F.15-535/Sec.III/92/D-379 dated 22.1.92 for the evaluation week during which senior Officers of the rank of Chief Engineers inspected the various works etc. and reviewed the functioning of the Divisions / Sub-Dns. / Circles. It was observed that there was lack of adequate of effective inspections, due to which not only the quality of works was found below standards but also the progress of works was behind the physical targets in a number of works.

The importance of inspections cannot be over emphasised and, therefore, instructions regarding the same were issued by this office from time to time in the past. Attention is, however, once again requested to the instructions laid down in the PWD Manual of Orders (Chapter 12, para 12:Inspections) and the instructions as under are hereby issued again for the guidance of the inspecting officers at various levels, enjoining that the same should be strictly adhered to.

1. INSPECTIONS :

The Government has attached great importance to inspections for effective supervision of the works, and, therefore all the field officers (ACE/SE/EE/AEn's) are expected to spend as much time as possible on inspections of their respective works. The above inspecting officers shall ensure that the works are executed according to the sanctioned estimates/contract agreements, proper quality control measures are adopted as per the circulars/orders issued and satisfy themselves that the system of management prevailing is efficient and economical and proper accounts and returns prepared. They shall also ensure that no works are carried out without sanctions and budget provisions and also that the necessary budget provision is fully utilised without any excess over the4 same keeping also in view the periodical and overall physical targets to be achieved. While on inspections, checking of Muster Rolls, Departmental Works, Bills oj Contractors should be carried out as detailed in Paras 32.5, 32.6 and 32.7 of Chapter 32 "Works" oj the PWD Manual of Orders.

2. DAY AND NIGHT HALTS :

The effective inspection on works and knowledge about at the local problems of the people of the area and the officials of the department, labour and contractors can be better solved by making more day and night halts at the site of works. The norms prescribed for day & night halts by officers of the rank of ACE to AEN vide PWD Manual of Orders paras 12.1.2 have been amended to 50% thereof; vide Government Order No. P.23(85)GA//90 dated 21.1.1991 and the same would be as under:-

Inspecting Officers	Day Halts	Night Halts
Addl. Chief Engineer	34	22
Superintending Engineer	34	22
Executive Engineer	42	28
Assistant Engineer	42	28

It is, therefore, directed that all inspecting Officers shall, henceforth make day and night halts as per norms and. whenever they proceed on inspections, they shall also make night halts in continuation with day halts and shall not return to their head quarters unless for compelling reasons, for which they shall obtain prior formal approval of the next higher authority in writing, failing which it will be treated as if they have not carried out the inspection. The immediate controlling authority concerned shall ensure that the compliance or otherwise, of these instructions is dully reflected in the Annual Confidential Reports of the Inspecting Officer. Reports in this behalf shall be submitted Quarterly as follows :-

Reporting Officer	Inspecting Officer	Report to whom
ACE Region	All SE's Circles under control	CE(PWD),CE(R-I)&CE(R-II)
SE Circle	All EE's Divisions under control	ACE Region concerned
EE Division	All AEN's Sub-Dns. under control	SE Circle concerned

The first such report for Quarter ending June, 1992, shall submitted by 31.7.1992 positively and, thereafter, for the subsequent Quarter, on the last day of the ensuing month of the Quarter.

3. Daily Diaries (Note Books) :

This Diary / Note Book is to be maintained by every Engineer down to the level of Junior Engineer and should contain dated original Memoranda on which the accounts are made, showing the gangs of labourers counted, check measurements taken of the work and materials received and any other information collected to assist in preparing the accounts and verifying the entries made in then reasons Measurements of works should be made in MB's and not herein, except for recorded re exceptional cases only. Bad works or materials, undue delay in carrying out the work ordered by competent authority, must be noted at once in the Note Books of the persons in immediate charge of the works, including report on the quality of work and the manner of its execution, showing also the date of inspection. Remarks should also be made regarding the qualifications of subordinates Artisans and others employed on the works and satisfactory / unsatisfactory operations of contractors. The Note Book will also form the basis of record of movements (time, dates and means of conveyance) for the purpose of preparation and check of T .A. Bills. Please also refer to para 26.4 of PWD Manual of Orders regarding the detailed instructions laid down.

4. Tour Programme :

Inspections should be well planned so as to cover the maximum work during the inspection and proper tour programmes should be prepared in advance by the Officers and a copy of the same should be sent to the next higher authority and the subordinate staff so that the latter and the contractors may be available at the time of inspection of the higher officers. This, however, does not debar an officer from touring without tour programme as in the Engineering profession, frequent and surprise inspections are also essential. Inspection reports detailing the inspections carried out, instructions given to the subordinates and other important points should be prepared and a copy of the same should be sent to the next higher officer for information and subordinate officers for compliance. A proper system should be evolved in each office to ensure timely compliance of inspection reports. The registers to watch the same shall be maintained in the office of the inspecting officer and the office inspected in PWD MF 13 & 14. The copies of inspection notes and Daily Diaries (Note Books) should be submitted with T.A. Bills every month. Previous reports should be referred to by the Inspecting Officer, and if it is found that any irregularity therein noticed has not been corrected, the same should be prominently noted as matters on which instructions have been issued but not attended to. The report should show briefly what steps have been taken to remedy the defects previously noticed.

5. **Test checking of Measurements & Countersigning of completion reports :**

The responsibility of test checking the measurements of works lies with the Executive Engineer but this does not debar the Superintending Engineer from test checking the measurements of any work which he likes to test check. The Superintending Engineer should inspect the work and issue certificate in the prescribed form before the final bills of contractors are passed as laid down in para 12.1.5, Chapter-12 and para- 33.24.2, Chapter-33 of the Manual of Orders. The Executive Engineers and Assistant Engineers should carry out test-checking of works as per the norms prescribed in para 32.5.4, Chapter-32 and checking of Muster rolls as per para 32.6, Chapter-32 of the PWD Manual of Orders.

6. **Inspection Books :**

It shall be ensured that an Inspection Book in the prescribed proforma is maintained at the site of each & every work by the Sub-ordinate Engineering staff incharge and the same shall be put up before the Inspection officer for recording his remarks. Every inspecting officer shall record under his dated initials in the Inspection Book his observations regarding the quality and progress of works along with instructions, if any, regarding alterations ordered by the competent authority and the action required to be taken in regard thereto, including remarks as to the performance of the sub-ordinate incharge of the work, artisans employed on the work and the contractor. On completion of the work, the Inspection Books shall be recorded in the Divisional Office. The other Superintending Engineer / Executive Engineer concerned will please ensure action as above. If no remarks are recorded in the Inspection Book, it shall be treated that the inspecting officer has not inspected the work inspite of these instructions.

7. **Monthly D.O. Letters :**

The Superintending Engineers incharge of Circles shall write every month invariably a D.O. letter to the Chief Engineer, PWD, with the copy of the Chief Engineer (Roads-I / II) and the Regional and Addl. Chief Engineer concerned mentioning therein, inter alia, the achievements made by the individual Divisions under their respective control in regard to the periodical and overall physical and financial targets laid down, the short comings with suggestions / action being taken to make up for the same and highlighting the action desired to be taken from the level of the Chief Engineer concerned / Govt. level. The first such D.O. letter (for the month of June 1992) by every Superintending Engineer incharge of Circle should be issued within the first week of July 1992 and, therefore, for every succeeding month, within the first week of the following month. This arrangement shall please be adhered to regularly and without fail, in the interest of Government work. In the matter requiring urgent attention of the Chief Engineer concerned/Government, however, the Superintending Engineers would be welcome to communicate with the Chief Engineer concerned demi-officially at any time, under intimation to the Regional Addl. Chief Engineer concerned, over and above the regular monthly D.O. letters to be sent by them as above.

It is enjoined upon all the officers of the Department, especially those who are in the field and are attach directly/indirectly responsible for execution of works, that they should spare no efforts/endeavor on their part to inspect and supervise the works effectively as expected of them as per the circulars/orders and to guide the to be Junior level officers and subordinates in particular in efficient and timely execution of works, maintaining the have prescribed standards or quality, and to avoid cost over-runs due to avoidable delays on account of laxity at the levels concerned. It is hereby made abundantly clear that no laxity or non-compliance of these instructions shall be tolerated and, in case or default, not only the defaulting officers/subordinates but also the higher supervisory officers shall have to be made to share responsibility for the same.

The contents of this Circular may please be formally brought to the notice of all the Assistant Engineers and subordinate Engineers (JEn's) for strict compliance.

Sd/-
(HARBINDER SINGH)
CHIEF ENGINEER-CUM
ADDL.SECY. TO GOVT.

SUB. : TEST CERTIFICATE FOR ROAD MATERIALS

REF. : ACE/F.16 Lab/L-547 dated 02.02.78

It has been observed that while framing the estimate for road works very little significance is attached to testing of materials. Many a times material is proposed to be collected from a quarry without 1 ensuring whether it conforms to the specified requirements. At a subsequent stage, when the material is found to be sub- standard, it calls for a change in quarry and the ultimate result is extra financial burden, which could have been foreseen. In such cases, legal complications for extra claims by contractors also crop in such situations can always be averted if materials are tested and their suitability is adjudged before hand.

In view of the above the above it is hereby enjoined upon all concerned that in future all estimates amounting to Rs. I.00 lacs or more, must be accompanied by certificate as detailed below :-

"Certified that I have inspected ……………………………………(Name of the quarry) located at ……………………………………..The quarry has large deposits of fresh rock and test result of the representative sample taken from this quarry indicate that the material does confirm to the standards laid down by Indian Roads Congress for sub- base/base course/bituminous work. It is further certified that no nearer quarry than this is feasible and that all the materials collected in past under famine relief operation or any other schema within economical leads have been accounted for in this estimate."

ASSISTANT ENGINEER **EXECUTIVE ENGINEER**

OFFICE OF THE CHIEF ENGINEER (B & R), PWD, RAJASTHAN, JAIPUR

STANDING ORDER No. 138

PURCHASE OF BUILDINGS

This department is charged with the duty of expressing opinion in connection with the purchase of the buildings with a view to deal with such cases in a uniform and a correct way be followed :-

(i) No building may be purchased for public purpose without the order of the Government.

(ii) The valuation of building sites and of buildings should be made in accordance with the following procedure :-

Valuation of Buildings :

The buildings may be divided in four categories :

1. Permanent (with life 80 to 100 years).

2. Permanent (with inferior specification 60 to 80 years)

3. Semi-permanent (with life 50 to 60 years)

4. Temporary (with life 10 years of less)

The cost of building should first be worked out on the present day rates by preparation of detailed sample estimates for representative portion of the building and then calculating the building or portion thereof with similar characters, the estimates will be exclusive of services viz. electric, water and sanitary fittings and cost of land.

After valuating the present day cost of the building the depreciated cost should be calculated with the formula :-

$$D = P \left\{ 1 - \frac{rd}{100} \right\}^n$$

Where D is the depreciated value.

P is the cost of present market rate.

rd is the fixed percentage of depreciation.

n is number of years of the building has been constructed.

The following may be assumed as life span of the various type of buildings :

(1) Buildings built of stone in lime masonry walls or bricks in lime masonry. Stone slabs or RCC roof cement concrete of flag stone flooring and teak wood joinery work.
Life 80 to 100 years,

(2) Building of slightly inferior specification such to as stone partly in lime mortar partly in mud or bricks in lime, masonry plastered walls, stone slabs roofing or terraced roofing on flag stone with ordinary cement or lime terrace and rough flag stone flooring and second class teak wood or other good quality timber joinery. Life 60 to 80 years,

(3) Building of semi permanent nature such as built partly of brick in lime mortar and partly built in mud mortar, wooden or A.C. or C.G.I sheet roofing, country wood joinery. Life 50 to 60 years

(4) Temporary building such as buildings in mud mortar, inferior specifications or structure with thatches. Life 10 year or less

The Executive Engineer should exercise his judgment carefully to decide as to which category the building in question belongs and to estimate the actual life with in the category. The rate of depreciation may be taken as –

100/life span of each category.

The amount of depreciation shall in no case be less than 10% of the present cost of the buildings, irrespective of the age.

The valuation arrived at will be exclusive of the cost of land water supply, electric and sanitary and other permanent fitting etc. and will apply to those buildings only which have been properly maintained.

If the repairs have been neglected in the past and the present conditions is bad or dilapidated suitable deductions should be made from the value as deducted above, for the neglected repairs not for ordinary repairs).

The present value of land and water supply, sanitary and electric fittings etc. should be added to the valuation of the buildings so arrived at the total valuation of the property the present value of land will be determined in consolation with the local revenue authorities.

A lump-sum amount may further be deducted representing such damages as may exist but which do not effect the life of the building.

If necessary, a further lump-sum representing the value of such features which exist and which have been included in the plinth area valuation but which represent no value to the Govt. (i.e. exceptionally thick wall unnecessary decoration and the like) may be deducted.

In case of sub-standard work, suitable reduction to account for inferior specification may be made by the Executive Engineer.

Valuation on plinth area basis :

Cases may also exist where it is not possible to estimate correctly the quality of different class of work under the above noted sub-heads for want of estimate of the works and its drawing according to which the work has actually executed. In such case, it would be possible to work out the valuation on the basis of plinth area rate of buildings. In doing so, the plinth area of a building should be ascertained & the present day cost construction of building of similar size & specification estimated at plinth area, rates, from the figure so arrived at the deduction due to depreciation, related repairs & other items, may be made.

VALUATION ON MONTHLY RENTAL OR MARKET VALUE BASIS

It some time happens that valuation worked out on either of the above two methods is apparently neither that of market value nor of the buildings in the locality. In such cases therefore the valuation has to be assessed depending on.

(i) Present monthly rent of the buildings.

(ii) If buildings is not rented one, then on its face market value that can be ascertained.

The valuation on the basis of rent would be 200 items the rent per month of the building exclusive of houses tax, if any.

For the second alternative of market of face value no hard and fast rules can be laid down as it will depend on the locality in which the building has been constructed, the purpose for which it was built and the purpose for which it could be utilized in the locality.

The market value may be defined as the price which will be received by a willing seller from the willing buyer in the open market for the particular property in its present condition with its particular advantage and its particular draw backs. It is not the same as the cost; nor marketed value be compared with the cost of rent statement. It can not be determined by the extreme prices in a rising or falling market in all cases, the volume of transactions and the general trend of price should be considered.

Before any building is purchased, the total present and future liabilities to Govt. should be determined and the probable cost in the following items be clearly shown :

(1) Any necessary dismantling of existing structure, and cleaning of site.

(2) The special repairs of additional new work required to make the building suitable for its future use and.

(3) Future annual maintenance.

The cost of (1) and (2) added to the proposed purchase price of the building, gives the total cost of acquision while (3) given the recurring liability.

Sale Deed

When the approval of the Govt. to the proposed purchase has been accorded the Executive Engineer, will prepare a site plan in triplicate and draft sale deed in proper form in consulation with the Collector and send it through the Superintending Engineer to the Chief Engineer. The Chief Engineer, will forward it to the Administrative Department in the Secretariat to have it approved from the legal remembrance, after this, the draft Sale Deed with plan will be returned, the signature of the vendor on the deed and on one copy of the site plan obtained and have the deed registered after signing it on behalf of the Govt.

The Collector will have registered title deed with the site plan recorded in safe custody with the Treasury Officer, inform the Executive Engineer accordingly and send him a copy of the deed and plan. The third copy of the deed and plan will be sent by the Collectorate to the Administrative Department concerned in the secretariat.

The Executive Engineer will maintain the register of such purchase of building in division in the form vide Appendix 'A'.

Sd/-
CHIEF ENGINEER, PWD,
Rajasthan, Jaipur.

No.F.9(10)8605/S/60 Dated : 17.2.1960

APPENDIX 'A'

Register showing purchase of Building, Land etc.

S. No.	Particular of deed with amount	Date on which the deed was registered	Name of treasury where the deed is recorded	No. & date of Collector's letter intimating the recording of the deed with Treasury	Remarks

STANDING ORDER No. 144

ASSESSMENT OF FAIR RENT OF BUILDINGS TAKEN ON RENT

BY VARIOUS DEPARTMETNS

It has been brought to the notice of the undersigned that the officers of the Department while issuing FRC in respect of Building do not assess the reasonable area of the built up space taken on Rent in accordance with the criteria laid down in the Govt. orders No.D.6274/F. (15)GA/A/59 dated 18.05.59 (Copy enclosed for ready reference).

It may, therefore please be ensured before issuing such certificate in future that the space being taken on rent by various Deptt. Is not is excess of space prescribed by the Govt. D.2674/F.5(15) FA/A/59 dt. Jaipur 18.05.1959.

In order to ensure uniformity in the construction of new Govt. Building for office purpose and for addition and alteration to the present Building and for taking private Building for office purpose, the Governor is pleased to lay down the following scale for the same :

160 Sq.ft.	Each for Gazetteer officers.
50 Sq. ft.	Each for Technical Staff (such as Draftsmen & Estimaters).
40 Sq. ft	Each for ministerial staff, Staff, Supdt. Head Clerk etc.

In addition 10% of accommodation allowed for ministerial staff will be admissible for records.

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/73/77/79/81

1. For determining the present day value of the buildings with a view to assess the Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings, rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD (B & R), Rajasthan.

2. This order supercedes, the Standing Order No. 151 & 160 issued earlier. The market value of building for the purpose of sale or purchase of the property shall be determined as per instructions contained in Standing Order No. 138.

3. As the rates of construction on plinth area basis differ from circle to circle the plinth area rates per sqm. given below are for Jaipur, Bikaner, Sri Ganganagar, Kota and Udaipur circles. For Ajmer & Jodhpur these may be reduced by 10% and 20% respectively in X3/73 and 5% & 15% respectively in X3 1977/79/81.

		Building Portion description		Plinth area rate in Rs.			
				5/9/1973	1/1/1977	1/6/1979	1/2/1981
	a.	Basement floor upto 2.5 m. height.	Sqm.	170	250	325	405
	b.	Ground floor over basement	Sqm.	180	280	350	435
	c.	Ground floor without basement	Sqm.	240	360	450	560
	d.	First floor	Sqm.	205	300	375	465
	e.	Second floor	Sqm.	220	320	400	500
	f.	Add for third & fourth floor (for each floor)	Sqm.	15	20	25	30
	g.	Mezzanine floor	Sqm.	70	100	125	150
	h.	Compound wall one meter high above ground level including ordinary gates	Rmt.	30	45	60	75
	i.	CGI/AC sheet shed closed on 3 sides with pucca floor	Sqm.	170	250	310	385
	j.	Pucca Out Houses garages with shutters	Sqm.	205	300	375	465
	k.	Plat-form	Sqm.	35	45	60	75
4		Electrical installation without ceiling fans.(percentage over item 3 (a) to 3(j)]		5%	5%	5%	5%
	a	Add extra for conduit wring		1%	2%	2%	2%
	B	Cost of ceiling fans to be added extra as per number of fans.					
5		Sanitary percentage over item 3(a) to 3 (f)		9%	10%	10%	10%
6		Water supply percentage over item 3 (a) to 3 (f)		3%	4%	4%	4%

Building Portion description		Plinth area rate in Rs.			
		5/9/1973	1/1/1977	1/6/1979	1/2/1981
7	Roads				
8	The above rates apply to the building having :				
(i)	A ceiling height of 3.50 m.				
(ii)	Ordinary Cement Floor.				
(iii)	Wall Plastered on both sides or pointed externally.				
(iv)	Walls and pillar masonry in lime or cement.				
(v)	First/Second class teak-wood or solid core/flush door and window shutters with ordinary iron fittings.				
(vi)	Ordinary electric wiring with light & ordinary power circuits and with moderate fittings as per norms fixed by PWD for type design.				
(vii)	Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc in each bath room as per norms fixed by PWD for type design.				
(viii)	Stone slab roofing without Beams/joists i.e. upto 3 meters.				
NOTE :	For superior fittings such as bath tubs, water heaters, water tank etc. as per actuals.				
9	For other specifications the following percentage shall be added/reduced to item 3(a) to 3(f) stated above :				
a.	Increase/Decrease in height above or below 3.5 meters by 30cms.	2.50%	2.50%	1.50%	1.50%
b.	Add for mosaic flooring with gray cement / unpolished rough dressed stone flooring.	5%	5%	5%	5%
c.	Add for mosaic flooring with white cement	7.50%	7.50%	7.50%	7.50%
d.	Add for polished stone flooring.	10%	10%	10%	10%
e	Add for high class finish with rich specification	5%	8%	8%	8%
f.	Less for lime floor instead of cement floor	1.50%	x	x	X

	Building Portion description	Plinth area rate in Rs.			
		5/9/1973	1/1/1977	1/6/1979	1/2/1981
g.	Add extra for first class select grade teak wood frame with teak facing ply flush door polished.	X	5%	5%	5%
h.	Add for wire gauge doors & windows.	3%	3%	3%	3%
i.	Less for poor finish.	5%	5%	5%	5%
j.	Add for RCC roof with lime terracing for spans upto 3 m	5%	X	X	X
k	Add for RCC roof with lime terracing/ stone slab roofing with RSJ or beams (actual percentage to be determined by working out a sample estimate).	8% to 15%	8% to 15%	8% to 15%	8% to 15%
l.	Add for item e.g. hot & cold water system expensive fixtures and fittings, glazed tiles and facilities to over and above those specified about as per actual estimate.	√	√	√	√
m.	Add also for lawns hedges etc. maintained.	X	√	√	√

10. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

11. After valuation of the present day cost of the building as thought it is new, the depreciated cost should be calculated by the formula given below :

$$D = P\left[1 - \frac{rd}{100}\right]^n$$

Where
D = Depreciated Value
rd = The fixed percentage of depreciation
P = The cost at the present market rate.
n = The number of years the building had been constructed.

12. The following may be assumed as the life span of the various type of buildings :

a. Building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.
Life 80 to 100 years

b. Building of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
Life 60 to 80 years

c. Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joists or AC or CGI sheet roofing and country wood joinery.
Life 50 to 60 years

d. Temporary building such as building in mud mortar and inferior specifications or structures with country tiles roof and thatch roof etc.
Life 10 years or less

13. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the minimum, life span according to the type of construction.

(ii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

(iii) If the exact age of the building can not be ascertained, the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed at 50 years.

14. The rate to depreciation be taken as (100/life span) for each category.

15. The cost of land should be worked out on the basis of rate obtained from or prescribed by UIT, Municipality, or any other authority having jurisdiction on the administration of land in that area (In case UIT, Municipality is unable to give current land rate, the same may be obtained from land and building taxation department). Land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

16. For buildings more than one storey :

a. The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.
b. The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.
c. The area under commercial use shall be given double weight-age of the area under residential use.

17. The rent calculated is exclusive of taxes.
18. This order shall take immediate effect.

Sd/-
CHIEF ENGINEER,
PWD, RAJASTHN,
JAIPUR.

No. P.9(15)(A/Cell-VI)/73/6512 Dated: 5-9-73

D-4512 12-1-77
D-2453 6-6-79
D-296 2-1-81

Note:
9 (i) Deleted in S.O.x/3-1977 & in subsequent orders.
9 (f 1) Deleted in S.O. x/3-1977 & in subsequent orders.
9 (f2) Added in S.O. x/3-1977 & in subsequent orders.
9 (k 1) Added in S.O. x/3-1977 & in subsequent orders.
13 (i) & (ii) Added in S.O. x/3-1979 & in subsequent orders.
15 with in ( )-do- x-3-1981.
16 (c) Added in S.O. –do- x3/-1979.

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1984

1. For determining the present day value of the buildings with a view to assess the fair rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD (B & R), Rajasthan.

2. This order supersedes, the Standing Order No. 151,X-3/1977/ X-3/1979 & X-3/1981 issued earlier. The market value of building for the purpose of the sale of purchase of the property shall be determined as per instructions contained in Standing Order No. 138.

3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which Standing Order No. 138 shall continue to be applied.

		Building Portion-description	Plinth area rate in rupees
	a.	Basement floor upto 2.5 m. height	525.00 per Sq.M.
	b.	Ground floor over basement	565.00per Sq.M.
	c.	Ground floor without basement	725.00 per Sq.M.
	d.	First floor	600.00 per Sq.M.
	e.	Second floor	650.00 per Sq.M.
	f.	Add for third & fourth floor	40.00 (for each floor)
	g.	Mezzanine floor	200.00 per Sq.M.
	h.	Compound wall one meter high above ground level including ordinary gates	95.00 per Sq.M.
	i.	CGI/AC sheet closed on 3 sides with pucca floor	500.00 per Sq.M.
	j.	Pucca Out Houses and garage with shutters	600.00 per Sq.M.
	k.	Platform	100.00 per Sq.M.
	l.	For Antitermite treatment	15.00 per Sq.M.
	m.	For mastic asphalt water proofing treatment of foundations (on ground floor area only)	110.00 per Sq.M.
4.		Electrical installation without cleaning fans. [percentage over item 3 (a) to (j)]	
	(a)	Add extra for conduit wiring.	5%
	(b)	Cost of ceiling fans to be added extra as per number of fans.	2%

		Building Portion-description	Plinth area rate in rupees
5.		Sanitary percentage over item 3(a) to 3(j) excepting item 3 (h)	10%
6.		Water supply percentage over item 3(a) to 3(j) excepting item 3(h).	4%
Note:		For superior fittings such as bath tubs, water heaters, water tank etc. as per actuals at current BSR.	
7.		Roads	At Current B.S.R.
8.		The above rates apply to the building having:	
	(i)	A ceiling height of 3.50.	
	(ii)	Ordinary Cement Floor.	
	(iii)	Wall Plastered on both sides or pointed externally.	
	(iv)	Walls and pillar masonry in lime or cement.	
	(v)	First / Second class teak-wood or solid core / flush door and window shutters with ordinary iron fittings.	
	(vi)	Ordinary electric wiring with light & ordinary power circuits and with moderate fittings as per norms fixed by P.W.D. for type design.	
	(vii)	Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design.	
	(viii)	Stone slab roofing without Beams/joists i.e. upto 3 meters. For RCC framed structures the following rates shall be applicable in place of rates of at (3 meters ht.) 3(a) to 3(f)	
	(a)	Basement floor (3 meters ht.)	Rs. 1350/- per sqm.
	(b)	Ground floor (3 meter ht.) over basement	Rs. 1000/- per sqm.
	(c)	Ground floor in buildings with out basement (3 meter ht.)	Rs. 1050/- per sqm.
	(d)	First floor (3 meter ht.)	Rs. 1050/- per sqm.
	(e)	Add extra for Second & every subsequent floor (3 meter ht.) over rate of item 9 (d).	Rs. 50/- per sqm.

	Building Portion-description	Plinth area rate in rupees
10.	For other specifications the following percentage shall be added/reduced to item 3(a) to 3(j) stated above :	
a.	Increase/Decrease in height above or below 3.5 meters by 30 cms.	1.5%
b.	Add for mosaic flooring with gray cement/unpolished rough dressed stone flooring.	5%
c.	Add for mosaic flooring with white cement.	7.5%
d.	Add for the polished stone flooring.	10%
e.	Add for high-class finish with rich specification.	8%
f.	Add extra for first class selected grade teak wood frame with teak facing ply flush door polished.	5%
g.	Add for wire gauge doors & windows.	3%
h.	Less for poor finish.	5%
i.	Add for RCC roof with lime terracing/stone slab roofing with RSJ or beams (actual percentage to be determined by working out a sample estimate in case of load bearing structures only) 8% to 15%.	
j.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles built earbobs & other built in in furniture, marble slabs and any other item or facilities not covered in the items specified above as per actual estimate.	
k.	Add also for lawns etc. maintained as per actual estimate.	

11. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

12. After valuation of the present day cost of the building as though it is new, the depreciated cost should be calculated by the formula given below :

D = P (1 - rd/100)n
Where -
D = Depreciated Value

rd = The fixed percentage for depreciation.

P = The cost at the present market rate.

n = The number of years the building had been constructed.

13. The following may be assumed as the life span of various type of building :-

   a. Building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.

   Life 80 to 100 years

   b. Building of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.

   Life 60 to 80 years

   c. Building of semi-permanent nature such as built partly or brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.

   Life 50 to 60 years

   d. Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roofs and thatch etc.

   Life 10 years or less

14. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the minimum, life span according in the type of construction.

   (ii) The Executive Engineers should exercise their judgment carefully to be decide as to which category the building in question belongs and to estimate the actual life within the category.

   (iii) If the exact age of the building can not be ascertained by, the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

15. The rate to depreciation be taken as (100/life span) for each category.

16. The cost of land should be worked out on the basis of rate obtained from or prescribed by UIT, Municipality, or any other authority having jurisdiction on the administration of land in that area. Land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

   "In case UIT/Municipality is unable to give current land rate, the same may be obtained from land and building taxation department)"

17. For buildings more than one storey :

   a. The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

   b. The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

   c. The area under commercial use shall be given double weight-age of the area under residential use.

18. The rent calculated is exclusive of taxes, is to be borne by the owner.

19. This order shall come in to force with immediate effect.

Sd/-
(M.M. SWAROOP)
CHIEF ENGINEER, PWD
RAJASTHAN, JAIPUR

Dated : 23rd July, 1984.

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1987

1. For determining the present day value of the buildings with a view to assess the fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD (B & R), Rajasthan.

2. This order supersedes, the Standing Order No. 151, 160, X-3/1973/ X-3/1979 & X-3/1981, X-3/1984 & X-3/1987 issued earlier. The market value of building for the purpose of the sale or purchase property shall be determined as per instructions contained in Standing Order No. 138.

3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which Standing Order No. 138 shall continue to be applied.

	Building Portion-description		**Plinth area rate in rupees**
a.	Basement floor upto 2.5 m. height		600.00 per Sqm.
b.	Ground floor over basement		650.00per Sqm.
c.	Ground floor without basement		850.00 per SqM.
d.	First floor		710.00 per Sqm.
e.	Second floor		770.00 per Sqm.
f.	Add for third & fourth floor		45.00 per sqm. (for each floor)
g.	Mezzanine floor		220.00 per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates		125.00 per Sqm.
i.	CGI/AC sheet closed on 3 sides with pucca floor		550.00 per Sqm.
j.	Pucca Out Houses and garage with shutters		650.00 per Sqm.
k.	Platform		120.00 per Sqm.
l.	For antitermite treatment		15/- per Sqm.
m.	For mastic asphalt water proofing treatment of foundation s (on ground floor area only)		110/- per Sqm.
4.	Electric installation without ceiling fans.		
		%age over item No. 3 (a) to 3(j) for wall bearing structures	%age over item no. 9(a) to (9e) for RCC framed structures.

		Building Portion-description		Plinth area rate in rupees
	(a)	Add extra for conduit wiring	5%	3%
	(b)	Cost of ceiling fans to be added extra as per number of fans.	2%	1.25%
5.		Sanitary Fittings	10%	6%
6.		Water Supply	4%	2.5%

7. Roads

8. The above apply to the building having :
   - (i) A ceiling height of 3.50 m. (wall bearing structures) 3.00m. framed structures.
   - (ii) Wall Plastered on both sides or pointed externally.
   - (iii) Ordinary Cement Floor.
   - (iv) Walls and pillar masonry in lime or cement.
   - (v) Second class teak-wood or solid core/flush door and window shutters with ordinary iron fittings and with ordinary paint.
   - (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.
   - (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank soak pit and sewer line within campus.
   - (viii) Stone slab roofing without Beams/joists i.e. upto 3 metres for wall bearing structures.

9. For RCC farmed structure the following rates shall be applicable in place of rates at 3(a) to 3(f)

a	Basement floor (3 mtr. ceiling height)	1500/- per sqm.
b	Ground floor (3 mtr. height) over basement.	1100/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	1150/- per sqm.
d	First floor (3 mtr. height)	1100/- per sqm.
e	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	55/- per sqm.

10. For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(b) to 4(e) stated above :

	Details of Specifications	%age over item No. 3(a) to (j) & 4(b), 4(d) & 4(e)	%age over item No. 4(a) & 4(c) for RCC framed structure
a	Increase/Decrease by every 30cms. in height above or below.	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%

	Details of Specifications	%age over item No. 3(a) to (j) & 4(b), 4(d) & 4(e)	%age over item No. 4(a) & 4(c) for RCC framed structure
c	Add for mosaic flooring with white cement flooring	7.5%	5%
d.	Add for fine polished stone flooring.	10%	6%
e.	Add for high-class finish with rich specifications and good condition off maintenance	8%	4.75%
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	5%	3%
g.	Add for wire gauge doors & windows.	3%	1.75%
h.	Less for poor finish.	7 to 10%	7 to 10%
i.	Add for RCC roof with lime terracing, stone slab roofing with RSJ or Beams (actual percentage to be determined by working out a bearing structures only).	8 to 15%	
j.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above as per actual estimate.		
k.	Add also for lawns etc. maintained as per actual.		

11. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

12. After valuation of the present day cost of the building as though it is new, the depreciated cost should be calculated by the formula given below :

$$D = P\left(1 - \frac{Rd}{100}\right)n$$

Where -

D = Depreciated Value

rd = The fixed percentage for depreciation.

P = The cost at the present market rate.

n = The number of years the building had been constructed.

13. The following may be assumed as the life span of the various type of buildings :

    a. RCC framed buildings, buildings built of stone in lime masonry wall bricks in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery
    Life 80 to 100 years

    b. Building of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
    Life 60 to 80 years

    c. Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
    Life 50 to 60 years

    d. Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roofs and thatch etc.
    Life 10 years or less

14. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the maximum, life span according to the type of construction.

    (ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

    (iii) The Executive Engineers should exercise their judgment carefully to be decide as to which category the building in question belongs and to estimate the actual life within the category.

15. The rate of depreciation be taken as:- (100/life span) for each category.

16. The cost of land should be worked out on the basis of rate obtained from or prescribed by UIT, Municipality, or any other authority having jurisdiction on the administration of land in that area. Land in excess of 4 times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

    "In case UIT/Municipality is unable to give current land rate, the same may be obtained from Land and Building Taxation Department or the Valuation Cell in the Income Tax Department."

17. For buildings more than one storey the distribution of cost of land shall be made as under :

    a. The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.
    b. The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.
    c. The area under commercial use shall be given double weight-age of the area under residential use.

18. Any taxes such as House tax, Land & Building taxes etc. is to be borne by the owner.

19. This order shall come in to force with immediate effect.

Sd/-
(M.C. SHARMA)
CHIEF ENGINEER, PWD
RAJASTHAN, JAIPUR

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1990

1. For determining the present day value of the buildings with a view to assess the fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD (B & R), Rajasthan.
2. This order supersedes, the Standing Order No. 151, 160, X-3/1973/ X-3/1979 & X-3/1981, X-3/1984 & X-3/1987 issued earlier. The market value of building for the purpose of the sale or purchase property shall be determined as per instructions contained in Standing Order No. 138.
3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which Standing Order No. 138 shall continue to be applied.

		Building Portion-description	**Plinth area rate in rupees**
	a.	Basement floor upto 2.5 m. height	850.00 per Sqm.
	b.	Ground floor over basement	950.00per Sqm.
	c.	Ground floor without basement	1150.00 per SqM.
	d.	First floor	1000.00 per Sqm.
	e.	Second floor	1050.00 per Sqm.
	f.	Add for third & fourth floor	15.00 per sqm. (for each floor)
	g.	Mezzanine floor	250.00 per Sqm.
	h.	Compound wall one meter high above ground level including ordinary gates	200.00 per Sqm.
	i.	CGI/AC sheet closed on 3 sides with pucca floor	650.00 per Sqm.
	j.	Pucca Out Houses and garage with shutters	800.00 per Sqm.
	k.	Platform	150.00 per Sqm.
4.		For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f) :	
	a	Basement floor (3 mtr. ceiling height)	2000/- per sqm.
	b	Ground floor (3 mtr. height) over basement.	1500/- per sqm.
	c.	Ground floor in buildings without basement (3 mtr. height)	2100/- per sqm.
	d	First floor (3 mtr. height)	1550/- per sqm.
	e	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	75/- per sqm.

5. For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(b) to 4(e) stated above :

		%age over item No. 3(a) to (j) & 4(b), 4(d) & 4(e)	**%age over item No. 4(a) & 4(c) for RCC framed structure**
a	Increase/Decrease by every 30cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement flooring	7.5%	5%
d.	Add for fine polished stone flooring.	(As per difference in rates in prevailing B.S.R.) (Actual area to be measured)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	8%	5%
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	5%	3% (on item No.4-c only)
g.	Add for wire gauge doors & windows.	3%	2% (on item No.4-c only)
h.	Less for poor finish.	7 to 10%	7 to 10%
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above as per actual estimate.		
j.	Add also for lawns etc. maintained as per actual.		
k.	Electrical installation without cleaning fans.	5%	3.5%
a.	Add extra for conduit wiring.	2%	1.5%
b.	Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary Fittings (except item 3h)	10%	7% (item No. 4-c only)
m.	Water supply (except item 3h)	4%	3% (item No.4-c only)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual at current BSR.

6. Roads At Current B.S.R.

7. The above rates apply to the building having :

   (i) A ceiling height of 3.20 m. (wall bearing structures)/ 3.00 m. framed structures.

   (ii) Wall Plastered on both sides or pointed externally.

   (iii) Ordinary Cement Floor.

   (iv) Walls and pillar masonry in lime or cement.

   (v) Second class teak-wood or solid core/flush door and window shutters with ordinary iron fittings and with ordinary paint.

   (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

   (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank soak pit and sewer line within campus.

   (viii) Stone slab roofing/R.C.C. slab roofing with Beams or joists.

8. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is new, the depreciated cost should be calculated by the formula given below :

$$D = P \left(1 - \frac{Rd}{100}\right)^n$$

Where -

D = Depreciated Value

rd = The fixed percentage for depreciation.

P = The cost at the present market rate.

n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

   a. RCC framed buildings, buildings built of stone in lime masonry wall bricks in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery
   Life 80 to 100 years

   b. Building of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
   Life 60 to 80 years

   c. Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
   Life 50 to 60 years

   d. Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roofs and thatch etc.
   Life 10 years or less

11. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the maximum, life span according to the type of construction.

    (ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

    (iii) The Executive Engineers should exercise their judgment carefully to be decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate of depreciation be taken as:- (100/life span) for each category.

13. The cost of land should be worked out on the basis of rate obtained from or prescribed by UIT, Municipality, or any other authority having jurisdiction on the administration of land in that area. Land in excess of 4 times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

    "In case UIT/Municipality is unable to give current land rate, the same may be obtained from Land and Building Taxation Department or the Valuation Cell in the Income Tax Department."

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a. The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b. The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c. The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building taxes etc. is to be borne by the owner.

16. This order shall come in to force with immediate effect.

Sd/-

(M.C. SHARMA)
CHIEF ENGINEER, PWD
RAJASTHAN, JAIPUR

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1993

1. For determining the present day value of the buildings with a view to assess the fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.
2. This order supersedes, the Standing Order No. 151, 160, X-3/1973/ X-3/1979 & X-3/1981, X-3/1984, X-3/1987 & X-3/1990 issued earlier. The market value of building for the purpose of the sale or purchase property shall be determined as per instructions contained in Standing Order No. 138.
3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which Standing Order No. 138 shall continue to be applied.

	Building Portion-description	Plinth area rate in rupees
a.	Basement floor upto 2.5 m. height	1200.00 per Sqm.
b.	Ground floor over basement	1350.00per Sqm.
c.	Ground floor without basement	1600.00 per SqM.
d.	First floor	1400.00 per Sqm.
e.	Second floor	1450.00 per Sqm.
f.	Add for third & fourth floor	100.00 each floor
g.	Mezzanine floor	350.00 per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	300.00 per Sqm.
i.	CGI/AC sheet closed on 3 sides with pucca floor	900.00 per Sqm.
j.	Pucca Out Houses and garage with shutters	1100.00 per Sqm.
k.	Platform	200.00 per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f) :
5.

a	Basement floor (3 mtr. ceiling height)	2800/- per sqm.
b	Ground floor (3 mtr. height) over basement.	2100/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	2950/- per sqm.
d	First floor (3 mtr. height)	2150/- per sqm.
e	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	100/- per sqm

5. For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above :

	Details of Specifications	Item 3(a) to 3(j) and No. 4 (b), 4(e) & 4(d)	Item No. 4 (a) and 4(c)
a	Increase/Decrease by every 30 cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement	7.5%	5%
d.	Add for fine polished stone flooring/marble flooring	(As per difference in rates in prevailing B.S.R.)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	upto 15%	upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above as per actual estimate.		(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 25/- per Sqm.	Rs. 25/- per Sqm.
k.	Electrical installation without ceiling fans.	5%	3.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Cost of ceiling fans to be added extra as per number of fans		
l.	Sanitary Fittings (except item 3(h) & 3(k)	10%	7%
m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding		(As per actual)
o.	Fire fighting system.		(As per actual)

Note :- For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads At Current B.S.R.

7. The above rates apply to the building having :

   (i) A ceiling height of 3.20 m. (wall bearing structures) 3.00 M framed structures.

   (ii) Wall plastered on both sides and pointed externally.

   (iii) Ordinary cement floor.

   (iv) Walls and pillar masonry in lime or cement.

   (v) Second class teak-wood or solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

   (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

   (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

   (viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

   Where -

   D = P (1 - rd/100)n

   D = Depreciated Value

   rd = The fixed percentage for depreciation.

   P = The cost at the present market rate.

   n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

   a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery. Life 80 to 100 years

   b Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery. Life 60 to 80 years

   c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery. Life 50 to 60 years

   d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc. Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the maximum life span according in the type of construction.

    (ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

    (iii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate to depreciation be taken as : 00/life span for each category.

13. The cost of land should be worked out by a committed consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction on the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building taxes etc. is to be borne by the owner

16. This order shall come in force with immediate effect 1-1-1993 and will not effect the cases assessed in the past.

Sd/-

(HARBINDER SINGH)
CHIEF ENGINEER, PWD

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1995

1. For determining the present day value of the buildings with a view to assess the Fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.
2. This order supersedes, the Standing Order No. 151, 160, X-3/1973/ X-3/1979 & X-3/1981, X-3/1984, X-3/1987 & X-3/1990, X-3/1993 issued earlier. The market value of building for the purpose of the sale or purchase of the property shall be determined as per instructions contained in Standing Order No. 138.
3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which, Standing Order No. 138 shall continue to be applied.

	Building Portion-description	**Plinth area rate in rupees**
a.	Basement floor upto 2.5 m. height	1360.00 per Sqm.
b.	Ground floor over basement	1525.00per Sqm.
c.	Ground floor without basement	1800.00 per SqM.
d.	First floor	1600.00 per Sqm.
e.	Second floor	1610.00 per Sqm.
f.	Add for third & fourth floor	115.00 per sqm.
g.	Mezzanine floor	400.00 per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	350.00 per RMT
i.	CGI/AC sheet closed on 3 sides with pucca floor	1000.00 per Sqm.
j.	Pucca Out Houses and garage with shutters	1250.00 per Sqm.
k.	Plat-form	230.00 per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f) :

a	Basement floor (3 mtr. ceiling height)	2800/- per sqm.
b	Ground floor (3 mtr. height) over basement.	2100/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	2950/- per sqm.
d	First floor (3 mtr. height)	2150/- per sqm.
e.	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	100/- per sqm.

5. For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above :

a	Increase/Decrease by every 30 cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement.	7.5%	5%
d.	Add for fine polished stone flooring/ marble flooring.	(As per difference in prevailing B.S.R.)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	upto 15%	Upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above.		(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 25/- per Sqm.	Rs. 25/- per Sqm.
k.	Electrical installation without ceiling fans.	5%	3.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary fittings (except item 3(h) & 3 (k)	10%	7%
m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding		(As per actual)
o.	Fire fighting system.		(As per actual)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads At Current B.S.R.

7. The above rates apply to the building having :

   (i) A ceiling height of 3.50 m. (wall bearing structures/ 3.00 M framed structures)

   (ii) Wall plastered on both sides and pointed externally.

   (iii) Ordinary cement floor.

   (iv) Walls and pillar masonry in lime or cement.

   (v) Second class teak-wood or solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

   (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

   (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

   (viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

   Where -
   D = P (1 - rd/100)n
   D = Depreciated Value
   rd = The fixed percentage for depreciation.
   P = The cost at the present market rate.
   n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

   a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.
   Life 80 to 100 years

   b. Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
   Life 60 to 80 years

   c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
   Life 50 to 60 years

   d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc.
   Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the maximum life span according in the type of construction.

    (ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by found out by any means, the same shall be assumed as 50 years.

    (iii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual ife within the category.

12. The rate to depreciation be taken as : 100/Life span for each category.

13. The cost of land should be worked out by a committee consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction or the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building taxes etc. is to be borne by the owner

16. This order shall come in force with immediate effect from 1-1-1993 and will not effect the cases assessed in the past.

Sd/-

(A.K. JAIN)
CHIEF ENGINEER, PWD

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/1997

1. For determining the present day value of the buildings with a view to assess the Fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.

2. This order supersedes, the Standing Order No. 151, 160, X-3/1973/ X-3/1979 & X-3/1981, X-3/1984, X-3/1987 and X-3/1990, X-3/1993, X-3/1995 issued earlier. The market value of building for the purpose of the sale or purchase of property shall be determined as per instructions contained in Standing Order No. 138.

3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which, Standing Order No. 138 shall continue to be applied.

	Building Portion-description	Plinth area rate in rupees
a.	Basement floor upto 2.5 m. height	1560/- per Sqm.
b.	Ground floor over basement	1750/- per Sqm.
c.	Ground floor without basement	2070/- per SqM.
d.	First floor	1840/- per Sqm.
e.	Second floor	1885/- per Sqm.
f.	Add for third & fourth floor	130/- each floor
g.	Mezzanine floor	460/- per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	400/- Rmt.
i.	CGI/AC sheet closed on 3 sides with pucca floor	1150/- per Sqm.
j.	Pucca Out Houses and garage with shutters	1435/- per Sqm.
k.	Plat-form	265/- per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f) :

a	Basement floor (3 mtr. ceiling height)	3220/- per sqm.
b	Ground floor (3 mtr. height) over basement.	2415/- per sqm.

c.	Ground floor in buildings without basement (3 mtr. height)	3390/- per sqm.
d	First floor (3 mtr. height)	2470/- per sqm.
e.	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	115/- per sqm.

5. For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above :

a	Increase/Decrease by every 30cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement.	7.5%	5%
d.	Add for fine polished stone flooring/ marble flooring.	(As per difference in prevailing B.S.R.) (Actual area to be measured)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	Upto 15%	Upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above.		(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 25/- per Sqm.	Rs. 25/- per Sqm.
k.	Electrical installation without ceiling fans.	5%	3.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary fittings (except item 3(h) & 3 (k)	10%	7%

m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding		(As per actual)
o.	Fire fighting system.		(As per actual)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads — At Current B.S.R.

7. The above rates apply to the building having :

   (i) A ceiling height of 3.20 m. (wall bearing structures/ 3.00 M framed structures)

   (ii) Wall plastered on both sides and pointed externally.

   (iii) Ordinary cement floor.

   (iv) Walls and pillar masonry in lime or cement.

   (v) Second class teak-wood of solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

   (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

   (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

   (viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8. If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

   Where -

   D = P (1 - rd/100)n

   D = Depreciated Value

   rd = The fixed percentage for depreciation.

   P = The cost at the present market rate.

   n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

   a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.

b. Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
Life 60 to 80 years

c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
Life 50 to 60 years

d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc.
Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc. the actual age of the building shall be taken subject to the maximum life span according in the type of construction.

(ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

(iii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate to depreciation be taken as : 100/Life span for each category.

13. The cost of land should be worked out by a committee consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction or the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building taxes etc. is to be borne by the owner

16. This order shall come in force with immediate effect from 1-4-1997 and will not effect the cases assessed in the past.

Sd/-

(H.N. SAXENA)
CHIEF ENGINEER, PWD

GOVERNMENT OF RAJASTHAN
PUBLIC WORKS DEPARTMENT
STANDING ORDER No. X-3/2006

1. For determining the present day value of the buildings with a view to assess the Fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.
2. This order supersedes, the Standing Order No. 151, 160, X-3/1973, X-3/1979, X-3/1981, X-3/1984, X-3/1987, X-3/1990, X-3/1993, X-3/1995 & X-3/1997 issued earlier. The market value of building for the purpose of the sale or purchase of property shall be determined as per instructions contained in Standing Order No. 138.
3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which, Standing Order No. 138 shall continue to be applied.

1 For masonry Structures:-

S.No.	Building Portion-description	Plinth area rate (in Rs.)
a.	Basement floor upto 2.5 m. height	2450/- per Sqm.
b.	Ground floor over basement	2850/- per Sqm.
c.	Ground floor without basement	3400/- per Sqm.
d.	First floor	3000/- per Sqm.
e.	Second floor	3100/- per Sqm.
f.	Add for third & fourth floor	210/- each floor
g.	Mezzanine floor	750/- per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	650/- Rmt.
i.	CGI/AC sheet closed on 3 sides with pucca floor	1800/- per Sqm.
j.	Pucca Out Houses and garage with shutters	2350/- per Sqm.
k.	Plat-form	400/- per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f):

a	Basement floor (3 mtr. ceiling height)	3000/- per sqm.
b	Ground floor (3 mtr. height) over basement.	3300/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	4000/- per sqm.
d	First floor (3 mtr. height)	3400/- per sqm.
e.	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	190/- per sqm.

5 For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above :

S. No.	Building Portion/ Description	For masonry structures	RCC framed structures
a	Increase/Decrease by every 30cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement.	7.5%	5%

S. No.	Building Portion/ Description	For masonry structures	RCC framed structures
d.	Add for fine polished stone flooring/ marble flooring.	(As per difference in prevailing B.S.R.)(Actual area to be measured)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	Upto 15%	Upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above.	(As per actual estimate)	(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 40/- per Sqm.	Rs. 40/- per Sqm.
k.	Electrical installation without ceiling fans.	5%	3.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary fittings (except item 3(h) & 3 (k)	10%	7%
m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding	(As per actual Estimate)	(As per actual)
o.	Fire fighting system.	(As per actual Estimate)	(As per actual)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads — At Current B.S.R.

7. The above rates apply to the building having :

   (i) A ceiling height of 3.20 m. (wall bearing structures/ 3.00 M framed structures)

   (ii) Wall plastered on both sides and pointed externally.

   (iii) Ordinary cement floor.

   (iv) Walls and pillar masonry in lime or cement.

   (v) Second class teak-wood of solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

   (vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

   (vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

   (viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8 If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

Where -
$D = P (1 - rd/100)^n$
D = Depreciated Value
rd = The fixed percentage for depreciation.
P = The cost at the present market rate.
n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.
Life 80 to 100 Years

b. Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
Life 60 to 80 years

c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
Life 50 to 60 years

d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc.
Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc., the actual age of the building shall be taken subject to the maximum life span according to the type of construction.
(ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.
(iii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate of depreciation be taken as : 100/Life span for each category.

13. The cost of land should be worked out by a committee consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction or the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration. Land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building, unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building tax etc., is to be borne by the owner.

16. This order shall come in force with immediate effect from 24.4.2006 and will not effect the cases assessed in the past.

-Sd-
(H.L. MEENA)
CHIEF ENGINEER
PWD Raj. Jaipur

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/2011

1. For determining the present day value of the buildings with a view to assess the Fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.

2. This order supersedes, the Standing Order No. 151, 160, X-3/1973, X-3/1979, X-3/1981, X-3/1984, X-3/1987, X-3/1990, X-3/1993, X-3/1995, X-3/1997 & A-3/2006 issued earlier. The market value of building for the purpose of the sale or purchase of property shall be determined as per instructions contained in Standing Order No. 138.

3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which, Standing Order No. 138 shall continue to be applied.

1 For masonry Structures:-

S.NO.	Building Portion-description	Plinth area rate (in Rs.)
a.	Basement floor upto 2.5 m. height	4050/- per Sqm.
b.	Ground floor over basement	4700/- per Sqm.
c.	Ground floor without basement	5625/- per Sqm.
d.	First floor	4950/- per Sqm.
e.	Second floor	5125/- per Sqm.
f.	Add for third & fourth floor	325/- each floor
g.	Mezzanine floor	1225/- per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	1050/- Rmt.
i.	CGI/AC sheet closed on 3 sides with pucca floor	2975/- per Sqm.
j.	Pucca Out Houses and garage with shutters	3875/- per Sqm.
k.	Plat-form	650/- per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f):

a	Basement floor (3 mtr. ceiling height)	4950/- per sqm.
b	Ground floor (3 mtr. height) over basement.	5450/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	6625/- per sqm.
d	First floor (3 mtr. height)	5625/- per sqm.
e.	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	300/- per sqm.

5 For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above :

S. No.	Building Portion/ Description	For masonry structures	RCC framed structures
a	Increase/Decrease by every 30cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement.	7.5%	5%
d.	Add for fine polished stone flooring/ marble flooring.	(As per difference in prevailing B.S.R.) (Actual area to be measured)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	Upto 15%	Upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above.	(As per actual estimate)	(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 40/- per Sqm.	Rs. 40/- per Sqm.
k.	Electrical installation without ceiling fans.	10%	8.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Add extra for Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary fittings (except item 3(h) & 3 (k)	10%	7%
m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding	(As per actual Estimate)	(As per actual)
o.	Fire fighting system.	(As per actual Estimate)	(As per actual)
p	External Glazing work, ACP, Aluminium work (Anodised / Powder Caoted), Pressed Steel Door frame work, Fibre / Acrylic/ Polycarbonate Sheet, Wall Panelling, False Ceiling work etc.	(As per actual Estimate)	(As per actual)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads Work at Current B.S.R.

7. The above rates apply to the building having :

(i) A ceiling height of 3.20 m. (wall bearing structures/ 3.00 M framed structures)

(ii) Wall plastered on both sides and pointed externally.

(iii) Ordinary cement floor.

(iv) Walls and pillar masonry in lime or cement.

(v) Second class teak-wood of solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

(vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

(vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

(viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8 If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

Where -

$D = P (1 - rd/100)^n$

D = Depreciated Value

rd = The fixed percentage for depreciation.

P = The cost at the present market rate.

n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.

Life 80 to 100 Years

b. Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.

Life 60 to 80 years

c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.

Life 50 to 60 years

d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc.

Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc., the actual age of the building shall be taken subject to the maximum life span according to the type of construction.

(ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

(iii) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate of depreciation be taken as : 100/Life span for each category.

13. The cost of land should be worked out by a committee consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction or the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration. Land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building, unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building tax etc., is to be borne by the owner.

16. This order shall come in force with immediate effect from 24.4.2006 and will not effect the cases assessed in the past.
17. The rate applicable for determining the Fair Rent Certificate (FRC), for Government/ Private Buildings shall be in accordance with the Circular No. CE/SE (DB)/D-1932 dated 20-01-2007.
18. Any taxes such as Property Tax, Development Tax, Sewerage Tax etc. is to be borne by the owner.
19. The order shall come into force with immediate effect and will not affect the cases assessed in the past.

Date :- 26-04-2011

-Sd-
(Chiranji Lal)
CHIEF ENGINEER
PWD Raj. Jaipur

TABLE OF DEPRECIATED COST OF BUILDINGS

$$D = P\left(1 - \frac{rd}{100}\right)^n$$

D = P x C

rd = RATE OF DEPRICATION

(a) Take rd 1% for lime and cement masonry.

(b) rd 2% for mud masonry.

n = Number of Years

P = Present Day Cost

D = Depreciated Cost.

Number of Years	Value of C		Number of Year	Value of C	
	rd 1 %	rd 2 %		rd 1 %	rd 2 %
1	2	3	4	5	6
1.	0.9900	0.9800	51.	0.5965	0.3558
2.	0.9800	0.9603	52.	0.5905	0.3487
3.	0.9700	0.9410	53.	0.5845	0.3417
4.	0.9603	0.9221	54.	0.5786	0.3348
5.	0.9506	0.9036	55.	0.5720	0.3281
6.	0.9410	0.8855	56.	0.5670	0.3215
7.	0.9315	0.8678	57.	0.5613	0.3151
8.	0.9221	0.8502	58.	0.5556	0.3087
9.	0.9128	0.8333	59.	0.5500	0.3026
10.	0.9036	0.8166	60.	0.5445	0.2965
11.	0.8945	0.8002	61.	0.5391	0.2905
12.	0.8855	0.7841	62.	0.5335	0.2847
13.	0.8766	0.7624	63.	0.5282	0.2790
14.	0.8678	0.7520	64.	0.5229	0.2734
15.	0.8590	0.7379	65.	0.5176	0.2680
16.	0.8502	0.7237	66.	0.5124	0.2626
17.	0.8418	0.7086	67.	0.5072	0.2572
18.	0.8333	0.6944	68.	0.5021	0.2522
19.	0.8250	0.6805	69.	0.4971	0.2471
20.	0.8166	0.6668	70.	0.4920	0.2421
21.	0.8082	0.6534	71.	0.4871	0.2372

Number of Years	Value of C		Number of Year	Value of C	
	rd 1 %	rd 2 %		rd 1 %	rd 2 %
22.	0.8002	0.6403	72.	0.4821	0.2325
23.	0.7921	0.6275	73.	0.4773	0.2278
24.	0.7841	0.6149	74.	0.4725	0.2232
25.	0.7762	0.6026	75.	0.467	0.2288
26.	0.7674	0.5905	76.	0.4630	0.2144
27.	0.7607	0.5786	77.	0.4583	0.2101
28.	0.7520	0.560	78.	0.4537	0.2059
29.	0.7454	0.5556	79.	0.4491	0.2018
30.	0.7379	0.5445	80.	0.4446	0.1977
31.	0.7305	0.5335	81.	0.4401	0.1937
32.	0.7231	0.5229	82.	0.4357	0.1899
33.	0.7158	0.5124	83.	0.4313	0.1861
34.	0.7086	0.5021	84.	0.4270	0.1823
35.	0.015	0.4920	85.	0.4227	0.1786
36.	0.6944	0.4821	86.	0.4184	0.1751
37.	0.6874	0.4725	87.	0.4142	0.1716
38.	0.6805	0.4630	88.	0.4101	0.1681
39.	0.6736	0.4537	89.	0.059	0.1647
40.	0.6668	0.4446	90.	0.4018	0.1614
41.	0.6601	0.4357	91.	0.3977	0.1582
42.	0.6534	0.4270	92.	0.3938	0.1550
43.	0.6468	0.4184	93.	0.3897	0.1519
44.	0.6403	0.4101	94.	0.3859	0.1489
45.	0.6339	0.4018	95.	0.4819	0.1459
46.	0.6275	0.3938	96.	0.3781	0.1430
47.	0.6212	0.3859	97.	0.3743	0.1401
48.	0.6149	0.3781	98.	0.3704	0.1373
49.	0.6067	0.3705	99.	0.3667	0.1345
50.	0.6026	0.3631	100.	0.3631	0.1318

GOVERNMENT OF RAJASTHAN

PUBLIC WORKS DEPARTMENT

STANDING ORDER No. X-3/2015

1. For determining the present day value of the buildings with a view to assess the Fair Rent of the residential/other ordinary buildings required to be hired by the Govt. Department and the standard rent of Govt. buildings rented to Central Govt./State Govt./other officers and Private parties, the following rules shall henceforth be observed. These rules are meant for use by the PWD, Rajasthan.

2. This order supersedes, the Standing Order No. 151, 160, X-3/1973, X-3/1979, X-3/1981, X-3/1984, X-3/1987, X-3/1990, X-3/1993, X-3/1995, X-3/1997, X-3/2006 & X-3/2011 issued earlier. The market value of building for the purpose of the sale or purchase of property shall be determined as per instructions contained in Standing Order No. 138.

3. The circular may be adopted for valuation of properties except for the purpose of sale and purchase of buildings by Govt. of Rajasthan for which, Standing Order No. 138 shall continue to be applied.

2 For masonry Structures:-

S.NO.	Building Portion-description	Plinth area rate (in Rs.)
a.	Basement floor up to 2.5 m. height	5830/- per Sqm.
b.	Ground floor over basement	6770/- per Sqm.
c.	Ground floor without basement	8100/- per Sqm.
d.	First floor	7130/- per Sqm.
e.	Second floor	7380/- per Sqm.
f.	Add for third & fourth floor	470/- each floor
g.	Mezzanine floor	1760/- per Sqm.
h.	Compound wall one meter high above ground level including ordinary gates	1510/- Rmt.
i.	CGI/AC sheet closed on 3 sides with pucca floor	4280/- per Sqm.
j.	Pucca Out Houses and garage with shutters	5580/- per Sqm.
k.	Plat-form	940/- per Sqm.

4. For RCC framed multistoried structures the following rates shall be applicable in place of rates at 3(a) to 3(f):

a	Basement floor (3 mtr. ceiling height)	7130/- per sqm.
b	Ground floor (3 mtr. height) over basement.	7850/- per sqm.
c.	Ground floor in buildings without basement (3 mtr. height)	9540/- per sqm.
d	First floor (3 mtr. height)	8100/- per sqm.
e.	Add extra for second & every subsequent floors (3 mtr. ht.) over rate of item 4 (d)	430/- per sqm.

5 For other specifications the following percentage shall be added/reduced in item 3(a) to 3(j) and 4(a) to 4(e) stated above:

S. No.	Building Portion/ Description	For masonry structures	RCC framed structures
a	Increase/Decrease by every 30cms. in height above or below	1.5%	1%
b	Add for mosaic flooring with gray cement or that of unpolished rough dressed stone flooring in place of C.C. floor.	5%	3%
c	Add for mosaic flooring with white cement.	7.5%	5%
d.	Add for fine polished stone flooring/ marble flooring.	(As per difference in prevailing B.S.R.) (Actual area to be measured)	
e.	Add for high-class finish with rich specifications and good condition off maintenance	Upto 15%	Upto 10% (Analysis to be worked out)
f.	Add extra for first class/selected grade teak wood frame with teak facing ply flush door polished.	10%	7%
g.	Add for wire gauge doors & windows & safety bars.	3%	2%
h.	Less for poor finish.	5 to 10%	5 to 10% (Analysis to be worked out)
i.	Add for items e.g. hot & cold water system expensive fixtures and fittings, glazed tiles, built in furniture, marble slabs and any other item or facilities not covered in the item specified above.	(As per actual estimate)	(As per actual estimate)
j.	Add also for lawns hedges etc. maintained.	Rs. 40/- per Sqm.	Rs. 40/- per Sqm.
k.	Electrical installation without ceiling fans.	5%	3.5%
(a)	Add extra for conduit wiring	2%	1.5%
(b)	Add extra for Cost of ceiling fans to be added extra as per number of fans.		
l.	Sanitary fittings (except item 3(h) & 3 (k)	10%	7%
m.	Water supply (except item 3(h) & 3(k)	4%	3%
n.	External cladding	(As per actual Estimate)	(As per actual)
o.	Fire fighting system.	(As per actual Estimate)	(As per actual)
p	External Glazing work, ACP, Aluminium work (Anodised / Powder Caoted), Pressed Steel Door frame work, Fibre / Acrylic/ Polycarbonate Sheet, Wall Panelling, False Ceiling work etc.	(As per actual Estimate)	(As per actual)

Note : For superior fittings such as bath tubs, water heaters etc. as per actual.

6. Roads Work at Current B.S.R.

7. The above rates apply to the building having :
   (i) A ceiling height of 3.20 m. (wall bearing structures/ 3.00 M framed structures)

   (ii) Wall plastered on both sides and pointed externally.

   (iii) Ordinary cement floor.

(iv) Walls and pillar masonry in lime or cement.

(v) Second class teak-wood of solid core/flush door and window shutters with ordinary iron fittings and ordinary paint.

(vi) Ordinary electric wiring with light & ordinary power circuits and with moderate fittings.

(vii) Good and adequate sanitary fittings such as Indian or English type W.C. with flushing cistern, wash basin, towel rail, etc. in each bath room as per norms fixed by P.W.D. for type design including septic tank, soak pit and sewer lines with in the campus.

(viii) Stone slab roofing/RCC slab roofing with Beams or joists.

8 If there are any variations in the above specifications then percentage to be added or reduced, should be worked out proportionately.

9. After valuation of the present day cost of the building as though it is now, the depreciated cost should be calculated by the formula given below :

Where -
$D = P\,(1 - rd/100)^n$
D = Depreciated Value
rd = The fixed percentage for depreciation.
P = The cost at the present market rate.
n = The number of years the building had been constructed.

10. The following may be assumed as the life span of the various type of buildings :

a RCC framed buildings, building built of stone in lime masonry walls or brick in lime masonry, stone slab or RCC roof, cement concrete or stone flag flooring and teak wood joinery.
Life 80 to 100 Years

b. Buildings of slightly inferior specifications such as stone in mud or bricks in mud masonry, lime plastered walls, stone slab roofing or terraced roofing or stone flags with ordinary or lime terrace and 2nd class teak wood or other equivalent quality, and timber joinery.
Life 60 to 80 years

c Building of semi-permanent nature such as built partly of brick in lime mortar and partly of brick in mud mortar, wooden joints or AC or CGI sheet roofing and country wood joinery.
Life 50 to 60 years

d Temporary building such as buildings in mud mortar and inferior specifications or structures with country tiles roof and thatch etc.
Life 10 years or less.

11. (i) If the age of the building is known or can be ascertained through local enquiry etc., the actual age of the building shall be taken subject to the maximum life span according to the type of construction.

(ii) If the exact age of the building can not be ascertained but the approximate age can be found out, then that age shall be adopted. If the age can not be found out by any means, the same shall be assumed as 50 years.

(iv) The Executive Engineers should exercise their judgment carefully to decide as to which category the building in question belongs and to estimate the actual life within the category.

12. The rate of depreciation be taken as : 100/Life span for each category.

13. The cost of land should be worked out by a committee consisting of the concerning S.E., T.A. to SE and Executive Engineer(s) after considering the rates obtained from UIT/ Municipality/Registration Deptt./any other authority having jurisdiction or the administration of land in that area. The committee should exercise its discretion carefully, taking all the facts like importance of the place etc. into consideration. Land in excess of four times the built-up area of the building should not be counted for determining the total cost of land and building, unless the entire land is specifically required.

14. For buildings more than one storey the distribution of cost of land shall be made as under :

    a The cost of built-up land area shall be distributed in proportion to the built-up area on individual floors.

    b The cost of open land shall be distributed in proportion to the actual use of land by different tenants on individual floors.

    c The area under commercial use shall be given double weight-age of the area under residential use.

15. Any taxes such as House tax, Land & Building tax etc., is to be borne by the owner.

16. This order shall come in force with immediate effect from 24.4.2006 and will not effect the cases assessed in the past.
17. The rate applicable for determining the Fair Rent Certificate (FRC), for Government/ Private Buildings shall be in accordance with the Circular No. CE/SE (DB)/D-1932 dated 20-01-2007.
18. Any taxes such as Property Tax, Development Tax, Sewerage Tax etc. is to be borne by the owner.
19. The order shall come into force with immediate effect and will not affect the cases assessed in the past.

Date :- 26-04-2011

-Sd-
(Chiranji Lal)
CHIEF ENGINEER
PWD Raj. Jaipur

TABLE OF DEPRECIATED COST OF BUILDINGS

$$D = P\left(1 - \frac{rd}{100}\right)^n$$

D = P x C

rd = RATE OF DEPRICATION

(a) Take rd 1% for lime and cement masonry.

(b) rd 2% for mud masonry.

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P = Present Day Cost

D = Depreciated Cost.

Number of Years	Value of C		Number of Year	Value of C	
	rd 1 %	rd 2 %		rd 1 %	rd 2 %
1	2	3	4	5	6
1.	0.9900	0.9800	51.	0.5965	0.3558
2.	0.9800	0.9603	52.	0.5905	0.3487
3.	0.9700	0.9410	53.	0.5845	0.3417
4.	0.9603	0.9221	54.	0.5786	0.3348
5.	0.9506	0.9036	55.	0.5720	0.3281
6.	0.9410	0.8855	56.	0.5670	0.3215
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18.	0.8333	0.6944	68.	0.5021	0.2522
19.	0.8250	0.6805	69.	0.4971	0.2471
20.	0.8166	0.6668	70.	0.4920	0.2421
21.	0.8082	0.6534	71.	0.4871	0.2372

Number of Years	Value of C		Number of Year	Value of C	
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29.	0.7454	0.5556	79.	0.4491	0.2018
30.	0.7379	0.5445	80.	0.4446	0.1977
31.	0.7305	0.5335	81.	0.4401	0.1937
32.	0.7231	0.5229	82.	0.4357	0.1899
33.	0.7158	0.5124	83.	0.4313	0.1861
34.	0.7086	0.5021	84.	0.4270	0.1823
35.	0.015	0.4920	85.	0.4227	0.1786
36.	0.6944	0.4821	86.	0.4184	0.1751
37.	0.6874	0.4725	87.	0.4142	0.1716
38.	0.6805	0.4630	88.	0.4101	0.1681
39.	0.6736	0.4537	89.	0.059	0.1647
40.	0.6668	0.4446	90.	0.4018	0.1614
41.	0.6601	0.4357	91.	0.3977	0.1582
42.	0.6534	0.4270	92.	0.3938	0.1550
43.	0.6468	0.4184	93.	0.3897	0.1519
44.	0.6403	0.4101	94.	0.3859	0.1489
45.	0.6339	0.4018	95.	0.4819	0.1459
46.	0.6275	0.3938	96.	0.3781	0.1430
47.	0.6212	0.3859	97.	0.3743	0.1401
48.	0.6149	0.3781	98.	0.3704	0.1373
49.	0.6067	0.3705	99.	0.3667	0.1345
50.	0.6026	0.3631	100.	0.3631	0.1318

SUB : SAFETY PRECAUTIONS -WHERE CONSTRUCTION WORKS ARE IN PROGRESS OR WHERE DIVERSION IS PROVIDED.

Ref No. F. 7(18) Sec.II/76 Dated 3rd May, 1978.

It has been observed that in many places where the road works either repairs or original works are in progress, necessary caution -boards are not displayed indicating that the labour is working on the road. Besides when the road is being out either partially or fully for laying of pipe-lines or constructing culverts, sign-boards well in advance of the face of the work should be displayed cautioning the traffic to slow down the speed. When the road pavement is not available for traffic, diversion should be provided are indication that the diversion is ahead should be given about 200' to 300' away from the line of obstruction. Another Board 'atleast 100' away from the work face showing the diversion road should be fixed on the road side. Barricading to prevent the traffic from entering the obstruction should be provided fully visible both in the day time and in the night should be displayed at all such points. Such lights should also be displayed where the stacks of stones are collected near the pavement, machinery or drums and other obstruction are left either near the pavement or on the berms.

All these instructions mentioned above have been issued from time to time in the past but unfortunately these are not being compiled with. I must bring to the notice of the Suptdg. Engineers that if these lapse are observed in future immediate disciplinary action should be taken against the Overseer & Assistant Engineer and the Executive Engineer should be charged for dereliction of duties and supervisory negligence.

The Superintending Engineers are requested to see that these instructions are fully compiled with and whenever the lapses or the lacuna is found immediate action against the concerned defaulter is taken under intimation to this office.

SUB: SAFETY ASPECTS IN EXCAVATION WORKS

REF: C.E. No. 3(7)-Lab/78/520 Dated 14-6-78

Excavation is one of the important phases of construction in any building activity. Due to in-sufficient attention to safety aspect, is some times becomes a major hazard and a cause of serious, accident.

While carrying out excavation work, precautions are to be taken.
(A) For safety of the workmen doing the excavation work including safety to the adjoining structures.

(B) For safety of the traffic passing near the site of excavation work.

(A) Precautions for safety of the workmen & adjoining structures :

(i) Before starting the excavation work a complete knowledge of the foundations of the adjoining structure & other underground structures e.g. sewers, water pipe line & electrical conduit system etc. is essential.

(ii) When the excavation is being carried out below the level of the foundation of adjoining structures, the trench should not be dug for a length of more than 3 m., in one stretch leaving a supporting section of stable earth of at least 2 m. in between.

(iii) All trenches in soil other then rock & hard compact soil where stable side slopes are not available & the depth is more than 1.5 m. shoring & timbering should be provided in accordance with the provision made in IS-3764 (Safety code for excavation work). Such shoring & timbering would be provided even in the hard compact soil if the depth exceeds 2 m.

(iv) Excavation material should be kept at least 2 m, away from the edge of the trenches. It should also not be dumped on the road or the berms in case the excavation is being done along the existing road.

(v) Material used in shoring and timbering work like ropes, planks and ballies etc. should be of good quality.

(B) For safety of the traffic passing near by the site of excavation :

(i) Temporary fencing should be done around the trench, where there is any like hood of public including the cattle using the area around the trenches.

(ii) Notice Boards, Danger Sign Boards and Red light should be provided on either side of the trench at least 45 metre before excavation.

For items A(i), A(ii) and A(v) the Assistant Engineer incharge of the work will be personally responsible and for items A(iv), B(i) and B(ii) the Junior Engineer/Sub-Engineer incharge of the work will be personally responsible. For the non-compliance of instructions given above, which may lead to accidents, this office will be left with no other choice except to suspend the defaulters and proceed to take disciplinary action against them irrespective of the fact whether the work is being done departmentally or through a contractor.

SAFE PERMISSIBLE LOAD ON MASONRY AND CONCRETE

TABLE – 1

S. No.	Masonry / Concrete	Safe permissible load in MT / Sqm.
1.	Brick Masonry in CM 1 : 3	100
2.	Brick Masonry in CM 1 : 4	88
3.	Brick Masonry in CM 1 : 6	50
4.	Brick Masonry in Lime Mortar 1 : 2	44
5.	Ashlar stone Masonry in Lime Mortar 1 : 3	99 to 110
6.	Rubble Masonry in Lime Mortar	30 to 50
7.	Cement concrete 1 :1½ : 3	510
8.	Cement concrete 1 :2 : 4	440
9.	Cement concrete 1 :3 : 6	280
10.	Lime Concrete	50 to 75

COMPRESSIVE STRENGTH OF MORTAR AND CONCRETE

TABLE – 2

S. No.	Mix	Compressive Strength in Kg/Cm^2	
		After 7 days curing	After 28 days curing
1.	Cement Sand Mortar 1:3	38	75 to 100
2.	Cement Sand Mortar 1:4	25	50 to 75
3.	Cement Sand Mortar 1:6	15	30 to 50
4.	Lime Sand Mortar 1:2	2.5	5 to 7
5.	Lime Surkhi Mortar 1:2	7.5	15 to 20
6.	Cement Concrete 1 : 4 : 8	38	75
7.	Cement Concrete 1 : 3 : 6	70	100
8.	Cement Concrete 1 : 2 : 4	105	150
9.	Cement Concrete 1 : 1½ : 3	140	200
10.	Cement Concrete 1 : 1 : 2	175	250

PROPERTIES OF MILD STEEL SQUARE AND ROUND BARS

TABLE-3

(0.7843KG./CM2/METRES)

Diameter or width mm	Weight per Meter		Sanctional Area		Perimeter	
	Square Kg.	Round Kg.	Square Cm2	Round Cm2	Square cm2	Round Cm2
05.0	0.20	0.15	0.25	0.20	2.0	1.57
05.5	0.24	0.19	0.30	0.24	2.2	1.77
06.0	0.28	0.22	0.36	0.28	2.4	1.88
07.0	0.38	0.30	0.49	0.38	2.8	2.20
08.0	0.50	0.39	0.64	0.50	3.2	2.51
09.0	0.64	0.50	0.81	0.64	3.6	2.83
10.0	0.78	0.62	1.00	0.79	4.0	3.14
11.0	0.95	0.75	1.21	0.95	4.4	3.46
12.0	1.13	0.89	1.44	1.13	4.8	3.77
13.0	1.54	1.21	1.96	1.54	5.6	4.40
16.0	2.01	1.58	2.56	2.01	6.4	5.03
18.0	2.54	2.00	3.24	2.54	7.2	5.65
20.0	3.14	2.47	4.00	3.14	8.0	6.28
22.0	3.80	2.98	4.84	3.80	8.8	6.91
25.0	4.91	3.85	6.25	4.91	10.0	7.85
28.0	6.15	4.82	7.84	6.16	11.2	8.80
32.0	8.04	6.31	10.24	8.04	12.8	10.05
36.0	10.17	7.99	12.96	10.18	14.40	11.31
40.0	12.56	9.86	16.00	12.67	16.00	12.57
45.0	15.90	12.49	20.25	15.90	18.00	14.14
50.0	19.62	15.41	25.00	19.64	20.0	15.71
56.0	24.62	19.34	31.36	24.63	22.4	17.59
63.0	31.16	24.47	39.69	31.17	25.2	19.78
71.0	39.47	31.08	50.41	39.49	28.4	22.31
80.0	50.24	39.36	64.0	50.27	32.0	25.13

PROPERTIES OF RIBBED OR TOR STEEL

TABLE – 4

Size (mm)	Area (Cm^2)	Weight per Metre (Kg.)	Perimeter (Cm)	Length per MT (Metre)	Minimum Bend Radius
06	0.283	0.222	1.89	4510	
08	0.503	0.395	2.51	2532	
10	0.785	0.617	3.14	1631	
12	1.131	0.888	3.77	1125	
14	1.539	1.208	4.40	829	4d
16	2.011	1.578	5.03	633	
18	2.545	2.000	5.65	500	
20	3.142	2.466	6.28	405	
22	3.801	2.980	6.91	336	
25	4.909	3.854	7.85	260	
28	6.157	4.830	8.80	207	
32	8.042	6.313	10.05	159	
36	10.179	7.990	11.31	125	
40	12.566	9.864	12.57	101	6d
50	19.635	15.410	15.71	65	

SUBSTITUTION AS TENSION REINFORCEMENT

TABLE – 5

Mild Steel (mm)	8	10	12	16	20	22	25	28	32	36	40	50
Ribbed or TOR Steel (mm)	6	8	10	12	16	18	20	22	25	28	32	36

SUBSTITUTION AS TENSION REINFORCEMENT

TABLE – 6

Mild Steel (mm)	12	16	20	22	25	28	32	36	40	50
Ribbed or TOR Steel (mm)	10	14	2x12	18	20	22	2x18	2x20	32	40

WEIGHTS OF CONSTRUCTION MATERIALS
TABLE – 7

S. No.	Materials	Weight in Kg.	Per	Remarks
Earth Soil Clay etc.				
1.	Earth alluvial undisturbed	1600	Cum.	
2.	Earth dry	1413 – 1850	Cum.	
3.	Earth Moist	1600 – 2000	Cum.	
4.	Clay wet compact	2080	Cum.	
5.	Clay Moist compact	1760	Cum.	
6.	Clay Dry compact	1440	Cum.	
7.	Clay Dry lumps	1040	Cum.	
8.	Gravel loose	1600	Cum.	
9.	Gravel rammed	1920 – 2160	Cum.	
10.	Shingle 3mm to 30mm	1460	Cum.	
11.	Sand dry clean	1540 – 1600	Cum.	
12.	Sand dry River	1840	Cum.	
13.	Sand wet	1760 – 2000	Cum.	
14.	Peat dry	580 – 640	Cum.	
15.	Sand dry compact	800	Cum.	
16.	Sand wet compact	1360	Cum.	
17.	Lime unslaked	870 – 1050		
Liquids			Cum.	
1.	Alcohol	780	Cum.	
2.	Creosote	1070	Cum.	
3.	Diesel oil	960	Cum.	
4.	Oil Linseed	930	Cum.	
5.	Oil Terpentine	860	Cum.	
6.	Petrol	690	Cum.	
7.	Varnish	960	Cum.	
8.	Water Fresh	1000	Cum.	
9.	Water Salty	1025	Cum.	
Building Stones				
1.	Basalt	2600	Cum.	
2.	Ginesis	2400 – 2690	Cum.	
3.	Graphic	2690 – 2800	Cum.	
4.	Laterite	2080 – 2400	Cum.	
5.	Lime stone	2400 – 2640	Cum.	
6.	Marble	7220	Cum.	
7.	Pumice	800 – 1200	Cum.	
8.	Quartzite	2640	Cum.	
9.	Sand stone	2240 – 2400	Cum.	
10.	Slate	2800	Cum.	
11.	Chalk	1600 – 1920	Cum.	

S. No.	Materials	Weight in Kg.	Per	Remarks
Metals and Alloys				
1.	Iron pig	7200	Cum.	
2.	Grey cast iron	7030 – 7130	Cum.	
3.	Iron wrought	7700	Cum.	
4.	Steel cast	7850	Cum.	
5.	Steel wrought mild	7800	Cum.	
6.	Copper cast	8790 – 8940	Cum.	
7.	Copper wrought	8840 – 8940	Cum.	
8.	Lead cast	11340	Cum.	
9.	Lead wrought	11360	Cum.	
10.	Mangansee	7400	Cum.	
11.	Mercury	13600	Cum.	
12.	Nickel	8280 – 8810	Cum.	
13.	Aluminium	2580 – 2710	Cum.	
14.	Aluminium wrought	2640 – 2800	Cum.	
15.	Steel black plates	7.9	Sqm. per mm thickness	
16.	Copper plates	8.69	Sqm. per mm thickness	
17.	Lead plates	11.0	Sqm. per mm thickness	
18.	Aluminium plates	2.8	Sqm. per mm thickness	
19.	Brasses (Red & White)	8190 – 8220	Cum.	
20.	Brasses Yellow	8440 – 8680	Cum.	
Timber				
1	Hardwood such as axle wood babool etc.	640-690	Cum	
2	Light wood such as fir, senal	400-480	Cum	
3	Medium wood such as Pine, Deodar, Chir. Mango	480-640	Cum	
4	Teak	625	Cum	
Bricks				
1	Common brunt clay bricks	1600-1920	Cum	
2	Engineering bricks	2160	Cum	
3	Brick blast	1200	Cum	
4	Brick dust (Surkhi)	1000	Cum	
Cement				
1	Ordinary and alumina	1440	Cum	
2	Repaid hardening	1050-1280	Cum	
Cement concrete Plain				
1	Areated	260	Cum	
2.	Brick Aggregate	1760-2160	Cum	
3.	Stone ballast	2240-2400	Cum	
Reinforcement Cement Concrete				
1	With 1% Steel	2310-2470	Cum	
2	With 2% Steel	2370-2530	Cum	

S. No.	Materials	Weight in Kg.	Per	Remarks
3	With 3% Steel	2560-2720	Cum	
4	Concrete with brick agg. and lime mortar	1920	Cum	
Masonry				
1	Brick masonry with common burnt clay bricks	1920	Cum	
2	Brick masonry with 7 mounted bricks	2400	Cum	
3	Stone masonry dry rubble	2000	Cum	
4	Stone masonry in mortar of lime, sand stone, granite etc.	2240-2640	Cum	
Roofing				
1	Country tilled (Double) including battens	120	Sqm	
2	Country tiled (Single)	70	Sqm	
3	Manglore tiled with battons	65	Sqm	
4	C.G.I. Sheet	10-13	Sqm	
5	A.C. Sheet	34	Sqm	
Mortar				
1	Cement	2030	Cum	
2	Lime	1600-1840	Cum	

USEFUL CONVERSION TABLES

TABLE 8

S. No	To Convert	Into	Multiply By	Divide By
1	Metres	Yards	1.09361	0.9144
2	Metres	Feet	3.28084	0.3048
3	Metres	Inches	39.3701	0.0254
4	Kilometre	Mile	0.62137	1.60934
5	Square Kilometre	Square Mile	0.386101	2.58999
6	Square Metre	Sq. Yards	1.19599	0.83613
7	Square Yards	Sq. Ft.	10.7659	0.092903
8	Square Metre	Sq. Inches	1550.00	0.00064516
9	Square Centimetres	Sq. Inches	0.1550	6.4516
10	Hectare (1000 Sq. M.)	Acres	247105	0.404686
11	Cubic Metres	Cubic Ft.	35.3147	0.028317
12	Cubic Metres	Gallons (IMP)	2.19969	0.00454609
13	Cubic Centimetres	Cubic Inches	0.061024	16.3871
14	Litre	Gallon	0.219975	4.54506
15	Kilograms	Pounds	2.2046	0.4535924
16	Quintal (100 Kg.)	Maunds	2.67923	0.373242
17	Quintal (100 Kg.)	Hundred Weight	1.96840	0.50802
18	Tonnes (Metric Ton)	Tons (British)	0.9892	1.01605
19	Tonnes (Metric Ton)	Maunds	26.7923	0.0373242
20	Kilometres per hour	Miles Per hour	0.62173	1.60934
21	Kilometres per hour	Ft. per Sec.	0.91134	1.09728
22	Gramme per Cub. Cm.	Pound Per cu. Inch	0.0361273	26.6799
23	Kilometres per Cum	Pound Per cu. Ft.	0.0624	16.018
24	Kilometres per litre	Pound Per cu. Ft.	62.426	0.0160189
25	Kilometres per Sqm	Pound Per Sq. Inch	14.2233	0.07.0307
26	Tonne/Per Sqm	Ton per Sq. foot	0.09143	10.937
27	Kilometres per Sqm	Pound Per Sq. Ft.	0.20428	4.8924
28	Kilometres per metre	Pound Per Ft.	0.61797	1.48816
29	Kilometres Metre	Ft Pound	7.231	0.1382

MATERIAL CONSUMPTION STATEMENT
QUANTITY OF MATERIALS REQUIRED FOR VARIOUS TYPE OF MASONRY FOR ONE CUM.

S. No.	Type of Masonry	Qty. of Stone / Bricks	Qty. of Mortar Cum
1.	R.R. Stone masonry for foundation / superstructure wall in mortars.	1.14 Cum	0.33
2.	Dry Stone R.R. retaining wall masonry	1.14 Cum	--
3.	Dry Stone C.R. retaining wall masonry	1.24 Cum	--
4.	R.R. Dry Stone Pitching	1.14 Cum	--
5.	C.R. Dry Stone Pitching	1.24 Cum	--
6.	R.R. Stone Pillar masonry in mortar	1.20 Cum	0.33
7.	Arch masonry in mortar	1.25 Cum	0.25
8	Cut Stone Arch masonry in mortar	1.25 Cum	0.25
9	Relieving Arch masonry in mortar	1.25 Cum	0.25
10	Brick masonry for foundation / superstructure in mortars size 22.9x11.1x7.0cm. with 1cm. thick mortar joint.	473Nos	0.237

QUANTITY OF MATERIALS FOR VARIOUS NOMINAL MIX (ONE CUM OF CONCRETE)

S. No.	Mix	Requirement of Cement			Sand	Coarse Aggregate
		Cum	Tonnes	Bags		
1	Cement Concrete M-20 (1:1.5:3)	0.283	0.400	8.00	0.425	0.85
2	Cement Concrete M-15 (1:2:4)	0.222	0.322	6.44	0.445	0.90
3	Cement Concrete (1:3:6)	0.156	0.220	4.40	0.470	0.94
4	Cement Concrete (1:4:8)	0.120	0.170	3.40	0.48	0.96
5	Cement Concrete (1:5:10)	0.098	0.140	2.80	0.49	0.98

MATEIAL CONSUMPION STATEMENT FOR VARIOUS TYPE OF MORTARS FOR ONE CUM

S. No.	Mix	Requirement of Cement			Lime Cum	Sand / Marble Powder
		Cum	Tonnes	Bags		
1	Lime Sand Mortar (1:2)	--	--	--	0.475	0.95
2	Lime Sand Mortar (1:3)	--	--	--	0.367	1.07
3	Cement Mortar (1:3)	0.357	0.510	10.20	--	1.07
4	Cement Mortar (1:4)	0.258	0.386	7.72	--	1.07
5	Cement Mortar (1:5)	0.214	0.308	6.16	--	1.07
6	Cement Mortar (1:6)	0.179	0.256	5.12	--	1.07
7	Cement Mortar (1:7)\	0.153	0.220	4.40	--	1.07
8	Cement Mortar (1:8)	0.134	0.193	3.86	--	1.07
9	Composite Mortar (Cement : Lime : Sand) 1:1:6	0.178	0.256	5.12	0.178	1.07
10	Composite Mortar (Cement : Lime : Sand) 1:2:9	0.190	0.173	3.46	0.24	1.07

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
RCC/BR Work :				
1.	R.B. work in cement mortar 1 : 2	Cum		3.00
2.	R.C.C. door and window frames :			
	(a) 100 x 75 mm size	Mtr.		0.06
	(b) 125 x 75 mm size	Mtr.	-	0.07
3.	R.C.C. shelves in 1 : 2 : 4 mix :			
	(a) 20mm thick	Sqm	-	0.13
	(b) 25mm thick	Sqm	-	0.16
	(c) 30 mm thick	Sqm	-	0.19
4.	Pre cast cement concrete jali in 1 : 2 : 4 for fixing of :			
	(a) 50mm thick	Sqm	-	0.03
	(b) 40mm thick	Sqm	-	0.03
	(c) 30mm thick	Sqm	-	0.03
	(d) 20mm thick	Sqm	-	0.03
Masonry :				
5.	R.R. stone masonry in :			
	(a) Cement mortar 1 : 3	Cum	-	3.40
	(b) Cement mortar 1 : 4	Cum	-	2.50
	(c) Cement mortar 1 : 6	Cum	-	1.60
	(d) Cement mortar 1 : 8	Cum	-	1.25
	(e) Cement lime sand mortar 1 : 2 : 9	Cum	-	1.10
	(f) Cement lime sand mortar 1 : 3 : 12	Cum	-	0.85
6.	Ashlar veneer facing in cement mortar 1 : 4 including Cement pointing in 1 : 3 cement mortar.	Cum	-	0.25
7.	Brick masonry in :			
	(a) Cement mortar 1 : 3	Cum	-	2.40
	(b) Cement mortar 1 : 4	Cum	-	1.80
	(c) Cement mortar 1 : 6	Cum	-	1.20
	(d) Cement mortar 1 : 8	Cum	-	0.90
	(e) Cement lime sand mortar 1 : 2 : 9	Cum	-	0.80
	(f) Cement lime sand mortar 1 : 3 : 12	Cum	-	0.60
8.	75mm thick brick pardi wall in :			
	(a) Cement mortar 1 : 4	Sqm	-	0.15
	(b) Cement mortar 1 : 6	Sqm	-	0.09
	(c) Cement mortar 1 : 8	Sqm	-	0.07
9.	112mm thick brick pardi wall in :			
	(a) Cement mortar 1 : 4	Sqm	-	0.21
	(b) Cement mortar 1 : 6	Sqm	-	0.15
	(c) Cement mortar 1 : 8	Sqm	-	0.10
	Damp Proof Course :			
10.	Laying damp proof coarse 25mm thick in C.M. 1:3.	Sqm	-	0.255
11.	Laying cement concrete M.15 (1 : 2 : 4) for damp proof or coping :			
	(a) 30mm thick	Sqm	-	0.192
	(b) 50mm thick	Sqm	-	0.320
	(c) 75mm thick	Sqm	-	0.480
	(d) 100mm thick	Sqm	-	0.640

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
	Stone Work			
12.	Fixing stone sills and coping in cement mortar 1 : 3	Sqm	-	0.10
13.	Fixing stone sills and coping in cement mortar 1 : 4	Sqm	-	0.08
14.	Fixing stone shelves in cement mortar 1 : 3	Sqm	-	0.02
15	Fixing Karauli stone steps in cement mortar 1 : 4	Sqm	-	0.08
	Roofing :			
16.	Stone slab roofing.	Sqm	-	0.01
17.	Brick kharanja in cement sand mortar 1 : 4 on edge.	Sqm	-	0.09
18.	Providing flat brick kharanja in cement mortar 1 : 6.	Sqm	-	0.07
19.	Providing stone slab covering over drain including filling of joints in cement sand mortar 1 : 3 with 35mm thick cement concrete flooring 1 : 2 : 4.	Sqm	-	0.43
20.	Making khurras 45 x 45 C.M. with minimum 5cm. thick cement concrete 1 : 2 : 4 over PVC sheet, finished with 12mm cement plaster 1 : 3 and a floating coat of cement.	Each	-	0.12
	Flooring :			
21.	Coarse stone paving 125mm to 150mm with pointing in cement mortar 1 : 3.	Sqm	-	0.25
22.	Coarse stone paving 200mm to 225mm with pointing in cement mortar 1 : 3.	Sqm	-	0.20
23.	Brick on edge flooring laid on 12mm thick mortar including filling joints with slurry etc. in :	Sqm		
	(a) Cement sand mortar 1 : 6	Sqm	-	0.26
	(b) Cement sand mortar 1 : 4	Sqm	-	0.37
25.	Providing and laying cement concrete flooring in M 10 grade :			
	(a) 25 mm thick.	Sqm	-	0.11
	(b) 30 mm thick.	Sqm	-	0.13
	(c) 38 mm thick.	Sqm	-	0.17
	(d) 50 mm thick.	Sqm	-	0.22
	(e) 75 mm thick.	Sqm	-	0.33
26.	40mm thick red oxide flooring under layer of 30mm thick cement concrete 1 : 2 : 4 and top layer of 10 mm thick plaster of cement red oxide mix and sand mortar 2 : 3 finished with a coat of neat cement red oxide mix etc.	Sqm	-	0.35
27.	40 mm thick polished cement concrete flooring under layer of 30mm 1 : 2 : 4 concrete and top layer 10 mm thick of mix 1 : 2 cement stone dust etc.	Sqm	-	0.40
28.	52mm thick cement concrete flooring with metallic concrete hardner topping under layer of 40mm thick 1 : 2 : 4 concrete and top layer of 12mm thick metallic concrete hardner consisting of mix 1 : 2 cement metallic hardening compound mixed in ratio 4 : 1 (4 cement 1 : part of metallic hardening compound by wt.) including slurry etc. complete.	Sqm	-	0.46

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
29.	Cement plaster skirting with cement mortar 1 : 3 finished with a floating coat of neat cement :			
	(a) 18 mm thick.	Sqm	-	0.21
	(b) 25 mm thick.	Sqm	-	0.28
30.	Red oxide plaster skirting with top layer of 6mm thick cement red oxide mix in mortar 1 : 3 (cement red oxide 1 : sand 3) finished with a floating coat of cement red oxide mix:			
	(a) 18 mm thick.	Sqm	-	0.20
	(b) 25 mm thick.	Sqm	-	0.26

S.No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg. (OPC)	Bags of 50 Kg. (White)
31.	40mm thick marble chips flooring under layer 34mm thick cement concrete 1 : 2 : 4 and top layer of 6mm thick marble chips laid in cement marble powder mix 3 : 1 by weight in proportion of 4 : 7 (4 marble cement powder : 7 marble chips) including cement slurry :				
	(a) With ordinary cement.	Sqm	-	0.34	-
	(b) With dark shade pigment with ordinary cement.	Sqm	-	0.33	-
	(c) Light shade pigment with white cement or without pigment in white cement.	Sqm	-	0.26	0.068
	(d) Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	-	0.30	0.034
32.	40mm thick marble chips flooring under layer 31mm thick cement concrete 1 : 2 : 4 and top layer of 9mm thick marble chips laid in cement marble powder mix 3 : 1 by weight in proportion of (4 marble cement powder : 6 marble chips) including cement slurry :				
	(a) With ordinary cement.	Sqm	-	0.37	-
	(b) With dark shade pigment with ordinary cement.	Sqm	-	0.36	-
	(c) Light shade pigment with white cement or without pigment in white cement.	Sqm	-	0.24	0.102
	(d) Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	-	0.30	0.051
33.	40mm thick marble chips flooring under layer 28mm thick cement concrete 1 : 2 : 4 and top layer of 12mm thick in cement marble powder mix 3 : 1 by weight in proportion of (4 cement marble powder mix : 5 marble chips) including cement slurry etc :				

S.No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg. (OPC)	Bags of 50 Kg. (White)
	(a) With ordinary cement.	Sqm	-	0.39	-
	(b) Dark shade pigment in ordinary cement.	Sqm	-	0.38	-
	(c) Light shade pigment in white cement or without pigment in white cement.	Sqm	-	0.21	0.136
	(d) Medium shade pigment with 50% white cement and 50% ordinary cement..	Sqm	-	0.30	0.068
34.	40mm thick marble chips flooring under layer 25mm thick cement concrete 1 : 2 : 4 and top layer of 15mm thick cement marble powder mix 3 : 1 by weight in proportion of cement marble powder mix marble chips by volume including cement slurry etc. complete :				
	(a) With ordinary cement.	Sqm	-	0.40	-
	(b) Dark shade pigment in ordinary cement.	Sqm	-	0.39	-
	(c) Light shade pigment in white cement or without pigment in white cement.	Sqm	-	0.22	0.170
	(d) Medium shade pigment with approximately 50% white cement and 50% ordinary cement.	Sqm	-	0.30	0.085
35.	Marble chips skirting 26mm thick under layer 20mm thick cement plaster 1 : 3 and top layer of 6mm thick marble chips :				
	(a) With ordinary cement.	Sqm	-	0.28	-
	(b) Dark shade pigment in ordinary cement.	Sqm	-	0.27	-
	(c) Light shade pigment in white cement or without pigment in white cement.	Sqm	-	0.20	0.052
	(d) Medium shade pigment with 50% white cement and 50% ordinary cement..	Sqm	-	0.24	0.026

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
36.	Crazy marble stone flooring including filling the gaps with cement marble chips mixture 3 : 1 by weight in the proportion of 4 : 7 (4 cement marble powder mix : 7 marble chips) by volume and under layer of 25mm thick cement concrete 1 : 2 and slurry etc. complete :			
	(a) With Irregular shaped crazy.	Sqm	-	0.24
	(b) With rectangular fine edged crazy.	Sqm	-	0.20

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
37.	Providing & fixing of pre-cast terrazo chequered or or embossed / ironite toppings tiles over 20 mm thick bed of :			
	(a) Cement mortar 1 : 6	Sqm	-	0.12
	(b) Cement mortar 1 : 4	Sqm	-	0.17
38.	Providing & fixing of Kota, Modak flooring over 20mm thick base of lime mortar jointing with cement mortar 1:3.	Sqm	-	0.18
39.	Providing & fixing of Kota, polished stone laid on 12mm thick cement mortar 1 : 3 and jointing with neat cement slurry.	Sqm	-	0.15
40.	Marble stone flooring over 20 mm thick base jointed with white cement slurry etc. complete on (All marble slabs):			
	(a) Base lime mortar	Sqm	-	W0.02
	(b) Base cement mortar 1 : 4	Sqm	-	0.15
41.	Marble slabs in risers steps skirting dado wall of pillars on 12 mm thick cement mortar 1 : 3 jointed with white cement.	Sqm	-	0.13
Plaster :				
42.	Cement plaster 12 mm thick :			
	(a) Cement sand 1 : 3	Sqm	-	0.15
	(b) Cement sand 1 : 4	Sqm	-	0.11
	(c) Cement sand 1 : 6	Sqm	-	0.07
	(d) Cement sand 1 : 8	Sqm	-	0.05
43.	20 mm thick cement plaster :			
	(a) Cement sand 1 : 3 ratio.	Sqm	-	0.23
	(b) Cement sand 1 : 4 ratio.	Sqm	-	0.17
	(c) Cement sand 1 : 6 ratio.	Sqm	-	0.11
	(d) Cement sand 1 : 8 ratio.	Sqm	-	0.08
44.	25 mm thick cement plaster :			
	(a) Cement sand 1 : 3 ratio.	Sqm	-	0.28
	(b) Cement sand 1 : 4 ratio.	Sqm	-	0.21
	(c) Cement sand 1 : 6 ratio.	Sqm	-	0.14
	(d) Cement sand 1 : 8 ratio.	Sqm	-	0.10
45.	Cement lime plaster 1 : 1 : 6 ratio :			
	(a) 25 mm thick	Sqm	-	0.14
	(b) 20 mm thick	Sqm	-	0.12
	(c) 12 mm thick	Sqm	-	0.07
46.	Cement lime plaster 1 : 2 : 9 ratio :			
	(a) 25 mm thick	Sqm	-	0.10
	(b) 20 mm thick	Sqm	-	0.08
	(c) 12 mm thick	Sqm	-	0.05
47.	6 mm cement plaster to ceiling:			
	(a) Cement sand 1 : 3 ratio.	Sqm	-	0.08
	(b) Cement sand 1 : 4 ratio.	Sqm	-	0.06
	(c) Cement sand 1 : 6 ratio.	Sqm	-	0.04
48.	Neat cement punning.	Sqm	-	0.04
49.	Fine finishing coat of cement sandla	Sqm	-	0.04

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
50.	Providing on stone masonry in cement mortar 1 : 3 :			
	(a) Flush or ruled or sunk of weather proof pointing.	Sqm	-	0.024
	(b) Raised and cut pointing	Sqm	-	0.04
51.	Pointing on brick or tile work in cement mortar 1 : 3 :			
	(b) Raised and cut pointing	Sqm	-	0.033
52.	Coursed Rubble masonry in cement mortar 1 : 6 in super structure.	Cum	-	1.50
53.	Plain Ashlar masonry in super structure in cement mortar 1 : 6 including pointing with cement mortar 1 : 2.	Cum	-	1.08
54.	Providing & fixing horizontal chhajja of stone in cement mortar 1 : 4 including pointing in cement mortar 1 : 2.	Sqm	-	0.10
55.	Providing & fixing EZ-7 frame embedded in cement concrete block 15 x 10 x 10 cm. size of 1 : 3 : 6.	Quintal	-	0.22
56.	Providing and fixing M.S. ring for fastening ropes of chick.	100 Nos.	-	0.10
57.	Fixing chowkhats in opening including embedding chowkhats in floor, cutting masonry for hold fasts embedded in C.C. 1 : 3 : 6 and making good the wall :			
	(a) Door chowkhats.	Each	-	0.24
	(b) Window chowkhats.	Each	-	0.12
	(c) Clear storey window.	Each	-	0.66
58.	Providing and fixing M.S. fan clamp in existing R.C.C. slab including cutting and making good.	Each	-	0.03
59.	Replacing sand stone slabs in roof laid in cement mortar 1 : 4 including necessary repairs and cement pointing.	Sqm	-	0.01
SANITARY :				
60.	Fixing of water closets (all type) along with flushing cisterns and brackets, telescopic flush pipe of bend with fittings and clamps, overflow pipes complete including cutting making walls and floors good.	Each	-	0.10
61.	Urinal basin with automatic flushing cistern.	Each	-	0.05
62.	Range of two urinals with one automatic flushing cistern.	Each	-	0.08
63.	Range of three urinals with one automatic flushing cistern.	Each	-	0.13
64.	Range of four urinals with one automatic flushing cistern.	Each	-	0.19
65.	Fixing long Pan pattern or Orissa pattern W.C. pan.	Each	-	0.05
66.	Fixing a pair of white glazed earthen ware or vitreous china foot rest.	Each	-	0.10

S. No.	Description of Item	Unit	Requirement of Cement Cum	Bags of 50 Kg.
67.	Fixing flat back or wall corner tube lipped front urinal basin.	Each	-	0.02
68.	Fixing laboratory basin with brackets, pillar taps etc., cutting and making wall good.	Each	-	0.05
69.	Fixing sink with brackets including cutting and making good the wall.	Each	-	0.05
70.	Fixing wash basin.	Each	-	0.03
71.	Fixing M.S. holder bat clamps or M.S. stays and clamps.	Each	-	0.005
72.	Fixing sand cast iron trap.	Each	-	0.025
73.	Providing and fixing wall face C.I. pipes including filling the jointing with spun yarn soaked neat cement slurry and cement mortar 1 : 2.	100 Mtr.	-	0.17
74.	Providing and fixing on wall face C.I. assessories for rain water pipe fittings, the joints with spun yarn soaked in neat cement slurry and cement mortar 1 : 2 :			
	(a) 75 mm dia.	Each	-	0.005
	(b) 100 mm dia.	Each	-	0.006
	(c) 150 mm dia.	Each	-	0.01

Unit Weight of Aluminium Sections

S. No.	Size of Aluminium Section		Unit Weight
Doors & Partitions			
1.	101.60 X 44.45 X 2.00 MM 63 X 38 X 2.0 MM 44.45 X 44.45 X 2.0 MM	83.50 X 44.45 X 2.0 MM 150.0 X 44.45 X 2.0 MM	1.64 Kg/M ± 5 %
2.			
3.			
4.			
5.			
Snap Beading			
1.	16.6 X 18.2 X 1.0 MM		
Sliding Windows			
1.	62.0 X 30.0 X 1.6 MM	62.0 X 30	
2.			
3.			
4.			
5.			
6.			
Hollow Sections			
Z inner			
Z outer			
Mullion			

Government of Rajasthan
Public Works Department

From :- Addl. Secretary to Govt.,
Public Works Department,
Rajasthan, Jaipur.

To : The Chief Engineer,
Public Works Department,
Rajasthan, Jaipur.

No. :F - Dated : /04/2007

Sub :- Forecast Estimate of Building.

Sir,

I am directed to convey the approval of the modification in delegation of power for forecast estimate of building works as per clause 31.7.3 of PWD Manual Vol. N. 2 of 1986 as detailed below with immediate effect.

Building Works :

S. No	Authority	Amount upto which forecast estimate can be issued	
		New Construction	Renovation / Repair & Maintenance Works
(i)	**Non-Residential Buildings**		
1.	Executive Engineer	Upto 25.00 Lacs.	Upto 5.00 Lacs.
2.	Superintending Engineer	Upto 30.00 Lacs.	Upto 7.50 Lacs.
3.	Additional Chief Engineer,	Upto 50.00 Lacs.	Upto 10.00 Lacs.
4.	Chief Engineer	Upto 100.00 Lacs.	Upto 20.00 Lacs.
5.	Government	Above 100.00 Lacs.	Above 20.00 Lacs.
(ii)	**Residential Buildings**		
1.	Executive Engineer	Upto 25.00 Lacs.	Upto 1.00 Lacs.
2.	Superintending Engineer	Upto 30.00 Lacs.	Upto 2.00 Lacs.
3.	Additional Chief Engineer,	Upto 50.00 Lacs.	Upto 3.00 Lacs.
4.	Chief Engineer	Upto 100.00 Lacs.	Upto 5.00 Lacs.
5.	Government	Above 100.00 Lacs.	Above 5.00 Lacs.

This bars the approval of Govt. in PWD vide Pr. Secretary ID No. 1504 dated 05-06-07

sd/-
Dy. Secretary (Works)

Copy to :- CE/SE (MB)/D-326 Dated : 12-06-2007

1- Ps. to Pr. Secretary, PWD, Rajasthan, Jaipur.
2- Ps. to Secretary, PWD, Rajasthan, Jaipur.

sd/-
Dy. Secretary (Works)

Phone | 0141-2227107 (O)
0141-5153222 Ext. 1587 (O)
Fax | 0141-2227107 (O)

सत्यमेव जयते

Dr. Dinesh Kumar Goyal I.A.S.
Principal Secretary to Government of Rajasthan
Public Works Department
डॉ.दिनेश कुमार गोयल आई.ए.एस.
शासन प्रमुख सचिव, राजस्थान सरकार
सार्वजनिक निर्माण विभाग

5022, Main Building, Secretariat,
Jaipur-302 005
5022, मुख्य भवन, सचिवालय,
जयपुर-302 005

क्रमांक :पीएस/पीएस–पीडब्ल्यूडी/10/D-728
दिनांक : 03.05.2010

भवन निर्माण में सुधार हेतु निर्देश

सार्वजनिक निर्माण विभाग द्वारा सम्पन्न किये जा रहे भवनों के निर्माण प्रक्रिया इत्यादि में सुधार हेतु विभाग द्वारा समय समय पर कदम उठाये गये हैं। भवनों के निर्माण कार्यो की समय पर एवं गुणवत्ता से निर्माण हेतु निम्न दिशा निर्देश जारी किये जाते हैं।

1. **प्रशासनिक स्वीकृति** :– प्रशासनिक विभाग द्वारा भवनों की उपयुक्त/अनुकूल भूमि उपलब्ध होने, बजट उपलब्ध कोने के पश्चात् प्रशासनिक स्वीकृति जारी की जावें। स्वीकृति से पूर्व जॉच सूची (परिशिष्ट –1) की पूर्ति किया जाना आवश्यक है।

2. प्रशासनिक विभाग द्वारा प्रस्तावित नये आवासीय/गैर आवासीय भवन अथवा मरम्मत के कार्यों का तकमीना प्रेषित करने के लिए सार्वजनिक निर्माण विभाग के निम्न अधिकारी अधिकृत हैं।

क्र. सं.	क्षेत्र के अधिकृत अधिकारी	नये आवासीय/ गैर आवासीय भवन (लागत रू. लाखों में)	परिवर्तन/परिवर्धन/मरम्मत के कार्य (लागत रू. लाखों में)	
			आवासीय भवन	गैर आवासीय भवन
1.	अधिशाषी अभियन्ता	20.00 तक	1.00 तक	5.00 तक
2.	अधीक्षण अभियन्ता	30.00 तक	2.00 तक	7.00 तक
3.	अति. मुख्य अभियन्ता	50.00 तक	3.00 तक	10.00 तक
4.	मुख्य अभियन्ता	100.00 तक	5.00 तक	20.00 तक
5.	राज्य सरकार	100.00 से अधिक	5.00 से अधिक	20.00 से अधिक

3. **भवन निर्माण की डिफेक्ट लायबिलिटी समय 5 वर्ष** :– संवेदकों द्वारा भवनों के निर्माण में गुणवत्ता सुनिश्चित करने के लिए आदेश दिनांक 07.10.2009 के अनुसार 10.00 लाख रुपये व अधिक के अनुबन्धों में डिफेक्ट लायबिलिटी समय 5 वर्ष रखा जावे। (परिशिष्ट–2)

4. **साधारण किफायती भवन/मितव्ययता से भवन निर्माण** :– भवन निर्माण में मितव्यता के संबंध में राजस्थान सरकार के दिनांक 13.07.09 को परिपत्र (परिशिष्ट – 3) की अनुपालना की जावे।

5. **विद्युत ऊर्जा को बचाने के उपाय** :– राज्य सरकार के परिपत्र दिनांक 08.11.07 (परिशिष्ट– 4) के अनुसार विद्युत ऊर्जा बचाने के विद्युत उपकरण, सोलन वाटर हीटर, सोलर स्ट्रीट लाईट, इत्यादि का उपयोग किया जावे।

6. निशक्तजन हेतु निर्माण :– निःशक्त जनों को भवन में सुगम आवागमन हेतु राज्य सरकार के आदेश दिनांक 05.02.10 (परिशिष्ट–5) के अनुसार भवनों में रेलिंग सहित रेम्प निर्माण कराया जावे, जिसका

स्लोप 1:10 रखा जावे (नक्शा परिशिष्ट–6)। निःशक्तजनों के लिए व्हील चेयर, शौचालय का निर्माण एवं ब्रेलपिपि संकेत/ऑडियो संकेत लिफ्ट आदि में कराया जावे।

7. **भवनों की खुदाई संबंधित सुरक्षा निर्देश :–** सार्वजनिक निर्माण विभाग के परिपत्र संख्या 1/2006 दिनांक 29.05.06 (परिशिष्ट–7) के अनुसार नींव खुदाई बाबत् निर्देशों की पालना की जावे।

8. **भवनों के निर्माण में "वर्षा जल पुनःर्भरण/संग्रहण संरचना" के निर्देश :–** वर्षा के पानी द्वारा भूगर्भ जलस्तर बढ़ाने के उद्देश्य से सार्वजनिक निर्माण विभाग के परिपत्र डी–355 दिनांक 04.06.05 (परिशिष्ट–8) के अनुसार 500 वर्ग मीटर व उससे बड़े भूखण्ड पर निर्माण के साथ जल पुनः र्भरण/संग्रहण संरचना का निर्माण कराया जावे।

9. **बी.एस.आर. में आई. एस. नम्बर अंकित करना :–** बी.एस.आर. में आईटमों में आई.एस. कोड के अनुसार निर्माण सामग्री का उपयोग सुनिश्चित करने के लिए आई. एस. कोड नं. अंकित किया जावे।

10. **कम्पनी का नाम :–** भवन निर्माण में गुणवत्ता सुनिश्चित करने के लिए निर्माण कार्य में ली गई फिटिंग की सप्लाई कम्पनी का नाम माप पुस्तिका में अंकित किया जावे।

(डॉ. दिनेश कुमार गोयल)
प्रमुख शासन सचिव

क्रमांक: मु.अ./अधि.अभि.(भवन)/डी-729 दिनांक 05-05-2010

प्रतिलिपि :–

1. प्रमुख शासन सचिव, माननीय मुख्यमंत्री महोदय, राजस्थान, जयपुर।
2. विशिष्ठ सहायक, माननीय सार्वजनिक निर्माण मंत्री महोदय, राजस्थान, जयपुर।
3. निजी सचिव, मुख्य सचिव, राजस्थान, जयपुर।
4. निजी सचिव, अतिरिक्त मुख्य सचिव, महामहिम राज्यपाल, राजभवन, जयपुर।
5. निजी सचिव, अतिरिक्त मुख्य सचिव, कृषि विभाग, राजस्थान, जयपुर।
6. निजी सचिव, अति0 मुख्य सचिव एवं निदेशक, हरिशचन्द्र माथुर लोक प्रशिक्षण संस्थान, जयपुर।
7. समस्त प्रमुख शासन सचिव/शासन सचिव, राजस्थान।
8. मुख्य अभियन्ता एवं अति0 सचिव, सार्वजनिक निर्माण विभाग, राजस्थान, जयपुर।
9. मुख्य अभियन्ता, एन.एच./भवन/पी.एम.जी.एस.वाई./एस.एस., सा0नि0वि0, जयपुर।
10. प्रबन्ध निदेशक, राजस्थान राज्य सड़क विकास एवं निर्माण निगम लि., जयपुर।
11. आयुक्त, जयपुर विकास प्राधिकरण, जयपुर/जोधपुर विकास प्राधिकरण, जोधपुर।
12. समस्त जिला कलक्टर।
13. अतिरिक्त मुख्य अभियन्ता, सार्वजनिक निर्माण विभाग (समस्त)।
14. अधीक्षण अभियन्ता, सार्वजनिक निर्माण विभाग (समस्त)। नगर वृत्त जयपुर
15. अधिशाषी अभियन्ता, सार्वजनिक निर्माण विभाग (समस्त)।
16. ए.सी.पी., सार्वजनिक निर्माण विभाग, जयपुर।

प्रमुख शासन सचिव

कार्यालय अधि. अभियन्ता
... जयपुर
क्रमांक 842 दिनांक 7-5-10

Copy to E.E. City Dn-I, II, III, Const & Secretariat Dn for information & compliance.

अधिक्षण अभियन्ता
सा0 नि0 वि0 शहर वृत्त
जयपुर

परिशिष्ट -- 1

Check list before issuing Administrative and Financial sanction
For building works to be executed by PWD

SN	Item	Adm. Department's Input	PWD Remarks
1	2	3	4
1	Name of Administrative Department/ Agency		
2	Name of work		
3	Location of work	Tehsil ____ Sub Div. ____ District	
4	Amount of this administrative sanction proposed (Rs. In lacs)		
5	Area of land already in possession for construction of building (sq.mtr)		
6	Whether the local PWD officer has certified the suitability of site	Yes / No	
7	Land area required (in sq.mtr.)		
8	Modification in existing building required	Yes / No	
9	Is land acquisition/allotment required ?	Yes / No	
10	Allotment Authority		
11	Letter for land allotment issued to (copy is to be enclosed)	Ref. No. ____ Date ____ Authority	
12	Location of proposed building has been marked on drawings, with complete dimensions (Enclosed)		
13	Drawings prepared and approved by the Administrative Department.	Yes / No.	
14	Detailed estimates are prepared.	Yes / No	
15	PWD Economy Circular No. F.(184) SE (B) / Circular/D-144 dated 13.7.2009 has been followed.	Yes / No	
16	Whether provision of "water harvesting system" has been made.	Yes / No	
17	Whether following of persons with Disabilities Act has been taken ramps. adoption of toilets for wheel chair users, Braille symbols & auditory signals in elevators or lifts.	Yes/No	
18	Area to be constructed (sq.mtr)		
19	Approximate cost per sq.mtr		
20	Budget provision year wise in case of stage construction (Rs. In lacs)	Year / Proposed Budget Total	
21	Administrative Sanction	No. Date	
22	Financial Sanction	No. Date	
23	Contact person in Administrative Department	Name ____ Post ____ Address ____ ____ Mobile	
24	Contact person for work in field	Deptt. ____ Name ____ Post ____ Address ____ Mobile	
25	Authority signing on behalf of Administrative Department	Name ____ Post ____ Contact No. ____ Signatures _ Date	

Signature ....
Designation ....

परिशिष्ट– 2

GOVERNMENT OF RAJASTHAN
PUBLIC WORKS DEPARTMENT

No. CE(Bldg.)DLP.(Bldg. Works)/D- 268 Dated 7|10|09

ORDER

In pursuance of amendment made by Finance Department (G&T Division) GoR vide circular no. 43/2009 dated 15.09.09, in PWF & AR Part- I and II regarding refund of Security Deposit (copy enclosed), Modified Agreement Format shall be applicable for all Road, Building and Bridge/ CD works having cost more than Rs. 10.00 lacs in Public Works Department with immediate effect.

2.The Defect Liability Period (DLP) for
(a) New Road/ Building/ Bridge/ CD works, Road widening, strengthening, up-gradation,
(b) Renewal and special repairs of roads and
(c) Special repair to Building/ Bridge/ CD works
costing more than Rs. 10.00 lacs, shall be Five Years. Special conditions in the Contract Agreement to this effect shall be as per the Annexure '1' which shall be part of modified agreement / tender document.

3. For National Highway works, the condition issued form time to time by GoR/ GoI will remain applicable.

4. This bears the approval of Govt. vide I.D. No.1233 dated 7.7.2009.

Encl:- Special Conditions of Contract
& F.D. Circular

(D.C. KATARA)
Addl. Secy. & Chief Engineer, PWD

No. CE(Bldgs)DLP(Bldg.works)D-268 Dated 7.10.09
Copy to the following for kind information / necessary action please:-
1. P.S. to Addl. Chief Secretary to CM, Govt. of Rajasthan, Jaipur.
2. Principal Secretary to CM, Govt. of Rajasthan, Jaipur.
3. P.S. to the Hon'ble Minister, PWD, Govt. of Rajasthan, Jaipur.
4. P.S. to Minister, Rural Development & Panchyat Raj, Govt. of India, New Delhi.
5. P.S. to Minister of State, MoRTH, Govt. of India, New Delhi.
6. P.S. to the Chief Secretary, Govt. of Rajasthan, Jaipur.
7. Principal Secretary, F.D., Govt. of Rajasthan, Jaipur.
8. Principal Secretary/ Secretary, PWD, Govt. of Rajasthan, Jaipur

9. Principle Secretaries/ Secretaries to GoR (All) for circulation to Organizations under their control.
10. Commissioner, Rajasthan Housing Board/ Avas Vikas Sansthan, Jaipur.
11 Commissioner, JDA Jaipur/ JoDA Jodhpur/ JNN Jaipur
12. Managing Director, RICCO Jaipur.
13. Administrator Rajasthan State Agriculture Marketing Board Jaipur
14. Chief Engineer (NH), PWD, Rajasthan, Jaipur. Please get issued the separate orders for defect liability period for road works of NHs.
15. Chief Engineer (Building/PMGSY/SS), Public Works Department, Jaipur.
16. Managing Director, RSRDCC, Jaipur with the directions to implement DLP in Building/bridge/CD work and road works with immediate effect accordingly.
17. F.A. /CAO (NH), PWD, Raj, Jaipur.
18. Chief Engineer, RHSDP, Medical Department Rajasthan, Jaipur
19. Chief Engineer, DPEP, Education Department Rajasthan, Jaipur
20. Addl. Chief Engineer, PWD Zone..........................(All)
21. Superintending Engineer, PWD, Circle.................(All).
22. TA-I/ SE (R)/ BOT/ TR/ PMGSY/ NH/ Buildings/ SS
23. Executive Engineers PWD, Div...........................(All)
24. ACP/Section-II/Joint L.R./DDS

Copy also for kind information to :

1. Secretary, MoRTH, Govt. of India, New Delhi.
2. Secretary, Rural Development & Panchyat Raj, Govt. of India, New Delhi.
3. Chairman, National Highways Authority of India, New Delhi.

Addl. Secy. & Chief Engineer, PWD

OFFICE OF THE CHIEF ENGINEER, P.W.D., RAJASTHAN, JAIPUR.

No. CE(Bldg)/DLP(Bldg Works)/D- Date: .10.2012

Office order

In partial modification to the Order no. CE(Bldg)/DLP(BldgWorks)/D-837 dated 10.09.2012, the Defect Liability Period (DLP) for New Building works and Special Repair to Building works shall be Three Years for the defects pertaining to Building Structure and other civil works. Special conditions in the Contract Agreement to this effect shall be as per Annexure '1' which shall be part of the modified agreement / tender document.

This bears the approval of Govt. ID No. 1664/M/PWD/12 dated 09.10.2012. This order shall be effective from date of issue.

Encl:-Special Conditions of Contract

sd/-
(HAZARI LAL MEENA)
Chief Engineer Cum Addl. Secy.,
PWD, Rajasthan Jaipur

No. CE(Bldg)/DLP(Bldg Works)/D- 980 Date: 12.10.2012

Copy submitted/ forwarded to the following for information and necessary action:-

1. P.S. to Pr. Secretary to CM, Govt. of Rajasthan, Jaipur.
2. P.S. to Hon'ble Minister, PWD, Govt. of Rajasthan, Jaipur.
3. P.S. to Hon'ble State Minister, PWD, Govt. of Rajasthan, Jaipur
4. D.S. to Chief Secretary, Govt. of Rajasthan, Jaipur.
5. P.S. to Principal Secretary, F.D., Govt. of Rajasthan, Jaipur.
6. P.S. to Principal Secretary, PWD, Govt. of Rajasthan, Jaipur.
7. Pr.Secy./Secretaries to GoR (All) for circulation to Organizations under their control..
8. P.S. to Secretary, PWD, Govt. of Rajasthan, Jaipur
9. Commissioner, Rajasthan Housing Board/Avas Vikas Sansthan, Jaipur.
10. Commissioner JDA Jaipur/JoDA Jodhpur/JNN Jaipur.
11. Managing Director, RICCO Jaipur.
12. Administrator, RSAMB, Jaipur.
13. Chief Engineer-----(All).
14. Managing Director, RSRDCC Jaipur with the direction to implement DLP accordingly.
15. FA, PWD Rajasthan, Jaipur
16. Addl Chief Engineer PWD Zone _____(All).
17. Chief Engineer, RHSDP, Medical Department Rajasthan, Jaipur.
18. Chief Engineer, DPEP, Education Department Rajasthan, Jaipur.
19. Superintending Engineer, PWD, Circle-........................(All).
20. TA-I/SE (Roads)/BOT/TR/PMGSY/NH/Buildings/SS.
21. Executive Engineer, PWD, Dn.-..................................(All).
22. ACP/Section-II/Joint L.R./DDS.
23. President, Contractor Association, Jaipur.

12/10/12
(HAZARI LAL MEENA)
Chief Engineer Cum Addl. Secy.,
PWD, Rajasthan Jaipur

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परिशिष्ट–३

...ERNMENT OF RAJASTHAN
...UBLIC WORKS DEPARTMENT

No.(184)SE(B)/Circulars/ Dated : 13 July, 2009

CIRCULAR

Subject: - Economy in Building Construction.

Construction of buildings should be ensured for its functional use with simple and economic construction practices. The following guidelines are given to achieve the economy in the constructions of all Government buildings including those constructed for Boards, Corporations and other bodies.

1. Selection of site should not be, in the filled up areas, in sloping ground with poor soil bearing capacity. Places close to water pond, pools etc. should be avoided.
2. Land for the proposed building should be checked to ensure that there are no overhead electric lines, underground major pipelines, or sewer lines.
3. Foundation sections should be designed economically with optimum use of soil bearing capacity. The R.C.C. frame should be conceived in such a manner that foundation section could be fully used.
4. Plinth height of building shall be fixed in such a manner that it should be functional. Unwanted raising of plinth should be avoided.
5. For Flooring in building, locally available materials, like Kota stone, should preferably be used. Expensive materials like marble, granite, wooden flooring, vitrified tiles etc. should be avoided.
6. Windows should preferably be strong for their functional requirements. Steel prefab windows should be used. Aluminum or high cost designer windows should be avoided.
7. Door frames in the building should have minimum wood use. Steel frames and flush doors should be maximally used.
8. Wooden wall paneling should not be used.
9. Interior and exterior fixtures and finish should be simple and durable with a better serviceability. High cost painting should be avoided.
10. False ceiling work should be avoided.
11. Number of electrical points should be kept minimum as per requirement.
12. Precast and costly ornamental electrical poles should not be used for campus lighting. In general, conventional type tubular poles should be used.
13. Any construction in deviation of the above guidelines will be permitted only with the prior approval of the Public Works Department .
14. This bears approval of F. D. vide I.D. No.100903051/PSF dated 9.7.2009

Principal Secretary, PWD

Copy to :-
1. P.S. to Principal Secretary-I/II to Hon'ble C.M.
2. P.S. to Hon'ble Minister, PWD
3. P.S. to Chief Secretary, Rajathan, Jaipur
4. All Principal Secretaries/Secretaries
5. All Boards/Corporations in Rajasthan, M.D. RSRDC Ltd
6. All Chief Engineers, PWD
7. All Additional Chief Engineer, PWD, Jaipur I, II/Ajmer/Bharatpur/ Bikaner/Jodhpur/Kota/Udaipur
8. All Suppdt. Engineers, ... for forward to concern S.E's.

Principal Secretary, PWD

परिशिष्ट – 4

ENERGY DEPARTMENT
NOTIFICATION
Jaipur, November 8, 2007

No. F.20(6) Energy/98:- In exercise of the powers conferred by Section 18 of the Energy Conservation Act, 2001 (Central Act No. 52 of 2001), the State Government, hereby issues the following directions the efficient use of energy and its conservation in the State of Rajasthan, namely:-

1. Mandatory use of Energy Efficient Lamps/compact Fluorescent Lamp (CFL):
(1) In all new buildings constructed in Government sector/Government aides sector Board and Corporation/Autonomous bodies incandescent lamps and conventional chokes for fluorescent lights shall not be used.

(2) In all existing Government sector/Government aided sector/Boards and Corporation /Autonomous bodies whenever defective incandescent lamps and chokes are replaced, shall be replaced by only compact fluorescent lamps (CFL) and Electronic chokes.

(3) The Distribution Companies shall make necessary modifications in the load demand notice within two months time from the date of this directions to promote the use of Compact Fluorescent lamps instead of conventional bulbs while releasing /sanctioning new connections /load in such buildings.

(4) All new buildings/institutions constructed in Government sector /Government aided sector/Boards and Corporation /Autonomous bodies shall also promote lighting luminaries with electronic ballast multi tap ballast/sensor and time controlled switching.

2. Mandatory use of Solar Water Heating Systems :
(1) The use of solar water heating systems shall be mandatory in the following categories of buildings, namely: -
(i) All Industrial buildings where hot water is required for processing.

(ii) All Private Hospitals and Nursing homes of 25 beds, and all Government Hospitals of 100 beds or more.

(iii) All Hotels and Resorts.

(2) Rajasthan Renewable Energy Corporation Limited will act as the State Nodal Agency to promote and review the specifications, quality of systems supplied and installation work of Solar Water Heating Systems in the State.
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3. Promotion of Energy Efficient Buildings Design :

(1) All the new buildings to be constructed in the Government /Government Aided sector shall incorporate Energy efficient buildings design concepts, computer energy simulation software like eQUEST, Energy plus etc. including Renewable Energy Technologies, with immediate effect.

(2) Chief Architect, Public Works Department shall ensure the incorporation of energy efficient buildings design concepts in all buildings to be constructed in Government /Government aided sector.

4. Promotion of Energy Efficient Air Conditioners:
Energy Efficient Air Conditioners (3 star and more star labeled as per BEE) shall only be used for Air Conditioning in all residential and non-residential Government buildings.

5. Exemptions:
These directions shall not apply for temporary lighting and display system for VVIP functions, Exhibitions, National Ceremonies etc.
These directions shall come into force with immediate effect.

By Order of the Governor,
Yaduvendra Mathur
Secretary to Government

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परिशिष्ट – 5

Government of Rajasthan
Office of the Chief Secretary

May kindly peruse the D.O. letter No. 13-01/2010-DD.III dated 31st January, 2010 of Shri Mukul Wasnik, Hon'ble Minister, Social Justice and Empowerment, Government of India addressed to Chief Minister, Rajasthan. As the UN Convention on the Rights of Persons with Disabilities has come into effect from 3.5.2008, the State Government has to take suitable action in regard to the provisions of UNCRPD.

Basically two actions are to be ensured in this regard:

(a) Persons with disabilities should have access, on an equal basis with others to buildings, roads, transportation and other indoor and outdoor facilities including schools, housing, medical facilities and work places;
(b) Information, communications and other services including electronic services and emergency services.

We have to make all the buildings of the State Government and all its offices accessible to the Disabled by the end of 2010-11. As per Section 46 of the concerned Act, ramps are to be provided in public places, toilet adapted for wheel chair users, ramps are to be provided in hospitals, primary health centers and other medical care and rehabilitation institutions and provision is to be made for Braille symbols and auditory signals on elevators or lifts.

It is also necessary that all web-sites of the State Government, as also their attached and subordinate offices, as well as public sector enterprises quickly comply with at least the mandatory level A of WCAG 2.0.A time bound action plan is to be chalked out for complete action in this regard so that web-sites of all organizations coming under the State Government are accessible to the Disabled within the financial year 2010-11. In case of any difficulty, IT Department may contact the Department of Information & Technology and the National Informatics Centre for any technical assistance in this regard.

I would suggest that Social Justice & Empowerment Department will be the Nodal Department to ensure the compliance of the first point i.e. "making all buildings accessible." to the Disabled. It will coordinate with other concerned departments particularly School Education, Medical & Health, Medical Education, PWD, UDH & LSG and Transport to ensure compliance. For point No. 2 i.e. "making of web-site accessible", IT Department will be the Nodal Department to ensure the compliance of the same.

Chief Secretary

Encl. A/a

1.Principal Secretary, Social Justice & Empowerment
2.Secretary, IT
U.O. Note No. 88 / CS / 9
Jaipur, dated : 05 February, 2010

F:\pwdlaptop\desktop\CE(Building) letters /Circulars

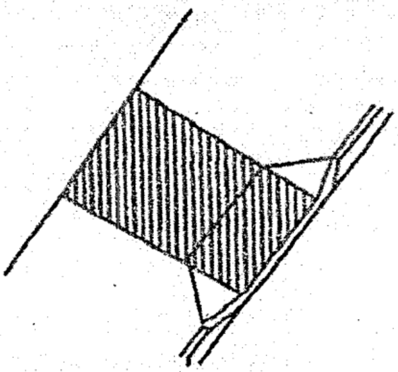
(2)

Copy also forwarded to the following for information & necessary action:

1. Principal Secretary, School Education
2. Principal Secretary, Medical & Health
3. Principal Secretary, Urban Dev. & Local Self Govt.
4. Principal Secretary, Medical Education
5. Principal Secretary, PWD
6. Secretary, Transport

Chief Secretary

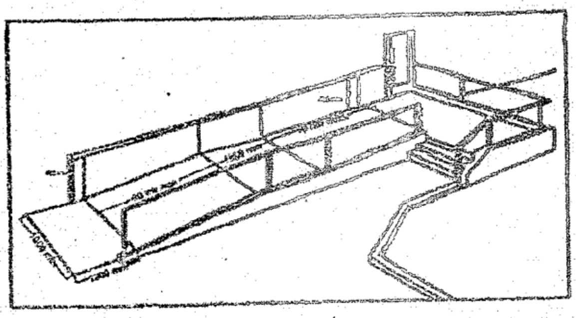
परिशिष्ट – 6



KERBS AND CROSSINGS

The pavement should be dropped to be flush with the road way at a gradient no greater than 1:12 on both sides of necessary and convenient crossing points.

The crossing points should also be highlighted by tactile paving/guiding blocks.



A ramp suitable for use by wheelchair users will be no steeper than 1:12 for a distance of up to 5m 1:15 for a distance of up to 10m. Thereafter a landing at least 1.5m long is required. The ramp should have a clear unobstructed width of 1000mm, a non-slip finish and continuous Handrails on both sides at 900mm above the surface of the ramp, it should also have landings 1.2m deep, clear of any door swing, at its head and foot and an up-stand of a least 100mm on any open side of the flight or landing. Wherever possible a ramp should be accompanied by a flight of easy going steps.

परिशिष्ट–7

— Most Important — (15)

OFFICE OF THE CHIEF ENGINEER, P.W.D. RAJASTHAN, JAIPUR

No. CE/PWD\SE(P&M)/Circular/D- 170 Dated:- 29-5-2006

CIRCULAR NO. 1 / 2006

Safety in Excavation work in Buildings

It has been observed that proper precautionary measures are not being adopted for safety at the time of excavation of foundation for the buildings which results in loss of lives and money.

For proper safety, the work should be got executed keeping in view, the provisions contained in National Building Code of India 2005 and relevant IS codes to avoid any mishappening. Specific care should be given on the following points as given in the various Codes as under:-

1. National Building Code of India 2005

(i) **Type of Strata**:-Adequate precautions depending on the type of strata met with during excavation (like quick sand, loose fills and loose boulder) shall be taken to protect the workmen during excavation.

(ii) **Overhang and Slopes**:-During any excavation, sufficient slopes to excavated sides by way of provision of steps or gradual slopes shall be provided to ensure the safety of men and machine working in the area.

(iii) **Fencing and Warning Signals**:-Where excavation is going on, for the safety of public and the workmen, fencing shall be erected, if there is likelihood of the public including cattle frequenting the area. Sufficient number of notice boards and danger sign lights shall be provided in the area to avoid any member of public from inadvertently falling into the excavation.

(iv) **Vibrations from nearby sources**:-Vibration due to adjacent machinery, vehicles, rail-roads, blasting, piling and other sources require additional precautions to be taken.

2. Handbook on Construction Safety Practices(SP 70 : 2001)

(i) **Minimum check and clear edge of trench**:-There is a tendency to dump the excavated material just on the edge of the trench where excavation is done manually. The material may slide back into the trench or apply additional load on shoring. A provision of clear berm of a width not less than one-third of the final depth of excavation is recommended. In areas where this width of the berm is not feasible, the reduced berm width of not less than 1 m. should be provided.

(ii) **Plant and Machinery**:-The excavating equipment should be parked at a distance of not less than the depth of the trench, or atleast 6 m. away from excavated sides for trenches deeper than 6 m.

3. Excavation work-Code of Safety(IS 3764 : 1992)

(i) Shoring and Timbering:-All trenches in soil more than 1.5 meter deep shall be securely shored and timbered.

These instructions should be followed strictly, otherwise concerned officer will be responsible for the consequences.

29/5/2009

Chief Engineer cum Addl. Secretary
PWD, Rajasthan, Jaipur.

No. CE/PWD\SE(P&M)/Circular/D- 170 Dated:- 29.5.06

Copy forwarded to the following for information and necessary action Please:-

1. P.S. to Hon'ble PWD Minister, Govt. of Rajasthan, Jaipur.
2. The Principal Secretary to Govt. PWD Rajasthan, Jaipur.
3. Addl. Chief Engineer PWD Zone..................................(All).
4. Superintending Engineer, PWD Circle ....(All)
5. Executive Engineer, PWD Dn. ....(All)
6. Sr. P.A. to CE/CE(R)/CE(NH)/All SE's & EE's in CE's Office, PWD, Jaipur.

F:\pwdlaptop\desktop\CE(Building) letters /Circulars

परिशिष्ट – 8

कार्यालय मुख्य [illegible] सार्वजनिक निर्माण विभाग, राजस्थान, जयपुर

क्रमांक सीई/एस ई (डी बी)/डी-335 दिनांक: 4-06-05

2

परिपत्र

विषय :- राजकीय भवनों में वर्षा जल पुनर्भरण संरचना ।

वर्षा के पानी द्वारा भू-गर्भ जलस्तर बढ़ाने के उद्देश्य से निर्देशित किया जाता है कि इस विभाग के अन्तर्गत 500 वर्ग मीटर अथवा इससे बड़े भूखण्डों पर निर्माण कराये जा रहे प्रस्तावित व वर्तमान भवनों में वर्षा के पानी द्वारा भू-गर्भ का जलस्तर बढ़ाने हेतु निम्न प्रावधान किया जाये।

(i) वर्तमान में प्रस्तावित/निर्माणाधीन 500 वर्ग मीटर से 1000 वर्ग मीटर तक के भूखण्डों में वर्षा के पानी को इकट्ठा करने के लिए 1मी. X1मी. X1मी. के दो पक्के गड्ढों का निर्माण किया जायेगा। पहले गड्ढे को बोल्डर व स्टोन बलास्ट से भरा जायेगा व इस पर लोहे की जाली लगाई जायेगी। पहले गड्ढे को दूसरे गड्ढे से जोड़ा जायेगा, जिसे बाद में 15 मीटर गहरे 6" व्यास वाले बोर वेल से जोड़ा जायेगा जिसमें छिद्रों वाला केसिंग पाइप लगा होगा। ऐसे दो बोरवेल आपस में 3 मीटर की दूरी पर खोदे जायेंगे। इसी प्रकार से 1000 वर्ग मीटर से बड़े प्लोट्स पर उपरोक्तानुसार प्रति 500 वर्ग मीटर क्षेत्रफल के अनुपात में विशेष प्रबन्ध किये जायें ताकि छतों का ज्यादातर पानी इन बोर वेल में पहुंचा कर जल रिसाव की व्यवस्था की जा सके।

एक हेक्टेयर से बड़े भूखण्ड में सक्षम अधिकारी द्वारा तय किये अनुसार वर्षा के पानी को इकट्ठा करने के लिए भूमिगत एक या एक से अधिक टैंक बनाये जाये तथा जल रिसाव हेतु बावड़ीनुमा अथवा सोक पिट जैसी अन्य व्यवस्था की जाये।

(2) वर्तमान भवनों में जो कि 500 वर्ग मीटर से बड़े भूखण्डों पर स्थित है, पुनर्भरण की व्यवस्था एक सुनिश्चित कार्यक्रम बना कर की जाये। सबसे पहले बड़े क्षेत्र पर स्थित बड़े भवनों को इस कार्य हेतु चुना जाये।

विभाग के सभी अतिरिक्त मुख्य अभियंता/अधीक्षण अभियंता/अधिशाषी अभियंता यह सुनिश्चित करेंगे कि इस व्यवस्था के लिए उचित प्रावधान करते हुए इसकी निर्धारित भवनों में पालना की जाये।

4/6/2005

(एच.एल. मीना)
मुख्य अभियन्ता,
सार्वजनिक निर्माण विभाग,
राजस्थान, जयपुर।

क्रमांक सीई/एस ई (डी बी)/डी-335 दिनांक: 4-06-05

प्रतिलिपि निम्न को सूचनार्थ एवं आवश्यक कार्यवाही हेतु प्रेषित है :-

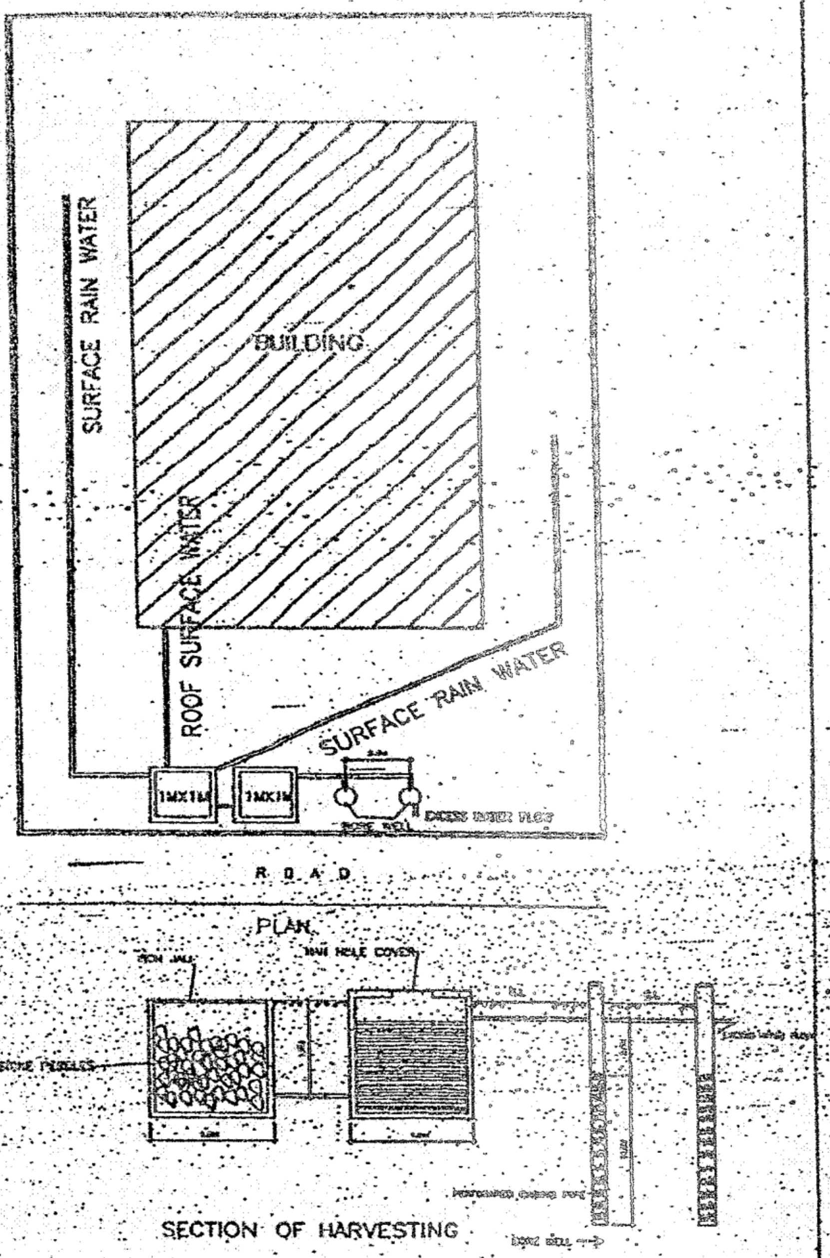
1. निजी सचिव, माननीय मंत्री महोदय, सा.नि.वि., राज., जयपुर।
2. प्रमुख शासन सचिव, सा.नि.वि., राज., जयपुर।
3. अतिरिक्त मुख्य अभियन्ता, सा.नि.वि., राज., (समस्त)
4. अधीक्षण अभियन्ता, सा.नि.वि., राज., (समस्त)
5. अधिशाषी अभियन्ता, सा.नि.वि., राज., (समस्त)
6. निजी सचिव, मुख्य अभियन्ता (पथ)/राष्ट्रीय राजमार्ग/समस्त अधीक्षण अभियन्ता व अधिशाषी अभियन्ता (मुख्य अभियन्ता कार्यालय)।

4/6/2005

(एच.एल. मीना)
मुख्य अभियन्ता,
सार्वजनिक निर्माण विभाग,
राजस्थान, जयपुर।

22

ARANGEMENT FOR RAIN WATER HARVESTING IN PUBLIC BUILDING



SKETCH IS NOT TO THE SCALE

EXECUTIVE ENGINEER (BUILDING CELL)
PUBLIC WORKS DEPARTMENT

कार्यालय मुख्य अभियन्ता, सार्वजनिक निर्माण विभाग, राजस्थान, जयपुर
क्रमांक: मु.अ./अ.अ.(भवन)/डी- दिनांक : .07.2012

परिपत्र

विषयः– भवन एवं पुलिया निर्माण कार्यों के सम्पादन के क्रम में।

प्रायः यह देखा गया है कि स्थानीय/क्षेत्रीय उत्पादकों द्वारा निर्मित स्टील/सरिया भवन एवं पुलिया निर्माण कार्यों में प्रयोग में लिया जाता है जिससे किए गये कार्य की गुणवत्ता एवं आयु पर विपरीत असर पड़ता है। अतः इस क्रम में सार्वजनिक निर्माण विभाग एवं मैसर्स आर.एस.आर.डी.सी.सी. के समस्त अभियंताओं को निर्देशित किया जाता है कि वर्तमान में निर्माणाधीन व भविष्य में निर्मित होने वाले समस्त भवन एवं पुलिया निर्माण कार्यों में केवल प्राथमिक उत्पादकों का स्टील/सरिया (आदिनांक को सेल/विजाग– आरआईएनएल/टाटा टिस्कोन/एस्सार/ जिन्दल विजय नगर) ही काम में लिया जावें। इसी क्रम में दिनांक 31.07.2012 के पश्चात आमंत्रित समस्त निविदाओं में तदनुसार संशोधन कर उसी अनुरूप निविदायें आमंत्रित किया जाना भी सुनिश्चित करें।

(हजारी लाल मीणा)
मुख्य अभियंता एवं अति० सचिव,
सा.नि.वि., राजस्थान जयपुर।

क्रमांकः मु.अ./अ.अ.(भवन)/डी– 577 दिनांक : 20.07.2012

प्रतिलिपि निम्नलिखित को सूचनार्थ एवं आवश्यक कार्यवाही हेतु प्रेषित हैं:–

1. विशिष्ठ सहायक, माननीय मंत्री महोदय, सा.नि.वि. राजस्थान, जयपुर।
2. प्रमुख शासन सचिव, सा.नि.वि. राजस्थान, जयपुर।
3. शासन सचिव, सा.नि.वि. राजस्थान, जयपुर।
4. मुख्य अभियंता, सा.नि.वि. (पथ/एन.एच./एस.एस./पी.एम.जी.एस.वाई.) राज. जयपुर।
5. प्रबन्धक, आर.एस.आर.डी.सी. लि., झालाना डूंगरी, जयपुर।
6. वित्तीय सलाहकार/मुख्य लेखाधिकारी, सार्वजनिक निर्माण विभाग, राज. जयपुर।
7. अति० मुख्य अभियंता, सा.नि.वि. सम्भाग–..............................(समस्त)।
8. अधीक्षण अभियंता, सा.नि.वि. वृत्त–..............................(समस्त)/भवन/पथ/एन.एच./एस.एस./ पी.एम.जी.एस.वाई.।
9. अधिशाषी अभियंता, सा.नि.वि. खण्ड–..............................(समस्त)।

3050 24/7/12
All EEs to comply strictly

3604 23/7/12

(हजारी लाल मीणा)
मुख्य अभियंता एवं अति० सचिव,
सा.नि.वि., राजस्थान जयपुर।

कार्यालय मुख्य अभियन्ता, सार्वजनिक निर्माण विभाग, राजस्थान, जयपुर

क्रमांक : CE/SE/(B)/PWD/D-1195 दिनांक : 05.07.2012

अतिरिक्त मुख्य अभियंता,
सार्वजनिक निर्माण विभाग
संभाग– I जयपुर.........(समस्त)।

अधीक्षण अभियंता,
सार्वजनिक निर्माण विभाग
वृत्त–.........................(समस्त)।

विषयः– भवन निर्माण कार्यों के अनुमानित तखमीनों में फायर फाईटिंग सिस्टम का प्रावधान करने बाबत्।

महोदय,

आपको विदित होगा कि नैशनल बिल्डिंग कोड 2005 में विभिन्न केटेगरी के भवनों में फायर फाईटिंग सिस्टम का प्रावधान किये जाने का कानूनी प्रावधान है। इसके बावजूद भी राज्य में निर्मित होने वाले राजकीय भवनों में फायर फाईटिंग सिस्टम लगाये जाने का अभाव पाया गया है जो कि एक गम्भीर चूक है जिससे आग लगने की स्थिति में जान माल की हानि होगी व राजकीय सम्पत्ति को भी नुकसान होगा। वर्तमान समय में ऐसी ही "आग" लगने की घटनाऐं घटित हुयी है जिसमें भारी जान माल की क्षति हुयी है। भविष्य में इस प्रकार की घटनाओं की पुनरावृत्ति न हो तथा वर्किंग स्थल इस प्रकार की आपदाओं से महफूज रहे, इस बाबत् राज्य स्तर पर मुख्य सचिव, राजस्थान, जयपुर के स्तर पर उच्च स्तरीय समीक्षा की गयी एवं यह निर्णित रहा कि भविष्य में राज्य में निर्मित होने वाले राजकीय भवनों के अनुमानित तखमीने स्वीकृति हेतु सम्बन्धित विभाग को प्रेषित करने से पूर्व उसमें "नैशनल बिल्डिंग कोड" 2005 के राजकीय भवन की केटेगरी अनुरूप फायर फाईटिंग सिस्टम/फायर एलार्म सिस्टम/एस्टीन्गूसर इत्यादि का प्रावधान किया जाना आवश्यक है।

अतः आप सभी को निर्देशित किया जाता है कि कि किसी भी विभाग द्वारा भवन निर्माण कार्यों हेतु वांछित अनुमानित तखमीने में नैशनल बिल्डिंग कोड 2005 के प्राविधित प्रावधानों अनुरूप फायर फाईटिंग सिस्टम की अनुमानित लागत का प्रावधान अवश्य जोड़ते हुए ही प्रस्तुत करें। यदि किसी स्तर पर इसमें चूक रहती है तो इस हेतु सम्बन्धित अधिकारी व्यक्तिशः उत्तरदायी रहेंगे।

भवदीय,
5/7/12
(हजारी लाल मीणा)
मुख्य अभियंता एवं अति० सचिव,
सा.नि.वि., राजस्थान जयपुर।

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17/7/12

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16/7/12

अधिशाषी अभियन्ता, एवं
[illegible] सा. नि. वि०
शहर वृत्त जयपुर

15/7
13/7/12

प्रतिलिपि अधीक्षण अभियंता वृत्त शहर जयपुर / सीकर / झुंझुनू को आवश्यक कार्यवाही हेतु प्रेषित है।

ACE Zone I, Jaipur

कार्यालय मुख्य अभियन्ता, सार्वजनिक निर्माण विभाग, राजस्थान, जयपुर
क्रमांक मु.अ./अ.अ.(भवन)/डी–   दिनांक :   .07.2012

परिपत्र

विषय:– समस्त निर्माण कार्यों के समापन के क्रम में।

समस्त निर्माण कार्यों को पूर्ण करने में विभिन्न प्रकार की कार्यवाहियां अपेक्षित होती है। डी.पी.आर. बनाने से पूर्व मौके पर उपयुक्त भूमि का चयन एवं उपलब्धता एवं तदनुसार विभाग द्वारा अधिग्रहण, साईट प्लान का अनुमोदन, आर.सी.सी. एवं वर्किंग डिजाइन एवं ड्राइंग तैयार करना इत्यादि गतिविधियों को पूर्ण एवं अंतिम रूप देना अनिवार्य होता है। विशेष परिस्थितियों में इन गतिविधियों के क्रियान्वयन में देरी होने के कारण सम्पूर्ण परियोजना की कार्य समाप्ति अवधि बढ़ जाती है एवं साथ ही प्रचलित बाजार दरों में इस अंतराल में हुई वृद्धि के कारण निर्माण लागत भी बढ़ जाने की स्थिति उत्पन्न हो जाती है। विभाग के सभी अतिरिक्त मुख्य अभियंता/अधीक्षण अभियंता/अधिशाषी अभियंता यह सुनिश्चित करेंगे कि अपरिहार्य कारणों/प्रशासनिक कारणों से ड्राईंग, डिजाइन तैयारी अथवा भूमि चयन/परिवर्तन में हुए विलम्ब के कारण तदनुसार कार्यादेश में निर्धारित समयावधि में वृद्धि को आवश्यकतानुसार राज्य सरकार के स्तर पर प्रस्तावित करते हुए स्वीकृत करायेंगे एवं उसके अनुरूप कोस्ट एस्केलेशन (यदि देय हो) भी स्वीकृत करायेंगे एवं कार्य की समयावधि (कार्य पूर्ण होने की तिथि) पूर्ण होने के उपरान्त भी यदि कार्य के किसी भाग में बाधा रहती है तो कार्य के उस भाग को अनुबन्धानुसार विड्रा किया जावेगा।

(हजारी लाल मीणा)
मुख्य अभियंता एवं अति0 सचिव,
सा.नि.वि. राजस्थान जयपुर।

क्रमांक मु.अ./अ.अ.(भवन)/डी– 529   दिनांक 12.07.2012

प्रतिलिपि निम्नलिखित को सूचनार्थ एवं आवश्यक कार्यवाही हेतु प्रेषित है:–

1. निजी सचिव, माननीय मंत्री महोदय, सा.नि.वि. राजस्थान, जयपुर।
2. प्रमुख शासन सचिव, सा.नि.वि. राजस्थान, जयपुर।
3. शासन सचिव, सा.नि.वि. राजस्थान, जयपुर।
4. मुख्य अभियंता, सा.नि.वि. (पथ/एन.एच./एस.एस./पी.एम.जी.एस.वाई.) राज. जयपुर।
5. वित्तीय सलाहकार/मुख्य लेखाधिकारी, सार्वजनिक निर्माण विभाग, राज. जयपुर।
6. अति0 मुख्य अभियंता, सा.नि.वि. संभाग.........................(समस्त)।
7. अधीक्षण अभियंता, सा.नि.वि. वृत्त.........................(समस्त)/भवन/पथ/एन.एच./एस.एस./पी.एम.जी.एस.वाई.।
8. अधिशाषी अभियंता, सा.नि.वि. खण्ड.........................(समस्त)।

A.EN

(हजारी लाल मीणा)
मुख्य अभियंता एवं अति0 सचिव,
सा.नि.वि. राजस्थान जयपुर।

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ANNEXURE - A

The following fittings shall be provided in Door's and Window's shutters :

S. No.	Particulars	Number of Fittings
(A)	**Single Leaf Door Shutters**	
(i)	Hinges of Size 100mm	3 Nos.
(ii)	Sliding Bolt one No. of length of 300mm of 12mm dia upto 35mm thick shutters and 16mm dia and 300m. length for above 35mm thick shutters	1 No.
(iii)	Handles 150mm	2 Nos.
(iv)	Tower Bolts 300mm	1 No.
(v)	Tower Bolts 150mm	1 No.
(vi)	Door Stopper	1 No.
(vii)	Back Stopper	1 No.
(B)	**Double Leaf Door Shutters**	
(i)	Hinges of Size 100mm	6 Nos.
(ii)	Sliding Bolt one No. of 350mm long 16mm dia	1 No.
(iii)	Handles 150mm	3 Nos.
(iv)	Tower Bolts 300mm	2 Nos.
(v)	Tower Bolts 150mm	2 Nos.
(vi)	Door Stopper	2 Nos.
(vii)	Back Stopper	2 Nos.
(C)	**Window shutters in each leaf**	
(i)	Hinges of Size 100mm	2 Nos.
(ii)	Handles 125mm	1 No.
(iii)	Tower Bolts 150mm	2 Nos.
(iv)	Door Stopper	1 No.